MRC National Institute for Medical Research

WHO INFLUENZA CENTRE

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Southern Hemisphere Vaccine Recommendation Meeting for 2014 influenza season.

Summary of NIMR Results.

Final Report for WHO meeting: 23-25 September 2013.

Influenza Activity in the European Region, October 2012 – September 2013

Weekly reporting on influenza activity in the Region started in week 40/2012. Activity started early compared to the 2011-2012 influenza season with levels of transmission increasing slowly from week 46/2012, and detections in week 05/2013 passing that seen at the peak (week 08/2012) for the 2011-2012 season (**Figure 1**). Approximately 61,000 influenza detections have been reported across the whole WHO Euro Region with type A detections (62%) predominating over type B (38%), but these relative proportions are slightly different for western (EU/EAA) countries (58% & 42%) and eastern countries (75% & 25%). Influenza B detections, as a percentage of total influenza detections across the Region, were 38% compared to 10% in the previous season (**Figure 2**). Within type A, the H1pdm09 subtype (66%) predominated over H3 viruses (34%) by a 2-fold margin across the Region (**Figures 1 & 2**): this represents a reversal of the situation in 2011-2012 (**Figure 2**). In terms of the intensity of influenza activity and the degree of geographic spread within countries there was evidence of west to east spread across the Region leading up to peak activities around week 05/2013 and decline thereafter (**Figure 3**). However, within individual countries the relative proportions of influenza type A/B and subtype H1/H3 detections showed considerable variation both in terms of total detections and timings of the waves and peaks of types A and B virus activities (**Figure 4**). Overall, influenza activity across the Region has been greater than that seen in the 2011-2012 season, with approximately 33% more detections (**Figure 2**), as shown in the UK through the HPA/Q-Surveillance National Syndromic Surveillance System based on a population sample of 20 x 10^6 in England and Wales: Influenza Like Illness (ILI) showed peak activity in weeks 52/2012-01/2013 but remained well within normal seasonal activity (**Figure 5**).

Antigenic and genetic characteristics of influenza viruses recovered from clinical specimens collected after 31 January 2013

Influenza positive specimens received at NIMR

Samples, viruses or clinical specimens, with collection dates after 2013-01-31 have been received from 35 countries in Europe, Africa, the Middle East and the Far East. Just over half (55%) were type A viruses with A(H1N1)pdm09 viruses predominating over A(H3N2) viruses by a ratio of 1.4:1 (**Table 1**). Of the type B viruses (45% of all specimens received) B/Yamagata lineage viruses predominated over those of the Victoria lineage by a ratio of 3:1. These proportions of influenza types/subtypes/lineages are similar to those reported within the WHO Euro Region (**Figures 1 & 2**). **Table 2** shows the isolation rates from specimens received with collection dates after 2013-01-31. Isolates from each type/subtype/lineage were recovered efficiently. **Table 3** summarises the number of viruses of the A(H1N1), A(H3N2) subtypes, and the two B/lineages that showed ≤2-, 4- and ≥8-fold reductions in HI titres with post-infection ferret antisera compared with the titres against their respective homologous vaccine/reference viruses. For A(H3N2) viruses a summary of the results of plaque-reduction neutralisation assays is also shown.

Influenza A(H1N1)pdm09 virus analysis.

H1N1pdm09 viruses were received from NICs in Europe, Africa (Algeria, Ghana, Madagascar and South Africa), the Middle-East (Jordan) and from Hong Kong SAR. **Table 4** summarises the antigenic results of the received viruses that were propagated successfully.

The vast majority (93%) of the H1N1 viruses collected after 2013-01-31 were antigenically similar to the vaccine virus (**Table 4**); eleven viruses (5% of the total tested) showed 4-fold reductions and five (2% of the total tested) ≥8-fold reductions in HI titre compared with the titre of the vaccine virus (A/California/7/2009) with its homologous post-infection ferret antiserum. The HI results for all viruses tested since the February 2013 Vaccine Composition Meeting (VCM) are shown in **Tables 5-1 to 5-17** and the HA1 amino acid substitutions/polymorphisms associated with 4- and ≥8-fold reductions in HI titre are indicated where known. Test viruses in each table are sorted by date of collection and those below the red horizontal lines are viruses with collection dates after 2013-01-31. Viruses for which gene sequences are included in phylogenetic trees are highlighted and, where known, the HA genetic group is indicated.

Figure 6 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. The HA genes cluster into eight designated genetic groups, with A/California/7/2009 representing group 1. However, viruses collected after 2013-01-31 fell into genetic groups 6 and 7 with group 6 viruses clustering in three subgroups, 6A-C (**Figure 6**). Both groups carry the substitutions **S185T** and **S203T** in **HA1** and **E47K** and **S124N** in **HA2** compared to A/California/7/2009. The two groups and three subgroups are defined by the following amino acid substitutions in **HA**1 and **HA2**:

(6) All viruses in **group 6** carry the substitution **D97N** in **HA1**,

6A) viruses in **subgroup 6A** also carry substitutions **H138R & V249L** in **HA1**, e.g. A/Hong Kong/5659/2012.

6B) viruses in **subgroup 6B** carry substitutions **K163Q, K283E & A256T** in **HA1** and **E172K** in **HA2**, e.g. A/Stockholm/15/2013.

6C) viruses in **subgroup 6C** carry substitutions **V234I & K283E** in **HA1** and **E172K** in **HA2**, e.g. A/Estonia/76677/2013.

(7) Viruses in **group 7** carry substitutions **S143G & A197T** in **HA1**, often with **S84G & K163I** in **HA1** and **V193A** in **HA2**, e.g. A/Norway/1675/2013.

Amino acid alignment of the HAs for a selected group of viruses is shown in **Figure 7**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted and the genetic group is shown in green.

Figure 8 illustrates the location of amino acid residues on an H1-HA structure, defining genetic group 7 and subgroups 6A, 6B and 6C. Also indicated are the location of residues that define a set of viruses that have emerged and shown limited circulation in countries of West Africa, notably Côte d'Ivoire, Ghana and Senegal.

Figure 9 shows a phylogenetic tree for NA genes of representative H1N1 viruses and **Figure 10** shows an amino acid alignment for a selected group of these viruses. The HA and NA phylogenetic trees are generally congruent.

Antiviral analysis.

Figure 25 shows a phylogenetic tree of representative H1N1 M genes. Amino acid substitutions in M1 and M2 are highlighted. All sequences retained coding for the S31N substitution in the M2 ion-channel protein, known to confer resistance to adamantanes.

Phenotypic testing of all viruses recovered from specimens with collection dates after 2013-01-31 were sensitive (normal inhibition) to the neuraminidase inhibitors zanamivir and oseltamivir (**Table 13**). A single previously untested virus collected in January 2013, A/IIV-Moscow/34/2013, (**Table 5-16**), carried NA-H275Y substitution and showed highly reduced inhibition to oseltamivir.

Conclusion.

The vast majority of A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus, A/California/7/2009.

Influenza A(H3N2) virus analysis.

Influenza A(H3N2) viruses have been received from Europe and Africa (Ghana, Madagascar and South Africa) and from Hong Kong SAR. As described in previous reports, these viruses have continued to be difficult to characterise antigenically by HI assay due to variable agglutination of red blood cells from guinea pig, turkey and humans. Those viruses with sufficient titre in HA assays using guinea pig red blood cells in the presence of 20nM oseltamivir, to circumvent the NA-mediated binding of H3N2 viruses to the red blood cells, were analysed by HI assay. Close to 80% of specimens received yielded viruses of sufficient HA titre to allow HI analysis in the presence of oseltamivir (**Table 2**). Virus neutralisation assays were used to complement HI tests; approximately 20% of the H3N2 viruses received were analysed in the plaque reduction neutralisation assay (PRNA).

A summary of the HI results is shown in **Table 6** and individual assay results are shown in **Tables 7-1 to 7-16.** Viruses for which gene sequences are included in phylogenetic trees are highlighted and, where known, the HA genetic group is indicated. Test viruses in each table are sorted by date of collection and those below the red horizontal lines are viruses with collection dates after 2013-01-31.

Approximately 90% of test viruses with collection dates after 2013-01-31 that were propagated in MDCK-SIAT1 cells reacted poorly in HI assays (≥8-fold decrease) with post-infection ferret antiserum raised against the previous egg-propagated vaccine virus, A/Victoria/361/2011, compared to the titre of the antiserum with the homologous virus. A similar low level of reactivity was seen with antiserum raised against the egg-propagated virus A/Hawaii/22/2012 (~4% of test viruses propagated in MDCK-SIAT1 cells reacted with this antiserum within 4-fold of the titre with the homologous egg-propagated virus). Somewhat better reactivity was seen with antiserum raised against the egg-propagated current vaccine virus, A/Texas/50/2012, which induced an antiserum with a high homologous titre (~5120); 50% of test viruses reacted within 4-fold of the titre with the homologous virus. However, only 10% of test viruses showed reactivity within 2-fold of the titre of the antiserum with homologous egg-propagated A/Texas/50/2012.

Ferret antisera raised against cell-propagated reference viruses A/Alabama/5/2010, A/Stockholm/18/2011, A/Berlin/93/2011, A/Athens/112/2012, A/Samara/73/2013 and A/Victoria/361/2011 recognised over 97% of test viruses propagated in MDCK-SIAT1 cells at titres within 4-fold of those for the antisera with their corresponding homologous viruses.

The results of PRNA performed with MDCK-SIAT1 cells are shown in **Tables 8-1 to 8-3.** Test viruses with collection dates after 2013-01-31 are below the red horizontal lines. Shown on the tables are the genetic groups in which, where known, the HA genes of the reference viruses and the test viruses fall. The results of PRNA were consistent with the results from HI assays done in the presence of oseltamivir. In PRNA none of the test viruses reacted within 4-fold of the titre of the homologous virus with antisera raised against the egg-propagated vaccine virus, A/Victoria/361/2011. However, all viruses reacted well with antiserum raised against cell-propagated A/Victoria/361/2011 virus, having titres within 4-fold of the titre with the homologous virus. Similarly, all test viruses gave titres within 4-fold of the reference viruses for antisera raised against cell-propagated A/Athens/112/2012 (genetic group 3B) and A/Samara/73/2013 (genetic group 3C.3). As in the HI assays, test viruses showed somewhat better reactivity with antiserum raised against the egg-propagated reference/vaccine virus A/Texas/50/2012 than with antiserum raised against the egg-propagated vaccine virus A/Victoria/361/2011 compared to their respective homologous viruses. Of the test viruses with collection dates after 2013-01-31 that were propagated in tissue culture approximately one third showed neutralisation titres within 4-fold of that of the homologous virus with antiserum raised against egg-propagated A/Texas/50/2012; however, only a single virus had a neutralisation titre within 2-fold of the titre of the antiserum with the homologous virus.

Previously seven genetic groups based on the HA gene have been defined for H3N2 viruses, two derived from A/Perth/16/2009 and five from A/Victoria/208/2009. Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 11**; the genes fell within the A/Victoria/208 genetic clade and predominantly into the genetic subgroups 3B and 3C of group 3. Amino acid substitutions that mark these genetic groups are:

Group 3 viruses all carry the amino acid substitutions **N145S, V223I** in **HA1** and **D158N** in **HA2** in addition the following subgroups are defined by substitutions:

(3A) **N144D** in **HA1** resulting in the loss of a potential glycosylation site, e.g. A/Stockholm/18/2011;

(3B) **A198S & N312S** in **HA1**, e.g. A/Athens/112/2012; some also carry the substitution **T30A** in **HA1**;

(3C) **S45N** (resulting in the gain of a potential glycosylation site), **T48I, S145N, A198S & N312S** in **HA1** and **N158D** in **HA2** (e.g. A/Victoria/361/2011). The 3C subgroup can be sub-divided into three:

(3C.1) **Q33R & N278K** in **HA1**, e.g. A/Berlin/93/2011, A/Texas/50/2012 and A/Hawaii/22/2012;

(3C.2) As 3C.1 plus **N145S** in **HA1**, e.g. A/Ireland/M28390/2013;

(3C.3) As 3C.2 plus **T128A** (resulting in the loss of a potential glycosylation site) **& R142G** in **HA1,** e.g. A/Samara/73/2013.

Note: in **Figure 11** the G186V substitutions in HA1 occur during adaptation of H3N2 viruses to passage in hens' eggs.

Figure 12 shows an HA amino acid alignment for a representative selection of the viruses used to generate the phylogeny (**Figure 11**). The two viruses used in vaccine production (A/Victoria/361/2011 and A/Texas/50/2012) are shown in red, glycosylation motifs are highlighted and the reference viruses are coloured blue. The genetic group/ to which each of the sequences belongs is also shown.

Figure 13 shows the location on the 2005 H3 HA structure of amino acid substitutions that define genetic groups 3A, 3B, 3C.1, 3C.2 (N145S in red) and 3C.3.

Figure 14 shows a NA phylogenetic tree for representative viruses with the HA genetic groups indicated. **Figure 15** shows a NA amino acid alignment for a representative selection of these viruses. As above, vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the genetic group to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of viruses with HA genes in genetic subgroup 3C do not cluster in the same manner as the HA sequences. It is noteworthy that the viruses of the genetic groups 3A, 3B and 3C all carry the dual substitutions **S367N** and **K369T** in their NA which results in the addition of a glycosylation site at residue **367**. The majority of recently collected viruses have lost a potential glycosylation site at residues **402**. **Figure 16** localises the predominant variable sites on the structure of an N2 NA along with amino residues that influence the ability of viruses to agglutinate red blood cells through the NA.

Antiviral resistance

Figure 26 shows a phylogenetic tree of representative H3N2 M genes. Amino acid substitutions in M1 and M2 are highlighted. All sequences retained coding for the S31N substitution in the M2 ion-channel protein, known to confer resistance to adamantanes.

Phenotypic testing of all 130 viruses recovered from specimens with collection dates after 2013-01-31 were sensitive (normal inhibition) to the neuraminidase inhibitors zanamivir and oseltamivir (**Table 13**).

Conclusion.

The vast majority of recent test viruses analysed show reduced reactivity with antisera raised against the egg-propagated vaccine viruses A/Victoria/361/2011 and A/Texas/50/2012 in both virus neutralisation and HI assays, with the latter being carried out in the presence of oseltamivir, but generally titres were higher with antisera raised against A/Texas/50/2012. Conversely, test viruses reacted well with antisera raised against reference viruses that are genetically similar to A/Victoria/361/2011 but propagated in cells. There is no evidence of antigenic drift from the cell-propagated reference viruses but the test viruses are antigenically distinguishable from egg-propagated viruses.

Influenza B viruses.

Influenza B viruses represented 45% of samples received with collection dates after 2013-01-31 (**Table 1**). As described above, B/Yamagata lineage viruses predominated over those of the B/Victoria lineage at a ratio of approximately 3:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from countries in Europe, Africa (Ghana and Senegal), the Middle East (Jordan) and Hong Kong SAR. The collection details and summary results are shown in **Table 9**.

HI results for the propagated viruses are shown in **Tables 10-1 to 10-8**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines are viruses with collection dates after 2013-01-31. Viruses for which gene sequences are included in phylogenetic trees are highlighted and, where known, the HA genetic group is indicated. Over 90% of cell-propagated viruses gave ≥8-fold reductions in HI titres, compared to the titres with the homologous viruses, with post-infection ferret antisera raised against the previously recommended prototype vaccine virus, egg-propagated B/Brisbane/60/2008, and the egg-propagated reference viruses B/Malta/636714/2011 and B/Johannesburg/3964/2012. All viruses showed reactivity within 4-fold of the titre of the homologous virus with antisera raised against viruses genetically closely related to B/Brisbane/60/2008 but propagated in cells (**Table 9**). As shown in **Tables 10-1 to 10-8**, these antisera were raised against B/Paris/1762/2008, B/Hong Kong/514/2009 and B/Odessa/3886/2010, and these viruses are considered as surrogate cell-propagated antigens representing the egg-propagated prototype strain. Most of the viruses that showed increased reactivity with antiserum raised against egg-propagated viruses carried mutations or polymorphisms in their HA gene sequences that cause loss of glycosylation at amino acid residue **197** of HA1, a glycosylation site that is generally lost on egg passage.

Phylogenetic analysis of the HA gene of representative B/Victoria lineage viruses is shown in **Figure 17** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 18**. The HA genes of the majority of recently collected viruses fall into the B/Brisbane/60 genetic clade in the subgroup designated clade 1A. **Figure 19** shows a phylogenetic tree for the NA gene and **Figure 20** shows a corresponding amino acid alignment for a selection of NA glycoproteins. Highlighted in the trees is a cluster of intra-clade reassortants between the HA gene from clade 1A and the NA gene of clade 3, exemplified by B/Dakar/15/2013 and B/Ghana/FS-718/2013. A second cluster of reassortants involving HA genes from clade 1B and NA genes of clade 4, as represented by B/St. Petersburg/27/2012 and B/Hong Kong/3514/2012 has not been detected in 2013. Another HA/NA reassortant, B/Norway/2090/2012, carries the HA gene from clade 1B and the NA gene of clade 1A; such viruses have not been detected in 2013.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from countries in Europe, Africa (Senegal, South Africa), the Middle East (Jordan) and Hong Kong SAR. The collection details and results are summarised in **Table 11**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 12-1 to 12-17.** Test viruses in each table are sorted by date of collection and those below the red horizontal lines are viruses with collection dates after 2013-01-31. The clade into which the HA falls is shown, where known, and viruses for which gene sequences are included in phylogenetic trees are highlighted. Post infection ferret antisera raised against the current, egg-propagated, vaccine virus B/Massachusetts/02/2012 recognised approximately 85% of viruses collected after 2013-01-31 at titres within 4-fold of their titres for the homologous virus. Similarly, a ferret antiserum raised against a cell-propagated cultivar of B/Massachusetts/02/2012 recognised over 90% of viruses collected since 2013-01-31 at titres within 4-fold of its titre with the homologous virus. Over 90% of test viruses were recognised by ferret antisera raised against two other viruses with HA genes belonging to the B/Massachusetts/02/2012 clade (Clade 2), B/Estonia/55669/2011 and B/Hong Kong/3577/2012, at titres within 4-fold of the titres of the antisera with their homologous viruses. In contrast, one third (33%) of test viruses were recognised by an antiserum raised against cell-propagated B/Novosibirsk/01/2012 (Clade 3) at titres lower than 4-fold of the titre of the antiserum with the homologous virus.

Many, but not all, viruses carrying clade 2 HA genes and falling within the B/Massachusetts/02/2012 clade could be differentiated antigenically from those carrying clade 3 HA genes, represented by B/Wisconsin/1/2010, by certain post-infection ferret antisera. However, not all antisera were able to antigenically differentiate between viruses falling in the alternate clades.

Figure 21 shows a phylogenetic analysis of the HA gene of representative B/Yamagata lineage viruses and an amino acid alignment of a selection of the HA glycoproteins is shown in **Figure 22**. The HA genes of just over 75% of the recently collected viruses fall into the B/Massachusetts/02/2012 clade (Clade 2), which includes the reference viruses B/Estonia/55669/2011 and B/Hong Kong/3577/2012. The remaining HA genes fall into the B/Wisconsin/1/2010 clade (Clade 3), which includes the reference viruses B/Stockholm/12/2011, B/Serbia/1894/2011 and B/Novosibirsk/1/2012. **Figure 23** shows a phylogenetic tree for NA genes and **Figure 24** shows a corresponding amino acid alignment for selected NA glycoproteins.

Antiviral Analysis.

No evidence for reduced sensitivity to the neuraminidase inhibitors zanamivir and oseltamivir was seen for influenza B viruses collected after 2013-01-31 **(Table 13).**

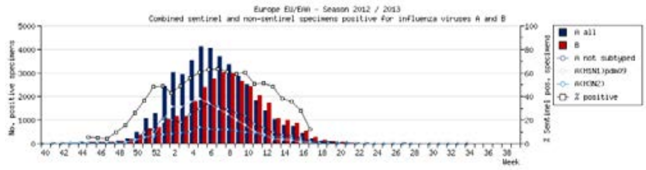
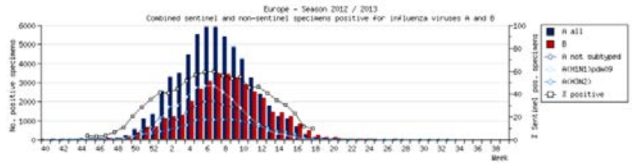
Conclusion.

Viruses of the B/Yamagata lineage have predominated. For viruses of the B/Victoria lineage most were antigenically similar to viruses closely related to B/Brisbane/60/2008. For viruses of the B/Yamagata lineage the majority was antigenically similar to B/Massachusetts/02/2012.

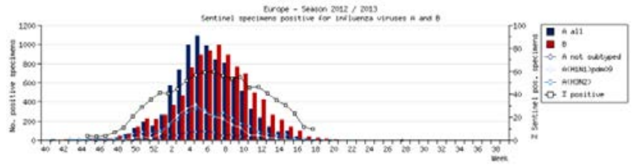
Figure 1. Influenza Activity in Europe: 2012-13 Season (weeks 40/12-34/13)

EU/EAA (ECDC)	WHO Euro

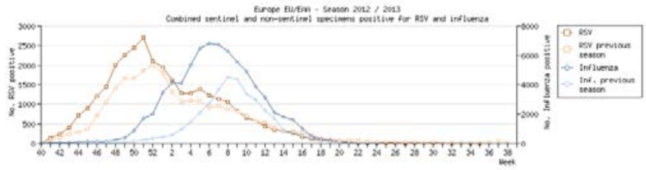
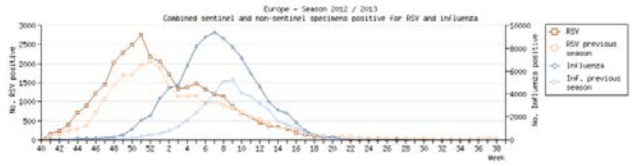
Total Influenza Detections

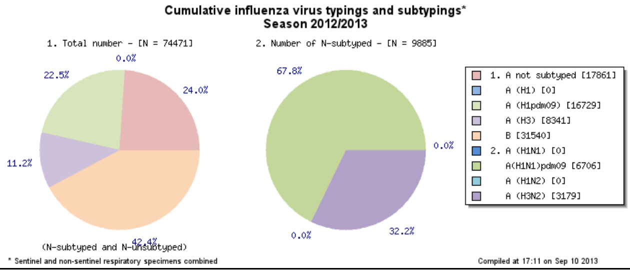
Sentinel Influenza Detections

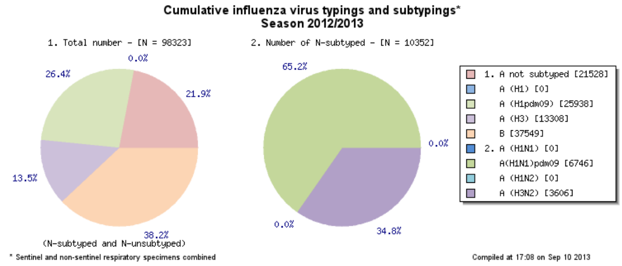



Past Season RSV & Influenza

Influenza Typing & Subtyping





Distribution of Influenza Detections

	Western (ECDC)	WHO Euro	Eastern
A	42931 (57.6%)	60774 (61.8%)	17843 (74.8%)
H1pdm09	16729 – 2.01x	25938 – 1.95x	9209 – 1.85x
H3*	8341	13308	4967
Not subtyped	17861	21528	3667
B	31540 (42.4%)	37549 (38.2%)	6009 (25.2%)
Total	74471	98323	23852

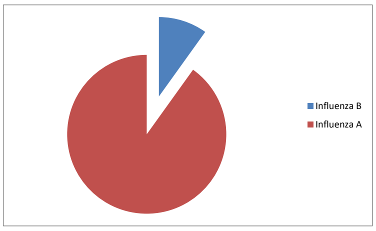
* Ratios for H1pdm09:H3 viruses are shown (x)

Figure 2. Comparison of 2011-12 & 2012-13 Influenza Seasons in Europe

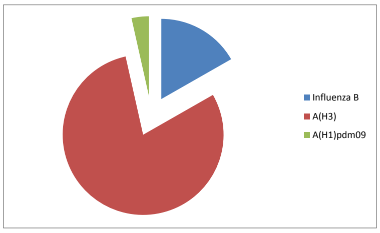
Influenza Detections 2011-12 season

	WHO Euro
Total	45267
A (subtype?)	18483
A(H1)pdm09	938 (4.2%)
A(H3)	21366 (95.8%)
Influenza B	4480 (9.9%)

Influenza A/B

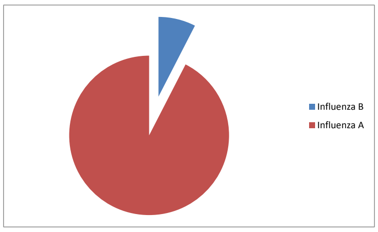


Subtyped influenza A

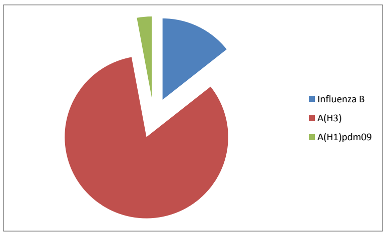


	ECDC (EU/EAA)
Total	35720
A (subtype?)	16975
A(H1)pdm09	548 (3.4%)
A(H3)	15505 (96.6%)
Influenza B	2692 (7.5%)

Influenza A/B



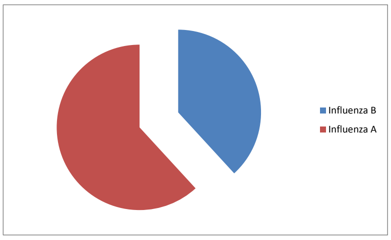
Subtyped influenza A



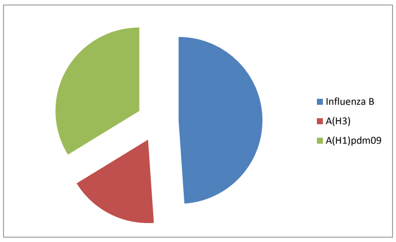
Influenza Detections 2012-13 season (to week 34/13)

	WHO Euro
Total	98323
A (subtype?)	21528
A(H1)pdm09	25938 (66.1%)
A(H3)	13308 (33.9%)
Influenza B	37549 (38.2%)

Influenza A/B

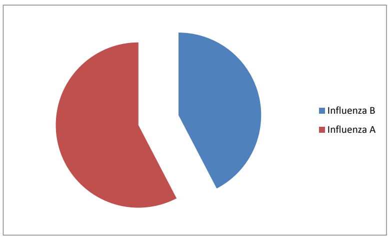


Subtyped influenza A



	ECDC (EU/EAA)
Total	74471
A (subtype?)	17861
A(H1)pdm09	16729 (66.7%)
A(H3)	8341 (33.3%)
Influenza B	31540 (42.4%)

Influenza A/B



Subtyped influenza A

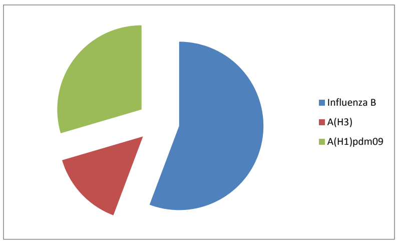


Figure 3. The decline of the 2012-13 Influenza Season in Europe from week 5/2013

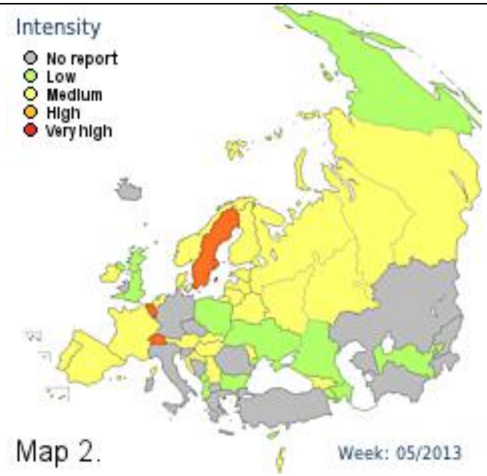
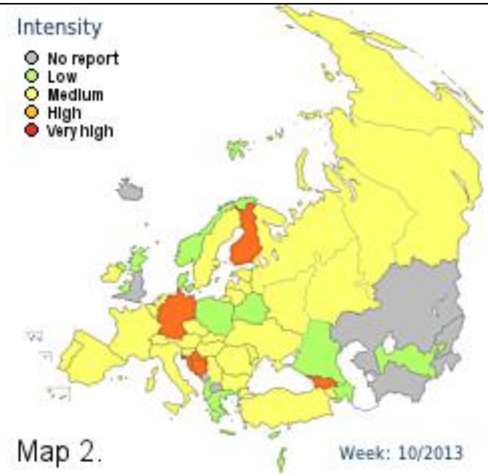
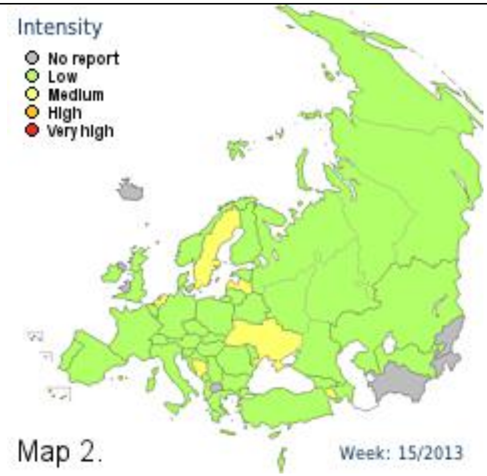
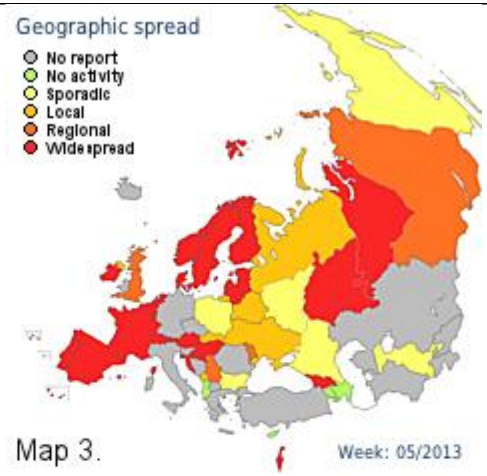
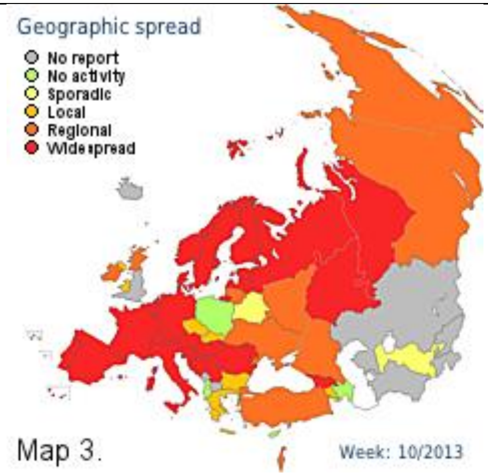
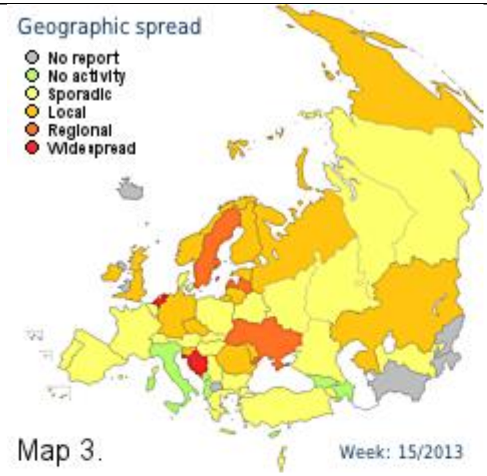
Week	05/2013	10/2013	15/2013
Intensity Of Influenza Activity			
Geographic Spread			

Figure 4. Patterns of Influenza Type/Subtype Circulation in WHO Euro Countries (Combined Surveillance)

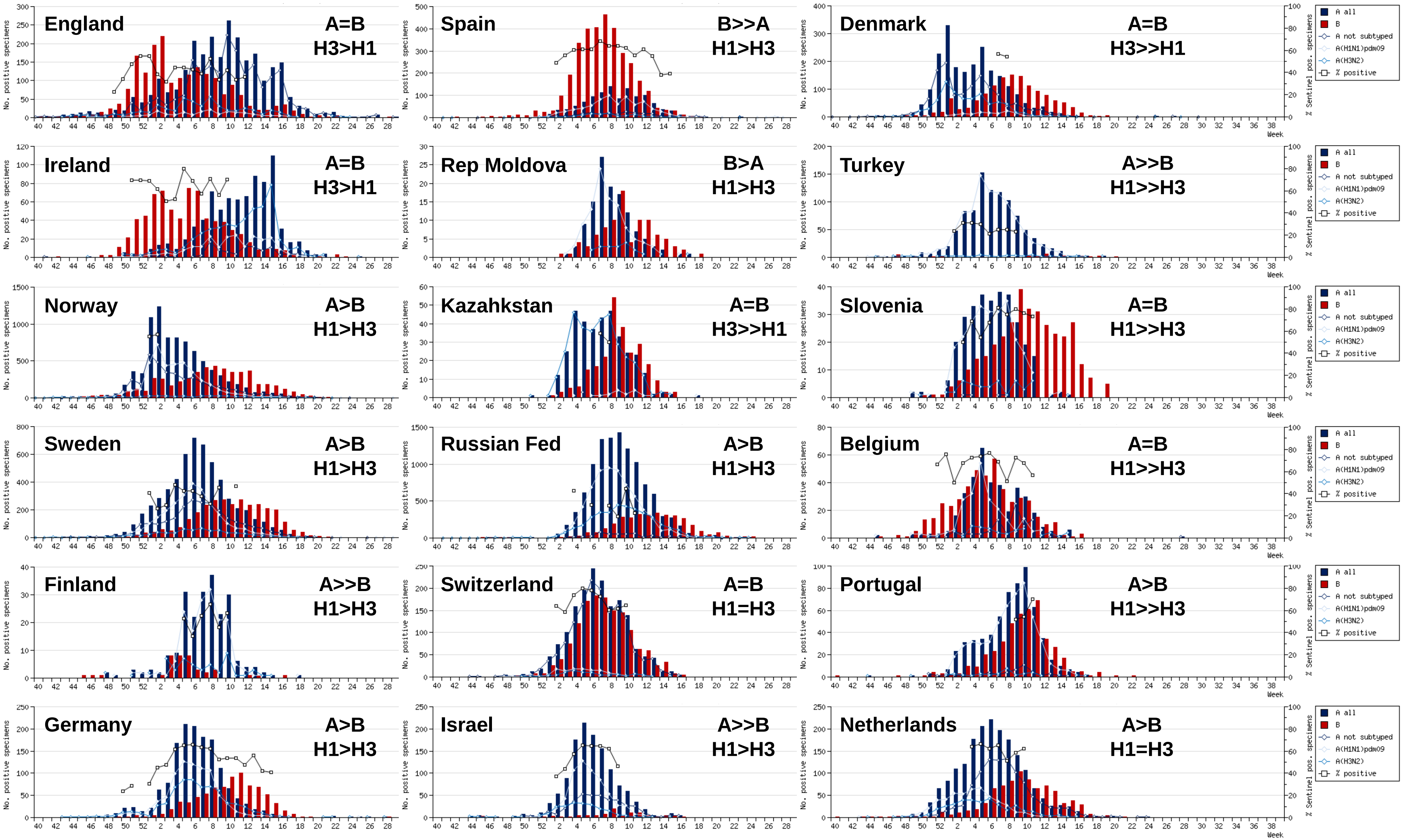
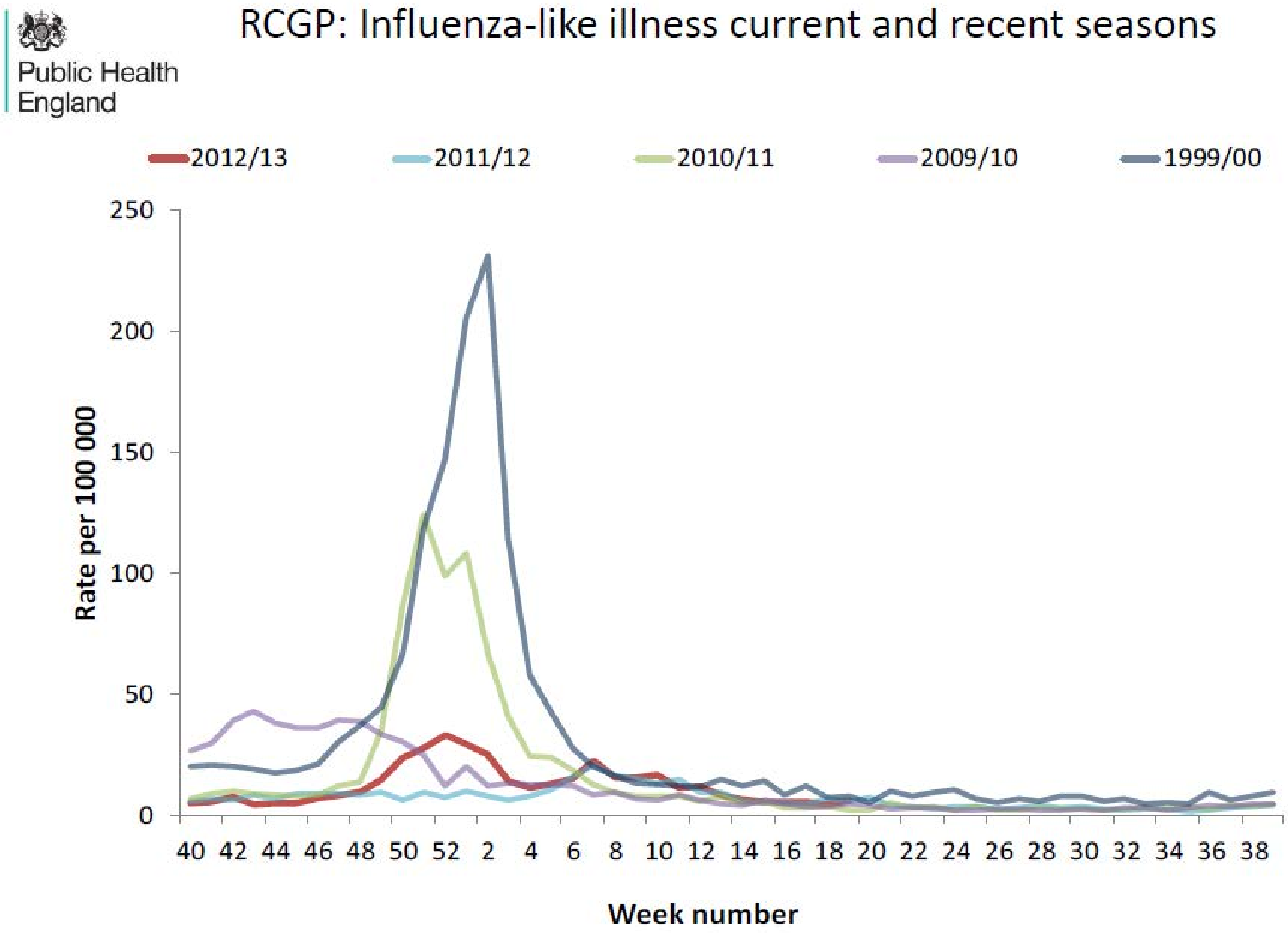


Figure 5. Influenza-Like Illness estimated from the UK PHE/Q-surveillance syndromic surveillance system (RCGP ILI consultation rates for England and Wales)



Note: Weekly Q-Surveillance national thresholds for ILI incidence rates are: <20 per 100,000 (baseline activity); 20-70 (normal seasonal activity); 70-130 (above average activity); >130 (exceptional activity). Taken from the PHE week 35/13 influenza report.

Table 1. Summary of clinical samples and isolates received, with collection dates after 2013-01-31, by country

MONTH	TOTAL RECEIVED	A	H1N1pdm09		H3N2		B	B Victoria lineage		B Yamagata lineage	
Country			Number received	Number propagated[1]	Number received	Number propagated[2]		Number received	Number propagated[1]	Number received	Number propagated[1]
FEBRUARY											
Algeria	3		3	0							
Austria	6		4	4				1	1	1	1
Belarus	2		1	1	1	1					
Belgium	12		2	2	2	2				8	8
Bulgaria	22		8	7	2	2				12	12
Czech Republic	4									4	4
Georgia	14		10	7				3	3	1	1
Ghana	1				1	0					
Greece	2		2	2							
Hong Kong	3				1	1				2	2
Hungary	12		6	4				3	3	3	2
Iceland	1									1	1
Italy	22		11	11	1	1		1	1	9	9
Jordan	10		3	2				4	4	3	3
Kazakhstan	9				1	in process	8				
Lithuania	45	18	7	3	11	7	6			3	3
Luxembourg	5		1	0	4	0					
Madagascar	5	1			2	2				2	1
Moldova	2				2	2					
Norway	1		1	1							
Poland	4									4	4
Portugal	10		4	4				4	4	2	2
Romania	13		8	8						5	5
Russia	25		10	10	7	7		1	1	7	6
Senegal	4							4	4		
Serbia	21	3	1	1	5	4		2	2	10	10
Slovakia	11		2	2	3	3				6	6
Slovenia	16		6	6	4	4		1	1	5	4
South Africa	1		1	1							
Spain	10		9	6	1	1					
Sweden	8				4	4				4	4
Ukraine	56		34	33	12	11				10	10
United Kingdom	8		3	3	2	2		1	1	2	2
MARCH											
Austria	7				1	1		1	1	5	5
Belarus	2				1	1		1	1		
Belgium	15		4	3	3	3				8	8
Bulgaria	2		1	1						1	1
Czech Republic	1									1	1
Estonia	22	4	7	7	6	6				5	5
Georgia	12	2	9	6						1	1
Ghana	2		1	0	1	0					
Hong Kong	5		5	5							
Hungary	11		1	1				1	1	9	3
Iceland	3				1	1				2	2
Ireland	11				11	4					
Italy	2		1	1						1	1
Jordan	25		1	1				17	17	7	7
Kazakhstan	3				3	in process					
Lithuania	38	2	2	in process	8	5	22			4	4
Luxembourg	1									1	1
Madagascar	12		3	2	1	1				8	6
Moldova	9		4	in process	2	in process	3				
Norway	22		6	6	9	9		1	1	6	6
Poland	7						1			6	6
Portugal	13		5	5				3	3	5	5
Romania	5		1	1	1	1				3	3
Russia	35		11	11	10	10		7	7	7	6
Senegal	6							5	5	1	1
Serbia	11	1	4	3	2	2	2			2	2
Slovakia	13		4	4	3	3		2	2	4	4
Slovenia	3							2	1	1	1
Spain	12		3	2	4	4		3	3	2	1
Sweden	1		1	1							
Ukraine	69		27	26	20	17	3			19	19
APRIL											
Belgium	3				2	2				1	1
Estonia	13	2	1	1	1	1		1	1	8	8
Ghana	8				8	6					
Hungary	4	1						2	2	1	1
Iceland	3		1	1	1	1				1	1
Ireland	5				5	3					
Jordan	25		1	1				14	14	10	10
Lithuania	4	3								1	1
Madagascar	15	2	6	5			3			4	4
Norway	16		3	3	4	4		1	1	8	8
Poland	1						1				
Portugal	2		1	1						1	1
Romania	2		1	1						1	1
Russia	2		1	1	1	1					
Senegal	7							5	5	2	2
Slovakia	1									1	1
Slovenia	2		1	1						1	1
South Africa	7		6	5	1	1					
Spain	3		1	0	1	1				1	1
Ukraine	12				2	1	1			9	9
MAY											
Austria	1									1	1
Ghana	1		1	1							
Iceland	3		1	1						2	2
Jordan	6							2	2	4	4
Madagascar	7		7	5							
Norway	4		1	1	2	2				1	1
South Africa	6		5	5						1	1
JUNE											
Ghana	8		1	0	7	7					
Madagascar	1		1	0							
South Africa	9		4	4	3	3				2	2
JULY											
Ghana	8		1	0	6	6		1	1		
	937	39	283	241	197	161	50	94	93	274	260
			30.2%		21.0%			10.0%		29.2%	
35 Countries			55%				45%				

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)
2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Table 2. Isolation rates for samples collected after 2013-01-31

	Clinical sample			Virus isolate		
	number processed	number propagated[1]	percent recovered	number processed	number propagated[1]	percent recovered
H1	77	42	55%	216	201	93%
H3	82	53	65%	128	109	85%
B Victoria	4	4	100%	98	89	91%
B Yamagata	31	30	97%	255	230	90%

1. Propagated to sufficient titre to perform HI assay

Table 3. Summary of isolates with collection dates after 2013-01-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays

	Number tested	≤2-fold[a]	4-fold[a]	≥8-fold[a]
H1[1]	243	227 (93%)	11 (5%)	5 (2%)
H3[2]	162	151 (93%)	10 (6%)	1 (1%)
H3[b]	34	23 (68%)	11 (32%)	0 (0%)
B/Victoria[3]	93	90 (97%)	3 (3%)	0 (0%)
B/Yamagata[4]	260	90 (35%)	127 (49%)	43 (16%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:
1. Compared to homologous titres for ferret antisera raised against vaccine virus A/California/7/2009
2. Compared to homologous titres for ferret antisera raised against tissue-culture grown A/Victoria/361/2011
b. Compared to homologous titres for ferret antisera raised against tissue-culture grown A/Victoria/361/2011 in plaque-reduction neutralisation
3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)
4. Compared to homologous titres for ferret antisera raised against vaccine virus B/Massachusetts/02/2012

Table 4. Summary of influenza A(H1N1)pdm09 samples analysed, collected after 2013-01-31

MONTH	H1N1pdm09				
			Fold reduction in HI titre		
Country	Number received	Number propagated[1]	≤2 fold[2]	4 fold[2]	≥8 fold[2]
FEBRUARY					
Algeria	3	0			
Austria	4	4	4		
Belarus	1	1		1	
Belgium	2	2	2		
Bulgaria	8	7	7		
Georgia	10	7	6	1	
Greece	2	2	2		
Hungary	6	4	2		2
Italy	11	11	11		
Jordan	3	2	2		
Lithuania	7	3	3		
Luxembourg	1	0			
Norway	1	1	1		
Portugal	4	4	3	1	
Romania	8	8	8		
Russia	10	10	10		
Serbia	1	1			1
Slovakia	2	2	2		
Slovenia	6	6	6		
South Africa	1	1	1		
Spain	9	6	6		
Ukraine	34	33	30	2	1
United Kingdom	3	3	3		
MARCH					
Belgium	4	3	3		
Bulgaria	1	1	1		
Estonia	7	7	7		
Georgia	9	6	6		
Ghana	1	0			
Hong Kong	5	5	5		
Hungary	1	1	1		
Italy	1	1	1		
Jordan	1	1	1		
Lithuania	2	in process			
Madagascar	3	2	2		
Moldova	4	in process	1	1	
Norway	6	6	6		
Portugal	5	5	3	2	
Romania	1	1	1		
Russia	11	11	11		
Serbia	4	3		2	1
Slovakia	4	4	4		
Spain	3	2	2		
Sweden	1	1	1		
Ukraine	27	26	26		
APRIL					
Estonia	1	1	1		
Iceland	1	1	1		
Jordan	1	1	1		
Madagascar	6	5	5		
Norway	3	3	3		
Portugal	1	1	1		
Romania	1	1	1		
Russia	1	1	1		
Slovenia	1	1	1		
South Africa	6	5	4	1	
Spain	1	0			
MAY					
Ghana	1	1	1		
Iceland	1	1	1		
Madagascar	7	5	5		
Norway	1	1	1		
South Africa	5	5	5		
JUNE					
Ghana	1	0			
Madagascar	1	0			
South Africa	4	4	4		
JULY					
Ghana	1	0			
30 Countries	283	241	227 93%	11 5%	5 2%

1. Propagated to sufficient titre to perform HI assays
2. Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

Of 15 viruses sequenced with >2 fold reduction in HI titre, 9 contained HA1 substitutions in the 153-155 region known to affect receptor binding/antigenicity, as a result of cell propagation

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-02-12)

				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	640	640	640	640	640	640	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	160	80	40	40	40	40	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	1280	640	160	80	160	160	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	2560	1280	2560	5120	2560	1280	1280	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	320	640	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	320	1280	640	1280	1280	640	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	1280	640	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	1280	1280	1280	1280	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	160	320	320	640	640	320	1280	1280	
TEST VIRUSES													
A/Blida/103/2012	6C	2012-10-30	C0/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	
A/Israel/9/2012		2012-11-06	Cx/MDCK1	640	320	640	640	1280	640	640	2560	1280	
A/Norway/2259/2012	6C	2012-11-09	MDCK2/MDCK3	640	320	160	80	80	160	80	80	80	N156D
A/Norway/2352/2012		2012-11-13	MDCK1/MDCK2	1280	1280	1280	1280	2560	2560	2560	5120	5120	
A/Jordan/30456/2012		2012-11-25	PI/SIAT2	640	320	640	640	1280	640	1280	2560	1280	
A/Norway/2362/2012		2012-11-26	MDCK1/MDCK2	1280	1280	1280	2560	2560	2560	2560	5120	5120	
A/Norway/2383/2012		2012-11-26	MDCK1/MDCK2	1280	1280	1280	1280	2560	2560	5120	5120	5120	
A/Norway/2388/2012	6C	2012-11-26	MDCK1/MDCK2	1280	640	640	320	80	320	160	160	320	N156D
A/Norway/2403/2012		2012-11-28	MDCK1/MDCK2	1280	1280	1280	1280	2560	1280	5120	5120	5120	
A/Norway/2411/2012	6C	2012-11-30	MDCK1/MDCK2	640	320	640	1280	1280	1280	1280	2560	2560	
A/Paris/1885/2012		2012-12-05	MDCK2/MDCK1	640	320	640	640	1280	1280	640	2560	2560	
A/Hong Kong/9204/2012	6A	2012-12-15	MDCK3/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	
A/Netherlands/529/2012	7	2012-12-16	MDCK2/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	
A/Centre/1976/2012		2012-12-17	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	5120	
A/Hong Kong/9259/2012		2012-12-18	MDCK3/MDCK1	1280	640	640	1280	1280	1280	1280	2560	2560	
A/Switzerland/7628329/2012		2012-12-18	MDCK2	1280	640	1280	1280	2560	1280	1280	5120	2560	
A/Switzerland/7628993/2012		2012-12-19	MDCK2	1280	1280	1280	1280	2560	1280	1280	5120	2560	
A/Switzerland/7628314/2012	7	2012-12-19	MDCK2	640	320	640	640	1280	640	640	2560	1280	
A/Switzerland/7628324/2012		2012-12-21	MDCK2	640	640	640	640	1280	640	640	2560	1280	
A/Switzerland/7628514/2012	6C	2012-12-21	MDCK2	320	640	640	320	640	320	320	640	640	
A/Jordan/30003/2012		2012-12-24	SIAT2	2560	1280	2560	2560	5120	2560	2560	5120	5120	
A/Hong Kong/62/2013	6C	2013-01-05	MDCK3/MDCK1	1280	640	1280	640	1280	1280	1280	2560	2560	
A/Parma/01/2013		2013-01-07	MDCK2/MDCK1	640	640	640	640	1280	640	1280	2560	1280	
A/Parma/06/2013		2013-01-07	MDCK2/MDCK1	640	640	640	1280	1280	1280	1280	2560	2560	
A/Netherlands/068/2013	6C	2013-01-07	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	
A/Parma/02/2013		2013-01-08	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	5120	
A/Israel/22/2013		2013-01-08	Cx/MDCK1	640	640	1280	640	1280	1280	1280	2560	2560	
A/Israel/23/2013		2013-01-08	Cx/MDCK1	1280	640	640	640	1280	1280	1280	2560	2560	
A/Parma/04/2013		2013-01-09	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	5120	
A/Israel/27/2013		2013-01-09	Cx/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	5120	
A/Trieste/02/2013		2013-01-10	Cx/MDCK1	1280	640	1280	1280	640	1280	1280	5120	5120	
A/Trieste/04/2013		2013-01-10	Cx/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Parma/03/2013	6C	2013-01-14	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	2560	
A/Trieste/03/2013	7	2013-01-14	Cx/MDCK1	640	1280	1280	1280	2560	1280	1280	5120	2560	
A/Dolj/131915/2013	6C	2013-01-14	MDCK2/MDCK1	1280	1280	1280	1280	1280	1280	2560	2560	2560	
A/Israel/26/2013		2013-01-15	Cx/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	
A/Israel/21/2013		2013-01-16	Cx/MDCK1	640	640	1280	1280	2560	1280	1280	5120	2560	
A/Turkey/579/2013	6C	2013-01-16	C0/MDCK2	1280	640	1280	1280	1280	1280	1280	2560	2560	
A/Estonia/74590/2013	7	2013-01-17	MDCK1/MDCK3	640	640	640	640	1280	1280	1280	2560	2560	
A/Turkey/605/2013	7	2013-01-18	C0/MDCK3	1280	640	1280	1280	2560	1280	2560	5120	5120	
A/Iasi/132200/2013	6C	2013-01-20	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	2560	2560	2560	
A/Turkey/616/2013	6C	2013-01-21	C0/MDCK3	1280	640	1280	1280	2560	2560	2560	5120	5120	
A/Dolj/132523/2013	6C	2013-01-21	MDCK1/MDCK1	1280	640	1280	1280	1280	1280	2560	2560	2560	
A/Dolj/132532/2013		2013-01-22	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560	

1. < = <40

Vaccine

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-02-26 + 2013-03-12)

				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	2560	1280	2560	1280	1280	2560	2560	2560	5120	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	160	80	40	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	320	160	320	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	2560	2560	5120	5120	2560	2560	5120	5120	320	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	640	640	1280	1280	640	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	2560	1280	2560	1280	2560	2560	2560	5120	5120	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	2560	1280	2560	1280	2560	2560	2560	5120	5120	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	640	320	640	640	1280	1280	640	2560	2560	
TEST VIRUSES													
A/Ghana/FS-1614/2012		2012-11-19	MDCK2	2560	1280	2560	2560	5120	5120	5120	5120	5120	
A/Ghana/FS-1615/2012	West African outlier	2012-11-26	MDCK2	2560	1280	2560	2560	5120	2560	2560	5120	5120	
A/Belgium/G944/2012	7	2012-12-11	MDCK2	1280	640	1280	1280	1280	1280	2560	5120	2560	
A/Ghana/DILI-1217/2012		2012-12-12	MDCK3	1280	320	1280	1280	1280	1280	2560	2560	2560	
A/Ghana/DILI-1218/2012	7	2012-12-12	MDCK2	1280	640	1280	640	2560	1280	2560	5120	2560	
A/Ghana/DILI-1219/2012		2012-12-12	MDCK2	1280	640	1280	640	2560	1280	2560	5120	2560	
A/Tehran/7359/2012	6C	2012-12-17	MDCK4/MDCK1	1280	640	1280	1280	1280	2560	2560	5120	5120	
A/Belgium/S0294/2012		2012-12-19	MDCK2	2560	640	1280	1280	2560	2560	2560	5120	5120	
A/Belgium/G992/2012		2012-12-20	MDCK3	1280	640	1280	640	2560	1280	1280	2560	2560	
A/Belgium/G1019/2012	6C	2012-12-26	MDCK2	2560	640	2560	2560	5120	2560	5120	5120	5120	
A/Belgium/S0318/2012		2012-12-27	MDCK4	640	640	1280	640	1280	1280	1280	5120	2560	
A/Belgium/S0334/2012		2012-12-27	MDCK3	640	640	640	640	1280	1280	1280	2560	1280	
A/Belgium/S0345/2012	6C	2012-12-28	MDCK3	640	320	1280	640	1280	1280	1280	5120	2560	
A/Ireland/00738/2012	6C	2012-12-31	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	
A/Belgium/S0007/2013	6C	2013-01-01	MDCK2	1280	640	2560	1280	2560	2560	1280	5120	5120	
A/Belgium/G0001/2013		2013-01-02	MDCK2	1280	640	640	2560	1280	1280	1280	2560	2560	
A/Belgium/S0003/2013		2013-01-02	MDCK2	1280	1280	1280	2560	2560	2560	2560	5120	5120	
A/Malta/MV14254/2013	7	2013-01-10	MDCK2	1280	640	1280	1280	1280	1280	1280	5120	2560	
A/Catalonia/5684S/2013	6C	2013-01-11	MDCK1	640	640	1280	320	640	640	640	640	1280	
A/Malta/MV14319/2013	7	2013-01-12	MDCK2	1280	640	640	640	1280	1280	1280	2560	2560	
A/Valencia/2S/2013		2013-01-12	MDCK2	640	640	1280	640	1280	1280	1280	5120	2560	
A/Belgium/S0091/2013	6C	2013-01-14	MDCK2	640	640	640	640	1280	1280	1280	2560	2560	
A/Belgium/S0093/2013	7	2013-01-14	MDCK2	1280	640	2560	1280	1280	2560	2560	5120	5120	
A/Catalonia/5737S/2013	6C	2013-01-22	MDCK1	1280	640	1280	1280	2560	2560	1280	5120	5120	

1. < = <40

Vaccine

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-04-16)

				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F05/10	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E1	1280	1280	1280	640	1280	1280	640	1280	1280	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	80	40	40	40	40	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	320	160	160	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	2560	2560	2560	5120	2560	2560	1280	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	640	640	1280	640	640	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	1280	2560	1280	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	1280	1280	1280	1280	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	1280	1280	1280	1280	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	160	320	320	640	640	640	1280	1280	
TEST VIRUSES													
A/Nablus/1/2012		2012-12-09	MDCK3	1280	640	640	2560	1280	1280	1280	2560	2560	
A/Jenin/2/2012	6C	2012-12-12	MDCK2	1280	640	1280	1280	2560	1280	1280	5120	2560	
A/Qalqilia/3/2012		2012-12-18	MDCK3	640	320	640	320	1280	640	640	1280	1280	
A/Nablus/4/2012		2012-12-20	MDCK3	1280	640	640	640	1280	1280	1280	2560	2560	
A/Tulkarem/28/2012	6C	2012-12-20	MDCK2	1280	640	640	640	2560	1280	1280	2560	2560	
A/Jericho/6/2012		2012-12-22	MDCK3	640	320	640	640	1280	1280	640	2560	2560	
A/Jenin/13/2012		2012-12-22	MDCK3	640	320	640	640	1280	640	640	2560	2560	
A/Jenin/9/2012		2012-12-23	MDCK3	1280	640	1280	1280	2560	1280	1280	2560	2560	
A/Jenin/11/2012		2012-12-23	MDCK3	640	640	640	640	1280	640	640	1280	1280	
A/Jenin/12/2012		2012-12-23	MDCK3	1280	640	640	640	1280	1280	1280	2560	2560	
A/Tubas/14/2012	6C	2012-12-23	MDCK2	2560	1280	1280	1280	2560	2560	2560	5120	5120	
A/Qalqilia/17/2012		2012-12-23	MDCK2	1280	640	1280	640	1280	1280	1280	2560	2560	
A/Tubas/15/2012		2012-12-24	MDCK3	640	320	640	640	1280	640	640	2560	1280	
A/Salfeet/16/2012	6C	2012-12-24	MDCK2	1280	640	1280	1280	2560	2560	1280	5120	5120	
A/Jericho/18/2012	6C	2012-12-24	MDCK3	160	640	320	160	80	160	80	80	160	G155E>G
A/Jenin/19/2012		2012-12-24	MDCK2	640	320	640	640	1280	640	1280	2560	2560	
A/Jenin/20/2012	6C	2012-12-24	MDCK2	640	1280	640	320	160	320	320	320	640	
A/Nablus/22/2012		2012-12-26	MDCK3	640	320	640	320	1280	640	640	1280	1280	
A/Tulkarem/25/2012		2012-12-26	MDCK2	1280	640	1280	1280	2560	1280	1280	2560	2560	
A/Beithlehem/33/2012	6C	2012-12-27	MDCK3	320	640	320	320	320	320	320	320	640	G155G>E
A/Nablus/23/2012	6C	2012-12-29	MDCK2	320	1280	640	160	160	320	320	160	320	K154K>E
A/Tulkarem/26/2012		2012-12-29	MDCK2	1280	640	1280	1280	5120	2560	1280	5120	2560	
A/Tulkarem/27/2012	6C	2012-12-29	MDCK2	1280	640	1280	640	2560	1280	1280	5120	2560	
A/Hebron/37/2012		2012-12-30	MDCK3	640	640	640	640	1280	640	640	1280	1280	
A/Qalqilia/38/2012	6C	2012-12-30	MDCK2	1280	640	640	1280	2560	1280	1280	5120	2560	
A/Nablus/46/2012	6C	2012-12-30	MDCK3	1280	640	640	640	2560	1280	1280	2560	2560	
A/Beithlehem/31/2012	6C	2012-12-31	MDCK2	1280	640	1280	640	1280	1280	1280	1280	2560	
A/Beithlehem/32/2012	6C	2012-12-31	MDCK2	1280	640	1280	640	2560	1280	1280	2560	2560	
A/Qalqilia/39/2012	6C	2012-12-31	MDCK2	1280	640	640	1280	2560	1280	1280	2560	2560	
A/Qalqilia/40/2012		2012-12-31	MDCK3	1280	640	1280	1280	2560	2560	1280	2560	2560	
A/Ramallah/41/2012		2012-12-31	MDCK3	1280	640	640	640	2560	2560	1280	2560	2560	
A/Ramallah/45/2012		2012-12-31	MDCK3	1280	320	640	640	2560	1280	1280	2560	2560	
A/Ramallah/47/2012		2012-12-31	MDCK3	1280	640	640	1280	2560	1280	1280	2560	2560	
A/Tulkarem/50/2012		2012-12-31	MDCK2	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Ramallah/42/2013		2013-01-01	MDCK3	1280	640	1280	1280	2560	2560	1280	2560	2560	
A/Ramallah/43/2013	6C	2013-01-01	MDCK2	320	320	320	320	1280	320	320	1280	1280	
A/Tulkarem/48/2013		2013-01-01	MDCK2	1280	640	640	1280	2560	1280	1280	2560	2560	
A/Tubas/44/2013	6C	2013-01-24	MDCK2	640	640	640	320	320	320	640	320	640	K154K>E, G155G>E
A/England/350/2013	7	2013-02-06	SIAT1/MDCK1	640	640	640	640	640	640	640	1280	1280	
A/England/353/2013	6C	2013-02-11	MDCK1/MDCK2	640	320	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/17/2013	6C	2013-02-13	C1/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	5120	
A/England/358/2013	6B	2013-02-18	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	5120	
A/St. Petersburg/11/2013	6C	2013-02-20	C1/MDCK2	640	320	640	640	1280	1280	1280	2560	2560	
A/St. Petersburg/24/2013	6B	2013-02-20	C1/MDCK2	1280	640	1280	1280	2560	2560	1280	2560	2560	
A/St. Petersburg/33/2013	6C	2013-02-20	C1/MDCK2	640	320	640	640	1280	1280	640	1280	1280	
A/St. Petersburg/38/2013	6C	2013-02-20	C1/MDCK1	1280	640	1280	1280	2560	2560	1280	2560	2560	

1. < = <40

Vaccine

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-04-23)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera: A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E2	640	640	640	320	320	320	640	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	160	80	40	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	320	160	160	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	2560	1280	2560	5120	2560	2560	2560	5120	5120	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	640	320	1280	640	640	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	640	640	1280	1280	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	2560	1280	640	1280	2560	2560	2560	5120	5120	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	2560	640	2560	1280	2560	2560	2560	5120	5120	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	640	160	320	640	1280	640	640	1280	1280	
TEST VIRUSES													
A/Jenin/5/2012		2012-12-12	MDCK4	640	320	640	640	1280	640	640	2560	2560	
A/Tubas/29/2012		2012-12-27	MDCK4	640	640	640	320	640	320	640	640	1280	
A/Ramallah/34/2012		2012-12-31	MDCK4	640	320	640	320	640	320	640	640	1280	
A/Ramallah/36/2012		2012-12-31	MDCK4	1280	320	640	640	2560	1280	1280	5120	2560	
A/Ramallah/49/2013		2013-01-01	MDCK3	1280	320	640	640	1280	1280	1280	2560	2560	
A/Nablus/51/2013		2013-01-02	MDCK3	1280	640	1280	640	1280	1280	1280	5120	2560	
A/Ramallah/52/2013		2013-01-02	MDCK3	640	160	640	640	1280	1280	1280	2560	2560	
A/Ramallah/53/2013		2013-01-02	MDCK3	640	320	640	640	1280	1280	1280	2560	1280	
A/Hebron/56/2013		2013-01-02	MDCK3	640	320	640	640	2560	1280	1280	2560	1280	
A/Hebron/54/2013		2013-01-03	MDCK3	640	320	640	640	1280	1280	1280	2560	2560	
A/Hebron/55/2013		2013-01-03	MDCK3	640	320	320	320	640	640	640	640	640	
A/Nablus/57/2013		2013-01-03	MDCK3	640	320	320	320	640	640	640	1280	1280	
A/Ramallah/59/2013		2013-01-03	MDCK3	1280	320	1280	640	2560	1280	1280	2560	2560	
A/Ramallah/58/2013		2013-01-04	MDCK3	1280	640	1280	1280	1280	2560	1280	2560	2560	
A/Ramallah/60/2013	6C	2013-01-04	MDCK3	640	640	1280	640	1280	640	640	1280	1280	
A/Hebron/63/2013		2013-01-04	MDCK2	1280	320	1280	640	1280	1280	1280	2560	2560	
A/Luxembourg/13/2013	7	2013-01-04	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560	
A/Ramallah/62/2013		2013-01-05	MDCK2	1280	320	1280	1280	1280	1280	1280	2560	2560	
A/Beithlehem/66/2013		2013-01-05	MDCK2	1280	320	1280	1280	1280	1280	1280	2560	2560	
A/Ramallah/67/2013		2013-01-05	MDCK2	1280	320	1280	640	1280	1280	1280	2560	2560	
A/Nablus/72/2013		2013-01-06	MDCK2	1280	640	1280	640	1280	1280	1280	2560	2560	
A/Ramallah/69/2013		2013-01-07	MDCK2	640	320	1280	320	1280	640	640	1280	1280	
A/Tulkarem/70/2013		2013-01-07	MDCK2	640	320	1280	320	640	1280	640	2560	1280	
A/Hebron/71/2013		2013-01-07	MDCK2	1280	320	1280	640	1280	1280	1280	2560	2560	
A/Tubas/73/2013		2013-01-07	MDCK2	640	640	1280	640	1280	1280	1280	2560	2560	
A/Ramallah/74/2013		2013-01-07	MDCK2	1280	640	1280	640	1280	1280	640	2560	2560	
A/Luxembourg/18/2013	6C	2013-01-07	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	2560	
A/Qalqilia/75/2013		2013-01-08	MDCK2	640	640	1280	640	1280	1280	1280	2560	2560	
A/Luxembourg/34/2013	6C	2013-01-08	MDCK1/MDCK1	1280	640	640	1280	2560	1280	1280	5120	2560	
A/Qalqilia/76/2013		2013-01-11	MDCK2	640	320	640	640	1280	640	640	1280	1280	
A/Qalqilia/77/2013		2013-01-11	MDCK2	1280	640	1280	1280	2560	2560	1280	5120	2560	
A/Ramallah/79/2013		2013-01-11	MDCK2	640	640	1280	640	640	640	640	640	640	
A/Ramallah/80/2013		2013-01-12	MDCK2	1280	640	1280	1280	2560	2560	1280	5120	5120	
A/Luxembourg/80/2013	6C	2013-01-16	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	2560	
A/Estonia/74816/2013	6C	2013-01-25	MDCK3	5120	2560	5120	2560	5120	5120	5120	5120	5120	
A/Estonia/74854/2013	6C	2013-01-28	MDCK2	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Estonia/74855/2013	6C	2013-01-28	MDCK2	2560	640	1280	1280	2560	2560	2560	5120	5120	
A/Bulgaria/260/2013		2013-02-08	C1/MDCK1	640	320	640	1280	1280	1280	1280	2560	2560	
A/Bulgaria/277/2013	6C	2013-02-08	C1/MDCK1	640	640	640	1280	1280	1280	640	2560	1280	
A/Florina/60/2013		2013-02-13	E2/E1	640	640	40	640	640	640	640	1280	1280	
A/Bulgaria/364/2013	6C	2013-02-14	C1/MDCK1	1280	320	1280	1280	2560	2560	1280	2560	5120	
A/Bulgaria/421/2013		2013-02-14	C1/MDCK1	1280	640	1280	1280	2560	2560	2560	2560	5120	
A/Thessaloniki/75/2013	7	2013-02-18	E2/E1	640	640	640	640	1280	1280	1280	2560	1280	
A/Bulgaria/477/2013		2013-02-23	C2/MDCK1	640	320	640	640	640	1280	640	2560	2560	
A/Bulgaria/492/2013		2013-02-23	C2/MDCK1	1280	640	1280	1280	2560	2560	1280	2560	2560	
A/Bulgaria/494/2013	6C	2013-02-23	C2/MDCK1	1280	320	640	640	1280	1280	1280	2560	2560	
A/Bulgaria/611/2013	6C	2013-03-04	C2/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-05-01 + 2013-05-08 + 2013-05-15)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E2	640	640	640	320	160	160	160	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	160	160	40	40	40	40	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	1280	160	80	160	160	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	2560	5120	2560	2560	1280	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	640	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	640	640	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	1280	1280	1280	1280	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	160	320	320	640	640	640	1280	1280	
TEST VIRUSES													
A/Luxembourg/628/2012		2012-12-13	MDCK1/SIAT2	640	160	640	640	1280	640	1280	2560	1280	
A/Jerusalem/65/2013		2013-01-03	MDCK3	640	640	640	320	640	640	640	1280	1280	
A/Hebron/64/2013		2013-01-05	MDCK3	640	640	640	320	640	640	640	1280	1280	
A/Ramallah/61/2013		2013-01-06	MDCK3	640	320	640	320	640	640	640	1280	1280	
A/Beithlehem/82/2013		2013-01-10	MDCK3	1280	320	1280	640	2560	1280	1280	5120	2560	
A/Luxembourg/62/2013		2013-01-11	MDCK1/MDCK2	320	160	320	320	640	640	640	640	640	
A/Ramallah/85/2013		2013-01-13	MDCK2	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Hebron/86/2013	6C	2013-01-15	MDCK2	320	320	640	320	640	640	640	1280	1280	
A/Beithlehem/87/2013		2013-01-15	MDCK2	640	640	1280	640	1280	1280	1280	2560	2560	
A/Jericho/88/201		2013-01-15	MDCK2	640	160	320	320	640	640	640	1280	1280	
A/Kochani/PD 4/2013	6C	2013-01-15	MDCK2	640	640	1280	1280	2560	1280	1280	2560	2560	
A/Skopje/AH 6/2013		2013-01-15	MDCK3	1280	1280	1280	1280	5120	2560	2560	5120	5120	
A/Beithlehem/89/2013		2013-01-16	MDCK2	640	640	640	640	1280	1280	1280	2560	1280	
A/Ohrid/SG 16/2013	6C	2013-01-16	MDCK4	320	320	320	160	160	160	160	160	640	K154K>E
A/Nablus/90/2013		2013-01-17	MDCK2	640	320	640	640	1280	640	640	2560	1280	
A/Hebron/92/2013	6C	2013-01-17	MDCK2	640	640	640	320	320	320	640	640	640	G155G>E
A/Ramallah/91/2013		2013-01-18	MDCK2	640	640	640	640	1280	1280	1280	2560	1280	
A/Beithlehem/93/2013	6C	2013-01-18	MDCK2	640	640	1280	1280	1280	1280	1280	2560	2560	
A/Jericho/94/2013		2013-01-18	MDCK2	640	320	640	640	1280	1280	1280	2560	2560	
A/Tulkarem/95/2013		2013-01-18	MDCK2	1280	640	1280	1280	2560	1280	2560	1280	2560	
A/Tetovo/HD 23/2013	6C	2013-01-18	MDCK3	1280	640	2560	2560	2560	2560	2560	5120	5120	
A/Hebron/96/2013		2013-01-20	MDCK2	320	640	640	320	640	640	640	640	1280	
A/Luxembourg/119/2013		2013-01-21	MDCK1/MDCK2	640	640	640	640	1280	1280	1280	2560	2560	
A/SkopjeNo/BM 40/2013		2013-01-22	MDCK2	640	640	1280	1280	2560	1280	1280	5120	2560	
A/Hebron/99/2013		2013-01-23	MDCK2	640	320	640	320	1280	640	640	1280	1280	
A/Ramallah/100/2013	6C	2013-01-23	MDCK3	640	1280	640	320	320	640	640	320	640	G155G=E
A/Macedonia/NA 44/2013	6C	2013-01-23	MDCK3	2560	1280	2560	2560	5120	2560	5120	5120	5120	
A/Luxembourg/186/2013	6B	2013-01-25	MDCK1/SIAT2	320	320	640	320	640	640	640	1280	1280	
A/Burgos/14/2013	6C	2013-01-30	MDCK2	2560	1280	2560	1280	2560	2560	2560	5120	5120	
A/Salamanca/17/2013	6C	2013-01-30	MDCK2	1280	1280	2560	1280	2560	1280	2560	2560	2560	
A/Salamanca/43/2013	6C	2013-02-13	MDCK3	1280	640	1280	640	1280	640	640	2560	2560	
A/Leon/70/2013	7	2013-02-18	MDCK3	320	320	640	640	640	320	640	2560	1280	
A/Segovia/59/2013	6C	2013-02-24	MDCK2	2560	1280	2560	2560	5120	1280	2560	5120	5120	
A/Salamanca/61/2013		2013-02-24	MDCK3	640	640	1280	1280	2560	640	1280	2560	5120	
A/Salamanca/63/2013		2013-02-24	MDCK3	1280	640	1280	1280	2560	640	1280	5120	2560	
A/Salamanca/83/2013	6C	2013-03-06	MDCK2	1280	640	2560	1280	2560	640	1280	2560	2560	
A/Leon/90/2013		2013-03-09	MDCK3	320	1280	5120	2560	5120	1280	2560	5120	5120	

Sequences in phylogenetic trees | 1. < = <40 | Vaccine

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-05-21)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E2	640	640	1280	320	320	320	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	40	40	80	80	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	320	160	160	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	1280	5120	1280	1280	1280	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	640	640	1280	640	1280	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	1280	1280	1280	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	640	640	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	2560	1280	2560	2560	2560	5120	5120	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	320	640	320	1280	1280	1280	2560	1280	
TEST VIRUSES													
A/Tehran/10538/2012	6C	2012-12-08	E2	640	320	640	320	640	320	640	1280	640	
A/Hormozgan/30318/2012	6A	2012-12-12	E2	320	320	320	320	640	320	640	1280	640	
A/Hormozgan/31167/2012	6C	2012-12-15	E2	640	640	640	640	1280	640	1280	2560	1280	
A/Tehran/30246/2012	6C	2012-12-17	E2	640	320	640	640	1280	1280	1280	1280	1280	
A/Tehran/8579/2012	6C	2012-12-17	E3	640	320	640	640	640	640	1280	1280	1280	
A/Tehran/7359/2012	6C	2012-12-17	E2	640	640	1280	640	1280	1280	640	2560	2560	
A/Plzen/39/2013		2012-12-19	MDCK4/MDCK1	320	640	320	160	160	160	320	160	320	
A/Plzen/18/2013	6C	2012-12-24	MDCK4/MDCK1	320	320	160	80	80	160	160	160	320	G155E>G
A/Tehran/35070/2013	6C	2013-01-02	E1	640	640	640	640	1280	640	1280	1280	1280	
A/Cheb/24/2013	6C	2013-01-11	MDCK4/MDCK1	1280	640	1280	640	1280	1280	1280	2560	2560	
A/Plzen/25/2013		2013-01-11	MDCK4/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Praha/26/2013	6C	2013-01-11	MDCK4/MDCK1	1280	320	1280	1280	1280	1280	1280	2560	2560	
A/Praha/28/2013		2013-01-11	MDCK4/MDCK1	640	320	640	640	1280	640	1280	2560	1280	
A/Hradec Kralove/29/2013	6C	2013-01-11	MDCK4/MDCK1	320	320	320	320	640	640	640	1280	1280	
A/Praha/30/2013	6C	2013-01-14	MDCK4/MDCK1	640	320	320	320	640	640	640	1280	1280	
A/Praha/31/2013	6C	2013-01-14	MDCK4/MDCK1	320	320	640	640	640	640	640	1280	1280	
A/Plzen/48/2013	6C	2013-01-23	MDCK4/MDCK1	640	640	640	640	1280	1280	1280	2560	1280	
A/Plzen/49/2013	6C	2013-01-28	MDCK4/MDCK1	640	320	640	640	640	1280	1280	1280	1280	
A/Valladolid/54/2013		2013-02-18	MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	

1. < = <40

Vaccine

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-05-24 + 2013-05-28)

				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E1	1280	1280	1280	640	1280	640	640	1280	1280	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	160	40	40	40	40	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	2560	1280	320	160	160	320	160	640	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	2560	2560	5120	1280	1280	1280	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK2	640	320	640	640	1280	1280	1280	2560	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	1280	1280	1280	2560	5120	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	1280	1280	1280	640	1280	1280	1280	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	640	640	1280	640	1280	1280	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	
TEST VIRUSES													
A/Annaba/G687/2012	6C	2012-10-30	MDCK2	1280	1280	1280	1280	2560	1280	2560	5120	2560	
A/Skopje/50/2013		2013-01-23	MDCK3	1280	1280	1280	1280	2560	1280	2560	5120	2560	
A/Skopje/54/2013	6C	2013-01-23	MDCK3	320	640	320	160	80	160	320	160	320	G155G=E
A/Skopje/59/2013		2013-01-24	MDCK3	1280	1280	1280	1280	2560	1280	2560	5120	2560	
A/Gostivar/19z/2013	6C	2013-01-24	MDCK3	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Strumica/27z/2013	7	2013-01-29	MDCK3	640	640	1280	1280	1280	1280	1280	5120	2560	
A/Slovenia/472/2013	6C	2013-01-31	MDCK1	1280	1280	1280	2560	2560	1280	2560	5120	2560	
A/Slovenia/505/2013	7	2013-02-01	MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	2560	
A/Slovenia/740/2013	6C	2013-02-14	MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	2560	
A/Slovenia/825/2013		2013-02-18	MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	2560	
A/Slovenia/870/2013		2013-02-21	MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	2560	
A/Slovenia/888/2013		2013-02-22	MDCK1	1280	1280	1280	1280	5120	1280	2560	5120	5120	
A/Slovenia/941/2013	6C	2013-02-25	MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	2560	
A/St. Petersburg/124/2013	6C	2013-03-12	C1/MDCK2	2560	2560	2560	2560	5120	2560	5120	5120	5120	
A/St. Petersburg/155/2013	6C	2013-03-20	C1/MDCK2	1280	1280	1280	640	1280	1280	1280	2560	2560	
A/Samara/83/2013	6C	2013-03-20	C1/MDCK1	1280	1280	2560	1280	5120	1280	2560	5120	2560	
A/Slovenia/1552/2013	6C	2013-04-05	MDCK1	1280	1280	1280	1280	2560	1280	2560	5120	2560	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-06-05)

				Haemagglutination inhibition titre[1]							
				Post infection ferret sera							
Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A
REFERENCE VIRUSES											
A/California/7/2009		2009-04-09	EP1/E2	320	320	640	160	160	160	320	160
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	160	80	40	<	40	40	40
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK2	320	640	640	160	80	160	80	160
A/Christchurch/16/2010	4	2010-07-12	E2/E2	640	1280	1280	5120	640	1280	2560	640
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	320	160	320	320	640	320	1280	640
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	640	640	640	1280	2560	1280
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	1280	640	640	1280	2560	1280
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	640	640	640	640	1280	1280	2560	2560
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	640	160	1280	640	1280	1280	2560	2560
TEST VIRUSES											
A/Salaj/132819/2013		2013-01-24	MDCK2/C1	640	320	320	640	1280	1280	1280	1280
A/Galati/133254/2013		2013-01-31	MDCK1/C1	640	320	640	640	1280	1280	2560	1280
A/Dolj/133772/2013	6C	2013-02-04	MDCK1/C1	640	320	640	640	1280	1280	1280	1280
A/Cluj/133922/2013	6C	2013-02-06	MDCK1/C1	640	640	1280	640	1280	1280	2560	2560
A/Iasi/134271/2013		2013-02-12	MDCK1/C1	1280	320	640	640	1280	1280	5120	2560
A/St. Petersburg/3/2013	6C	2013-02-13	E2/E1	640	320	640	640	640	640	2560	1280
A/St. Petersburg/7/2013		2013-02-13	E2/E1	640	320	640	320	640	640	2560	1280
A/Timis/134705/2013	6C	2013-02-15	MDCK2/C1	320	320	640	640	640	640	1280	1280
A/Sibiu/134998/2013	6C	2013-02-15	MDCK1/C1	640	640	1280	640	1280	1280	2560	2560
A/Iasi/134898/2013		2013-02-18	MDCK1/C1	640	320	320	640	640	1280	2560	1280
A/St. Petersburg/9/2013		2013-02-20	E2/E1	640	320	640	320	640	640	2560	1280
A/St. Petersburg/20/2013	6C	2013-02-20	E1/E1	640	640	640	640	640	640	1280	640
A/St. Petersburg/16/2013		2013-02-20	E1/E1	640	640	640	640	640	640	1280	1280
A/Constanta/135659/2013		2013-02-21	MDCK1/C1	640	320	640	640	1280	1280	2560	1280
A/Iasi/135857/2013		2013-02-28	MDCK1/C1	640	320	640	320	640	640	1280	1280
A/Iasi/137598/2013	6C	2013-03-18	MDCK1/C1	1280	640	640	1280	1280	1280	5120	2560
A/Vaslui/139682/2013	6C	2013-04-02	MDCK1/C1	640	320	640	640	1280	1280	2560	1280

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-06-12)

				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	1280	1280	1280	640	1280	640	640	1280	640	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	80	40	80	80	80	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	320	160	160	160	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	2560	5120	2560	2560	1280	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	160	640	640	1280	640	640	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	1280	2560	1280	1280	2560	5120	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	2560	2560	2560	1280	2560	2560	5120	5120	5120	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	1280	640	2560	1280	2560	1280	1280	5120	2560	
TEST VIRUSES													
A/Tbilisi/21/2013	6C	2013-01-10	MDCK2	640	320	640	640	1280	640	640	2560	1280	
A/Belgrade/112/2013		2013-01-11	C4/MDCK1	320	320	320	160	160	320	320	160	320	
A/Batumi/62/2013		2013-01-12	MDCK2	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Belgrade/125/2013	6C	2013-01-15	C4/MDCK1	160	320	320	640	320	320	320	640	320	G155G>E
A/Serbia/NS-212/2013		2013-01-18	MDCK2	640	640	1280	1280	2560	1280	1280	2560	2560	
A/Tbilisi/70/2013		2013-01-18	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Belgrade/231/2013	6C	2013-01-18	C4/MDCK1	640	640	640	2560	1280	1280	1280	2560	1280	
A/Serbia/NS-211/2013		2013-01-18	E1/E1	640	640	1280	640	1280	1280	1280	2560	2560	
A/Ukraine/5772/2013		2013-01-21	C2/MDCK1	640	320	640	640	1280	1280	2560	2560	2560	
A/Serbia/NS-223/2013	7	2013-01-24	MDCK2	320	320	320	640	640	640	640	1280	1280	
A/Serbia/NS-226/2013		2013-01-25	MDCK2	1280	640	1280	1280	2560	1280	1280	5120	2560	
A/Ukraine/5726/2013		2013-01-25	C2/MDCK1	640	320	640	320	640	640	640	1280	2560	
A/Ukraine/5727/2013		2013-01-25	C2/MDCK1	640	320	640	640	1280	1280	2560	2560	2560	
A/Serbia/NS-237/2013	6C	2013-01-28	MDCK2	320	160	640	640	1280	640	640	2560	1280	
A/Ukraine/5686/2013		2013-01-28	C2/MDCK1	320	320	320	320	640	640	640	1280	1280	
A/Ukraine/5773/2013	6C	2013-01-30	C2/MDCK1	320	320	640	640	640	640	1280	2560	1280	
A/Ukraine/5776/2013		2013-02-01	C2/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	
A/Ukraine/5774/2013	6C	2013-02-03	C2/MDCK1	1280	640	1280	640	2560	1280	2560	2560	2560	
A/Sabac/600/2013		2013-02-04	C3/MDCK1	160	640	320	640	640	640	640	1280	1280	
A/Ukraine/5730/2013	6C	2013-02-05	C2/MDCK1	320	160	320	160	320	320	320	640	1280	
A/Ukraine/5731/2013		2013-02-08	C2/MDCK1	640	640	640	640	1280	1280	2560	2560	2560	
A/Ukraine/5785/2013		2013-02-12	C2/MDCK1	640	320	640	640	640	640	640	1280	1280	
A/Akhaltsikhe/442/2013		2013-02-14	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Ukraine/5793/2013		2013-02-19	C2/MDCK1	640	320	1280	640	1280	1280	1280	2560	1280	
A/Akhaltsikhe/546/2013	6C	2013-02-20	MDCK2	640	320	1280	1280	1280	1280	1280	2560	2560	
A/Ukraine/5797/2013		2013-02-20	C2/MDCK1	640	320	640	640	1280	640	160	2560	1280	
A/Ukraine/5794/2013	7	2013-02-20	C2/MDCK1	160	80	160	160	160	160	160	320	320	
A/Ukraine/5798/2013	6C	2013-02-20	C2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/Ukraine/5799/2013	6C	2013-02-20	C2/MDCK1	320	320	640	640	640	640	640	1280	1280	
A/Ukraine/5800/2013		2013-02-20	C2/MDCK1	640	320	640	640	640	640	640	1280	1280	
A/Ukraine/5796/2013		2013-02-21	C2/MDCK1	640	320	640	640	1280	1280	640	2560	1280	
A/Ukraine/5795/2013		2013-02-21	C2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/Ukraine/5802/2013		2013-02-23	C2/MDCK1	640	320	640	320	640	640	640	1280	1280	
A/Ukraine/5801/2013	6C	2013-02-24	C2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/Ukraine/5803/2013	7	2013-02-24	C2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	
A/Ukraine/5805/2013		2013-02-24	C2/MDCK1	1280	640	1280	1280	2560	5120	1280	5120	2560	
A/Ukraine/5673/2013		2013-02-25	C2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	
A/Ukraine/5804/2013		2013-02-26	C2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	
A/Ukraine/5806/2013		2013-02-26	C2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	
A/Ukraine/5807/2013	6C	2013-02-27	C2/MDCK1	640	640	1280	640	1280	1280	1280	2560	1280	
A/Ukraine/5808/2013	7	2013-02-28	C2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Moldova/317/2013	6C	2013-03-04	MDCK1/MDCK1	640	1280	1280	640	1280	640	640	1280	1280	
A/Moldova/335/2013	6C	2013-03-05	MDCK2	320	320	640	640	1280	640	640	1280	1280	
A/Belgrade/1519/2013	6C	2013-03-09	C3/MDCK1	160	80	320	320	320	320	80	160	1280	
A/St. Petersburg/135/2013		2013-03-12	E2/E1	640	320	640	640	640	640	640	1280	1280	
A/St. Petersburg/137/2013		2013-03-12	E2/E1	640	320	640	640	1280	1280	1280	2560	1280	
A/Kaliningrad/3/2013		2013-03-12	E1/E1	640	640	640	640	1280	1280	1280	2560	2560	
A/N.Novgorod/2/2013		2013-03-12	E1/E1	640	640	1280	640	1280	1280	1280	2560	2560	
A/Belgrade/1772/2013		2013-03-19	C3/MDCK1	320	320	640	320	320	640	320	1280	1280	
A/Belgrade/1773/2013	6C	2013-03-19	C3/MDCK1	320	320	320	320	320	640	320	640	1280	G155G>E
A/St. Petersburg/172/2013		2013-03-20	E2/E1	640	640	640	640	1280	1280	1280	2560	2560	
A/St. Petersburg/103/2013		2013-03-20	E2/E1	640	640	640	640	1280	1280	1280	2560	2560	
A/St. Petersburg/94/2013		2013-03-20	E2/E1	640	640	640	640	1280	1280	1280	2560	2560	
A/St. Petersburg/73/2013	6C	2013-03-20	E2/E1	1280	640	1280	1280	1280	1280	1280	2560	2560	
A/Saratov/1/2013	6C	2013-04-03	E3/E1	640	640	1280	640	1280	1280	1280	2560	2560	

Sequences in phylogenetic trees 1. < = <40 Vaccine

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-06-18)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	640	1280	1280	320	320	320	320	640	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	160	80	40	80	80	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	1280	640	160	80	160	160	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	320	1280	1280	5120	1280	1280	1280	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	320	320	640	640	1280	1280	640	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	640	1280	1280	1280	1280	1280	5120	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	1280	1280	1280	1280	1280	1280	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	640	1280	1280	1280	2560	2560	2560	5120	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK2/MDCK4	640	320	1280	1280	1280	1280	1280	5120	2560	
TEST VIRUSES													
A/Chiatura/187/2013		2013-02-04	MDCK3	640	320	640	640	640	640	640	640	640	
A/Gardabani/222/2013		2013-02-05	MDCK3	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Hungary/02/2013	6C	2013-02-12	MDCK2/E2/MDCK1	320	160	160	160	40	80	40	40	80	N156D
A/Ukraine/5883/2013		2013-02-16	C2/MDCK1	320	320	640	640	640	640	640	2560	1280	
A/Chiatura/505/2013	6C	2013-02-19	MDCK4	160	320	320	80	80	80	80	80	160	K154E
A/Ukraine/5885/2013		2013-02-19	C2/MDCK1	640	640	1280	640	1280	1280	1280	2560	2560	
A/Ukraine/5888/2013	6C	2013-02-20	C2/MDCK1	320	320	320	320	320	320	320	640	640	
A/Batumi/723/2013		2013-02-27	MDCK3	1280	640	1280	640	1280	1280	1280	2560	2560	
A/Ukraine/5927/2013		2013-02-27	C1/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/Ukraine/5903/2013	6C	2013-02-28	C2/MDCK1	640	320	640	640	1280	640	1280	2560	1280	
A/Ukraine/5901/2013		2013-03-01	C2/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560	
A/Ukraine/5899/2013		2013-03-04	C2/MDCK1	640	320	640	640	1280	1280	640	2560	1280	
A/Tbilisi/749/2013	6C	2013-03-05	MDCK3	640	320	640	640	1280	640	640	1280	1280	
A/Ukraine/5898/2013		2013-03-05	C2/MDCK1	320	320	640	320	640	640	640	1280	1280	
A/Ukraine/5908/2013		2013-03-06	C2/MDCK1	320	320	640	320	640	640	640	1280	1280	
A/Ukraine/5929/2013		2013-03-09	C1/MDCK1	1280	320	320	320	1280	640	1280	1280	1280	
A/Ukraine/5930/2013	6C	2013-03-09	C1/MDCK1	320	320	320	320	640	640	640	1280	1280	
A/Ukraine/5911/2013	6C	2013-03-11	C2/MDCK1	320	160	320	640	640	640	640	1280	1280	
A/Ukraine/5912/2013	6C	2013-03-11	C2/MDCK1	320	320	640	320	1280	640	1280	2560	1280	
A/Kobuleti/860/2013	6C	2013-03-12	MDCK3	640	320	640	640	640	1280	1280	1280	320	
A/Ukraine/5839/2013		2013-03-15	C1/MDCK1	320	160	640	320	640	640	640	1280	1280	
A/Ukraine/5923/2013		2013-03-19	C1/MDCK1	320	320	320	640	640	640	640	1280	1280	
A/Ukraine/5924/2013	6C	2013-03-21	C1/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/Keda/958/2013		2013-03-22	MDCK3	640	640	640	640	1280	1280	1280	2560	1280	
A/Tskaltubo/948/2013	6C	2013-03-25	MDCK3	640	640	640	640	640	1280	1280	2560	1280	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-06-25 + 2013-07-02)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
				Haemagglutination inhibition titre[1] Post infection ferret antisera									
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E2	640	640	640	320	160	320	160	320	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	80	40	80	80	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	1280	320	160	160	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	2560	5120	2560	2560	2560	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	320	640	640	1280	1280	640	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	1280	2560	2560	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	2560	1280	1280	2560	2560	2560	5120	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	1280	1280	1280	1280	1280	2560	5120	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	640	640	1280	1280	1280	1280	1280	2560	2560	
TEST VIRUSES													
A/Parma/13/2013		2013-01-17	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	
A/Ukraine/181/2013		2013-01-21	MDCK1/MDCK1	320	160	320	320	640	640	640	1280	1280	
A/Perugia/7/2013		2013-01-21	MDCK2/MDCK1	640	320	640	640	1280	640	640	1280	1280	
A/Trieste/07/2013		2013-01-21	Cx/MDCK1	320	160	320	320	640	640	640	1280	1280	
A/Perugia/18/2013	7	2013-01-26	MDCK1/MDCK1	640	640	1280	640	1280	640	1280	1280	1280	
A/Ukraine/91/2013		2013-01-31	MDCK1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	
A/Parma/28/2013		2013-02-02	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/Firenze/9/2013		2013-02-04	MDCK2/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	
A/Perugia/29/2013	7	2013-02-04	MDCK1/MDCK1	320	320	320	320	640	640	640	1280	640	
A/Hungary/16/2013	6C	2013-02-11	MDCK2/E6/MDCK1	160	40	80	40	<	40	<	40	40	K153E
A/Ukraine/161/2013		2013-02-12	MDCK2/MDCK1	320	320	640	640	640	640	640	1280	1280	
A/Ukraine/141/2013	6C	2013-02-12	MDCK2/MDCK2	320	320	320	320	640	320	640	640	640	
A/Trieste/22/2013	6C	2013-02-12	Cx/MDCK1	640	320	320	160	160	320	160	160	160	N156D>N
A/Ukraine/173/2013	6C	2013-02-13	MDCK2/MDCK1	640	640	1280	640	2560	1280	1280	2560	2560	
A/Hungary/17/2013	6C	2013-02-15	MDCK1/E5/MDCK1	80	40	40	<	<	<	<	<	<	K153E
A/Ukraine/156/2013	6C	2013-02-17	MDCK2/MDCK1	640	640	1280	1280	2560	1280	1280	2560	2560	
A/Parma/52/2013	6C	2013-02-18	MDCK1/MDCK1	640	640	640	640	1280	1280	640	2560	1280	
A/Hungary/03/2013	6C	2013-02-20	MDCK1/E2/MDCK3	320	160	320	320	640	640	640	1280	1280	
A/Ukraine/5928/2013	6C	2013-02-27	C1/MDCK2	640	320	640	640	1280	1280	1280	1280	1280	
A/Ukraine/237/2013		2013-02-28	MDCK1/MDCK3	640	320	640	640	1280	1280	640	2560	1280	
A/Ukraine/5931/2013		2013-03-02	C1/MDCK2	320	160	320	320	640	640	320	640	1280	
A/Ukraine/220/2013		2013-03-02	MDCK1/MDCK2	640	320	640	640	1280	1280	1280	2560	2560	
A/Ukraine/5926/2013		2013-03-05	C1/MDCK2	1280	320	640	640	1280	1280	1280	2560	2560	
A/Kmelnitsk/352/2013		2013-03-05	MDCK1/SIAT2	640	320	640	640	1280	1280	1280	2560	2560	
A/Kmelnitsk/354/2013	6C	2013-03-06	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560	
A/Kmelnitsk/355/2013		2013-03-09	MDCK1/SIAT2	640	320	640	640	1280	1280	1280	2560	2560	
A/Ukraine/5913/2013		2013-03-11	C2/MDCK2	320	160	320	320	640	640	640	640	640	
A/Kmelnitsk/357/2013		2013-03-11	MDCK1/SIAT2	320	160	320	320	640	640	640	1280	1280	
A/Dmanisi/823/2013	6C	2013-03-12	MDCK4	1280	640	640	320	640	640	640	640	640	
A/Khobi/828/2013	6C	2013-03-12	MDCK5	320	1280	640	160	160	160	320	160	320	G155E>G
A/Ukraine/309/2013		2013-03-13	MDCK1/MDCK1	640	640	640	640	2560	1280	1280	2560	2560	
A/Kmelnitsk/359/2013		2013-03-15	MDCK1/MDCK1	320	320	640	320	640	640	640	1280	640	
A/Kmelnitsk/359/2013		2013-03-15	MDCK1/SIAT2	320	320	640	320	640	640	640	1280	1280	
A/Hungary/25/2013	6C	2013-03-18	MDCK2/E2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	
A/Ukraine/5932/2013	6C	2013-03-19	C1/MDCK2	640	320	640	640	1280	1280	1280	2560	2560	
A/Ukraine/348/2013		2013-03-20	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Ukraine/312/2013	6C	2013-03-26	MDCK1/MDCK1	320	160	320	320	640	1280	320	640	640	
A/Ukraine/344/2013		2013-03-26	MDCK1/MDCK1	1280	640	640	640	2560	2560	1280	2560	2560	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-07-09)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
				Haemagglutination inhibition titre[1] — Post infection ferret antisera									
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E2	640	640	640	320	320	320	320	320	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	80	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	320	160	160	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	1280	5120	1280	1280	1280	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	320	640	640	2560	1280	1280	2560	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	640	640	640	1280	1280	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	2560	1280	2560	1280	2560	2560	2560	5120	5120	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	320	640	640	1280	640	640	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
TEST VIRUSES													
A/Iceland/28/2012	6A	2012-12-14	MDCK3	160	640	320	160	160	160	160	160	320	G155E>G
A/Iceland/30/2012	6C	2012-12-26	MDCK1/MDCK1	1280	640	1280	640	1280	1280	1280	2560	2560	
A/Iceland/03/2013		2013-01-04	MDCK1/MDCK1	320	160	640	320	640	640	640	1280	1280	
A/Iceland/13/2013		2013-01-10	MDCKx/MDCK1	640	160	640	320	1280	640	640	1280	1280	
A/Stockholm/4/2013	6C	2013-01-11	MDCK2/MDCK1	640	640	640	320	160	320	320	160	640	G155E>G
A/Stockholm/5/2013	7	2013-01-19	MDCK2/MDCK1	320	640	320	320	640	640	640	1280	1280	
A/Pavia/28/2013	6A	2013-02-02	Cx/MDCK2	1280	640	1280	320	640	640	640	1280	1280	
A/Zugdidi/304/2013		2013-02-07	MDCK2	320	640	640	320	640	640	640	1280	640	
A/Roma/15/2013	7	2013-02-08	Cx/MDCK1	640	320	640	640	1280	640	640	2560	1280	
A/Trencin/406/2013	6C	2013-02-08	MDCK1/MDCK1	640	160	640	320	1280	640	1280	2560	1280	
A/Pavia/47/2013		2013-02-12	Cx/MDCK1	640	320	640	320	1280	640	640	2560	1280	
A/Komarno/457/2013	6C	2013-02-12	MDCK1/MDCK1	640	320	640	320	1280	640	640	1280	640	
A/Perugia/39/2013	6C	2013-02-18	MDCK2/MDCK1	640	320	640	320	1280	640	640	1280	640	
A/Trieste/27/2013		2013-02-22	Cx/MDCK1	640	640	1280	640	2560	1280	1280	2560	2560	
A/Trieste/30/2013		2013-02-26	Cx/MDCK1	1280	640	640	640	640	640	640	1280	640	
A/Bratislava/732/2013		2013-03-01	MDCK2/MDCK1	640	160	320	320	640	640	640	1280	1280	
A/Perugia/44/2013		2013-03-06	MDCK1/MDCK1	640	320	640	320	1280	640	640	1280	640	
A/Ukraine/271/2013		2013-03-09	MDCK2/MDCK3	320	320	640	320	640	640	640	1280	1280	
A/Levice/846/2013	6C	2013-03-12	MDCK1/MDCK1	1280	640	1280	640	1280	1280	1280	2560	2560	
A/Bratislava/872/2013		2013-03-13	MDCK1/MDCK1	320	320	640	320	640	640	640	1280	640	
A/Bratislava/873/2013	6C	2013-03-13	MDCK1/MDCK1	320	320	320	320	640	640	640	1280	640	
A/Stockholm/15/2013	6B	2013-03-26	MDCK0/MDCK1	640	640	640	1280	1280	1280	1280	2560	2560	
A/Iceland/55/2013		2013-04-22	MDCK1/MDCK1	640	640	1280	2560	1280	1280	1280	2560	2560	
A/Iceland/58/2013		2013-05-07	MDCK1/MDCK1	640	320	1280	640	2560	2560	1280	2560	2560	
A/Firenze/12/2013			MDCK2/MDCK1	640	320	640	320	640	640	640	1280	1280	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-07-16 + 2013-07-24)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E2	640	320	640	160	320	320	320	640	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	80	80	80	80	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	1280	640	160	80	160	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	1280	5120	1280	1280	1280	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	160	640	320	1280	1280	1280	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	2560	1280	2560	2560	2560	5120	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	640	640	1280	1280	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	
TEST VIRUSES													
A/Santarem PT/1/2013	6C	2013-01-10	SIAT1/MDCK1	640	1280	640	160	160	320	640	320	1280	G155E>G
A/Coimbra PT/10/2013		2013-01-10	SIAT1/MDCK1	1280	640	2560	640	2560	2560	5120	5120	5120	
A/Leiria PT/14/2013		2013-01-21	SIAT1/MDCK1	1280	1280	2560	2560	2560	2560	2560	5120	5120	
A/Castelo Branco PT/15/2013		2013-01-23	SIAT1/MDCK1	1280	640	1280	1280	1280	1280	2560	5120	2560	
A/Braga PT/19/2013		2013-01-23	SIAT1/MDCK1	2560	1280	1280	1280	2560	2560	2560	5120	2560	
A/Castelo Branco PT/36/2013		2013-01-26	SIAT1/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Belgium/S0299/2013	6C	2013-02-04	MDCK2	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Belgium/G0534/2013	6C	2013-02-05	MDCK2	1280	640	1280	1280	2560	1280	1280	5120	2560	
A/Lisboa PT/43/2013		2013-02-05	SIAT1/MDCK3	320	320	640	320	640	640	640	1280	1280	
A/Aveiro PT/61/2013	6C	2013-02-06	SIAT3/MDCK1	1280	640	1280	1280	1280	1280	2560	5120	2560	
A/Lisboa PT/79/2013		2013-02-08	SIAT2/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Minsk/1172/2013	3	2013-02-09	MDCK2/MDCK1	160	320	160	40	40	80	80	80	80	G155E
A/Castelo Branco PT/85/2013	6C	2013-02-22	SIAT2/MDCK1	160	640	320	80	80	160	160	80	160	G155E
A/Lisboa PT/137/2013	6C	2013-03-01	SIAT4/MDCK1	160	640	320	80	80	80	160	80	160	G155E
A/Castelo Branco PT/130/2013	6C	2013-03-03	SIAT1/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	2560	
A/Coimbra PT/102/2013	6C	2013-03-04	SIAT1/MDCK1	640	320	640	640	1280	1280	2560	2560	2560	
A/Norway/1675/2013	7	2013-03-06	MDCK2/MDCK1	320	320	320	320	640	640	640	1280	640	
A/Belgium/G1041/2013	6C	2013-03-18	MDCK2	640	640	640	640	1280	640	1280	1280	1280	
A/Belgium/G1042/2013		2013-03-18	MDCK3/MDCK1	320	160	320	160	640	640	640	1280	640	
A/Portalegre PT/142/2013	6C	2013-03-22	SIAT1/MDCK1	160	640	320	160	80	160	160	160	320	G155E>G
A/Belgium/G1061/2013	6C	2013-03-22	MDCK2	640	320	320	320	160	320	640	320	320	G155G>E, N156N>R
A/Hong Kong/1737/2013	6C	2013-03-22	MDCK2/MDCK1	640	320	640	640	1280	640	1280	2560	1280	
A/Hong Kong/1722/2013	6C	2013-03-23	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	
A/Hong Kong/1743/2013	6C	2013-03-24	MDCK2/MDCK2	640	320	640	320	1280	640	1280	2560	1280	
A/Hong Kong/1720/2013	6C	2013-03-25	MDCK2/MDCK1	640	320	640	640	1280	640	1280	2560	1280	
A/Norway/1954/2013	6C	2013-03-25	MDCK2/MDCK1	640	320	640	320	1280	640	1280	2560	1280	
A/Norway/1960/2013	6C	2013-03-27	MDCK1/MDCK1	320	160	320	320	640	640	640	1280	1280	
A/Madeira PT/135/2013	6C	2013-03-28	SIAT3/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	
A/Hong Kong/1894/2013	6C	2013-03-30	MDCK2/MDCK1	1280	640	1280	1280	1280	1280	2560	5120	2560	
A/Acores PT/139/2013	6C	2013-04-01	SIAT1/MDCK1	320	320	640	640	1280	640	1280	2560	1280	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-14. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-07-30)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	640	640	640	320	320	320	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	40	40	80	80	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	1280	320	160	320	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	2560	5120	1280	2560	1280	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	320	640	640	1280	1280	1280	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	320	320	640	320	640	640	1280	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	2560	1280	2560	2560	2560	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	640	640	1280	640	640	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	320	160	320	320	640	640	640	1280	1280	
TEST VIRUSES													
A/Ghana/DILI-0007/2013	7	2013-01-04	C1/MDCK1	1280	640	1280	640	2560	2560	2560	5120	2560	
A/Norway/1299-1/2013	6C	2013-02-14	MDCK1/MDCK2	640	160	640	320	320	320	320	320	320	
A/Norway/1702/2013	6C	2013-03-02	MDCK1/MDCK1	320	160	640	320	640	640	640	1280	1280	
A/Norway/1774/2013		2013-03-04	MDCK1/MDCK2	640	320	640	640	1280	640	1280	2560	1280	
A/Estonia/76520/2013		2013-03-08	MDCK1/MDCK1	640	640	1280	640	2560	1280	1280	2560	2560	
A/Estonia/76626/2013	6C	2013-03-12	MDCK2/MDCK1	320	640	320	160	160	320	320	160	320	
A/Estonia/76745/2013		2013-03-14	MDCK2/MDCK1	640	320	640	320	1280	1280	1280	2560	1280	
A/Norway/1939/2013	6C	2013-03-19	MDCK2/MDCK2	320	320	640	320	1280	1280	1280	2560	1280	
A/Mahajanga/1171/2013	6C	2013-03-25	MDCK1/MDCK1	640	640	1280	640	2560	2560	2560	5120	2560	
A/Madagascar/01123/2013	6C	2013-03-25	MDCK2	1280	640	1280	640	1280	1280	2560	2560	2560	
A/Norway/1983/2013		2013-04-01	MDCK1/MDCK2	640	640	640	640	1280	1280	1280	2560	2560	
A/Norway/2060/2013	6C	2013-04-05	MDCK2/MDCK2	640	320	640	640	1280	1280	1280	2560	1280	
A/Estonia/77389/2013	6C	2013-04-08	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560	
A/Norway/2197/2013	7	2013-04-12	MDCK2/MDCK1	640	320	640	640	1280	640	1280	2560	1280	
A/Madagascar/01535/2013		2013-04-24	MDCK2	1280	640	1280	640	1280	1280	1280	2560	2560	
A/Madagascar/01729/2013		2013-04-30	MDCK2	640	320	640	320	1280	640	640	2560	1280	
A/Madagascar/01774/2013	6C	2013-05-03	MDCK2	1280	640	1280	640	1280	1280	2560	2560	2560	
A/Madagascar/02064/2013	6C	2013-05-24	MDCK2	1280	640	1280	1280	2560	1280	2560	5120	2560	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-15. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-08-06)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	320	320	320	320	640	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	80	40	80	80	40	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	160	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E2	1280	1280	2560	5120	1280	1280	1280	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	320	640	640	1280	640	1280	2560	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	640	1280	1280	1280	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	2560	1280	2560	1280	2560	5120	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	320	640	320	640	640	640	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK1	1280	640	1280	1280	2560	2560	2560	2560	5120	
TEST VIRUSES													
A/Austria/713625/2013		2013-01-08	SIAT1/MDCK1	640	320	640	320	640	640	640	1280	1280	
A/South Africa/187/2013		2013-01-16	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	
A/Austria/718287/2013		2013-02-05	SIAT1/MDCK1	640	320	640	320	1280	640	1280	1280	1280	
A/South Africa/826/2013	6C	2013-02-19	MDCK2/MDCK1	1280	640	640	640	1280	1280	1280	5120	2560	
A/Austria/721159/2013	6C	2013-02-21	SIAT1/MDCK1	640	320	640	1280	1280	1280	1280	2560	2560	
A/Estonia/76534/2013		2013-03-11	MDCK1/SIAT2	640	320	640	320	1280	640	1280	2560	1280	
A/Estonia/76677/2013	6C	2013-03-13	MDCK2/MDCK1	1280	320	1280	640	1280	1280	1280	5120	2560	
A/Estonia/76773/2013	6C	2013-03-15	MDCK2/MDCK2	640	320	640	320	640	640	640	1280	1280	
A/Estonia/76993/2013		2013-03-21	MDCK2/SIAT2	640	320	640	320	1280	640	640	1280	640	
A/Madagascar/01430/2013	6C	2013-04-15	MDCK2	1280	320	640	640	1280	640	1280	2560	1280	
A/Madagascar/01452/2013		2013-04-18	MDCK3	1280	320	640	640	1280	640	1280	2560	2560	
A/Madagascar/01708/2013	6C	2013-04-29	MDCK3	1280	640	1280	640	2560	1280	1280	2560	2560	
A/South Africa/2118/2013	6C	2013-04-29	MDCK2/MDCK1	640	640	1280	1280	1280	1280	1280	2560	2560	
A/South Africa/2226/2013	6C	2013-04-30	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	
A/Ghana/FS-0464/2013	7	2013-05-02	MDCK3	640	640	640	640	1280	1280	1280	2560	2560	
A/Norway/2417/2013	6B	2013-05-03	MDCK2	1280	640	1280	640	1280	1280	1280	2560	2560	
A/South Africa/2303/2013	6C	2013-05-06	MDCK2/MDCK1	640	640	320	160	80	160	320	160	320	G155G>E
A/South Africa/2360/2013	6C	2013-05-07	MDCK1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	
A/South Africa/2392/2013		2013-05-08	MDCK2/MDCK1	1280	640	640	320	1280	1280	1280	1280	1280	
A/South Africa/2464/2013	6C	2013-05-13	MDCK1/MDCK1	1280	1280	1280	640	1280	1280	1280	5120	2560	
A/Madagascar/01909/2013	6C	2013-05-14	MDCK2	640	320	640	320	1280	640	1280	2560	1280	
A/Madagascar/01938/2013	6C	2013-05-17	MDCK3	640	640	640	160	640	640	640	640	640	
A/Madagascar/02007/2013	6C	2013-05-23	MDCK2	1280	640	1280	640	1280	1280	1280	5120	2560	
A/Madagascar/02001/2013	6C	2013-05-24	MDCK2	1280	320	1280	640	1280	1280	1280	2560	1280	
A/South Africa/3626/2013	6B	2013-06-06	MDCK2/MDCK1	1280	640	640	640	1280	1280	1280	2560	1280	
A/South Africa/3686/2013	6B	2013-06-10	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560	
A/South Africa/4706/2013	6C	2013-06-26	MDCK1/MDCK1	1280	640	1280	640	1280	1280	1280	2560	2560	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-16. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-08-16 + 2013-08-14)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	1280	640	640	320	320	320	320	640	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	80	40	80	80	80	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	160	160	320	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	1280	1280	1280	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	160	640	320	1280	1280	640	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E2	1280	640	1280	640	1280	1280	1280	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E2	1280	640	1280	1280	1280	1280	1280	2560	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	640	320	640	640	640	1280	1280	
TEST VIRUSES													
A/IIV-Moscow/13/2013		2013-01-11	MDCK2/MDCK2	1280	320	640	320	320	640	640	1280	640	
A/Jordan/30021/2013	6C	2013-01-03	SIAT1/MDCK1	640	320	640	640	640	640	640	1280	1280	
A/Poland/14/2013	6C	2013-01-03	MDCKx/MDCK2	640	640	640	320	1280	1280	1280	1280	1280	
A/Jordan/30052/2013		2013-01-08	SIAT1/MDCK1	640	640	640	640	1280	640	1280	1280	1280	
A/Poland/16/2013	6C	2013-01-09	Ex/E1	1280	640	640	320	160	320	320	320	320	
A/Jordan/30044/2013		2013-01-13	SIAT1/MDCK1	640	320	640	320	1280	640	640	1280	1280	
A/Jordan/30055/2013		2013-01-22	SIAT1/MDCK1	640	320	640	640	1280	640	640	1280	1280	
A/Jordan/30056/2013		2013-01-22	SIAT1/MDCK1	640	320	640	640	1280	640	640	1280	1280	
A/Jordan/30062/2013		2013-01-22	SIAT1/MDCK1	640	320	640	320	1280	640	640	1280	1280	
A/Jordan/30069/2013		2013-01-22	SIAT1/MDCK1	1280	640	1280	640	2560	1280	1280	2560	2560	
A/Jordan/30058/2013		2013-01-24	SIAT1/MDCK1	640	640	640	640	1280	640	640	1280	1280	
A/Jordan/30082/2013		2013-01-24	SIAT2/MDCK1	1280	640	640	640	1280	1280	1280	1280	1280	
A/IIV-Moscow/15/2013		2013-01-24	MDCK1/MDCK1	640	320	320	160	320	320	320	640	1280	
A/IIV-Moscow/22/2013	6C	2013-01-25	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	
A/Austria/717240/2013		2013-01-28	SIAT1/MDCK2	640	320	320	320	640	640	640	1280	2560	
A/IIV-Moscow/34/2013	6C	2013-01-28	MDCK1/MDCK1	320	320	640	160	320	320	320	320	1280	
A/IIV-Moscow/23/2013		2013-01-29	MDCK1/MDCK1	320	160	320	320	640	640	640	1280	2560	
A/Jordan/30088/2013		2013-01-30	SIAT1/MDCK1	640	320	640	320	1280	640	640	1280	1280	
A/Jordan/30089/2013		2013-01-30	SIAT1/MDCK1	1280	640	1280	640	2560	1280	1280	2560	2560	
A/Austria/717999/2013		2013-02-01	SIAT1/MDCK2	640	320	640	640	1280	1280	1280	2560	5120	
A/Austria/718985/2013	6C	2013-02-08	SIAT2/MDCK2	640	320	640	320	1280	640	640	2560	2560	
A/Jordan/30114/2013		2013-02-11	SIAT1/MDCK1	1280	640	1280	640	2560	1280	1280	2560	2560	
A/Jordan/30121/2013		2013-02-11	SIATx/MDCK1	1280	640	1280	640	2560	1280	1280	2560	2560	
A/Jordan/30203/2013	6C	2013-03-19	SIAT1/MDCK1	1280	640	640	640	1280	1280	1280	2560	2560	
A/Jordan/30326/2013	6C	2013-04-01	SIAT1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	
A/South Africa/1817/2013		2013-04-16	E3/E1	640	320	640	320	640	640	640	1280	1280	
A/South Africa/1857/2013	6C	2013-04-18	E2/E1	320	320	640	320	640	640	640	1280	2560	
A/South Africa/2226/2013	6C	2013-04-30	E1/E1	1280	1280	1280	1280	2560	1280	2560	5120	5120	
A/South Africa/2865/2013		2013-05-22	E1/E1	640	640	640	640	1280	1280	1280	2560	2560	
A/South Australia/17/2013	6B	2013-05-23	E3/E1	640	640	640	640	1280	640	640	1280	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E1	640	320	640	640	1280	1280	640	1280	2560	
A/South Africa/2273/2013		unknown	E1/E1	640	640	1280	640	1280	1280	1280	2560	5120	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Table 5-17. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-09-03)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 Group 4	A/HK 3934/11 F21/11 Group 3	A/Astrak 1/11 F22/11 Group 5	A/St. P 27/11 F23/11 Group 6	A/St. P 100/11 F24/11 Group 7	A/HK 5659/12 F30/12 Group 6A	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	640	320	640	160	320	160	160	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	160	40	40	40	80	40	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	80	160	160	320	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E2/E1	320	1280	1280	5120	1280	1280	1280	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	160	320	320	1280	640	640	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	320	640	320	320	640	640	640	1280	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	320	640	320	640	640	640	1280	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	640	320	2560	2560	1280	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	1280	320	640	640	1280	2560	1280	1280	2560	
TEST VIRUSES													
A/Lithuania/5509/2013	6C	2013-02-18	MDCK3	1280	640	640	640	1280	1280	1280	2560	2560	
A/Lithuania/5630/2013		2013-02-20	MDCK3	640	320	320	160	320	320	320	320	320	
A/Lithuania/6526/2013		2013-02-26	MDCK3	1280	640	1280	640	1280	1280	1280	5120	2560	
A/Dominican Republic/7293/2013	6C	2013-05-01	C2/MDCK1	640	160	320	320	640	640	640	1280	1280	
A/Dominican Republic/7293/2013	6C	2013-05-01	E3/E1	320	320	320	320	640	640	640	1280	1280	
A/Fukuoka-C/8/2013	7	2013-05-08	MDCK2/MDCK1/MDCK1	320	320	320	320	640	640	640	1280	1280	
A/Bolivia/559/2013	6B	2013-06-08	C2/MDCK1	640	320	640	640	1280	1280	640	1280	1280	
A/Bolivia/559/2013	6B	2013-06-08	E4/E1	640	640	640	640	1280	640	640	1280	1280	

Sequences in phylogenetic trees

1. < = <40

Vaccine

Figure 6. Phylogenetic comparison of influenza A(H1N1)pdm09 HA genes

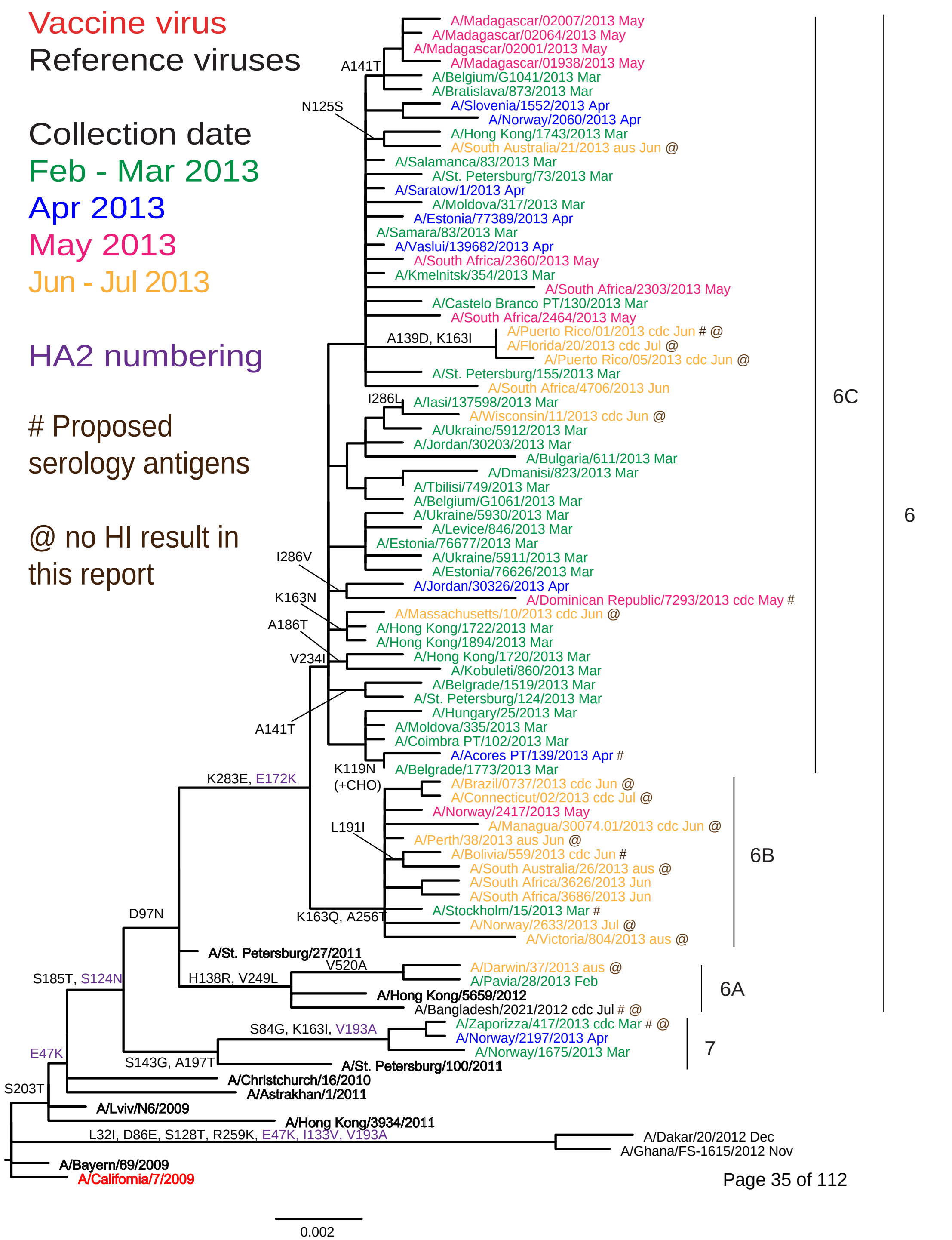


Figure 7. HA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group (Out = outlier): HA1/HA2 boundary: Potential N-linked glycosylation site

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                                    10        20        30        40        50        60        70        80        90        100
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009           DTLCIGYHANNSTDTVDTVLEKNVTVTHSVNLLEDKHNGKLCKLRGVAPLHLGKCNIAGWILGNPECESLSTASSWSYIVETPSSDNGTCYPGDFIDYEE
A/Bayern/69/2009              ..................................................................................S.................
A/Dakar/20/2012               ...............................I..................................................S..E.............. Out
A/Hong Kong/3934/2011         ..................................................................................S................. 3
A/Lviv/N6/2009                ..................................................................................S.................
A/Astrakhan/1/2011            ..................................................................................S.............N... 5
A/Christchurch/16/2010        ..................................................................................S..........N...... 4
A/St. Petersburg/100/2011     ..................................................................................S................. 7
A/Norway/2197/2013            ..................................................................................SG................ 7
A/Hong Kong/5659/2012         ..................................................................................S.............N... 6A
A/Pavia/28/2013               .........................I........................................................S.............N... 6A
A/Darwin/37/2013 aus          ..................................................................................S.............N... 6A
A/St. Petersburg/27/2011      ..................................................................................S.............N... 6
A/Stockholm/15/2013           ..............................................I...................................S.............N... 6B
A/South Africa/3686/2013      ..................................................................................S.............N... 6B
A/Norway/2417/2013            ..................................................................................S.............N... 6B
A/Acores PT/139/2013          ..................................................................................S.............N... 6C
A/Kobuleti/860/2013           ..................................................................................S.............N... 6C
A/Hong Kong/1894/2013         ........X.........................................................................S.............N... 6C
A/Jordan/30326/2013           ..................................................................................S.............N... 6C
A/Levice/846/2013             ..............................................I...................................S.............N... 6C
A/Tbilisi/749/2013            ..................................................................................S.............N... 6C
A/Iasi/137598/2013            ..................................................................................S.............N... 6C
A/South Africa/4706/2013      ..................................................................................S.............N... 6C
A/Puerto Rico/01/2013 cdc     ..................................................................................S.............N... 6C
A/South Africa/2464/2013      ..................................................................................S.............N... 6C
A/Vaslui/139682/2013          ..................................................................................S.............N... 6C
A/Estonia/77389/2013          ..................................................................................S.............N... 6C
A/Bratislava/873/2013         ..................................................................................S.............N... 6C
A/Madagascar/02064/2013       ..................................................................................S.............N... 6C

                                   110       120       130       140       150       160       170       180       190       200
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009           LREQLSSVSSFERFEIFPKTSSWPNHDSNKGVTAACPHAGAKSFYKNLIWLVKKGNSYPKLSKSYINDKGKEVLVLWGIHHPSTSADQQSLYQNADAYVF
A/Bayern/69/2009              ......................................................E.............................................
A/Dakar/20/2012               ...........................T........................................................................
A/Hong Kong/3934/2011         ..................X..............T......S.........................................P.................
A/Lviv/N6/2009                ....................................................................................................
A/Astrakhan/1/2011            ....................................................................................................
A/Christchurch/16/2010        ........................D..................................T........................................
A/St. Petersburg/100/2011     ..........................................G.........................................T...........T...
A/Norway/2197/2013            ..........................................G...................I.....................T...........T...
A/Hong Kong/5659/2012         ..................N..................R..............................................T...............
A/Pavia/28/2013               .....................................R..............................................T...............
A/Darwin/37/2013 aus          .....................................R..............................................T.....I.........
A/St. Petersburg/27/2011      ....................................................................................T...............
A/Stockholm/15/2013           ..............................................................Q.....................T...............
A/South Africa/3686/2013      ..............................................................Q.....................TT..............
A/Norway/2417/2013            ..............................................................Q.....................T...............
A/Acores PT/139/2013          ..................N.................................................................T.Y.............
A/Kobuleti/860/2013           ....................................................................................TT..............
A/Hong Kong/1894/2013         ..............................................................N.....................T...............
A/Jordan/30326/2013           .......................................................................K............T...............
A/Levice/846/2013             ....................................................................................T...............
A/Tbilisi/749/2013            ....................................................................................T...............
A/Iasi/137598/2013            .......X............................................................................T...............
A/South Africa/4706/2013      ....................................................................................T...............
A/Puerto Rico/01/2013 cdc     ......................................D.......................I.....................T...............
A/South Africa/2464/2013      ....................................................................................T...............
A/Vaslui/139682/2013          ....................................................................................T...............
A/Estonia/77389/2013          ....................................................................................T...............
A/Bratislava/873/2013         ......................................T.T...........................................T...............
A/Madagascar/02064/2013       ........................................T...........................................T...............

                                   210       220       230       240       250       260       270       280       290       300
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009           VGSSRYSKKFKPEIAIRPKVRDQEGRMNYYWTLVEPGDKITFEATGNLVVPRYAFAMERNAGSGIIISDTPVHDCNTTCQTPKGAINTSLPFQNIHPITI
A/Bayern/69/2009              ....................................................................................................
A/Dakar/20/2012               ..........................................................K...A.....................................
A/Hong Kong/3934/2011         ..T..................N........................................................................V.....
A/Lviv/N6/2009                ..T..................G..............................................................................
A/Astrakhan/1/2011            ..T.K.........XV................................L...................................................
A/Christchurch/16/2010        ..T..................N...........................A..................................................
A/St. Petersburg/100/2011     ..T...................R.............................................................................
A/Norway/2197/2013            ..T.................................................................................................
A/Hong Kong/5659/2012         ..T.............................................L...................................................
A/Pavia/28/2013               ..T...........X.................................L...................................................
A/Darwin/37/2013 aus          ..T.............................................L...................................................
A/St. Petersburg/27/2011      ..T...................R.............................................................................
A/Stockholm/15/2013           ..T....................................................T..........................E.................
A/South Africa/3686/2013      ..T....................................................T..........................E.................
A/Norway/2417/2013            ..T....................................................T..........................E.................
A/Acores PT/139/2013          ..T..............................I................................................E.................
A/Kobuleti/860/2013           ..T..............................I................................................E.................
A/Hong Kong/1894/2013         ..T..............................I................................................E.................
A/Jordan/30326/2013           ..T..............................I......................................Y.........E..V..............
A/Levice/846/2013             ..T.K............................I................................................E.................
A/Tbilisi/749/2013            ..T..............................I................................................E.................
A/Iasi/137598/2013            ..T..............................I................................................E..L..............
A/South Africa/4706/2013      ..T..............................I................................................E.................
A/Puerto Rico/01/2013 cdc     ..T..............................I................................................E.................
A/South Africa/2464/2013      ..T............................A.I................................................E.................
A/Vaslui/139682/2013          ..T..............................I................................................E.................
A/Estonia/77389/2013          ..T..............................I................................................E.................
A/Bratislava/873/2013         ..T..............................I................................................E.................
A/Madagascar/02064/2013       ..T..............................I................................................E.................
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                                     310       320       330       340       350       360       370       380       390       400
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009        GKCPKYVKSTKLRLATGLRNIPSIQSRGLFGAIAGFIEGGWTGMVDGWYGYHHQNEQGSGYAADLKSTQNAIDEITNKVNSVIEKMNTQFTAVGKEFNHL
A/Bayern/69/2009           ....................V...............................................................................
A/Dakar/20/2012            .........................................................................K..........................
A/Hong Kong/3934/2011      ....................V...............................................................................
A/Lviv/N6/2009             ....................V...............................................................................
A/Astrakhan/1/2011         ....................V....................................................K..........................
A/Christchurch/16/2010     ....................V....................................................K..........................
A/St. Petersburg/100/2011  ....................V....................................................K..........................
A/Norway/2197/2013         ....................V....................................................K..........................
A/Hong Kong/5659/2012      ....................V....................................................K..........................
A/Pavia/28/2013            ....................V....................................................K..........................
A/Darwin/37/2013 aus       ....................V....................................................K..........................
A/St. Petersburg/27/2011   ....................V....................................................K..........................
A/Stockholm/15/2013        ....................V....................................................K..........................
A/South Africa/3686/2013   ....................V....................................................K..........................
A/Norway/2417/2013         ....................V....................................................K..........................
A/Acores PT/139/2013       ....................V....................................................K..........................
A/Kobuleti/860/2013        ....................V....................................................K..........................
A/Hong Kong/1894/2013      ....................V....................................................K..........................
A/Jordan/30326/2013        ....................V....................................................K..........................
A/Levice/846/2013          ....................V....................................................K..........................
A/Tbilisi/749/2013         ....................V....................................................K..........................
A/Iasi/137598/2013         ....................V....................................................K..........................
A/South Africa/4706/2013   ....................V....................................................K..........................
A/Puerto Rico/01/2013 cdc  ....................V....................................................K..........................
A/South Africa/2464/2013   ....................V....................................................K..........................
A/Vaslui/139682/2013       ....................V....................................................K..........................
A/Estonia/77389/2013       ....................V....................................................K..........................
A/Bratislava/873/2013      ....................V....................................................K..........................
A/Madagascar/02064/2013    ....................V....................................................K..........................
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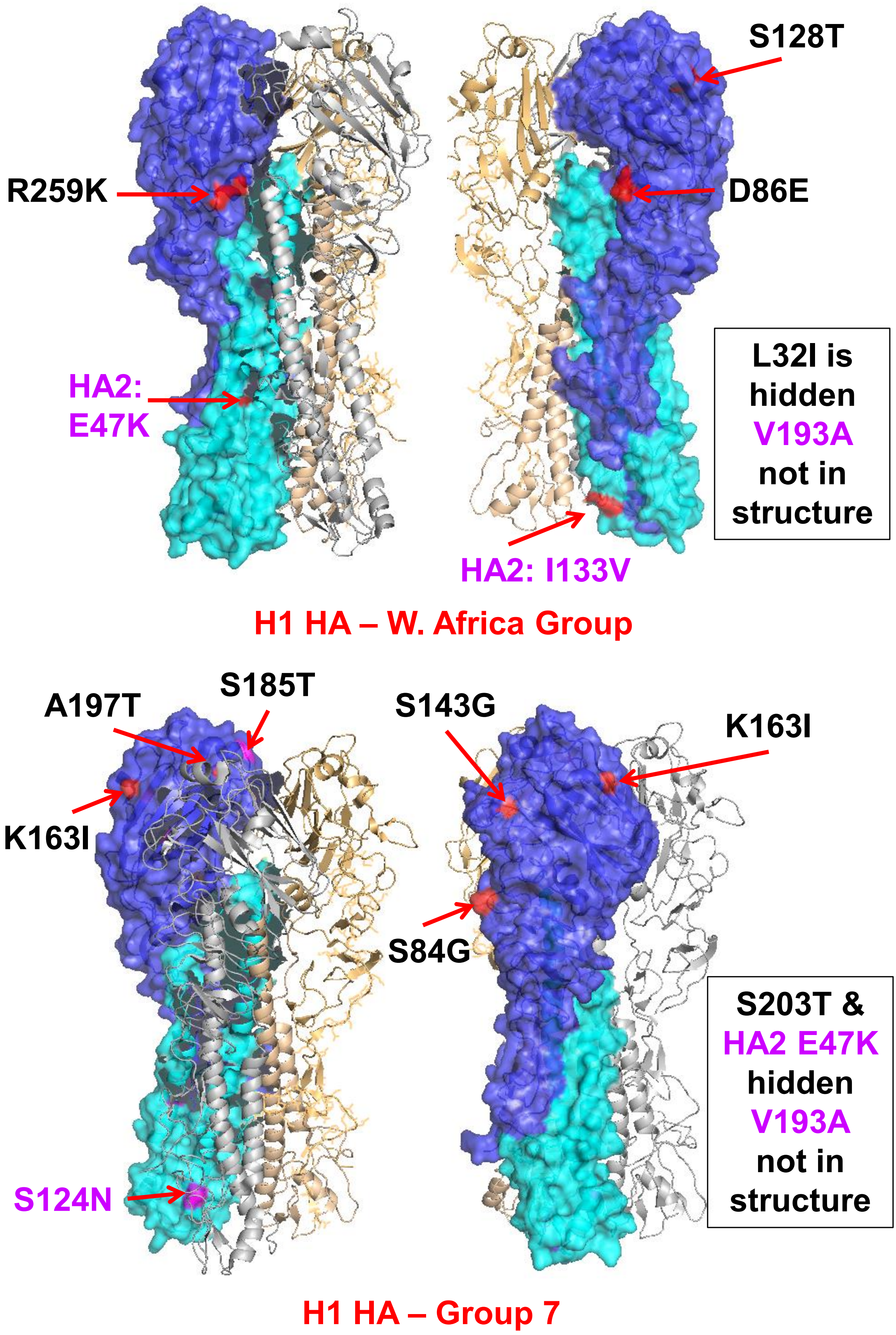
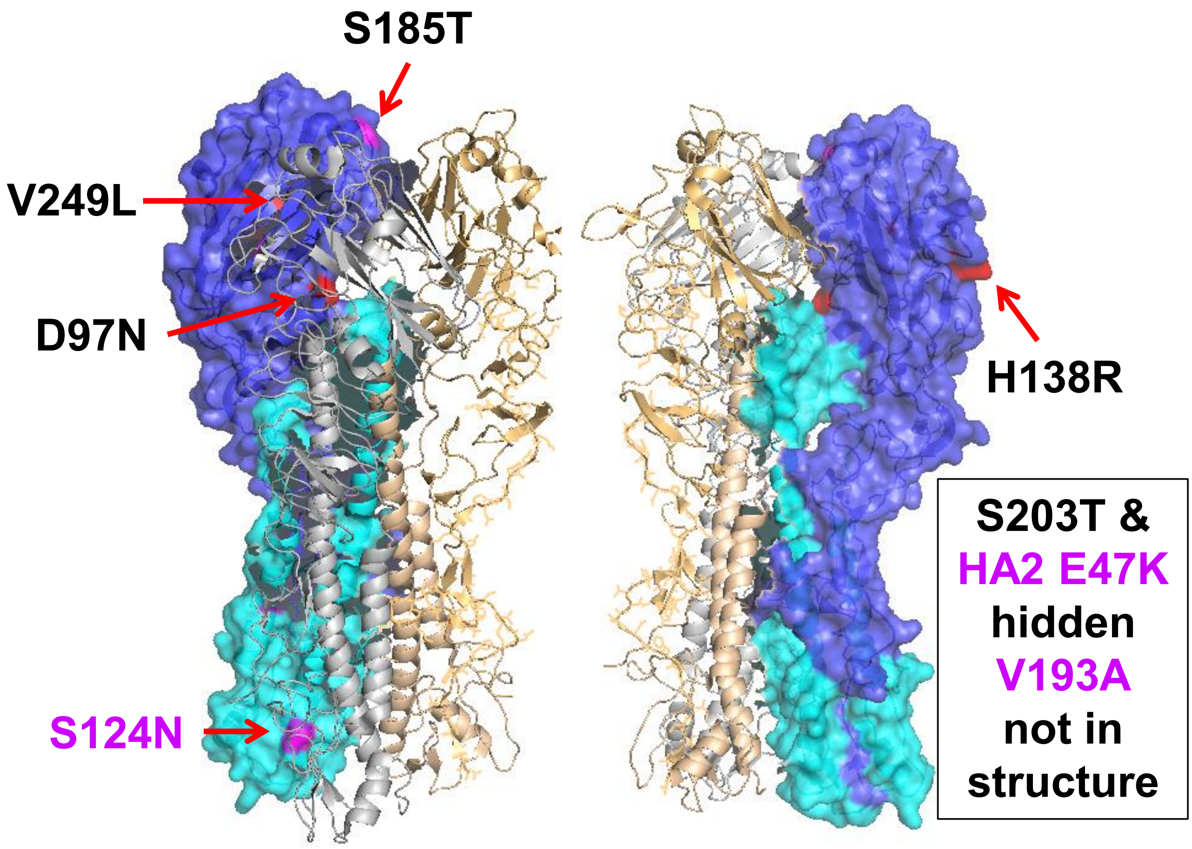
                                     410       420       430       440       450       460       470       480       490       500
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009        EKRIENLNKKVDDGFLDIWTYNAELLVLLENERTLDYHDSNVKNLYEKVRSQLKNNAKEIGNGCFEFYHKCDNTCMESVKNGTYDYPKYSEEAKLNREEI
A/Bayern/69/2009           ....................................................................................................
A/Dakar/20/2012            ...........................................................V........................................
A/Hong Kong/3934/2011      ..............................................................................I.....................
A/Lviv/N6/2009             ....................................................................................................
A/Astrakhan/1/2011         ....................................................................................................
A/Christchurch/16/2010     ....................................................................................................
A/St. Petersburg/100/2011  ..................................................N.................................................
A/Norway/2197/2013         ..................................................N.........................................S.......
A/Hong Kong/5659/2012      ..................................................N.................................................
A/Pavia/28/2013            ..................................................N.................................................
A/Darwin/37/2013 aus       ..................................................N.................................................
A/St. Petersburg/27/2011   ..................................................N.................................................
A/Stockholm/15/2013        ..................................................N...............................................K.
A/South Africa/3686/2013   ..................................................N...............................................K.
A/Norway/2417/2013         ..................................................N...............................................K.
A/Acores PT/139/2013       ..................................................N...............................................K.
A/Kobuleti/860/2013        ..................................................N...T...........................................K.
A/Hong Kong/1894/2013      ..................................................N...............................................K.
A/Jordan/30326/2013        ..................................................N...............................................K.
A/Levice/846/2013          ..................................................N...............................................K.
A/Tbilisi/749/2013         ..................................................N...............................................K.
A/Iasi/137598/2013         ..................................................N...............................................K.
A/South Africa/4706/2013   ..................................................N...............................................K.
A/Puerto Rico/01/2013 cdc  ....................................X.............N...............................................K.
A/South Africa/2464/2013   ..................................................N...............................................K.
A/Vaslui/139682/2013       ..................................................N...............................................K.
A/Estonia/77389/2013       ..................................................N...............................................K.
A/Bratislava/873/2013      ..................................................N...............................................K.
A/Madagascar/02064/2013    ..................................................N...............................................K.
```

```
                                     510       520       530       540
                           ....|....|....|....|....|....|....|....|....|....
A/California/7/2009        DGVKLESTRIYQILAIYSTVASSLVLVVSLGAISFWMCSNGSLQCRICI
A/Bayern/69/2009           .................................................
A/Dakar/20/2012            ...................A.............................
A/Hong Kong/3934/2011      .................................................
A/Lviv/N6/2009             .................................................
A/Astrakhan/1/2011         .................................................
A/Christchurch/16/2010     .................................................
A/St. Petersburg/100/2011  .................................................
A/Norway/2197/2013         ...................A.............................
A/Hong Kong/5659/2012      .................................................
A/Pavia/28/2013            ...................A....I........................
A/Darwin/37/2013 aus       ...................A.............................
A/St. Petersburg/27/2011   .................................................
A/Stockholm/15/2013        .................................................
A/South Africa/3686/2013   .................................................
A/Norway/2417/2013         .................................................
A/Acores PT/139/2013       .................................................
A/Kobuleti/860/2013        .................................................
A/Hong Kong/1894/2013      .................................................
A/Jordan/30326/2013        .................................................
A/Levice/846/2013          .................................................
A/Tbilisi/749/2013         .................................................
A/Iasi/137598/2013         .................................................
A/South Africa/4706/2013   .................................................
A/Puerto Rico/01/2013 cdc  .................................................
A/South Africa/2464/2013   .........V.......................................
A/Vaslui/139682/2013       .................................................
A/Estonia/77389/2013       .................................................
A/Bratislava/873/2013      .................................................
A/Madagascar/02064/2013    .................................................
```

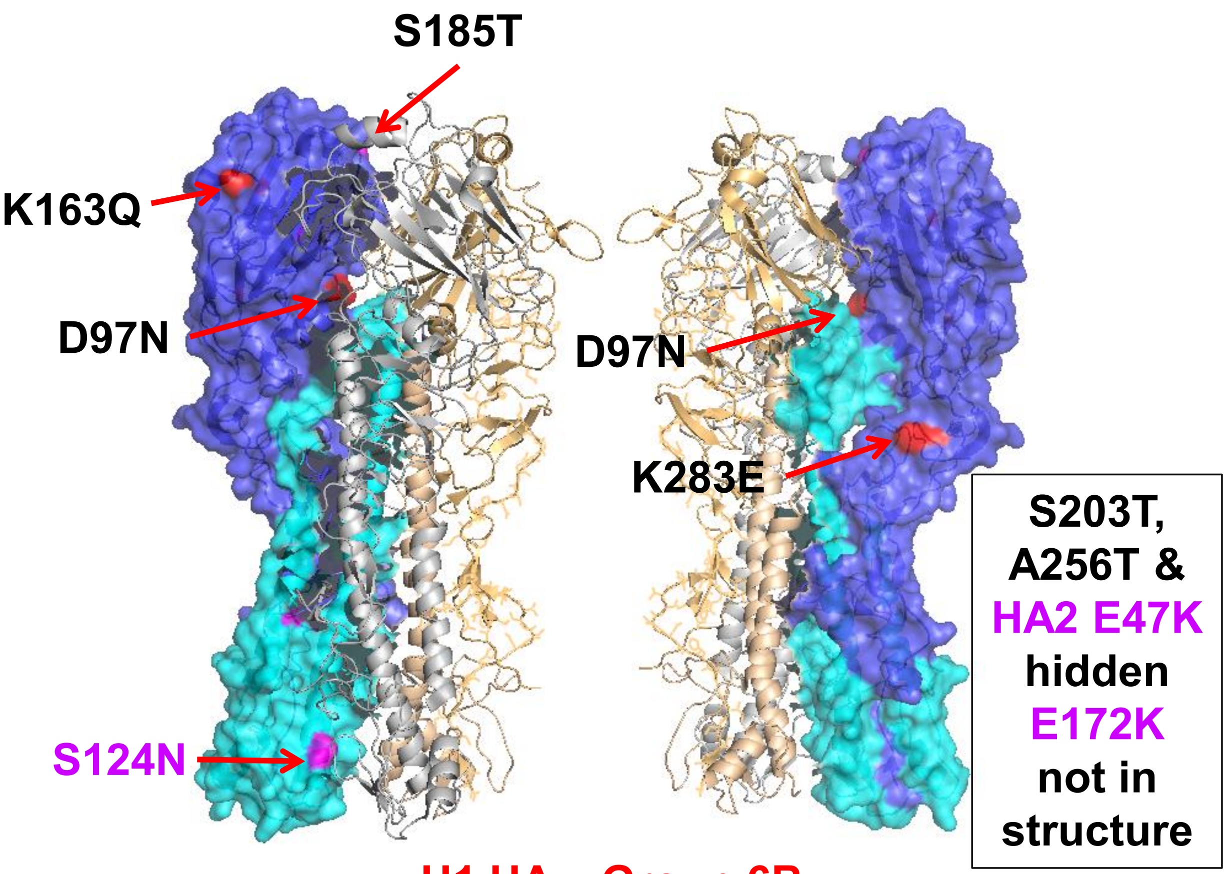
Vaccine virus: Reference virus: Genetic group (Out = outlier): HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 8. H1-HA locations of genetic group defining amino acid substitutions





H1 HA – Group 6A



H1 HA – Group 6B,
Group 6C: V234I hidden, no change at 163 or 256

Figure 9. Phylogenetic comparison of influenza A(H1N1)pdm09 NA genes

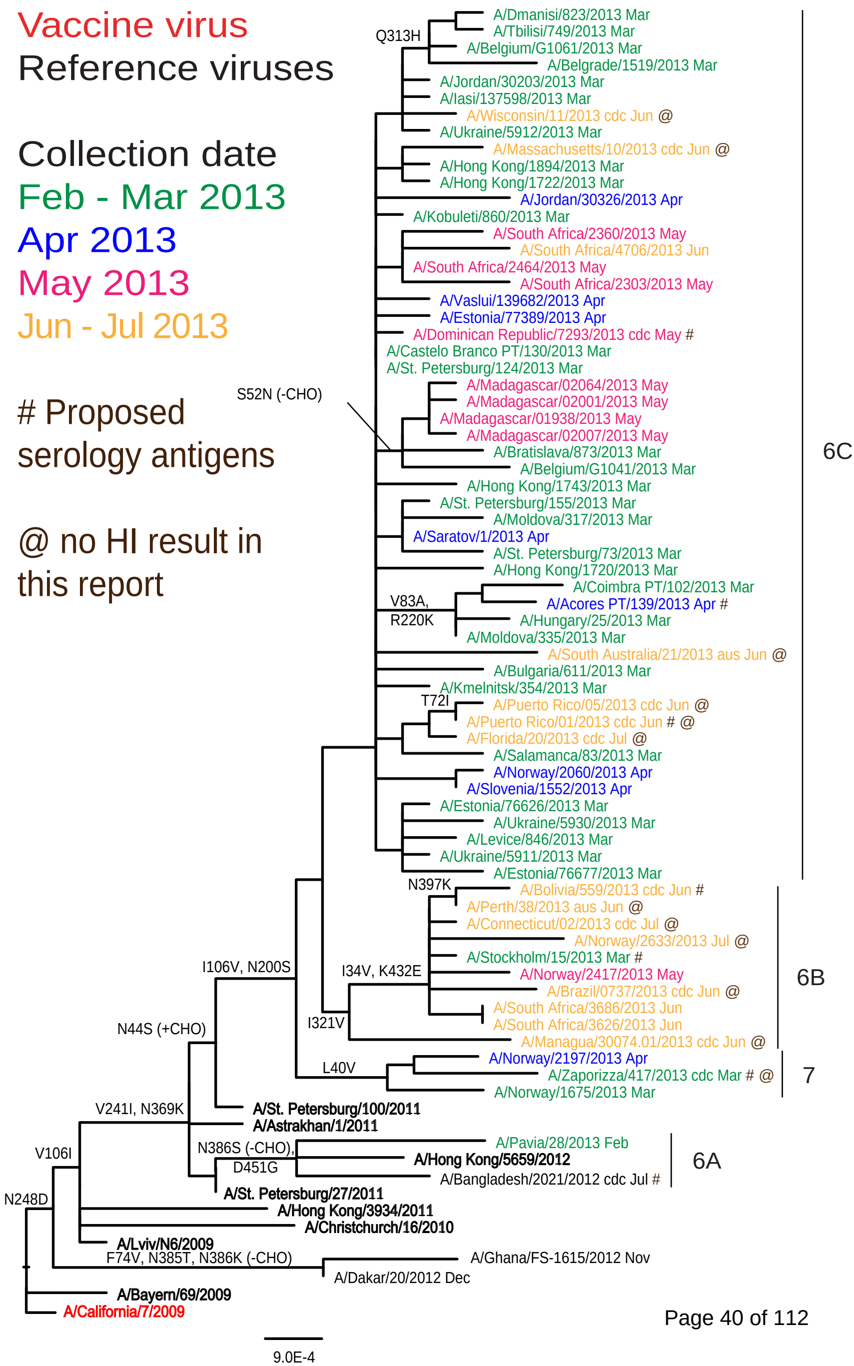


Figure 10. NA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group (Out = outlier): Potential N-linked glycosylation site

```
                                     10        20        30        40        50        60        70        80        90       100
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         MNPNQKIITIGSVCMTIGMANLILQIGNIISIWISHSIQLGNQNQIETCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLAGNSSLCPVSGWAIY
A/Bayern/69/2009            ....................................................................................................
A/Dakar/20/2012             .........................................................................V.......................... Out
A/Hong Kong/3934/2011       .............................................T...................................................... 3
A/Lviv/N6/2009              ....................................................................................................
A/Astrakhan/1/2011          .................................................................................................... 5
A/Christchurch/16/2010      ...I..........I..................................................................................... 4
A/St. Petersburg/100/2011   ...........................................S........................................................ 7
A/Norway/2197/2013          .......................................V...S........................................................ 7
A/Hong Kong/5659/2012       ..................I................................................................................. 6A
A/Pavia/28/2013             ..................V................................................................................. 6A
A/St. Petersburg/27/2011    .................................................................................................... 6
A/Stockholm/15/2013         .................................V.........S.........................N.............................. 6C
A/South Africa/3686/2013    .................................V.........S........................................................ 6C
A/Norway/2417/2013          .................................V.........S........................................................ 6C
A/Acores PT/139/2013        ...........................................S......................................A................. 6C
A/Kobuleti/860/2013         ...........................................S........................................................ 6C
A/Hong Kong/1894/2013       ...........................................S........................................................ 6C
A/Jordan/30326/2013         ...........................................S........................................................ 6C
A/Levice/846/2013           ...........................................S........................................................ 6C
A/Tbilisi/749/2013          ...........................................S........................................................ 6C
A/Iasi/137598/2013          ...........................................S........................................................ 6C
A/South Africa/4706/2013    ..................I........................S........................................................ 6C
A/Puerto Rico/01/2013       ...........................................S...........................I............................ 6C
A/South Africa/2464/2013    ...........................................S........................................................ 6C
A/Vaslui/139682/2013        ...........................................S........................................................ 6C
A/Estonia/77389/2013        ...........................................S........................................................ 6C
A/Bratislava/873/2013       ...........................................S.......N................................................ 6C
A/Madagascar/02064/2013     ...........................................S.......N................................................ 6C

                                    110       120       130       140       150       160       170       180       190       200
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         SKDNSVRIGSKGDVFVIREPFISCSPLECRTFFLTQGALLNDKHSNGTIKDRSPYRTLMSCPIGEVPSPYNSRFESVAWSASACHDGINWLTIGISGPDN
A/Bayern/69/2009            ....................................................................................................
A/Dakar/20/2012             ....................................................................................................
A/Hong Kong/3934/2011       .....I..............................................................................................
A/Lviv/N6/2009              .....I..............................................................................................
A/Astrakhan/1/2011          .....I..............................................................................................
A/Christchurch/16/2010      .....I..................................................................................S...........
A/St. Petersburg/100/2011   .....I..............................................................................................
A/Norway/2197/2013          ........................................................................K..........................S
A/Hong Kong/5659/2012       .....I..............................................................................................
A/Pavia/28/2013             .....I..............................................................................................
A/St. Petersburg/27/2011    .....I..............................................................................................
A/Stockholm/15/2013         ...................................................................................................S
A/South Africa/3686/2013    ...................................................................................................S
A/Norway/2417/2013          ........................................................................K..........................S
A/Acores PT/139/2013        ......................................................H............................................S
A/Kobuleti/860/2013         ...................................................................................................S
A/Hong Kong/1894/2013       ...................................................................................................S
A/Jordan/30326/2013         ...................................................................................................S
A/Levice/846/2013           .....................M.............................................................................S
A/Tbilisi/749/2013          ...................................................................................................S
A/Iasi/137598/2013          ...................................................................................................S
A/South Africa/4706/2013    ...................................................................................................S
A/Puerto Rico/01/2013       ..................................................X................................................S
A/South Africa/2464/2013    ...................................................................................................S
A/Vaslui/139682/2013        ...................................................................................................S
A/Estonia/77389/2013        ...................................................................................................S
A/Bratislava/873/2013       ...................................................................................................S
A/Madagascar/02064/2013     ...................................................................................................S

                                    210       220       230       240       250       260       270       280       290       300
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         GAVAVLKYNGIITDTIKSWRNNILRTQESECACVNGSCFTVMTDGPSNGQASYKIFRIEKGKIVKSVEMNAPNYHYEECSCYPDSSEITCVCRDNWHGSN
A/Bayern/69/2009            .....................................................................................G..............
A/Dakar/20/2012             ...............................................D....................................................
A/Hong Kong/3934/2011       ...............................................D....................................................
A/Lviv/N6/2009              ...............................................D....................................................
A/Astrakhan/1/2011          ........................................I......D....................................................
A/Christchurch/16/2010      ...............................................D....................................................
A/St. Petersburg/100/2011   ........................................I......D....................................................
A/Norway/2197/2013          ........................................I......D....................................................
A/Hong Kong/5659/2012       ........................................I......D....................................................
A/Pavia/28/2013             ........................................I......D...............I....I...............................
A/St. Petersburg/27/2011    ........................................I......D....................................................
A/Stockholm/15/2013         ........................................I......D....................................................
A/South Africa/3686/2013    ........................................I......D....................................................
A/Norway/2417/2013          ........................................I......D....................................................
A/Acores PT/139/2013        ...................K....................I......D....................................................
A/Kobuleti/860/2013         ........................................I......D....................................................
A/Hong Kong/1894/2013       ........................................I......D....................................................
A/Jordan/30326/2013         ........................................I......D....................................................
A/Levice/846/2013           ........................................I......D....................................................
A/Tbilisi/749/2013          ........................................I......D....................................................
A/Iasi/137598/2013          ........................................I......D....................................................
A/South Africa/4706/2013    ........................................I......D....................................................
A/Puerto Rico/01/2013       ........................................I......D....................................................
A/South Africa/2464/2013    ........................................I......D....................................................
A/Vaslui/139682/2013        ........................................I......D....................................................
A/Estonia/77389/2013        ........................................I......D....................................................
A/Bratislava/873/2013       ........................................I......D....................................................
A/Madagascar/02064/2013     ........................................I......D....................................................
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                                  310       320       330       340       350       360       370       380       390       400
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009      RPWVSFNQNLEYQIGYICSGIFGDNPRPNDKTGSCGPVSSNGANGVKGFSFKYGNGVWIGRTKSISSRNGFEMIWDPNGWTGTDNNFSIKQDIVGINEWS
A/Bayern/69/2009         ....................................................................................................
A/Dakar/20/2012          ....................................................................................TK..............
A/Hong Kong/3934/2011    ............R................................................................................I......
A/Lviv/N6/2009           ....................................................................................................
A/Astrakhan/1/2011       ....................................................................K...............................
A/Christchurch/16/2010   ........................................................................I...........................
A/St. Petersburg/100/2011 ........................................................I...........K...............................
A/Norway/2197/2013       ....................................................................K...............................
A/Hong Kong/5659/2012    ....................................................................K................S..............
A/Pavia/28/2013          ....................................................................K................S..............
A/St. Petersburg/27/2011 ....................................................................K...............................
A/Stockholm/15/2013      ....................V...............................................K...............................
A/South Africa/3686/2013 ....................V...............................................K...............................
A/Norway/2417/2013       ....................V...............................................K...............................
A/Acores PT/139/2013     ....................................................................K...............................
A/Kobuleti/860/2013      ....................................................................K...............................
A/Hong Kong/1894/2013    ....................................................................K...............................
A/Jordan/30326/2013      ............K.......................................................K...............................
A/Levice/846/2013        ....................................................................K...............................
A/Tbilisi/749/2013       ............H.......................................................K...............................
A/Iasi/137598/2013       ....................................................................K...............................
A/South Africa/4706/2013 .............................................I......................K...............................
A/Puerto Rico/01/2013    ....................................................................K...............................
A/South Africa/2464/2013 ....................................................................K...............................
A/Vaslui/139682/2013     ....................................................................K...............................
A/Estonia/77389/2013     ....................................................................K...............................
A/Bratislava/873/2013    ....................................................................K...............................
A/Madagascar/02064/2013  ....................................................................K...............................

                                  410       420       430       440       450       460
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/California/7/2009      GYSGSFVQHPELTGLDCIRPCFWVELIRGRPKENTIWTSGSSISFCGVNSDTVGWSWPDGAELPFTIDK
A/Bayern/69/2009         .....................................................................
A/Dakar/20/2012          .....................................................................
A/Hong Kong/3934/2011    ..................................................................V..
A/Lviv/N6/2009           .....................................................................
A/Astrakhan/1/2011       .....................................................................
A/Christchurch/16/2010   .....................................................................
A/St. Petersburg/100/2011 .....................................................................
A/Norway/2197/2013       .....................................................D...............
A/Hong Kong/5659/2012    ..................................................G..................
A/Pavia/28/2013          ..................................................G..................
A/St. Petersburg/27/2011 .....................................................................
A/Stockholm/15/2013      ...............................E.....................................
A/South Africa/3686/2013 ...............................E.....................................
A/Norway/2417/2013       ...............................E.....................................
A/Acores PT/139/2013     .....................................................................
A/Kobuleti/860/2013      .....................................................................
A/Hong Kong/1894/2013    .....................................................................
A/Jordan/30326/2013      .....................................................V...............
A/Levice/846/2013        .....................................................................
A/Tbilisi/749/2013       .....................................................................
A/Iasi/137598/2013       .....................................................................
A/South Africa/4706/2013 .....................................................................
A/Puerto Rico/01/2013    .....................................................................
A/South Africa/2464/2013 .....................................................................
A/Vaslui/139682/2013     .....................................................................
A/Estonia/77389/2013     .....................................................................
A/Bratislava/873/2013    .....................................................................
A/Madagascar/02064/2013  .....................................................................
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Vaccine virus: Reference virus: Genetic group (Out = outlier): Potential N-linked glycosylation site

Table 6. Summary of influenza A(H3N2) samples analysed, collected after 2013-01-31

MONTH	H3N2								
			Fold reduction in HI titre						
			A/Victoria/361/2011 antisera				A/Texas/50/2012 antisera		
Country	Number received	Number propagated[1]	≤2 fold[2]	4 fold[2]	≥8 fold[2]		≤2 fold[3]	4 fold[3]	≥8 fold[3]
FEBRUARY									
Belarus	1	1	1					1	
Belgium	2	2	2						2
Bulgaria	2	2	2				1		1
Ghana	1	0							
Hong Kong	1	1	1					1	
Italy	1	1	1						1
Kazakhstan	1	in process							
Lithuania	11	7	7					6	1
Luxembourg	4	0							
Madagascar	2	2	2					1	1
Moldova	2	2	2						2
Russia	7	7	6	1				4	3
Serbia	5	4	4						4
Slovakia	3	3	2	1				2	1
Slovenia	4	4	4				1	3	
Spain	1	1	1						1
Sweden	4	4		4				1	3
Ukraine	12	11	11				1	5	5
United Kingdom	2	2	2						2
MARCH									
Austria	1	1	1					1	
Belarus	1	1	1						1
Belgium	3	3	3						3
Estonia	6	6	6				3	2	1
Ghana	1	0							
Iceland	1	1	1				1		
Ireland	11	4	4					3	1
Kazakhstan	3	in process							
Lithuania	8	5	5				1	1	3
Madagascar	1	1	1					1	
Moldova	2	in process	1						1
Norway	9	9	9					1	8
Romania	1	1	1						1
Russia	10	10	9	1				1	9
Serbia	2	2	2						2
Slovakia	3	3	3				3		
Spain	4	4	4						4
Ukraine	20	17	16	1			1	9	7
APRIL									
Belgium	2	2	2						2
Estonia	1	1	1						1
Ghana	8	6	6					6	
Hungary									
Iceland	1	1	1						1
Ireland	5	3	1	2				1	2
Norway	4	4	4					2	2
Russia	1	1	1						1
South Africa	1	1	1					1	
Spain	1	1	1						1
Ukraine	2	1	1				1		
MAY									
Norway	2	2	1		1				2
JUNE									
Ghana	7	7	7				1	1	5
South Africa	3	3	3					1	2
JULY									
Ghana	6	6	6				1	4	1
	197	161	151	10	1		15	59	88
27 Countries			93%	6%	1%		9%	37%	54%

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

2. Compared to homologous titres for ferret antisera raised against tissue-culture grown A/Victoria/361/2011

2. Compared to homologous titres for ferret antisera raised against egg grown A/Texas/50/2012 vaccine virus

Table 7-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-02-15

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
				Haemagglutination inhibition titre[1] — Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	160	160	160	320	320	320	320	160
A/Victoria/208/2009		2009-06-02	E3/E2	640	2560	640	1280	2560	2560	5120	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	80	160	160	160	160	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	40	80	40	320	80	80	640	160	320	320	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	320	640	1280	640	1280	640	1280	2560	640
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	160	160	640	2560	1280	320	160	2560	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	160	160	320	320	320	1280	640	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	80	80	160	320	160	160	1280	640	640	1280	640
A/Athens/112/2012	3B	2012-02-01	SIAT8	40	40	80	160	80	80	640	320	640	640	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	640	1280	640	1280	1280	1280	2560	1280	1280	5120	2560
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	1280	320	640	1280	1280	2560	640	1280	5120	5120
TEST VIRUSES														
A/Christchurch/512/2012	3B	2012-07-05	E6/E2	640	2560	640	2560	5120	2560	5120	2560	5120	5120	5120
A/Cameroon/12V-4093/2012	3C.3	2012-09-05	MDCK2/SIAT1	40	40	40	160	80	80	320	320	320	320	160
A/Cameroon/12V-4221/2012		2012-09-14	MDCK2/SIAT1	40	40	40	320	80	40	320	160	320	320	160
A/Cameroon/12V-4389/2012		2012-09-19	MDCK2/SIAT1	40	40	40	320	80	40	320	160	320	320	160
A/Cameroon/12V-4653/2012		2012-09-25	MDCK1/SIAT1	<	<	<	160	40	80	320	160	160	160	160
A/Cameroon/12V-4651/2012		2012-09-26	MDCK1/SIAT1	80	160	160	640	160	160	640	320	640	640	640
A/Cameroon/12V-4647/2012		2012-09-27	MDCK1/SIAT1	40	160	80	320	80	160	640	320	640	320	320
A/Cameroon/12V-4648/2012		2012-09-27	MDCK1/SIAT1	<	<	40	160	80	80	160	160	160	320	160
A/Cameroon/12V-4666/2012		2012-10-02	MDCK1/SIAT1	40	40	40	640	80	40	320	160	320	640	160
A/Cameroon/12V-4660/2012	3C.3	2012-10-02	MDCK1/SIAT1	40	80	160	320	160	160	640	320	640	640	640
A/Cameroon/12V-4661/2012		2012-10-04	MDCK1/SIAT1	160	160	320	1280	640	320	1280	320	1280	5120	1280
A/Cameroon/12V-4685/2012	3C.3	2012-10-05	MDCK1/SIAT1	80	160	320	320	320	320	1280	640	640	1280	640
A/Cameroon/12V-4585/2012		2012-10-05	MDCK1/SIAT1	<	<	80	80	80	80	640	320	160	320	160
A/Cameroon/12V-4601/2012		2012-10-08	MDCK1/SIAT1	40	40	40	320	160	160	640	320	160	320	320
A/Cameroon/12V-4580/2012	3C.3	2012-10-10	MDCK1/SIAT1	<	40	40	160	40	80	160	160	160	320	160
A/Jordan/30502/2012	3C.1	2012-12-10	SIAT2	40	80	160	160	160	160	640	640	640	640	160
A/Netherlands/555/2012	3C.3	2012-12-17	MDCK2/SIAT1	40	160	160	320	160	40	320	320	320	640	320
A/Jordan/30536/2012		2012-12-18	SIAT2	<	40	40	80	40	40	320	320	320	320	160
A/Jordan/30539/2012	3C.3	2012-12-19	SIAT2	<	40	40	80	40	40	320	160	160	320	160
A/Netherlands/533/2012	3C.3	2012-12-20	MDCK2/SIAT1	40	160	160	320	320	160	640	320	640	640	320
A/Hong Kong/15/2013	3C.3	2012-12-25	MDCK2/SIAT1	<	160	40	160	160	40	320	160	320	320	160
A/Hong Kong/43/2013	3C.3	2013-01-02	MDCK2/SIAT1	<	40	40	160	160	40	320	160	320	160	160
A/Hong Kong/146/2013	3C.3	2013-01-11	MDCK2/SIAT1	40	40	40	160	80	40	320	160	320	160	160
A/Astrakhan/1/2013	3C.2	2013-01-28	C3/SIAT1	80	160	160	320	320	320	640	320	640	640	320
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Table 7-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-02-19 + 2013-02-26

				Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
Viruses		Collection Date	Passage History	A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens 112/12	A/Texas 50/12	A/Hawaii 22/12
				F35/11	F7/10	F27/10	F28/11	F15/11	Egg F35/12	T/C F11/12	T/C F34/12	F16/12	F36/12	F37/12
	Genetic group					group 5	group 3A	group 6	group 3C.1	group 3C.1	group 3C.1	group 3B	group 3C.1	group 3C.1
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	1280	80	160	160	160	160	320	160	640	160	160
A/Victoria/208/2009		2009-06-02	E3/E2	1280	5120	1280	2560	2560	2560	5120	2560	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	40	<	80	160	160	80	320	320	320	160	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	40	80	80	320	160	80	640	320	640	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	640	1280	640	1280	2560	1280	2560	2560	2560	5120	2560
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	160	160	640	2560	1280	640	160	5120	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	80	80	80	320	320	160	640	640	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	160	80	160	320	320	320	1280	640	640	1280	640
A/Athens/112/2012	3B	2012-02-01	SIAT7	40	80	80	160	160	160	640	320	640	640	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	320	640	320	640	640	640	2560	640	1280	5120	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	640	1280	320	1280	1280	1280	2560	1280	1280	5120	5120
TEST VIRUSES														
A/Cameroon/12V-3993/2012		2012-08-27	MDCK2/SIAT2	<	<	<	160	40	80	320	160	320	320	160
A/Cameroon/12V-4269/2012		2012-09-24	MDCK1/SIAT2	<	40	40	160	80	80	320	320	160	320	80
A/Cameroon/12V-4270/2012		2012-09-24	MDCK1/SIAT2	<	40	40	320	80	160	320	320	320	320	160
A/Cameroon/12V-4655/2012		2012-09-25	MDCK1/SIAT2	<	<	<	320	80	80	320	320	320	320	160
A/Cameroon/12V-4272/2012	3A	2012-09-25	MDCK1/SIAT2	<	80	40	160	80	80	320	80	160	160	40
A/Cameroon/12V-4412/2012		2012-09-25	MDCK1/SIAT2	<	40	40	320	80	160	640	320	320	640	160
A/Cameroon/12V-4265/2012		2012-09-25	MDCK1/SIAT2	<	40	40	160	80	80	640	320	320	640	160
A/Cameroon/12V-4652/2012		2012-09-26	MDCK1/SIAT2	40	40	40	320	80	160	640	320	320	640	160
A/Cameroon/12V-4663/2012		2012-10-01	MDCK1/SIAT2	<	<	<	320	40	80	160	160	160	320	160
A/Cameroon/12V-4665/2012		2012-10-02	MDCK1/SIAT2	40	40	40	320	80	80	320	320	320	640	160
A/Cameroon/12V-4664/2012	3A	2012-10-02	MDCK1/SIAT2	<	<	<	160	40	40	160	160	160	320	40
A/Cameroon/12V-4658/2012		2012-10-05	MDCK1/SIAT2	<	<	40	160	80	80	320	320	320	320	40
A/Cameroon/12V-4562/2012		2012-10-05	MDCK1/SIAT3	<	40	40	160	160	80	640	320	640	640	320
A/Cameroon/12V-4659/2012		2012-10-06	MDCK1/SIAT2	<	40	<	160	80	80	320	320	160	320	80
A/Cameroon/12V-5052/2012		2012-10-10	MDCK1/SIAT2	80	80	160	640	320	320	1280	1280	1280	1280	640
A/Cameroon/12V-5049/2012		2012-10-12	MDCK1/SIAT2	<	40	40	160	80	80	640	320	640	640	320
A/Cameroon/12V-5050/2012		2012-10-12	MDCK1/SIAT2	80	160	160	320	320	320	2560	1280	1280	1280	640
A/Cameroon/12V-5051/2012		2012-10-12	MDCK1/SIAT2	<	<	40	320	160	160	640	640	640	640	320
A/Cameroon/12V-5064/2012		2012-10-22	MDCK1/SIAT2	40	160	160	320	160	160	640	320	640	640	320
A/Cameroon/12V-5065/2012		2012-10-22	MDCK1/SIAT2	40	80	80	640	160	160	640	320	640	640	320
A/Cameroon/12V-5066/2012		2012-10-22	MDCK1/SIAT2	<	40	40	160	80	80	640	320	320	640	160
A/Algiers/107/2012	3C.3	2012-11-07	SIAT2	40	80	40	320	80	80	640	320	320	640	160
A/Ghana/FS-1633/2012	3C.3	2012-12-12	SIAT2	<	40	80	320	160	160	640	640	640	640	320
A/Glasgow/RVL5/2012	3C.3	2012-12-21	SIAT3	<	80	40	160	160	160	640	320	640	640	320
A/Almaty/2871/2013	3C.3	2013-01-17	SIAT2	<	40	<	160	80	80	640	160	320	640	160
A/Taraz/2918/2013	3C.3	2013-01-17	SIAT2	<	40	<	160	80	80	640	160	320	640	160
A/Taraz/2921/2013		2013-01-18	SIAT2	<	80	40	160	160	160	320	320	320	640	160
A/Taraz/2922/2013	3C.3	2013-01-18	SIAT2	<	40	40	160	80	160	640	320	320	640	160
A/Almaty/2878/2013		2013-01-20	SIAT2	<	80	40	640	160	160	640	320	640	640	320
A/Almaty/2880/2013	3C.3	2013-01-20	SIAT2	<	80	<	80	160	80	640	320	640	640	160
A/Almaty/2901/2013		2013-01-22	SIAT2	<	40	40	160	80	80	640	320	320	640	160
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Table 7-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-03-01

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	80	80	160	80	320	160	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	640	2560	640	1280	2560	2560	5120	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	160	80	160	160	320	160	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	160	160	640	320	160	1280	640	640	1280	640
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	320	640	1280	640	2560	1280	1280	2560	1280
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	320	80	640	2560	640	320	160	2560	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT3	160	160	160	320	320	320	640	640	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	160	160	320	320	320	320	1280	640	1280	1280	640
A/Athens/112/2012	3B	2012-02-01	SIAT7	80	160	160	320	320	160	640	320	640	1280	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	640	1280	320	1280	1280	1280	2560	640	1280	5120	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	1280	320	640	1280	1280	1280	640	1280	5120	2560
TEST VIRUSES														
A/Moramanga/3423/2012	3C.1	2012-05-24	MDCK2/SIAT1	160	160	160	320	320	160	1280	640	640	1280	320
A/Antananarivo/3487/2012		2012-05-31	MDCK2/SIAT1	80	80	80	320	160	160	640	320	640	1280	320
A/Mahajanga/3628/2012	3B	2012-06-01	MDCK1/SIAT1	40	160	80	320	320	160	640	320	640	640	320
A/Tsiroanomandidy/3540/2012	3B	2012-06-04	MDCK2/SIAT1	40	80	80	320	160	160	640	640	640	640	320
A/Tsiroanomandidy/3542/2012		2012-06-04	MDCK1/SIAT1	160	160	160	320	320	320	640	640	1280	2560	640
A/Antananarivo/3575/2012		2012-06-11	MDCK2/SIAT1	160	160	160	320	320	320	320	320	320	320	320
A/Antananarivo/3669/2012	3B	2012-06-13	MDCK1/SIAT1	<	<	<	80	80	80	320	640	640	640	640
A/Antananarivo/3635/2012	3B	2012-06-14	MDCK1/SIAT1	160	160	160	640	320	320	1280	640	1280	2560	640
A/Cameroon/12V-4611/2012		2012-10-09	MDCK1/SIAT4	80	160	160	640	320	320	640	320	640	1280	640
A/Cameroon/12V-5054/2012	3A	2012-10-11	MDCK1/SIAT2	80	160	160	640	320	320	640	320	640	1280	640
A/Cameroon/12V-5056/2012	3A	2012-10-15	MDCK1/SIAT2	40	80	80	320	160	160	640	160	320	640	320
A/Cameroon/12V-4995/2012		2012-10-17	MDCK1/SIAT1	160	320	320	1280	640	320	1280	640	1280	2560	640
A/Cameroon/12V-4989/2012	3C.2	2012-10-19	MDCK1/SIAT1	160	160	160	640	320	320	1280	640	1280	1280	640
A/Cameroon/12V-4990/2012		2012-10-20	MDCK1/SIAT1	80	160	160	320	320	320	1280	320	640	1280	640
A/Cameroon/12V-4996/2012		2012-10-22	MDCK1/SIAT1	80	160	160	320	320	320	640	640	640	1280	640
A/Cameroon/12V-4982/2012		2012-10-24	MDCK1/SIAT1	80	80	160	320	320	320	1280	640	320	1280	640
A/Cameroon/12V-5131/2012		2012-10-25	MDCK1/SIAT1	160	160	160	320	640	320	1280	640	1280	1280	640
A/Cameroon/12V-5140/2012		2012-10-25	MDCK1/SIAT1	160	160	160	640	320	320	1280	640	640	1280	640
A/Cameroon/12V-5143/2012		2012-10-29	MDCK1/SIAT1	80	160	160	320	320	320	1280	640	1280	1280	640
A/Cameroon/12V-5144/2012		2012-10-29	MDCK1/SIAT1	80	160	320	640	320	320	1280	640	1280	2560	640
A/Cameroon/12V-5135/2012		2012-10-30	MDCK1/SIAT1	80	160	80	320	320	160	640	640	640	1280	640
A/Cameroon/12V-5136/2012	3C.2	2012-10-31	MDCK1/SIAT1	160	160	160	640	320	320	1280	640	1280	2560	640
A/Cameroon/12V-5330/2012		2012-11-01	MDCK1/SIAT1	160	160	160	640	320	320	640	640	1280	1280	320
A/Cameroon/12V-5335/2012		2012-11-05	MDCK1/SIAT1	160	320	320	640	640	320	1280	640	1280	2560	640
A/Cameroon/12V-5334/2012	3C.2	2012-11-05	MDCK1/SIAT1	160	160	160	640	320	320	1280	320	640	1280	320
A/Cameroon/12V-5496/2012		2012-11-06	MDCK1/SIAT1	160	160	160	640	320	320	1280	1280	640	2560	640
A/Cameroon/12V-5502/2012		2012-11-08	MDCK1/SIAT1	160	160	160	320	320	320	2560	640	640	1280	320
A/Cameroon/12V-5507/2012	3C.2	2012-11-08	MDCK1/SIAT1	160	160	160	640	320	320	1280	640	1280	2560	640
A/Cameroon/12V-5498/2012	3C.2	2012-11-08	MDCK1/SIAT1	80	160	160	320	320	160	2560	640	640	640	640
A/Glasgow/RVL2/2012	3C.3	2012-12-19	SIAT3	40	80	80	160	160	160	640	320	320	320	320
A/Switzerland/7628499/2012	3C.2	2012-12-21	SIAT5	40	80	40	160	160	160	320	160	320	640	320
A/Glasgow/RVL6/2012	3C.3	2012-12-21	SIAT4	40	80	80	160	160	160	640	320	640	640	320
A/Glasgow/RVL7/2012	3C.3	2012-12-21	SIAT3	40	160	80	320	160	320	640	320	640	640	320
A/Glasgow/RVL8/2012	3C.3	2012-12-21	SIAT4	40	160	160	320	320	320	1280	320	640	1280	320
A/Ireland/00367/2013	3C.3	2013-01-02	SIAT4	<	40	40	160	160	160	320	160	320	640	160
A/Malta/MV720824/2013	3C.3	2013-01-15	MDCK3	40	160	160	160	320	320	1280	640	1280	1280	640
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Sequences in phylogenetic trees

Table 7-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-03-05

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
				A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 Egg F35/12	A/Berlin 93/11 T/C F11/12	A/Vic 361/11 T/C F34/12	A/Athens 112/12 F16/12	A/Texas 50/12 F36/12	A/Hawaii 22/12 NIBSC F102/12
	Genetic group					group 5	group 3A	group 6	group 3C.1	group 3C.1	group 3C.1	group 3B	group 3C.1	group 3C.1
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	160	160	160	320	320	320	320	160
A/Victoria/208/2009		2009-06-02	E3/E2	640	1280	640	1280	1280	1280	2560	1280	1280	5120	1280
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	40	40	80	80	160	80	160	160	320	320	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	40	80	80	320	160	80	640	320	320	640	80
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	320	1280	1280	640	1280	1280	1280	2560	640
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	320	160	640	2560	640	320	160	2560	320
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	160	160	320	320	320	1280	640	640	1280	320
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	160	80	160	320	160	160	640	320	640	1280	320
A/Athens/112/2012	3B	2012-02-01	SIAT7	160	160	160	640	160	160	640	320	640	1280	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	640	1280	320	1280	640	640	2560	640	1280	5120	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	640	320	640	1280	640	1280	640	1280	2560	2560
TEST VIRUSES														
A/Antsirabe/3734/2012		2012-06-18	MDCK2/SIAT1	40	80	80	160	40	80	160	160	320	320	80
A/Antsirabe/3733/2012		2012-06-19	MDCK1/SIAT1	40	80	160	320	320	320	640	320	640	1280	160
A/Antananarivo/3892/2012		2012-07-09	MDCK2/SIAT1	40	80	80	160	160	160	640	320	320	640	160
A/Antananarivo/4906/2012	3B	2012-10-09	MDCK1/SIAT1	<	80	<	80	<	<	160	80	160	160	<
A/Cameroon/12V-5310/2012		2012-10-30	MDCK1/SIAT1	80	160	160	640	320	320	640	640	640	1280	320
A/Cameroon/12V-5314/2012		2012-11-01	MDCK1/SIAT1	80	80	80	320	160	160	640	320	640	1280	320
A/Cameroon/12V-5316/2012	3C.2	2012-11-01	MDCK1/SIAT1	<	<	<	80	80	80	160	160	160	160	80
A/Cameroon/12V-5295/2012		2012-11-02	MDCK1/SIAT1	40	40	80	320	160	160	320	320	320	640	160
A/Cameroon/12V-5320/2012		2012-11-03	MDCK1/SIAT1	80	80	160	320	160	320	640	320	640	640	320
A/Cameroon/12V-5321/2012		2012-11-03	MDCK1/SIAT1	40	80	80	320	160	320	640	640	640	640	320
A/Cameroon/12V-5324/2012		2012-11-06	MDCK1/SIAT1	80	80	160	320	160	320	640	640	640	1280	320
A/Cameroon/12V-5325/2012		2012-11-06	MDCK1/SIAT1	80	80	80	320	160	160	640	320	640	640	320
A/Cameroon/12V-5638/2012		2012-11-15	MDCK1/SIAT1	40	80	80	320	160	160	640	320	640	640	160
A/Cameroon/12V-5653/2012		2012-11-15	MDCK1/SIAT1	40	80	80	320	160	160	640	640	640	1280	320
A/Cameroon/12V-5657/2012	3C.2	2012-11-18	MDCK1/SIAT1	<	<	<	80	40	40	160	160	160	320	80
A/Cameroon/12V-5655/2012	3C.2	2012-11-19	MDCK1/SIAT1	<	40	<	160	40	80	320	160	320	320	80
A/Cameroon/12V-5640/2012		2012-11-21	MDCK1/SIAT1	80	80	80	640	320	320	640	640	320	1280	160
A/Cairo/60/2012		2012-12-09	C1/SIAT1	80	80	160	320	320	160	640	320	640	1280	160
A/Cairo/63/2012	3B	2012-12-09	C1/SIAT1	160	80	80	160	160	160	640	320	320	640	160
A/Cairo/64/2012		2012-12-09	C1/SIAT1	320	160	160	320	320	320	1280	640	640	1280	320
A/Cairo/68/2012		2012-12-09	C1/SIAT1	80	80	160	320	320	160	640	320	640	1280	160
A/Cairo/75/2012		2012-12-09	C1/SIAT1	320	160	160	320	320	320	1280	640	640	1280	320
A/Cairo/76/2012		2012-12-09	C1/SIAT1	320	160	160	320	320	320	640	640	640	1280	320
A/Cairo/77/2012		2012-12-09	C1/SIAT1	320	160	160	320	320	320	1280	640	1280	1280	320
A/Cairo/93/2012	3B	2012-12-13	C1/SIAT1	320	160	320	640	320	320	1280	640	640	2560	320
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Table 7-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-03-08

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
				Haemagglutination inhibition titre[1] – Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	80	160	160	160	320	160	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	640	2560	640	1280	2560	2560	2560	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	80	160	160	160	160	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	160	160	640	320	320	640	640	640	640	640
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	320	640	1280	320	1280	640	1280	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	320	160	640	2560	640	320	320	2560	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	80	160	320	160	320	640	640	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	160	80	160	320	320	320	640	640	640	640	640
A/Athens/112/2012	3B	2012-02-01	SIAT7	160	160	160	640	320	320	640	640	1280	1280	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	320	1280	320	1280	1280	1280	2560	640	1280	5120	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	640	160	640	640	640	1280	640	1280	2560	2560
TEST VIRUSES														
A/Cameroon/12V-4226/2012		unknown	MDCK2/SIAT1	80	80	80	320	160	160	640	320	320	1280	320
A/Cameroon/12V-2580/2012	3A	2012-05-29	MDCK1/SIAT1	<	40	80	160	80	160	320	160	320	320	160
A/Moramanga/3694/2012		2012-06-12	MDCK1/SIAT2	320	160	320	640	320	320	1280	640	1280	640	640
A/Cameroon/12V-3841/2012		2012-08-10	MDCK1/SIAT1	80	80	80	320	160	160	320	320	320	1280	160
A/Cameroon/12V-3991/2012		2012-08-28	MDCK2/SIAT1	80	80	160	640	160	160	640	320	320	1280	320
A/Cameroon/12V-4090/2012		2012-08-31	MDCK2/SIAT1	40	80	40	320	160	160	320	320	320	320	320
A/Cameroon/12V-3990/2012		2012-09-03	MDCK2/SIAT1	80	80	80	640	160	160	640	320	640	640	320
A/Cameroon/12V-4091/2012		2012-09-05	MDCK2/SIAT1	40	40	80	320	80	160	640	320	320	640	160
A/Cameroon/12V-4219/2012		2012-09-12	MDCK2/SIAT1	40	80	80	320	160	160	320	160	640	640	320
A/Cameroon/12V-4223/2012	3A	2012-09-15	MDCK2/SIAT1	40	40	40	320	80	160	320	160	320	320	80
A/Cameroon/12V-4222/2012	3A	2012-09-17	MDCK2/SIAT1	<	40	<	160	40	80	160	80	160	160	40
A/Cameroon/12V-4268/2012		2012-09-21	MDCK1/SIAT2	80	160	80	640	320	320	1280	640	640	1280	320
A/Cameroon/12V-4569/2012		2012-10-01	SIAT2	80	160	80	640	320	160	1280	640	640	1280	640
A/Cameroon/12V-5141/2012		2012-10-25	SIAT2	40	80	80	320	320	160	640	320	640	1280	320
A/Cameroon/12V-5649/2012		2012-11-02	MDCK1/SIAT1	80	160	80	640	320	320	640	640	640	1280	320
A/Cameroon/12V-5647/2012		2012-11-07	MDCK1/SIAT1	40	80	80	320	160	160	640	320	640	640	320
A/Cameroon/12V-6099/2012		2012-11-16	MDCK1/SIAT1	80	160	160	640	320	320	1280	640	640	2560	640
A/Cameroon/12V-5639/2012		2012-11-16	SIAT2	40	80	80	320	160	160	640	320	640	640	160
A/Cameroon/12V-5918/2012	3C.2	2012-11-28	SIAT2	80	80	160	320	160	160	640	640	640	1280	320
A/Cameroon/12V-6178/2012		2012-12-12	MDCK1/SIAT1	40	40	40	320	80	160	640	320	320	640	320
A/Aktobe/2941/2013		2013-01-21	SIAT2	40	80	80	320	160	160	640	320	640	1280	160
A/Aktobe/2935/2013	3C.3	2013-01-22	SIAT2	<	40	40	160	80	80	320	160	160	320	160
A/Almaty/2926/2013		2013-01-23	SIAT2	40	80	80	320	160	160	640	640	640	640	320
A/Almaty/2927/2013		2013-01-23	SIAT2	40	160	80	320	320	160	640	640	640	1280	320
A/Almaty/2929/2013	3C.3	2013-01-24	SIAT2	40	40	40	320	80	160	640	160	640	640	320
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Table 7-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-03-22

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1	A/Trieste 1/13 NIBSC F15/13 group 3C.3	A/Neth 555/12 NIBSC F16/13 group 3C.3
				Haemagglutination inhibition titre[1] — Post infection ferret antisera												
REFERENCE VIRUSES																
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	160	160	160	320	320	320	320	160	80	40
A/Victoria/208/2009		2009-06-02	E3/E2	1280	2560	640	1280	2560	2560	2560	1280	2560	5120	2560	640	640
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	40	40	80	160	80	160	160	160	320	320	80	40	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	160	80	320	160	320	640	320	640	640	320	320	80
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	320	1280	2560	640	2560	640	1280	2560	640	320	160
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	160	1280	160	160	640	1280	640	320	160	2560	640	160	80
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT3	160	160	160	320	320	640	1280	640	640	2560	640	320	160
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	160	160	80	320	160	160	640	320	640	640	320	160	80
A/Athens/112/2012	3B	2012-02-01	SIAT8	80	160	160	320	160	160	640	320	640	640	320	320	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	320	1280	640	1280	2560	640	1280	5120	1280	640	160
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	640	320	640	1280	640	1280	640	1280	5120	2560	320	160
A/Trieste/1/2013	3C.3	2013-01-14	Cx/SIAT5	<	80	80	160	80	160	640	320	640	640	320	160	80
A/Netherlands/555/2012	3C.3	2012-12-17	MDCK2/SIAT3	40	160	160	160	80	160	640	320	640	640	320	320	160
TEST VIRUSES																
A/Cameroon/12V-4257/2012		2012-09-07	MDCK1/SIAT3	80	80	160	320	160	160	640	320	640	640	320	320	320
A/Cameroon/12V-4224/2012		2012-09-12	MDCK2/SIAT2	40	80	80	320	160	160	640	320	640	640	320	160	320
A/Cairo/96/2012		2012-12-13	C1/SIAT1	320	160	160	320	320	320	1280	640	640	2560	640	320	80
A/Cairo/132/2012	3B	2012-12-18	C1/SIAT1	320	160	320	320	320	320	1280	640	640	640	640	320	160
A/Cairo/133/2012		2012-12-18	C1/SIAT1	320	160	320	640	320	320	1280	640	640	1280	640	320	160
A/Cairo/134/2012	3B	2012-12-18	C1/SIAT1	640	320	320	320	320	320	1280	640	1280	2560	640	320	160
A/Cairo/148/2013		2013-01-02	C1/SIAT1	320	160	320	320	320	320	640	640	640	1280	320	320	160
A/Cairo/152/2013	3B	2013-01-02	C1/SIAT1	320	160	160	320	160	320	640	320	640	1280	640	320	160
A/Cairo/153/2013	3B	2013-01-02	C1/SIAT1	320	160	320	320	320	640	640	320	640	1280	640	320	160
A/Cairo/157/2013		2013-01-09	C1/SIAT1	80	160	160	320	320	320	1280	320	640	1280	320	160	80
A/Cairo/158/2013	3B	2013-01-09	C1/SIAT1	320	160	320	320	160	160	320	320	640	1280	320	320	80
A/Cairo/171/2013	3C.1	2013-01-14	C1/SIAT1	160	160	320	320	320	320	1280	320	640	1280	320	320	80
A/Catalonia/5674S/2013	3C.2	2013-01-15	SIAT1	40	80	80	160	80	80	320	320	640	640	160	160	80
A/Catalonia/5689S/2013	5	2013-01-17	SIAT1	160	160	320	640	640	320	2560	320	1280	2560	320	320	80
									Vaccine 2012-13				Vaccine 2013-14			

1. < = <40

Table 7-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-03-26 + 2013-04-17 + 2013-05-14

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] – Post infection ferret antisera: A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	80	160	80	160	320	320	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	1280	2560	640	2560	2560	2560	5120	1280	2560	5120	5120
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	80	160	160	160	160	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	80	80	320	80	320	640	320	320	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	320	1280	1280	1280	2560	1280	2560	5120	1280
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	160	640	160	160	320	2560	640	320	160	2560	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	40	40	40	160	40	80	320	160	320	320	160
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	160	80	80	320	160	320	640	640	640	1280	320
A/Athens/112/2012	3B	2012-02-01	SIAT6	80	80	80	160	80	160	640	320	640	640	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	320	1280	1280	1280	2560	1280	2560	5120	2560
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	1280	320	640	1280	1280	2560	1280	1280	5120	5120
TEST VIRUSES														
A/Moscow/84/2012	3C.2	2012-12-25	Cx/SIAT1	40	80	40	160	80	80	640	320	320	640	320
A/England/119/2013	3C.3	2013-01-09	MDCK2/SIAT1	40	80	80	160	80	160	640	160	160	640	320
A/England/104/2013	3C.3	2013-01-09	MDCK2/SIAT1	40	40	80	160	40	160	640	640	640	640	320
A/Valencia/1S/2013	3C.2	2013-01-10	SIAT2	<	40	40	160	80	80	320	160	320	640	160
A/Serbia/NS-200/2013	3C.3	2013-01-14	SIAT2	80	<	80	160	80	80	1280	320	640	640	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	SIAT3	80	<	80	160	80	80	640	160	320	160	160
A/Catalonia/2070282NS/2013	3C.2	2013-01-22	SIAT2	<	80	40	160	80	80	320	320	320	640	160
A/Catalonia/5742S/2013	3C.3	2013-01-23	SIAT2	<	40	40	160	80	80	320	320	320	640	160
A/Catalonia/2071057NS/2013	3C.3	2013-01-24	SIAT2	<	40	40	160	80	80	320	160	320	640	160
A/England/188/2013	3C.3	2013-01-29	SIAT1/SIAT1	40	40	40	320	40	160	640	640	640	640	640
A/England/256/2013	3C.2	2013-01-30	MDCK1/SIAT1	40	40	40	160	40	160	1280	640	640	640	320
A/England/280/2013	3C.3	2013-01-31	MDCK1/SIAT1	40	80	80	320	160	640	1280	640	640	1280	640
A/Bulgaria/270/2013	3C.3	2013-02-04	C2/SIAT1	320	160	160	640	160	320	1280	640	1280	2560	640
A/Bulgaria/253/2013	3C.3	2013-02-08	C2/SIAT1	40	80	80	320	80	160	640	320	640	640	320
A/England/279/2013	3C.3	2013-02-09	SIAT1/SIAT1	40	40	40	160	40	160	640	640	640	640	320
A/St. Petersburg/4/2013	3C.2	2013-02-13	C1/SIAT1	40	80	80	320	80	320	640	640	640	640	320
A/St. Petersburg/6/2013	3C.1	2013-02-13	C1/SIAT1	40	80	160	320	160	160	640	640	640	1280	320
A/Valladolid/39/2013	3C.1	2013-02-14	SIAT2	320	<	160	320	160	160	1280	640	640	640	1280
A/England/308/2013	3C.3	2013-02-15	SIAT1/SIAT1	<	40	40	160	40	80	640	320	320	320	320
A/Novosibirsk/77/2012	3C.3	2013-02-20	C4/SIAT1	80	160	160	640	160	320	1280	640	1280	1280	1280
A/Astrakhan/2/2013	3C.3	2013-02-20	C1/SIAT1	<	40	40	80	40	40	320	160	160	320	160
A/Astrakhan/5/2013	3C.3	2013-02-20	C1/SIAT1	40	80	80	320	80	160	1280	640	640	1280	640
A/St. Petersburg/36/2013	3C.2	2013-02-20	C2/SIAT1	40	40	80	320	80	160	1280	640	640	1280	640
A/Novosibirsk/78/2012	3C.3	2013-02-20	C4/SIAT1	40	80	40	160	40	160	320	320	320	640	320
A/Burgos/96/2013	3C.3	2013-03-12	SIAT2	160	<	80	640	160	80	640	320	640	640	2560
A/Valladolid/95/2013	3C.3	2013-03-14	SIAT2	160	<	80	320	160	160	640	320	640	640	320
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Sequences in phylogenetic trees

Table 7-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-05-21

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F11/13 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	80	80	80	80	320	160	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	640	2560	640	1280	1280	1280	2560	2560	1280	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	40	<	80	80	80	80	160	160	320	160	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	80	40	160	80	80	320	160	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	320	640	1280	640	2560	1280	1280	2560	640
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	640	640	160	160	320	2560	640	640	160	2560	640
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	80	80	320	160	160	640	320	640	640	320
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT3	160	80	80	320	160	160	640	320	640	640	320
A/Athens/112/2012	3B	2012-02-01	SIAT7	80	80	80	320	160	160	640	320	640	640	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	640	1280	320	1280	640	1280	2560	640	2560	5120	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	640	320	640	640	640	1280	640	1280	2560	5120
TEST VIRUSES														
A/Zamora/10/2012		2012-01-19	SIAT3	<	80	40	160	80	80	320	320	320	320	160
A/Spain/13831/2012		2012-02-16	SIAT3	80	160	160	320	160	160	640	320	640	1280	320
A/Spain/13916/2012		2012-02-17	SIAT3	80	80	80	160	80	160	320	320	320	640	160
A/Spain/14701/2012		2012-02-21	SIAT3	40	80	80	160	80	40	320	320	320	640	160
A/Spain/14758/2012	3C.1	2012-02-21	SIAT3	40	80	80	320	160	160	640	320	320	640	160
A/Spain/20322/2012		2012-03-10	SIAT3	40	160	160	320	160	160	640	320	640	640	320
A/Spain/23634/2012		2012-03-22	SIAT3	80	160	160	320	160	160	1280	640	640	1280	320
A/Zamora/87/2012	3B	2012-04-03	SIAT3	<	80	40	160	80	80	320	320	320	640	160
A/Plzen/17/2013	3C.3	2013-01-03	MDCK4/SIAT1	<	40	40	160	40	80	320	160	320	320	160
A/Tulkarem/83/2013	3C.3	2013-01-08	SIAT3	<	40	40	160	40	80	320	160	320	320	160
A/Plzen/22/2013	5	2013-01-10	MDCK4/SIAT1	80	160	160	320	160	160	640	320	320	640	320
A/Slovenia/218/2013	3C.3	2013-01-17	MDCK1/SIAT1	<	80	40	160	40	160	320	320	320	640	160
A/Skopje/DZ K 33/2013	3C.2	2013-01-21	SIAT3	40	80	40	160	80	80	320	160	320	320	160
A/Almaty/2944/2013		2013-01-24	SIAT3	<	40	40	80	40	80	320	160	320	320	160
A/Almaty/2946/2013		2013-01-25	SIAT3	<	40	40	80	40	80	320	160	320	320	160
A/Slovenia/388/2013	3C.3	2013-01-25	MDCKx/SIAT1	40	80	80	160	80	160	320	320	640	640	160
A/Ramallah/84/2013	3C.2	2013-01-26	SIAT3	<	80	80	160	80	80	320	320	320	640	320
A/Almaty/2952/2013		2013-01-27	SIAT3	<	80	40	160	40	80	640	320	320	640	160
A/Almaty/2955/2013		2013-01-27	SIAT3	<	80	40	160	40	80	320	320	320	640	160
A/Almaty/2956/2013		2013-01-27	SIAT3	<	80	80	160	80	80	640	320	320	640	160
A/Almaty/2958/2013	3C.3	2013-01-27	SIAT3	<	40	40	160	40	80	320	320	320	320	160
A/Slovenia/466/2013	3C.3	2013-01-30	MDCKx/SIAT1	40	80	40	160	80	160	320	320	320	640	160
A/Slovenia/549/2013	3C.2	2013-02-04	MDCK1/SIAT1	40	80	40	160	80	160	320	160	320	640	160
A/Slovenia/709/2013	3C.2	2013-02-12	MDCK1/SIAT1	160	640	320	640	160	640	1280	640	1280	2560	1280
A/Slovenia/741/2013	3C.2	2013-02-14	MDCKx/SIAT1	40	160	80	320	80	160	640	320	640	1280	640
A/Slovenia/760/2013	3C.3	2013-02-15	MDCKx/SIAT1	<	80	80	320	80	160	640	320	640	1280	640
A/Spain/18719/2013		2013-03-03	SIAT4	40	80	80	320	160	160	640	320	640	1280	320
A/St. Petersburg/69/2013	3C.2	2013-03-12	C2/SIAT1	40	80	40	160	40	160	320	320	640	640	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT1	80	160	80	320	160	320	640	320	640	1280	320
A/Kaliningrad/1/2013	3C.2	2013-03-12	C1/SIAT1	80	160	80	320	160	320	640	320	640	640	320
A/St. Petersburg/125/2013		2013-03-12	C1/SIAT1	<	80	40	160	40	160	320	320	320	640	160
A/St. Petersburg/106/2013		2013-03-12	C1/SIAT1	40	80	80	160	80	160	640	320	640	640	320
A/St. Petersburg/86/2013		2013-03-13	C2/SIAT1	<	80	80	160	80	160	320	320	320	640	160
A/Valladolid/98/2013	3C.1	2013-03-18	SIAT3	40	160	80	320	160	160	640	320	320	640	320
A/St. Petersburg/170/2013	3C.2	2013-03-20	C2/SIAT1	<	40	<	80	<	40	160	160	160	320	160
A/Valladolid/118/2013	3C.2	2013-04-02	SIAT3	<	80	80	160	80	80	320	320	320	320	160
A/Novosibirsk/32/2013	3C.3	2013-04-03	C1/SIAT1	40	80	40	160	80	160	320	320	320	640	160
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Sequences in phylogenetic trees

Table 7-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-06-07 + 2013-06-11

				Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F14/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	80	80	80	80	320	320	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	320	1280	320	640	1280	1280	2560	1280	1280	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	80	160	320	160	160	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	80	80	160	80	160	320	320	320	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	640	2560	1280	1280	2560	1280
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	320	80	640	2560	640	640	160	2560	640
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	80	80	320	320	160	160	640	640	640	1280	320
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	40	80	80	160	80	160	640	640	640	320	160
A/Athens/112/2012	3B	2012-02-01	SIAT7	80	80	160	320	160	320	640	640	1280	1280	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	320	640	320	1280	640	1280	2560	1280	1280	5120	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	640	320	640	1280	640	2560	1280	1280	5120	5120
TEST VIRUSES														
A/Firenze/13/2013	3C.3	unknown	MDCK2/SIAT1	<	80	80	160	80	160	640	640	640	640	320
A/Pozarevac/6330/2013	3C.3	2012-12-21	C2/SIAT1	<	<	<	80	<	40	160	160	160	160	80
A/Perugia/13/2013		2013-01-28	MDCK1/SIAT1	<	80	80	160	80	160	640	640	640	640	320
A/Ukraine/5809/2013		2013-02-13	C2/SIAT1	40	40	80	320	80	160	640	640	640	640	160
A/Ukraine/5887/2013		2013-02-13	C2/SIAT1	40	80	160	160	40	160	640	640	320	640	320
A/Ukraine/5810/2013		2013-02-17	C2/SIAT1	40	80	160	320	160	320	1280	1280	640	1280	320
A/Ukraine/5811/2013		2013-02-17	C2/SIAT1	80	160	160	640	160	320	1280	1280	640	2560	320
A/Ukraine/5884/2013		2013-02-19	C2/SIAT1	80	160	320	160	320	320	640	640	320	1280	160
A/Ukraine/5890/2013		2013-02-20	C2/SIAT1	40	40	160	320	160	160	640	640	320	1280	160
A/Moldova/242/2013		2013-02-22	MDCK2/SIAT1	40	80	40	320	80	80	320	320	320	320	160
A/Belgrade/1166/2013	3C.2	2013-02-24	C1/SIAT2	<	40	40	160	80	80	640	640	320	640	160
A/Ukraine/5892/2013		2013-02-24	C2/SIAT1	40	80	160	320	80	160	1280	640	640	1280	320
A/Ukraine/5893/2013		2013-02-25	C2/SIAT1	40	80	320	320	80	320	1280	1280	640	1280	320
A/Ukraine/5896/2013	3C.1	2013-03-04	C2/SIAT1	80	80	320	320	160	320	1280	1280	640	1280	320
A/Timis/136369/2013		2013-03-04	MDCK3/SIAT1	<	80	80	320	80	160	640	640	640	640	160
A/Ukraine/5837/2013	3C.2	2013-03-06	C2/SIAT1	<	<	<	80	<	80	640	640	320	640	160
A/Ukraine/5909/2013	3C.2	2013-03-06	C2/SIAT1	<	40	40	160	80	80	640	640	320	320	320
A/Moldova/326/2013	3C.3	2013-03-07	MDCK2/SIAT1	<	80	40	160	80	80	320	320	160	320	160
A/Ukraine/288/2013	3C.3	2013-03-13	SIAT1/SIAT1	<	40	40	80	40	80	640	320	320	320	320
A/Leskovac/1653/2013	3C.3	2013-03-13	C2/SIAT1	<	40	<	160	<	80	320	320	160	320	160
A/Leskovac/1655/2013		2013-03-13	C1/SIAT1	<	40	40	160	40	80	320	320	160	320	160
A/Ukraine/316/2013	3C.3	2013-03-20	MDCK1/SIAT1	<	80	80	160	80	160	640	640	640	640	320
A/Ukraine/315/2013		2013-03-23	MDCK1/SIAT1	<	80	80	160	80	160	640	640	320	640	320
A/Ukraine/5840/2013	3C.2	2013-03-25	C1/SIAT1	<	<	<	80	<	80	320	320	320	640	160
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Sequences in phylogenetic trees

Table 7-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-06-14 + 2013-06-18 + 2013-06-25

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F14/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
				Haemagglutination inhibition titre[1] — Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	320	40	80	80	80	80	320	320	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	320	1280	320	640	1280	1280	2560	1280	1280	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	80	160	320	160	160	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	80	80	160	80	160	320	320	320	640	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	640	2560	1280	1280	2560	1280
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	320	80	640	1280	640	640	160	2560	640
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	80	80	320	320	160	160	640	640	640	1280	320
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	40	80	80	160	80	160	640	640	640	320	160
A/Athens/112/2012	3B	2012-02-01	SIAT7	80	80	160	320	160	320	640	640	1280	1280	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	320	640	320	1280	640	1280	2560	1280	1280	2560	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	640	320	640	1280	640	2560	1280	1280	5120	5120
TEST VIRUSES														
A/Iceland/27/2012		2012-11-20	MDCK2/SIAT1	40	80	80	160	80	160	640	320	320	320	320
A/Odessa/175/2013		2013-01-09	MDCK2/SIAT1	40	80	160	160	160	160	640	640	640	640	320
A/Iceland/14/2013		2013-01-10	MDCK1/SIAT1	40	80	80	320	80	160	640	640	640	640	320
A/Ukraine/5725/2013		2013-01-21	C2/MDCK1/SIAT1	40	40	40	80	80	160	640	320	320	640	80
A/Ukraine/181/2013		2013-01-21	MDCK1/SIAT1	40	40	160	160	160	160	320	640	320	320	160
A/Parma/27/2013		2013-01-25	MDCK2/SIAT2	<	80	80	160	80	160	640	320	640	640	320
A/Perugia/14/2013		2013-01-27	MDCK2/SIAT2	<	40	80	160	80	160	640	320	640	640	320
A/Trieste/12/2013	3C.3	2013-01-28	SIAT2	<	80	80	160	80	160	1280	320	640	640	320
A/Iceland/30/2013	3C.3	2013-01-28	MDCKx/SIAT1	<	40	40	160	40	80	320	320	320	320	160
A/Belgrade/649/2013		2013-02-05	C2/SIAT3	<	40	40	160	40	160	640	320	320	640	320
A/Belgrade/650/2013		2013-02-05	C2/SIAT3	40	40	40	160	80	160	640	320	640	640	320
A/Belgrade/648/2013	3C.2	2013-02-05	C2/SIAT2	<	<	<	80	40	80	640	320	320	320	320
A/Ukraine/5812/2013		2013-02-17	C2/SIAT2	<	40	40	80	40	80	640	320	320	640	320
A/Milano/28/2013	3C.3	2013-02-19	MDCK1/SIAT3	<	40	40	160	80	80	640	320	320	320	160
A/Ukraine/211/2013		2013-02-20	SIAT1/SIAT2	<	40	80	160	80	160	1280	320	640	640	320
A/Moldova/235/2013		2013-02-21	MDCK2/SIAT3	<	<	<	160	80	160	640	320	640	640	320
A/Dnipropetrovsk/257/2013	3C.2	2013-03-01	MDCK1/SIAT2	<	40	80	160	80	320	1280	320	640	640	320
A/Ukraine/5906/2013	3C.2	2013-03-05	C2/SIAT3	<	40	40	160	80	40	640	320	320	640	160
A/Dnipropetrovsk/361/2013	3C.2	2013-03-06	MDCK1/SIAT1	40	80	80	160	40	160	640	320	320	640	320
A/Ukraine/275/2013	3C.2	2013-03-08	SIAT1/SIAT2	<	40	<	160	80	160	640	320	640	640	320
A/Ukraine/263/2013		2013-03-11	SIAT1/SIAT2	<	40	<	160	80	160	640	320	640	640	320
A/Dnipropetrovsk/390/2013	3C.2	2013-03-15	MDCK1/SIAT1	40	80	80	320	80	160	640	640	640	640	320
A/Dnipropetrovsk/364/2013	3C.2	2013-03-15	MDCK1/SIAT1	40	40	80	160	80	160	640	640	320	320	320
A/Dnipropetrovsk/369/2013		2013-03-22	MDCK1/SIAT1	40	40	80	320	80	160	320	640	640	640	320
A/Kmelnitsk/477/2013		2013-03-25	MDCK1/SIAT1	80	160	160	320	160	320	1280	1280	320	1280	640
A/Ukraine/466/2013	3C.2	2013-04-18	MDCK1/SIAT2	40	160	160	320	160	320	1280	640	1280	1280	320
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Sequences in phylogenetic trees

Table 7-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-06-28 + 2013-07-02 + 2013-07-09

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F11/12 group 3C.1	A/Vic 361/11 T/C F14/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
				Haemagglutination inhibition titre[1] – Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	80	160	160	160	320	320	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	640	5120	640	1280	1280	2560	5120	2560	1280	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	<	80	80	80	80	160	320	160	160	40
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	80	40	320	80	160	320	320	320	320	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	640	1280	1280	640	2560	1280	1280	2560	1280
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	320	160	640	2560	1280	640	320	2560	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	80	160	160	320	160	320	1280	1280	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	80	160	160	320	320	160	1280	1280	640	640	320
A/Athens/112/2012	3B	2012-02-01	SIAT7	40	40	80	160	80	80	320	640	640	320	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	640	1280	640	1280	640	1280	5120	1280	1280	5120	2560
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	640	1280	640	640	1280	1280	2560	2560	1280	5120	5120
TEST VIRUSES														
A/Iceland/25/2012	3C.2	2012-10-17	MDCK1/SIAT3	40	80	160	320	160	160	640	640	640	640	320
A/Iceland/25/2012	3C.2	2012-10-17	SIAT2	160	160	80	320	80	160	1280	640	640	1280	320
A/Iceland/16879/2012		2012-12-31	SIAT3	80	80	160	320	80	160	640	640	640	640	320
A/Iceland/02/2013	3C.2	2013-01-02	MDCKx/SIAT3	40	80	80	320	80	160	640	640	640	640	320
A/Iceland/04/2013	3C.3	2013-01-04	SIAT2	160	160	160	320	160	160	1280	1280	640	1280	640
A/Madeira PT/12/2013	3C.2	2013-01-07	SIAT4/SIAT1	80	160	160	320	160	160	640	320	640	1280	320
A/Stockholm/1/2013		2013-01-13	MDCK1/SIAT1	40	80	80	320	160	160	640	320	640	1280	320
A/Acores PT/11/2013	3C.3	2013-01-19	SIAT1/SIAT1	40	80	80	160	160	160	640	320	640	640	320
A/Stockholm/12/2013		2013-01-21	MDCK0/SIAT1	80	80	160	160	320	160	640	320	640	640	320
A/Iceland/29/2013	3C.2	2013-01-24	MDCK1/SIAT3	40	40	40	160	40	160	640	320	640	640	320
A/Stockholm/10/2013	3C.2	2013-01-30	MDCK1/SIAT1	40	80	80	160	80	160	640	320	640	640	320
A/Nitra/360/2013	3C.3	2013-02-05	MDCK1/SIAT1	80	160	160	320	160	320	1280	640	1280	1280	640
A/Stockholm/11/2013	3C.2	2013-02-09	MDCK1/SIAT1	40	80	80	160	80	160	640	320	640	640	320
A/Bratislava/530/2013		2013-02-15	MDCK2/SIAT1	80	160	160	320	320	320	1280	1280	640	1280	640
A/Stockholm/9/2013	3C.3	2013-02-17	MDCK1/SIAT1	80	80	80	320	160	160	640	320	640	1280	320
A/Galanta/592/2013	3C.3	2013-02-21	MDCK1/SIAT1	40	80	80	160	80	160	640	320	640	640	640
A/Stockholm/8/2013	3C.3	2013-02-22	MDCK0/SIAT1	<	80	80	160	80	160	640	320	320	640	320
A/Stockholm/7/2013	3C.3	2013-02-23	MDCK1/SIAT1	<	40	40	160	80	160	320	320	320	320	160
A/Kmelnitsk/350/2013	3C.2	2013-02-25	MDCK2/MDCK1	160	80	160	320	160	160	640	640	640	640	320
A/Bratislava/811/2013	3C.3	2013-03-07	MDCK1/SIAT1	80	320	320	640	320	640	1280	1280	1280	2560	640
A/Iceland/4311/2013	3C.3	2013-03-13	SIAT2	320	160	160	320	160	320	1280	1280	1280	2560	640
A/Bratislava/887/2013	3C.3	2013-03-14	MDCK1/SIAT1	80	160	320	320	160	160	1280	1280	1280	1280	640
A/Nove Zamky/897/2013	3C.3	2013-03-15	MDCK1/SIAT1	320	640	640	1280	640	1280	2560	2560	2560	2560	2560
A/Dnipropetrovsk/365/2013		2013-03-18	MDCK1/SIAT2	<	40	40	160	80	80	320	320	320	320	160
A/Iceland/56/2013	3C.3	2013-04-26	SIAT2	80	80	80	160	80	80	640	640	320	640	320
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Sequences in phylogenetic trees

Table 7-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-07-17 + 2013-07-23

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F17/12 group 3C.1	A/Vic 361/11 T/C F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	320	<	80	80	80	80	160	160	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	640	2560	1280	1280	2560	2560	5120	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	40	<	80	80	80	80	320	160	320	160	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	<	80	40	160	80	80	320	160	160	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	640	1280	1280	1280	1280	1280	1280	2560	1280
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	640	320	160	640	1280	1280	320	160	2560	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	320	160	320	320	320	320	1280	640	1280	2560	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	80	160	160	320	160	160	640	320	640	1280	320
A/Athens/112/2012	3B	2012-02-01	SIAT6	80	80	80	320	160	160	640	320	640	640	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	320	1280	320	1280	1280	640	2560	640	1280	5120	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	1280	320	640	1280	640	1280	640	1280	5120	5120
TEST VIRUSES														
A/Belgium/S0932/2013		2012-12-20	SIAT2	40	80	80	160	160	160	320	320	320	640	320
A/Hong Kong/209/2013	3C.2	2013-01-16	MDCK2/SIAT1	<	40	40	80	40	80	320	160	320	320	160
A/Hong Kong/392/2013	3C.2	2013-01-28	MDCK2/SIAT1	<	80	80	160	80	80	320	160	320	640	160
A/Hong Kong/419/2013	3C.3	2013-01-29	MDCK2/SIAT1	<	80	80	160	80	80	320	160	320	640	160
A/Belgium/G0508/2013		2013-02-04	SIAT2	<	40	40	160	80	80	320	160	320	320	160
A/Belgium/G0528/2013	3C.3	2013-02-04	SIAT2	<	40	40	160	80	80	320	160	320	320	160
A/NosyBe/429/2013	3B	2013-02-11	MDCK1/SIAT1	80	160	160	320	320	320	640	640	640	1280	320
A/Minsk/1262/2013	3C.3	2013-02-22	MDCK3/SIAT1	80	160	160	320	160	320	1280	640	640	1280	640
A/Maevatanana/563/2013	3B	2013-02-25	MDCK1/SIAT1	<	40	40	160	80	80	320	320	320	320	160
A/Hong Kong/1036/2013	3C.3	2013-02-25	MDCK2/SIAT1	80	320	320	320	320	320	1280	640	640	1280	640
A/Norway/1693/2013		2013-03-06	MDCK2/SIAT1	<	80	80	160	80	160	640	320	320	640	320
A/Minsk/1431/2013	3C.2	2013-03-06	MDCK3/SIAT1	<	<	40	160	40	80	640	160	320	320	80
A/Norway/1665/2013		2013-03-07	MDCK2/SIAT1	40	80	40	160	80	80	640	160	320	640	320
A/Norway/1683/2013		2013-03-07	MDCK2/SIAT1	40	80	80	160	80	160	640	320	320	640	320
A/Norway/1736/2013		2013-03-07	MDCK2/SIAT1	80	80	80	160	80	160	320	320	320	640	320
A/Norway/1780/2013		2013-03-09	MDCK1/SIAT1	80	160	160	320	160	160	1280	320	640	1280	320
A/Maevatanana/974/2013	3B	2013-03-11	MDCK1/SIAT1	160	160	160	320	320	320	1280	640	640	1280	320
A/Estonia/76614/2013	3C.2	2013-03-11	MDCK1/SIAT1	160	160	320	640	320	320	1280	640	1280	2560	640
A/Estonia/76663/2013	3C.2	2013-03-13	MDCK2/SIAT1	<	40	80	160	80	160	320	320	320	320	160
A/Estonia/76676/2013	3C.3	2013-03-13	MDCK2/SIAT1	160	160	320	640	640	320	1280	640	1280	2560	640
A/Norway/1894/2013		2013-03-17	MDCK1/SIAT1	40	80	80	320	80	80	640	320	320	640	320
A/Norway/1940/2013		2013-03-18	MDCK2/SIAT1	40	80	40	160	80	160	320	160	320	320	160
A/Belgium/G1034/2013	3C.3	2013-03-18	SIAT2	<	80	80	160	160	160	320	320	320	640	320
A/Norway/1861/2013	3C.2	2013-03-19	MDCK1/SIAT1	40	80	80	160	80	80	320	320	320	640	320
A/Norway/1905/2013	3C.2	2013-03-24	MDCK2/MDCK1	<	<	<	160	<	80	320	160	160	320	160
A/Belgium/G1067/2013	3C.3	2013-03-25	SIAT2	40	80	80	160	160	160	640	320	320	640	320
A/Belgium/G1071/2013		2013-03-26	SIAT2	<	80	40	160	80	80	320	160	320	320	160
A/Norway/2010/2013	3C.3	2013-04-07	MDCK2/SIAT1	40	80	80	160	80	160	320	320	320	640	160
A/Norway/2160/2013	3C.3	2013-04-07	MDCK1/SIAT1	40	160	80	320	80	160	640	320	640	640	320
A/Norway/2200/2013	3C.2	2013-04-10	MDCK1/SIAT1	80	160	160	640	160	320	1280	640	640	1280	640
A/Belgium/S0923/2013	3C.3	2013-04-10	SIAT2	<	80	40	160	80	80	320	160	320	320	160
A/Belgium/S0930/2013	3C.2	2013-04-11	SIAT2	<	80	40	160	80	80	320	160	320	320	160
A/Norway/2255/2013	3C.3	2013-04-23	MDCK1/SIAT1	80	160	160	320	320	320	1280	640	640	1280	320
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Sequences in phylogenetic trees

Table 7-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-07-30

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F17/12 group 3C.1	A/Vic 361/11 T/C F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Hawaii 22/12 F37/12 group 3C.1
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	160	160	160	320	320	320	320	80
A/Victoria/208/2009		2009-06-02	E3/E2	640	2560	1280	1280	2560	2560	2560	1280	2560	5120	2560
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	40	<	80	160	160	80	320	160	320	160	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	80	80	160	80	80	320	160	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	640	1280	1280	640	1280	1280	1280	2560	1280
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	1280	640	320	640	2560	1280	640	320	5120	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	160	160	320	320	160	640	640	640	1280	320
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	80	160	160	320	160	160	640	640	640	640	320
A/Athens/112/2012	3B	2012-02-01	SIAT6	80	80	80	160	80	160	640	320	640	640	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	320	1280	640	1280	1280	1280	1280	640	1280	2560	1280
A/Hawaii/22/2012	3C.1	2012-07-09	E4/E1	320	1280	640	640	1280	640	1280	640	1280	5120	5120
TEST VIRUSES														
A/Estonia/77020/2013		2013-03-25	MDCK2/SIAT1	40	80	80	320	80	160	640	320	640	640	320
A/Estonia/77060/2013	3C.2	2013-03-25	MDCK2/SIAT1	80	160	160	320	160	160	640	320	1280	640	320
A/Estonia/76844/2013		2013-03-18	MDCK2/SIAT1	80	160	160	320	160	160	1280	640	640	1280	640
A/Estonia/77365/2013	3C.3	2013-04-05	MDCK2/SIAT1	<	80	80	160	80	80	320	320	320	320	320
A/Ghana/DILI-495/2013	3C.3	2013-06-28	C1/SIAT1	80	160	160	320	160	160	1280	640	1280	1280	640
A/Ghana/DILI-498/2013	3C.3	2013-07-01	C1/SIAT1	80	160	160	320	160	160	1280	640	640	1280	640
									Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Table 7-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-08-07 + 2013-08-08

				Haemagglutination inhibition titre[1]									
				Post infection ferret antisera									
Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F17/12 group 3C.1	A/Vic 361/11 T/C F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Samara 73/13 F24/13 group 3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	320	80	80	80	80	160	160	160	160	80
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	80	80	80	80	160	160	160	160	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	<	40	160	80	80	320	160	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	320	640	1280	640	1280	640	1280	2560	640
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	160	160	80	320	1280	640	320	160	2560	320
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT4	80	160	160	160	160	640	640	640	640	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	80	160	320	160	160	640	320	640	640	640
A/Athens/112/2012	3B	2012-02-01	SIAT6	80	80	320	160	160	640	320	640	640	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	320	320	640	640	640	2560	640	1280	2560	640
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	160	320	160	640	640	320	320	640	640
TEST VIRUSES													
A/Austria/716774/2013	3C.2	2013-01-25	SIAT1/SIAT1	40	80	160	80	80	640	320	640	640	640
A/Austria/726749/2013		2013-03-21	SIAT1/SIAT1	40	80	160	80	80	640	320	640	640	640
A/Ghana/DARI-0063/2013	3C.3	2013-04-04	SIAT2	40	160	160	160	160	640	320	320	640	320
A/Ghana/FS-0373/2013	3C.2	2013-04-04	SIAT2	40	80	160	80	80	320	160	320	640	320
A/Ghana/DILI-0311/2013		2013-04-17	SIAT2	40	80	160	80	160	640	320	320	640	320
A/South Africa/1857/2013	3C.2	2013-04-18	MDCK2/SIAT1	40	160	320	160	320	640	320	640	640	640
A/Ghana/DILI-0314/2013		2013-04-18	SIAT2	<	80	160	80	80	320	160	320	640	320
A/Ghana/DILI-0319/2013	3C.2	2013-04-19	SIAT2	40	80	160	80	80	320	160	320	640	320
A/Ghana/DILI-0321/2013	3C.2	2013-04-22	SIAT2	40	80	160	160	160	640	320	320	640	640
A/Ghana/DILI-0436/2013		2013-06-10	SIAT2	<	80	160	80	160	320	320	320	640	320
A/South Africa/3941/2013	3C.2	2013-06-12	SIAT1/SIAT1	40	80	160	80	80	320	160	320	320	320
A/Ghana/DILI-0443/2013	3C.3	2013-06-13	SIAT2	<	80	160	80	80	320	160	320	320	320
A/Ghana/DILI-0464/2013		2013-06-20	SIAT2	<	80	160	80	80	320	160	320	320	320
A/Ghana/DILI-0465/2013		2013-06-20	SIAT2	<	80	160	80	80	320	160	320	320	320
A/Ghana/DILI-0467/2013	3C.3	2013-06-20	SIAT2	<	40	80	40	80	320	160	160	320	320
A/Ghana/DILI-0470/2013		2013-06-21	SIAT2	40	80	160	80	160	320	160	320	640	320
A/South Africa/4655/2013	3C.3	2013-06-25	SIAT1/SIAT1	<	40	160	80	80	320	160	320	320	320
A/South Africa/4671/2013	3C.3	2013-06-25	MDCK2/SIAT1	40	80	160	160	160	640	320	640	640	640
A/Ghana/DARI-0098/2013	3C.2	2013-07-04	SIAT2	40	80	160	160	160	640	320	320	640	320
A/Ghana/DILI-505/2013		2013-07-04	SIAT2	40	80	160	160	160	640	320	320	640	320
A/Ghana/DILI-515/2013	3C.3	2013-07-08	SIAT2	<	80	160	80	80	320	160	320	320	320
A/Ghana/DILI-516/2013		2013-07-09	SIAT2	40	80	160	80	160	640	320	320	640	640
A/Ghana/DILI-518/2013	3C.3	2013-07-09	SIAT2	40	80	160	80	160	640	320	320	640	640
								Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Table 7-15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-08-13 + 2013-08-20 + 2013-08-23

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera: A/Perth 16/09 F35/11	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F17/12 group 3C.1	A/Vic 361/11 F14/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Samara 73/13 F24/13 group 3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	1280	160	160	160	160	640	640	640	640	160
A/Alabama/5/2010	5	2010-07-13	MK1/C2/SIAT2	<	160	160	160	80	320	320	320	320	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	40	160	640	160	320	640	640	640	1280	640
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	1280	1280	640	1280	1280	1280	5120	640
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	320	320	160	640	2560	1280	640	160	5120	320
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	160	320	320	320	1280	640	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	80	160	320	320	160	640	640	640	1280	640
A/Athens/112/2012	3B	2012-02-01	SIAT8	80	160	160	160	160	640	640	1280	1280	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	640	1280	1280	640	2560	1280	1280	5120	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	640	320	320	1280	640	1280	2560	1280
TEST VIRUSES													
A/Brisbane/1/2013		2013-01-03	E6/E1	160	160	640	640	640	1280	640	640	2560	640
A/Tomsk/04/2013		2013-01-24	MDCK2/SIAT1	<	40	160	40	80	320	160	320	320	320
A/Tomsk/05/2013		2013-01-24	MDCK2/SIAT1	40	80	160	80	80	320	160	320	320	320
A/Lithuania/5271/2013		2013-02-05	SIAT2	40	160	320	320	160	640	640	640	1280	640
A/Lithuania/5273/2013	3C.3	2013-02-05	SIAT2	<	40	160	160	80	320	320	320	640	320
A/Lithuania/5274/2013		2013-02-05	SIAT2	40	160	640	320	160	1280	1280	1280	1280	1280
A/Lithuania/5810/2013	3C.3	2013-02-19	SIAT2	40	160	320	160	160	640	640	640	1280	640
A/Lithuania/6105/2013		2013-02-24	SIAT2	40	160	320	160	160	1280	1280	1280	1280	640
A/Lithuania/6376/2013		2013-02-26	SIAT2	80	160	320	160	320	1280	1280	1280	1280	640
A/Lithuania/6385/2013	3C.3	2013-02-27	SIAT2	40	160	320	160	160	640	640	640	1280	640
A/Lithuania/7241/2013	3C.3	2013-03-06	SIAT2	40	80	320	160	160	640	640	640	640	640
A/Lithuania/7849/2013	3C.3	2013-03-08	SIAT2	40	80	320	160	160	640	640	640	640	640
A/Moscow/105/2013	3C.2	2013-03-12	MDCK2/SIAT1	80	160	320	80	80	640	320	320	640	320
A/Vladivostok/73/2013	3C.3	2013-03-30	MDCK2/SIAT1	40	160	160	80	160	320	160	320	320	320
A/Norway/2418/2013	3C.2	2013-05-03	SIAT2	<	40	160	80	80	320	80	320	320	320
A/Norway/2514/2013		2013-05-03	SIAT2	40	80	160	160	160	320	320	320	640	320
A/Victoria/505/2013	3C.3	2013-06-17	E4/E1	320	320	1280	1280	640	1280	1280	1280	5120	1280
A/Victoria/504/2013		2013-06-17	E4/E1	160	160	640	640	640	640	640	640	5120	640
								Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Table 7-16. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir) 2013-08-30 + 2013-09-03 + 2013-09-06

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F35/11	A/Ala 5/10 F27/10 group 5	A/Stock 18/11 F28/11 group 3A	A/Iowa 19/10 F15/11 group 6	A/Vic 361/11 Egg F35/12 group 3C.1	A/Berlin 93/11 T/C F17/12 group 3C.1	A/Vic 361/11 F34/12 group 3C.1	A/Athens 112/12 F16/12 group 3B	A/Texas 50/12 Egg F36/12 group 3C.1	A/Samara 73/13 F24/13 group 3C.3
				Haemagglutination inhibition titre[1] — Post infection ferret antisera									
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	640	160	160	160	160	320	320	320	320	160
A/Alabama/5/2010	5	2010-07-13	MK1/C1/SIAT2	40	160	160	160	160	320	160	320	320	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	80	320	160	160	640	320	640	640	640
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	1280	2560	640	1280	1280	2560	5120	1280
A/Victoria/361/2011	3C	2011-10-24	E3/E2	160	320	160	640	1280	640	320	160	2560	320
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	160	320	320	160	1280	640	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT2	160	320	320	320	320	640	640	640	1280	640
A/Athens/112/2012	3B	2012-02-01	SIAT8	80	80	160	160	160	640	320	640	640	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	640	1280	1280	1280	2560	1280	1280	2560	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	320	640	320	320	1280	640	640	1280	1280
TEST VIRUSES													
A/New York/39/2012	3C.3	2012-10-20	C2/SIAT1	40	160	160	160	160	640	320	640	640	640
A/New York/39/2012	3C.3	2012-10-20	E4/E1	320	640	640	640	640	1280	640	1280	2560	640
A/American Samoa/4786/2013	3C.2	2013-02-22	C2/SIAT1	40	40	160	160	80	320	320	320	320	320
A/American Samoa/4786/2013	3C.2	2013-02-22	E4/E1	640	640	1280	2560	1280	2560	1280	2560	5120	2560
A/Ireland/M19748/2013	3C.3	2013-03-04	SIAT2	40	160	320	160	160	640	320	1280	640	640
A/Lithuania/6934/2013	3C.3	2013-03-04	SIAT3	40	80	320	80	160	640	320	640	640	640
A/Ireland/M20427/2013	3C.3	2013-03-05	SIAT3	<	80	160	80	80	640	320	640	640	640
A/Ireland/M20189/2013	3C.3	2013-03-05	SIAT3	40	80	160	80	80	640	320	640	640	640
A/Lithuania/8003/2013	3C.3	2013-03-12	SIAT3	<	40	160	80	80	320	320	320	320	640
A/Lithuania/8115/2013	3C.3	2013-03-16	SIAT2	80	160	320	160	320	640	640	640	1280	1280
A/Ireland/M26695/2013	3C.3	2013-03-26	SIAT3	40	40	160	80	40	320	320	320	320	320
A/Ireland/M28426/2013	3C.2	2013-04-02	SIAT3	<	40	80	40	40	320	160	320	320	320
A/Ireland/M28390/2013	3C.2	2013-04-02	SIAT3	<	40	80	40	40	320	160	320	320	320
A/Ireland/M28859/2013	3C.3	2013-04-03	SIAT2	40	160	320	160	160	640	320	640	640	640
A/Yokohama/153/2013	3C.3	2013-04-12	MDCK3/MDCK1/SIAT1	<	80	320	80	80	320	320	640	640	320
A/Chiba-C/39/2013	3C.2	2013-04-12	MDCK2/MDCK1/SIAT2	<	<	<	<	<	<	<	40	160	40
A/Osaka/32/2013	3C.2	2013-05-01	MDCK2/MDCK1/SIAT1	<	<	80	40	40	160	160	320	320	320
A/Kanagawa/141/2013	3C.3	2013-05-09	MDCK2/MDCK1/SIAT1	<	80	160	80	80	320	320	640	640	640
								Vaccine 2012-13				Vaccine 2013-14	

1. < = <40

Sequences in phylogenetic trees

Table 8-1. Antigenic analyses of influenza A(H3N2) viruses - Plaque Reduction Neutralisation[2] (MDCK-SIAT1) 2013-08-15

Viruses	Genetic group	Collection Date	Passage History	A/Athens 112/12 T/C F16/12 group 3B	A/Vic 361/11 Egg F35/12 group 3C.1	A/Vic 361/11 T/C F14/12 group 3C.1	A/Texas 50/12 Egg F36/12 group3C.1	A/Samara 73/13 T/C F24/13 group3C.3
				Neutralisation titre[1] – Post infection ferret sera				
REFERENCE VIRUSES								
A/Athens/112/2012	3B	2012-02-01	SIAT4	320	40	160	640	320
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	160	5120	320	2560	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	40	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	640	320	640	2560	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	320	160	160	640	320
TEST VIRUSES								
A/Iceland/25/2012	3C.2	2012-10-17	MDCK1/SIAT3	80	40	80	160	160
A/Iceland/25/2012	3C.2	2012-10-17	SIAT2	640	160-320	160-320	640	320
A/Iceland/02/2013	3C.2	2013-01-02	MDCKx/SIAT3	640	160-320	320	1280	320
A/Iceland/04/2013	3C.3	2013-01-04	SIAT2	320	160	320	640	320
A/Stockholm/1/2013		2013-01-13	MDCK1/SIAT1	160	80	160	320	160
A/Stockholm/12/2013		2013-01-21	MDCK0/SIAT1	160	80	160	320	80
A/Iceland/29/2013	3C.2	2013-01-24	MDCK1/SIAT3	80	40-80	80	320	160
A/Stockholm/10/2013	3C.2	2013-01-30	MDCK1/SIAT1	320	80	160	640	320
A/Nitra/360/2013	3C.3	2013-02-05	MDCK1/SIAT1	320	160	160	640	320
A/Bratislava/530/2013		2013-02-15	MDCK2/SIAT1	320	160	320	640	320
A/Galanta/592/2013	3C.3	2013-02-21	MDCK1/SIAT1	160	40-80	80-160	320	160
A/Bratislava/811/2013	3C.3	2013-03-07	MDCK1/SIAT1	320	160	160-320	640-1280	320
A/Bratislava/887/2013	3C.3	2013-03-14	MDCK1/SIAT1	160	80	80-160	160	160
A/Nove Zamky/897/2013	3C.3	2013-03-15	MDCK1/SIAT1	160	80	80-160	320	160
					Vaccine 2012-13		Vaccine 2013-14	

1. <=<40   2. 80% reduction

Sequences in phylogenetic trees

Table 8-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation[2] (MDCK-SIAT1) 2013-08-22

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post infection ferret sera A/Athens 112/12 T/C F16/12 group 3B	A/Vic 361/11 Egg F35/12 group 3C.1	A/Vic 361/11 T/C F14/12 group 3C.1	A/Texas 50/12 Egg F36/12 group3C.1	A/Samara 73/13 T/C F24/13 group3C.3
REFERENCE VIRUSES								
A/Athens/112/2012	3B	2012-02-01	SIAT4	320	40	160	640	320
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	160	5120	320	2560	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	40	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	640	320	640	2560	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	320	160	160	640	320
TEST VIRUSES								
A/Madeira PT/12/2013	3C.2	2013-01-07	SIAT4/SIAT1	320	160	160	640	320
A/Acores PT/11/2013	3C.3	2013-01-19	SIAT1/SIAT1	320-640	160	160	320-640	160-320
A/Stockholm/11/2013	3C.2	2013-02-09	MDCK1/SIAT1	80	40	80	160	160
A/Stockholm/9/2013	3C.3	2013-02-17	MDCK1/SIAT1	320	160	160	320-640	160
A/Stockholm/8/2013	3C.3	2013-02-22	MDCK0/SIAT1	640	160	160-320	320-640	320
A/Stockholm/7/2013	3C.3	2013-02-23	MDCK1/SIAT1	160	40-80	80	160	80-160
A/Norway/1693/2013		2013-03-06	MDCK2/SIAT1	160	80	80	160	80
A/Norway/1665/2013		2013-03-07	MDCK2/SIAT1	160	40	80	320	40
A/Norway/1683/2013		2013-03-07	MDCK2/SIAT1	160	40	80	160	160
A/Norway/1736/2013		2013-03-07	MDCK2/SIAT1	160	80	80	160	160
A/Norway/1780/2013		2013-03-09	MDCK1/SIAT1	320	160	160	640	160
A/Norway/1894/2013		2013-03-17	MDCK1/SIAT1	160	80	80-160	320	160
A/Norway/1861/2013	3C.2	2013-03-19	MDCK1/SIAT1	160-320	80	80-160	320	160
					Vaccine 2012-13		Vaccine 2013-14	

1. <=<40 2. 80% reduction

Sequences in phylogenetic trees

Table 8-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation[2] (MDCK-SIAT1) 2013-08-29

Viruses	Genetic group	Collection Date	Passage History	A/Athens 112/12 T/C F16/12 group 3B	A/Vic 361/11 Egg F35/12 group 3C.1	A/Vic 361/11 T/C F14/12 group 3C.1	A/Texas 50/12 Egg F36/12 group3C.1	A/Samara 73/13 T/C F24/13 group3C.3
				Neutralisation titre[1] — Post infection ferret sera				
REFERENCE VIRUSES								
A/Athens/112/2012	3B	2012-02-01	SIAT4	320	40	160	640	320
A/Victoria/361/2011	3C.1	2011-10-24	E3/E2	160	5120	320	2560	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	40	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E1	640	320	640	2560	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	320	160	160	640	320
TEST VIRUSES								
A/Tomsk/04/2013		2013-01-24	MDCK2/SIAT1	80-160	40	40-80	80-160	80
A/Bulgaria/270/2013	3C.3	2013-02-04	C2/SIAT1	160-320	160	160	640	160
A/Nosy Be/429/2013	3B	2013-02-11	MDCK1/SIAT1	160-320	160	160	640	160
A/Maevatanana/563/2013	3B	2013-02-25	MDCK1/SIAT1	160	80	80	320	160
A/Maevatanana/974/2013	3B	2013-03-11	MDCK1/SIAT1	160	80	80	160	80-160
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT1	160-320	160	160	640	320
A/Moscow/105/2013	3C.2	2013-03-12	MDCK2/SIAT1	160	80	80	160-320	80-160
A/Estonia/76663/2013	3C.2	2013-03-13	MDCK2/SIAT1	80-160	40	80	80-160	80-160
A/Estonia/76676/2013	3C.3	2013-03-13	MDCK2/SIAT1	160	80	160	320	160
A/Norway/1940/2013		2013-03-18	MDCK2/SIAT1	160	80	80-160	320	80
A/Norway/1905/2013	3C.2	2013-03-24	MDCK2/MDCK1	160-320	80-160	160	320	160-320
A/Norway/2010/2013	3C.3	2013-04-07	MDCK2/SIAT1	160	40	80	160	160
A/Norway/2160/2013	3C.3	2013-04-07	MDCK1/SIAT1	160-320	80	80-160	160-320	160
A/Norway/2200/2013	3C.2	2013-04-10	MDCK1/SIAT1	320	160	160	320	160
A/Norway/2255/2013	3C.3	2013-04-23	MDCK1/SIAT1	320	80-160	160	320	160
A/Estonia/76614/2013	3C.2	2013-04-29	MDCK1/SIAT1	320	160	160	640	320
A/Norway/2418/2013	3C.2	2013-05-03	SIAT2	320-640	160	160-320	640	160
A/Norway/2514/2013		2013-05-03	SIAT2	320-640	320	320	640	320

1. <=<40 2. 80% reduction

Sequences in phylogenetic trees

Vaccine 2012-13 (A/Vic 361/11 Egg F35/12 column)

Vaccine 2013-14 (A/Texas 50/12 Egg F36/12 column)

Figure 11. Phylogenetic comparison of influenza A(H3N2) HA genes

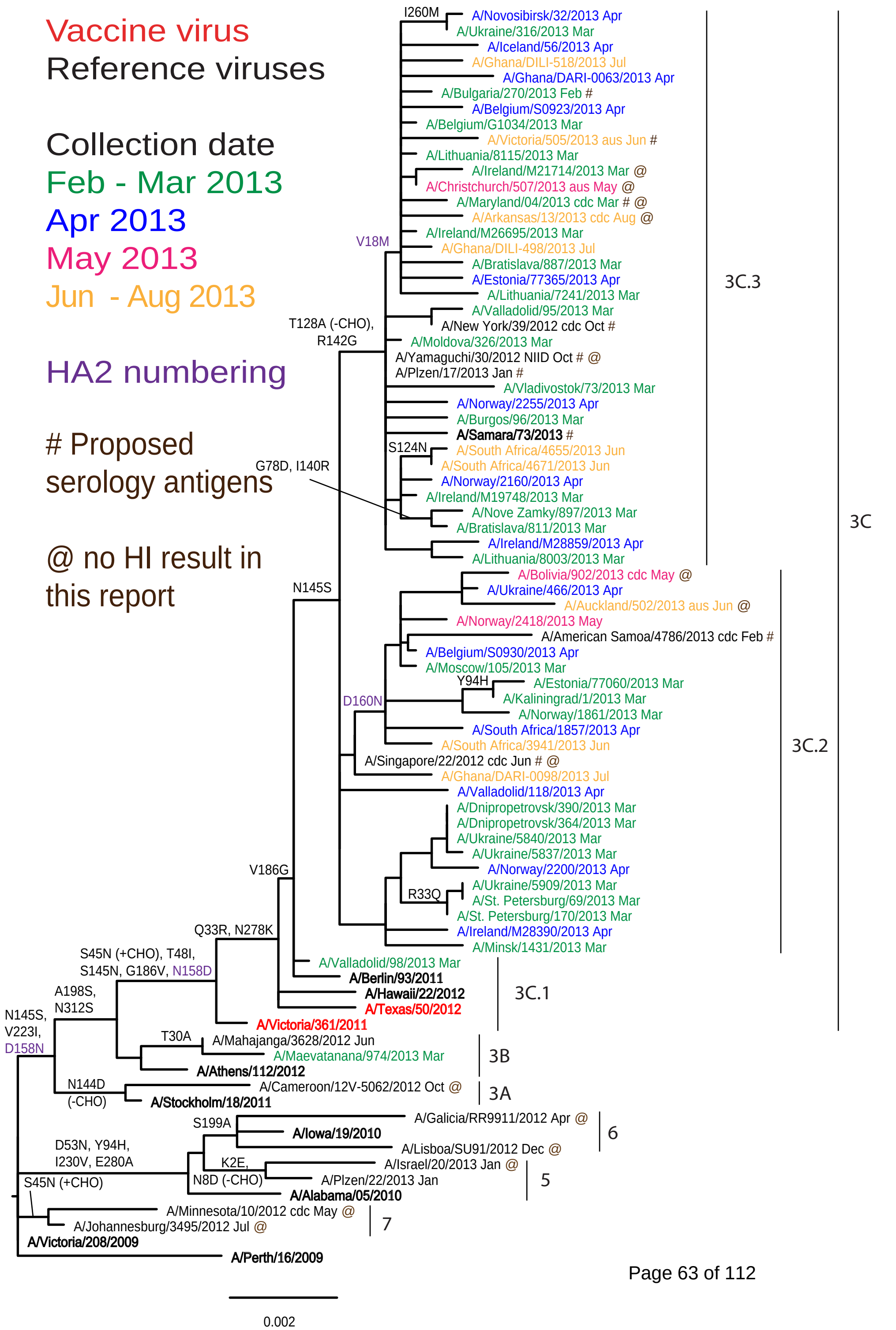

Figure 12. HA protein alignment for A(H3N2) viruses.

Vaccine viruses: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                   10        20        30        40        50        60        70        80        90       100
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009           QKLPGNDNSTATLCLGHHAVPNGTIVKTITNDQIEVTNATELVQSSSTGEICDSPHQILDGKNCTLIDALLGDPQCDGFQNKKWDLFVERSKAYSNCYPY
A/Victoria/208/2009       .............................................................E......................................
A/Johannesburg/3495/2012  ............................................N................E...................................... 7
A/Alabama/05/2010         ....................................................N........E................................H...... 5
A/Israel/20/2013          .E.....D............................................N........E................................H...... 5
A/Iowa/19/2010            ....................................................N........E................................H...... 6
A/Stockholm/18/2011       .............................................................E...................................... 3A
A/Athens/112/2012         .....................................................R.......E...................................... 3B
A/Maevatanana/974/2013    ....V....................A...................................E...................................... 3B
A/Victoria/361/2011       ............................................N..I.............E...................................... 3C.1
A/Texas/50/2012           ................................R...........N..I.............E...................................... 3C.1
A/Hawaii/22/2012          ................................R...........N..I.............E...................................... 3C.1
A/Berlin/93/2011          ................................R...........N..I.............E...................................... 3C.1
A/Valladolid/98/2013      ................................R...........N..I.............E...................................... 3C.1
A/Ireland/M28390/2013     ................................R...........N..I.............E...................................... 3C.2
A/St. Petersburg/170/2013 ............................................N..I.............E...................................... 3C.2
A/Norway/2200/2013        ................................R...........N..I.............E...................................... 3C.2
A/Ghana/DARI-0098/2013    ................................R...........N..I.............E...................................... 3C.2
A/South Africa/3941/2013  ................................R...........N..I.............E...................................... 3C.2
A/Estonia/77060/2013      ................................R...........N..I.............E................................H...... 3C.2
A/Ukraine/466/2013        ................................R...........N..I.............E...................................... 3C.2
A/Bratislava/811/2013     ................................R...........N..I.............E...............D...................... 3C.3
A/South Africa/4655/2013  ................................R...........N..I.............E...................................... 3C.3
A/Samara/73/2013          ................................R...........N..I.............E...................................... 3C.3
A/Plzen/17/2013           ................................R...........N..I.............E...................................... 3C.3
A/Moldova/326/2013        ................................R...........N..I.............E...................................... 3C.3
A/Lithuania/7241/2013     ..I.............................R...........N..I.............E...................................... 3C.3
A/Bulgaria/270/2013       ................................R...........N..I.............E...................................... 3C.3
A/Iceland/56/2013         ................................R...........N..I.............E................................H...... 3C.3
A/Novosibirsk/32/2013     ................................R...........N..I.............E...................................... 3C.3

                                  110       120       130       140       150       160       170       180       190       200
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009           DVPDYASLRSLVASSGTLEFNNESFNWTGVTQNGTSSACIRRSKNSFFSRLNWLTHLNFKYPALNVTMPNNEQFDKLYIWGVLHPGTDKDQIFLYAQASG
A/Victoria/208/2009       ...........................................N......................................H.................
A/Johannesburg/3495/2012  ...........................................N......................................H.................
A/Alabama/05/2010         ...........................................N......................................H.................
A/Israel/20/2013          ...........................................N......................................H.................
A/Iowa/19/2010            ...........................................N......................................H..X.........T..A.
A/Stockholm/18/2011       ...........................................DS.....................................H.................
A/Athens/112/2012         ...........................................NS.....................................H..............S..
A/Maevatanana/974/2013    ...........................................N......................................H..............S..
A/Victoria/361/2011       ...........................................N...........R..........................H..V...........S..
A/Texas/50/2012           ...........................N...............N......................................H..V...........P..
A/Hawaii/22/2012          ...........................................N............S.........................H..V...........S..
A/Berlin/93/2011          ...........................................N...........R..........................H..............S..
A/Valladolid/98/2013      ...........................................N......................................H....E.........S..
A/Ireland/M28390/2013     .....T.....................................NS.....................................H..............S..
A/St. Petersburg/170/2013 ...........................................NS.....................................H..............S..
A/Norway/2200/2013        ...........................................NS.....................................H..............S..
A/Ghana/DARI-0098/2013    .......................N...................NS.....................................H..............S..
A/South Africa/3941/2013  ...........................................NS.....................................H..............S..
A/Estonia/77060/2013      ...........................................NS.....................................H..............S..
A/Ukraine/466/2013        ...........................................NS.....................................H..............S..
A/Bratislava/811/2013     ...........................A...........R.G.NS.....................................H..............S..
A/South Africa/4655/2013  .......................N...A.............G.NS.....................................H..............S..
A/Samara/73/2013          ...........................A...........K.G.NS.....................................H..............S..
A/Plzen/17/2013           ...........................A.............G.NS.....................................H..............S..
A/Moldova/326/2013        ...........................A.............G.NS.....................................H..............S..
A/Lithuania/7241/2013     ...........................A.............G.NS.....................................H..............S..
A/Bulgaria/270/2013       ...........................A.............G..S.....................................H..............S..
A/Iceland/56/2013         ...........................A.............G.NS.....................................H..............S..
A/Novosibirsk/32/2013     ...........................A.............G.N......................................H..............S..

                                  210       220       230       240       250       260       270       280       290       300
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009           RITVSTKRSQQTVSPNIGSRPRVRNIPSRISIYWTIVKPGDILLINSTGNLIAPRGYFKIRSGKSSIMRSDAPIGKCNSECITPNGSIPNDKPFQNVNRI
A/Victoria/208/2009       ...........A.I......................................................................................
A/Johannesburg/3495/2012  ...........A.I.............................................................................G........
A/Alabama/05/2010         ......K....A.I.X........D....V.................................................A....................
A/Israel/20/2013          ...........A.I...............V.................................................A....................
A/Iowa/19/2010            ...........A.I...............V.................................................A....................
A/Stockholm/18/2011       ...........A.I........I.............................................................................
A/Athens/112/2012         ...........A.I........I.............................................................................
A/Maevatanana/974/2013    ...........A.I........I.............................................................................
A/Victoria/361/2011       ...........A.I........I.............................................................................
A/Texas/50/2012           ...........A.I....F...I.......................................................K.....................
A/Hawaii/22/2012          ...........A.I........I..S....................................................K.....................
A/Berlin/93/2011          ...........A.I........I.......................................................K.....................
A/Valladolid/98/2013      ...........A.I........I.......................................................K.....................
A/Ireland/M28390/2013     ...........A.I........I.......................................................K.....................
A/St. Petersburg/170/2013 ...........A.I......X.I.......................................................K.....................
A/Norway/2200/2013        ...........A.I........I.......................................................K.....................
A/Ghana/DARI-0098/2013    ...........A.I........I.......................................................K.....................
A/South Africa/3941/2013  ...........A.I........I.......................................................K.....................
A/Estonia/77060/2013      ...........A.I........I.......................................................K.....................
A/Ukraine/466/2013        ...........A.I........I.......................................................K.....................
A/Bratislava/811/2013     ...........A.I........I.......................................................K.....................
A/South Africa/4655/2013  ...........A.I........I.......................................................K.....................
A/Samara/73/2013          ...........A.I........I.......................................................K.....................
A/Plzen/17/2013           ...........A.I........I.......................................................K.....................
A/Moldova/326/2013        ...........A.I........I.......................................................K.....................
A/Lithuania/7241/2013     ...........A.I........I.......................................................K.....................
A/Bulgaria/270/2013       ...........A.I........I.......................................................K.....................
A/Iceland/56/2013         ...........A.I........I.......................................................K.....................
A/Novosibirsk/32/2013     ...........A.I........I....................................M..................K.....................
```

```
                              310       320       330       340       350       360       370       380       390       400
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009          TYGACPRYVKQNTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKEFS
A/Victoria/208/2009      ....................................................................................................
A/Johannesburg/3495/2012 ....................................................................................................
A/Alabama/05/2010        ....................................................................................................
A/Israel/20/2013         ....................................................................................................
A/Iowa/19/2010           ....................................................................................................
A/Stockholm/18/2011      ....................................................................................................
A/Athens/112/2012        ...........S........................................................................................
A/Maevatanana/974/2013   ...........S........................................................................................
A/Victoria/361/2011      ...........S........................................................................................
A/Texas/50/2012          ...........S........................................................................................
A/Hawaii/22/2012         ...........S..................................M.....................................................
A/Berlin/93/2011         ...........S........................................................................................
A/Valladolid/98/2013     ...........S........................................................................................
A/Ireland/M28390/2013    ...........S........................................................................................
A/St. Petersburg/170/2013 ...........S........................................................................................
A/Norway/2200/2013       ...........S........................................................................................
A/Ghana/DARI-0098/2013   ...........S........................................................................................
A/South Africa/3941/2013 ...........S........................................................................................
A/Estonia/77060/2013     ...........S........................................................................................
A/Ukraine/466/2013       ...........S........................................................................................
A/Bratislava/811/2013    ...........S........................................................................................
A/South Africa/4655/2013 ...........S........................................................................................
A/Samara/73/2013         ...........S........................................................................................
A/Plzen/17/2013          ...........S........................................................................................
A/Moldova/326/2013       ...........S........................................................................................
A/Lithuania/7241/2013    ...........S..................................M.....................................................
A/Bulgaria/270/2013      ...........S..................................M.....................................................
A/Iceland/56/2013        ...........S..................................M.....................................................
A/Novosibirsk/32/2013    ...........S..................................M.....................................................

                              410       420       430       440       450       460       470       480       490       500
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009          EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSEMNKLFEKTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF
A/Victoria/208/2009      ....................................................................................................
A/Johannesburg/3495/2012 ....................................................................................................
A/Alabama/05/2010        ....................................................................................................
A/Israel/20/2013         ....................................................................................................
A/Iowa/19/2010           ....................................................................................................
A/Stockholm/18/2011      ......................................................................................N.............
A/Athens/112/2012        ......................................................................................N.............
A/Maevatanana/974/2013   ......................................................................................N.............
A/Victoria/361/2011      ....................................................................................................
A/Texas/50/2012          ....................................................................................................
A/Hawaii/22/2012         ....................................................................................................
A/Berlin/93/2011         ....................................................................................................
A/Valladolid/98/2013     ....................................................................................................
A/Ireland/M28390/2013    ....................................................................................................
A/St. Petersburg/170/2013 ....................................................................................................
A/Norway/2200/2013       ....................................................................................................
A/Ghana/DARI-0098/2013   ....................................................................................................
A/South Africa/3941/2013 ........................................................................................N...........
A/Estonia/77060/2013     ........................................................................................N...........
A/Ukraine/466/2013       ........................................................................................N...........
A/Bratislava/811/2013    ....................................................................................................
A/South Africa/4655/2013 .....V..............................................................................................
A/Samara/73/2013         ....................................................................................................
A/Plzen/17/2013          ....................................................................................................
A/Moldova/326/2013       ....................................................................................................
A/Lithuania/7241/2013    .................V..................................................................................
A/Bulgaria/270/2013      ....................................................................................................
A/Iceland/56/2013        ....................................................................................................
A/Novosibirsk/32/2013    ....................................................................................................

                              510       520       530       540       550
                         ....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009          QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI
A/Victoria/208/2009      ..................................................
A/Johannesburg/3495/2012 ..................................................
A/Alabama/05/2010        ..................................................
A/Israel/20/2013         ..................................................
A/Iowa/19/2010           ..................................................
A/Stockholm/18/2011      ..................................................
A/Athens/112/2012        ..................................................
A/Maevatanana/974/2013   .........E........................................
A/Victoria/361/2011      ..................................................
A/Texas/50/2012          ..................................................
A/Hawaii/22/2012         ..................................................
A/Berlin/93/2011         .....................T............................
A/Valladolid/98/2013     ..................................................
A/Ireland/M28390/2013    ..................................................
A/St. Petersburg/170/2013 ..................................................
A/Norway/2200/2013       ..................................................
A/Ghana/DARI-0098/2013   ..................................................
A/South Africa/3941/2013 H.................................................
A/Estonia/77060/2013     ..................................................
A/Ukraine/466/2013       ..................................................
A/Bratislava/811/2013    ..................................................
A/South Africa/4655/2013 ..................................................
A/Samara/73/2013         ..................................................
A/Plzen/17/2013          ..................................................
A/Moldova/326/2013       ..................................................
A/Lithuania/7241/2013    ..................................................
A/Bulgaria/270/2013      ..................................................
A/Iceland/56/2013        ..................................................
A/Novosibirsk/32/2013    ..................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 13. H3-HA locations of genetic group defining amino acid substitutions

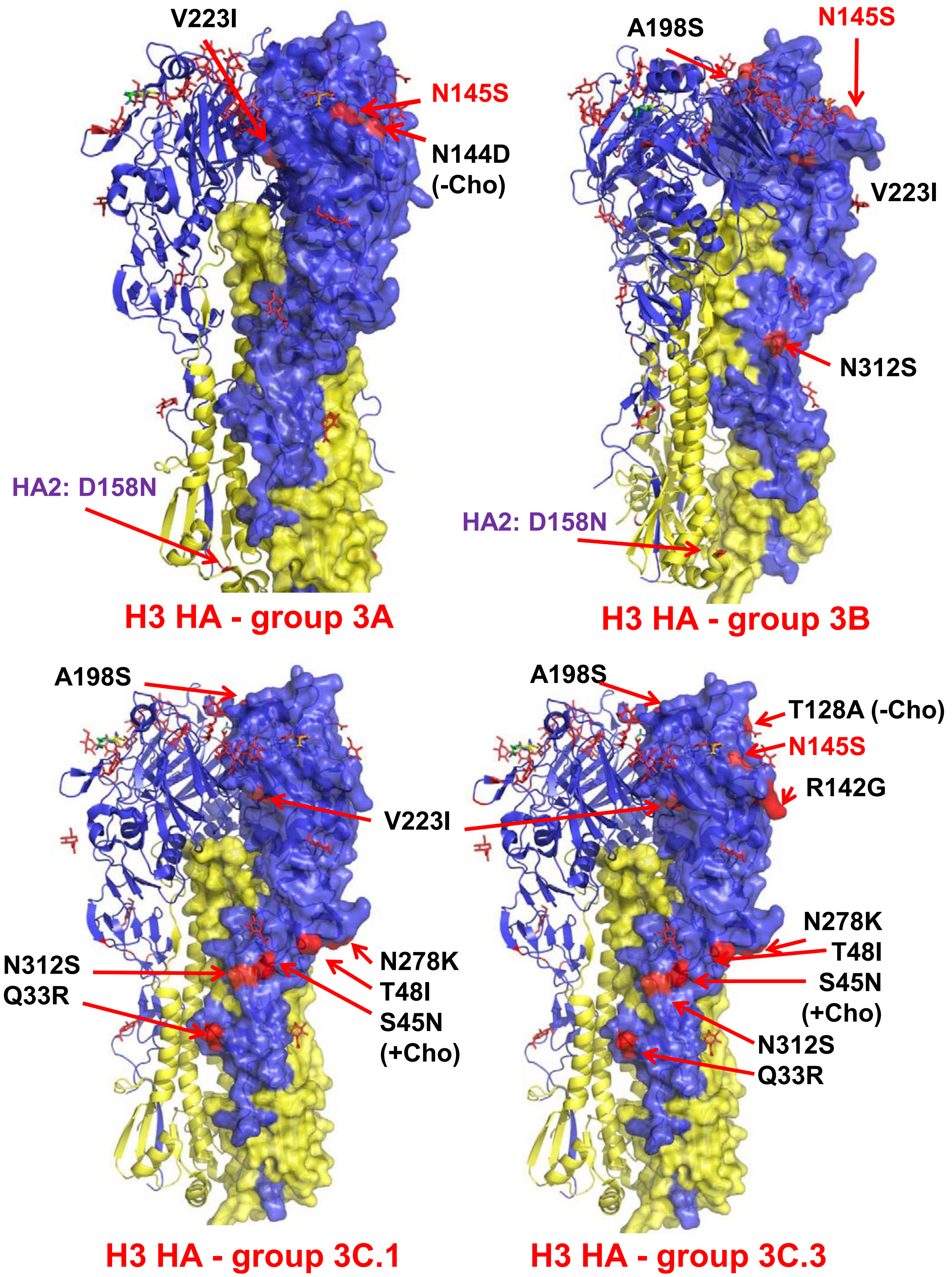


Figure 14. Phylogenetic comparison of influenza A(H3N2) NA genes

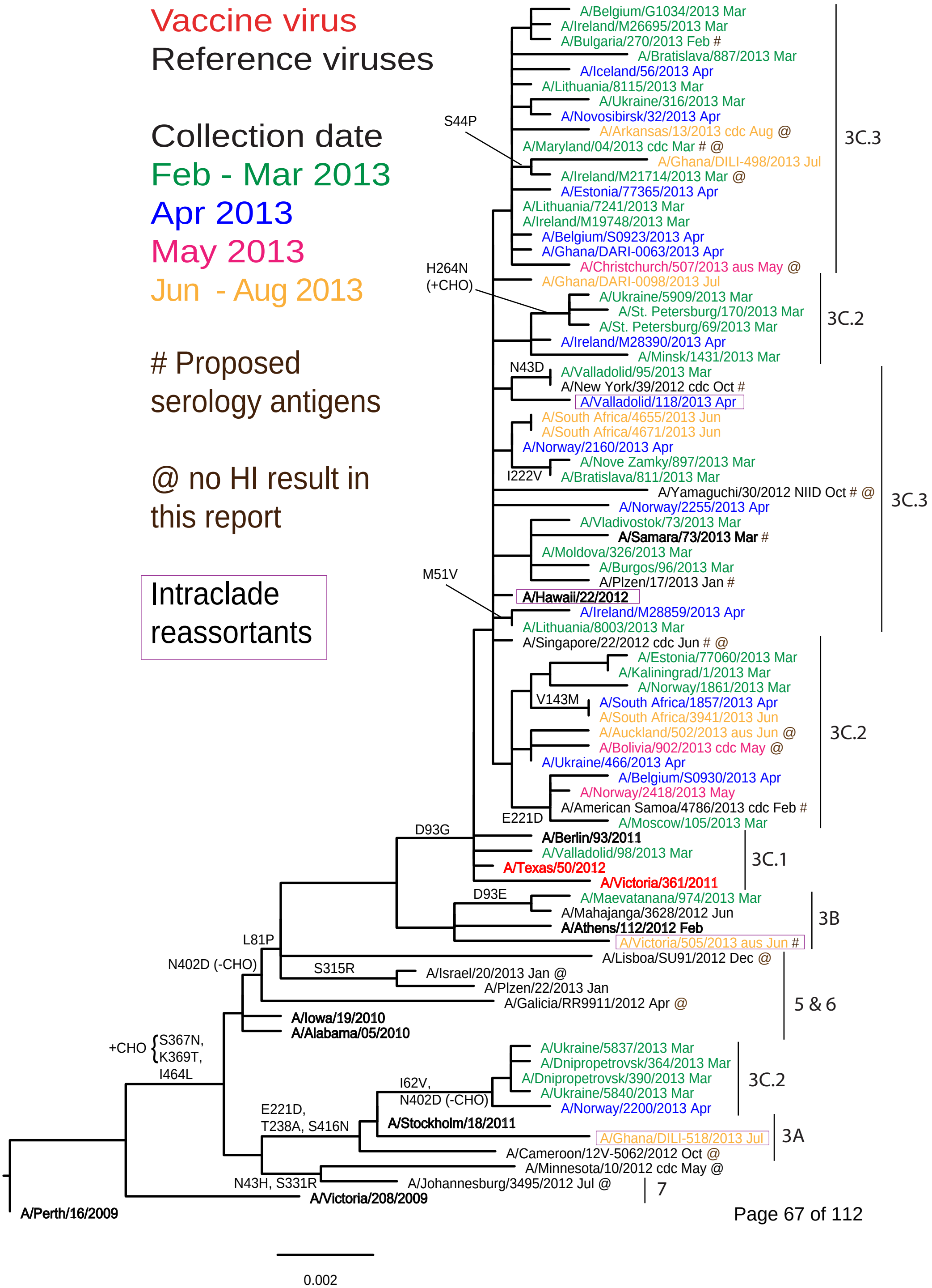


Figure 15. NA protein alignment for A(H3N2) viruses.

Vaccine viruses: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                    10        20        30        40        50        60        70        80        90       100
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009            MNPNQKIITIGSVSLTISTICFFMQIAILITTVTLHFKQYEFNSPPNNQVMLCEPTIIERNITEIVYLTNTTIEKEICPKLAEYRNWSKPQCDITGFAPF
A/Victoria/208/2009        .........................T.............C............................................................
A/Johannesburg/3495/2012   ..........................................H......................................................... 7
A/Alabama/05/2010          .................................................................................................... 5/6
A/Israel/20/2013           ................................................................................P................... 5/6
A/Iowa/19/2010             .................................................................................................... 5/6
A/Stockholm/18/2011        ........................................................................X........................... 3A
A/Athens/112/2012          ................................................................................P.K................. 3B
A/Maevatanana/974/2013     ................................................................................P...........E....... 3B
A/Victoria/361/2011        ................................................................................P...........G....... 3C.1
A/Texas/50/2012            ................................................................................P...........G....... 3C.1
A/Hawaii/22/2012           ................................................................................P...........G....... 3C.3
A/Berlin/93/2011           ................................................................................P...........G....... 3C.1
A/Valladolid/98/2013       ................................................................................P...........G....... 3C.1
A/Ireland/M28390/2013      ................................................................................P...........G....... 3C.2
A/St. Petersburg/170/2013  ................................................................................P...........G....... 3C.2
A/Norway/2200/2013         .............................................................V...................................... 3C.2
A/Ghana/DARI-0098/2013     ................................................................................P...........G....... 3C.2
A/South Africa/3941/2013   ................................................................................P...........G....... 3C.2
A/Estonia/77060/2013       ................................................................................P...........G....... 3C.2
A/Ukraine/466/2013         ................................................................................P...........G....... 3C.2
A/Bratislava/811/2013      ................................................................................P...........G....... 3C.3
A/South Africa/4655/2013   ................................................................................P...........G....... 3C.3
A/Samara/73/2013           ...............................................................K................P...........G....... 3C.3
A/Plzen/17/2013            ............................................................................M...P...........G....... 3C.3
A/Moldova/326/2013         ................................................................................P...........G....... 3C.3
A/Lithuania/7241/2013      ................................................................................P...........G....... 3C.3
A/Bulgaria/270/2013        ................................................................................P...........G....... 3C.3
A/Iceland/56/2013          ................................................................................P...........G....... 3C.3
A/Novosibirsk/32/2013      ................................................................................P...........G....... 3C.3

                                   110       120       130       140       150       160       170       180       190       200
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009            SKDNSIRLSAGGDIWVTREPYVSCDPDKCYQFALGQGTTLNNVHSNNTVRDRTPYRTLLMNELGVPFHLGTKQVCIAWSSSSCHDGKAWLHVCITGDDKN
A/Victoria/208/2009        ....................................................................................................
A/Johannesburg/3495/2012   ....................................................................................................
A/Alabama/05/2010          ....................................................................................................
A/Israel/20/2013           ..........................................................X.........................................
A/Iowa/19/2010             ....................................................................................................
A/Stockholm/18/2011        ....................................................................................................
A/Athens/112/2012          ....................................................................................................
A/Maevatanana/974/2013     ....................................................................................................
A/Victoria/361/2011        ................................................................X...................................
A/Texas/50/2012            .........................................................H..........................................
A/Hawaii/22/2012           ....................................................................................................
A/Berlin/93/2011           ................................................................................................N...
A/Valladolid/98/2013       ....................................................................................................
A/Ireland/M28390/2013      ....................................................................................................
A/St. Petersburg/170/2013  ..........................................................X.........................................
A/Norway/2200/2013         ..........................................................X.........................................
A/Ghana/DARI-0098/2013     ................................................................................................E...
A/South Africa/3941/2013   ..........................................M.........................................................
A/Estonia/77060/2013       .......................................................X..X.........................................
A/Ukraine/466/2013         ....................................................................................................
A/Bratislava/811/2013      ....................................................................................................
A/South Africa/4655/2013   ....................................................................................................
A/Samara/73/2013           ....................................................................................................
A/Plzen/17/2013            ....................................................................................................
A/Moldova/326/2013         ....................................................................................................
A/Lithuania/7241/2013      ....................................................................................................
A/Bulgaria/270/2013        ....................................................................................................
A/Iceland/56/2013          ....................................................................................................
A/Novosibirsk/32/2013      .........................................................HX.........................................

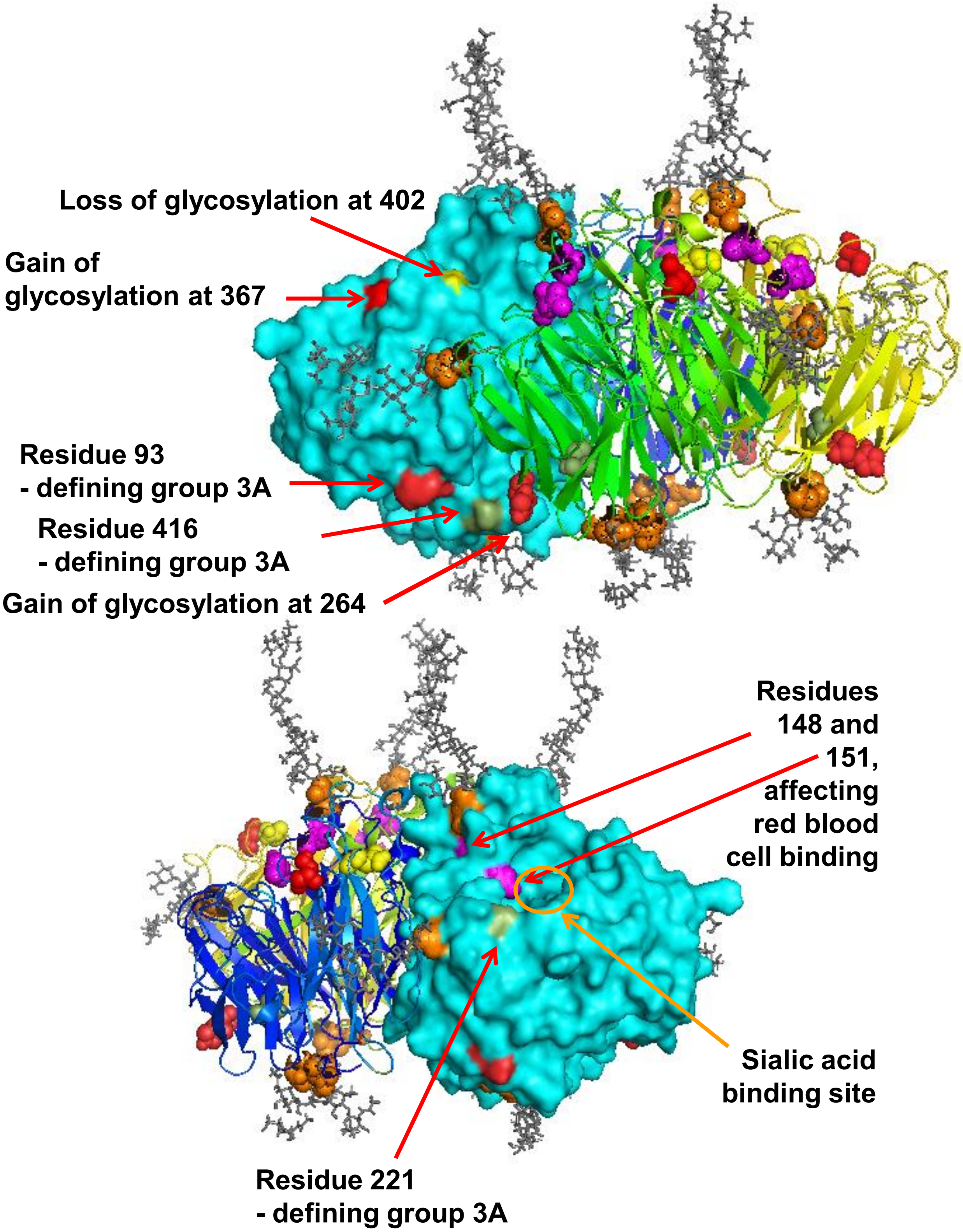
                                   210       220       230       240       250       260       270       280       290       300
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009            ATASFIYNGRLVDSVVSWSKEILRTQESECVCINGTCTVVMTDGSASGKADTKILFIEEGKIVHTSTLSGSAQHVEECSCYPRYPGVRCVCRDNWKGSNR
A/Victoria/208/2009        ....................................................................................................
A/Johannesburg/3495/2012   ....................................................................................................
A/Alabama/05/2010          ....................................................................................................
A/Israel/20/2013           ....................................................................................................
A/Iowa/19/2010             ...................Q................................................................................
A/Stockholm/18/2011        ....................D................A..............................................................
A/Athens/112/2012          .............................................................V......................................
A/Maevatanana/974/2013     ....................................................................................................
A/Victoria/361/2011        .........................................................K..........................................
A/Texas/50/2012            ....................................................................................................
A/Hawaii/22/2012           ....................................................................................................
A/Berlin/93/2011           ....................................................................................................
A/Valladolid/98/2013       ....................................................................................................
A/Ireland/M28390/2013      ....................................................................................................
A/St. Petersburg/170/2013  ...............................................................N....................................
A/Norway/2200/2013         ....................D................A..............................................................
A/Ghana/DARI-0098/2013     ....................................................................................................
A/South Africa/3941/2013   ....................................................................................................
A/Estonia/77060/2013       ....................................................................................................
A/Ukraine/466/2013         ....................................................................................................
A/Bratislava/811/2013      .....................V..............................................................................
A/South Africa/4655/2013   ....................................................................................................
A/Samara/73/2013           ....................................................................................................
A/Plzen/17/2013            ....................................................................................................
A/Moldova/326/2013         ....................................................................................................
A/Lithuania/7241/2013      ....................................................................................................
A/Bulgaria/270/2013        ....................................................................................................
A/Iceland/56/2013          ....................................................................................................
A/Novosibirsk/32/2013      ....................................................................................................
```

```
                                       310       320       330       340       350       360       370       380       390       400
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009                 PIVDINIKDHSIVSSYVCSGLVGDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVWMGRTISEKSRLGYETFKVIEGWSNPKSKLQINRQVIVDR
A/Victoria/208/2009             ....................................................................................................
A/Johannesburg/3495/2012        ..............................R...................................N.T...............................
A/Alabama/05/2010               ..................................................................N.T...............................
A/Israel/20/2013                ..............R...................................................N.T...............................
A/Iowa/19/2010                  ..................................................................N.T......................M........
A/Stockholm/18/2011             ..................................................................N.T...............................
A/Athens/112/2012               ..................................................................N.T...............................
A/Maevatanana/974/2013          .....................................S............................N.T...............................
A/Victoria/361/2011             ............................T.....................................N.T...............................
A/Texas/50/2012                 ..................................................................N.T...............................
A/Hawaii/22/2012                ..................................................................N.T...............................
A/Berlin/93/2011                ..................................................................N.T...............................
A/Valladolid/98/2013            ..................................................................N.T...............................
A/Ireland/M28390/2013           ..................................................................N.T...............................
A/St. Petersburg/170/2013       ..................................................................N.T......................M........
A/Norway/2200/2013              ..................................................................N.T......................T........
A/Ghana/DARI-0098/2013          ..................................................................N.T...............................
A/South Africa/3941/2013        ..................................................................N.T...............................
A/Estonia/77060/2013            ..................................................................N.T...............................
A/Ukraine/466/2013              ..................................................................N.T...............................
A/Bratislava/811/2013           ..................................................................N.T...............................
A/South Africa/4655/2013        ..................................................................N.T...............................
A/Samara/73/2013                ..................................................................N.T...............................
A/Plzen/17/2013                 ..................................................................N.T...........................V...
A/Moldova/326/2013              ..................................................................N.T...............................
A/Lithuania/7241/2013           ..................................................................N.T...............................
A/Bulgaria/270/2013             ............................D.....................................N.T...............................
A/Iceland/56/2013               ...........V......................................................N.T...............................
A/Novosibirsk/32/2013           ..................................................................N.T...............................

                                       410       420       430       440       450       460
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/Perth/16/2009                 GNRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTSGTYGTGSWPDGADINLMPI
A/Victoria/208/2009             .....................................................................
A/Johannesburg/3495/2012        ...............................................................L.....
A/Alabama/05/2010               ...............................................................L.....
A/Israel/20/2013                .D.............................................................L.....
A/Iowa/19/2010                  ...............................................................L.....
A/Stockholm/18/2011             ...............N...............................................L.....
A/Athens/112/2012               .D.............................................................LK....
A/Maevatanana/974/2013          .D.............................................................L.....
A/Victoria/361/2011             .D.............................................................L.....
A/Texas/50/2012                 .D.............................................................L.....
A/Hawaii/22/2012                .D.............................................................L.....
A/Berlin/93/2011                .D.............................................................L.....
A/Valladolid/98/2013            .D.............................................................L.....
A/Ireland/M28390/2013           .D.............................................................L.....
A/St. Petersburg/170/2013       .D.............................................................L.....
A/Norway/2200/2013              .D.............N...............................................L.....
A/Ghana/DARI-0098/2013          .D.............................................................L.....
A/South Africa/3941/2013        .D.............................................................L.....
A/Estonia/77060/2013            .D.............................................................L.....
A/Ukraine/466/2013              .D.............................................................L.....
A/Bratislava/811/2013           .D.............................................................L.....
A/South Africa/4655/2013        .D.............................................................L.....
A/Samara/73/2013                .D........................T....................................L.....
A/Plzen/17/2013                 .D.............................................................L.....
A/Moldova/326/2013              .D.............................................................L.....
A/Lithuania/7241/2013           .D.............................................................L.....
A/Bulgaria/270/2013             .D.............................................................L.....
A/Iceland/56/2013               .D.............................................................L.....
A/Novosibirsk/32/2013           .D.............................................................L.....
```

Vaccine viruses: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 16. N2-NA locations of genetic group defining & N-linked glycosylation site modifying amino acid substitutions



N2 NA, conserved (gold) and variable (loss: yellow, gain: red) glycosylation sites, residues affecting red blood cell binding (magenta) & group-defining substitutions.

Table 9. Summary of influenza B (Victoria-lineage) samples analysed, collected after 2013-01-31

MONTH	B Victoria lineage				
			Fold reduction in HI titre		
Country	Number received	Number propagated[1]	≤2 fold[2]	4 fold[2]	≥8 fold[2]
FEBRUARY					
Austria	1	1	1		
Georgia	3	3	3		
Hungary	3	3	3		
Italy	1	1	1		
Jordan	4	4	4		
Portugal	4	4	4		
Russia	1	1	1		
Senegal	4	4	4		
Serbia	2	2	2		
Slovenia	1	1	1		
United Kingdom	1	1	1		
MARCH					
Austria	1	1	1		
Belarus	1	1	1		
Hungary	1	1		1	
Jordan	17	17	17		
Norway	1	1	1		
Portugal	3	3	3		
Russia	7	7	7		
Senegal	5	5	5		
Slovakia	2	2	2		
Slovenia	2	1	1		
Spain	3	3	3		
APRIL					
Estonia	1	1	1		
Hungary	2	2		2	
Jordan	14	14	14		
Norway	1	1	1		
Senegal	5	5	5		
MAY					
Jordan	2	2	2		
JULY					
Ghana	1	1	1		
17 Countries	94	93	90 97%	3 3%	0 0%

1. Propagated to sufficient titre to perform HI assay
2. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

Table 10-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-02-13 + 2013-02-19 + 2013-02-27

				Haemagglutination inhibition titre										
					Post infection ferret sera[1]									
Viruses	Genetic group	Collection date	Passage History	B/Bris[2] 60/08 Sh 523	B/Mal 2506/05 F37/11	B/Eng 393/08 F05/11 1A	B/Bris 60/08 F22/12 1A	B/Paris 1762/08 F11/09 1A	B/HK 514/09 F13/10 1B	B/Odessa 3886/10 F19/11 1B	B/Malta 636714/11 F33/11 1A	B/Jhb 3964/12 F01/13 1A	B/For V2367/12 F04/13 1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	320	40	80	<	<	<	80	160	80	
B/England/393/2008	1A	2008-08-29	E1/E2	1280	40	320	160	40	20	20	160	160	160	
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	320	320	80	40	40	320	320	320	
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	1280	<	40	20	80	40	40	20	40	80	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	2560	20	20	20	80	80	80	20	40	80	
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	1280	20	80	80	20	40	40	40	80	160	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	320	160	40	20	40	320	160	320	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E1	2560	160	320	320	80	40	40	320	640	320	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK2	2560	20	80	40	20	40	40	40	160	320	
TEST VIRUSES														
B/Antananarivo/3968/2012		2012-07-17	MDCK1/MDCK2	2560	<	10	10	80	40	40	10	40	80	
B/Maevatanana/4280/2012	1A	2012-08-07	MDCK2/MDCK1	2560	<	10	20	80	40	40	10	40	40	
B/Maevatanana/4280/2012	1A	2012-08-07	MDCK2/MDCK1	2560	<	20	20	80	40	40	10	80	80	
B/Tsiroanomandidy/4477/2012	1B	2012-08-29	MDCK1/MDCK2	2560	<	20	40	80	40	40	20	80	160	
B/Reims/1624/2012	1A	2012-10-22	MDCK2/MDCK1	2560	<	10	20	40	40	40	20	40	160	
B/Caen/1623/2012		2012-10-24	MDCK2/MDCK1	2560	10	20	40	80	80	80	40	80	160	
B/Blida/113/2012	1A	2012-11-08	C0/MDCK1	5120	<	20	40	80	80	80	40	80	160	
B/Hong Kong/3608/2012	1A	2012-11-09	MDCK2/MDCK1	2560	10	20	20	40	40	40	20	80	160	
B/Norway/2402/2012	1A	2012-11-27	MDCK1/MDCK1	2560	<	20	20	40	40	40	20	40	160	
B/Norway/2389/2012		2012-11-28	MDCK1/MDCK1	2560	80	160	160	40	40	40	160	640	320	
B/Hong Kong/3613/2012		2012-11-28	MDCK2/MDCK1	5120	10	20	40	80	80	80	40	80	160	
B/Andalucia/359/2012	1A	2012-12-03	E1/E2	1280	80	160	320	80	40	40	160	320	320	197/199 DET
B/Andalucia/358/2012	1A	2012-12-04	E1/E1	2560	320	160	320	40	40	80	160	320	320	197/199 NEA
B/Hong Kong/6/2013	1A	2013-01-07	MDCK2/MDCK1	2560	10	20	40	80	80	80	40	80	160	
B/Iasi/131732/2013	1A	2013-01-15	MDCK2/MDCK1	2560	10	20	20	40	80	80	40	80	160	
B/Petropavlovsk/9/2013	1A	Jan 2013	MDCKx/MDCK1	1280	40	160	160	20	20	20	80	160	320	197/199 NEA
B/Taraz/83/2013	1A	Jan 2013	MDCKx/MDCK1	2560	10	40	20	80	40	40	20	80	80	
							Vaccine*							

1. < = <10; 2. hyperimmune sheep serum

Sequences in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 10-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-03-12 + 2013-04-16 + 2013-05-14 + 2013-05-29

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre										
					Post infection ferret sera[1]									
				B/Bris[2] 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Eng 393/08 F05/11 1A	B/Bris 60/08 F22/12 1A	B/Paris 1762/08 F07/11 1A	B/HK 514/09 F13/10 1B	B/Odessa 3886/10 F19/11 1B	B/Malta 636714/11 F33/11 1A	B/Jhb 3964/12 F01/13 1A	B/For V2367/12 F04/13 1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	320	40	80	10	<	<	80	160	80	
B/England/393/2008	1A	2008-08-29	E1/E2	2560	80	320	320	80	40	40	160	320	160	
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	320	320	80	40	40	320	320	320	
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	10	40	40	80	40	40	20	40	80	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	5120	<	20	40	160	80	80	20	80	80	
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	2560	40	160	160	40	40	40	80	160	160	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	320	320	80	20	40	320	320	320	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E1	5120	160	640	640	80	40	40	320	640	640	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK2	2560	10	80	160	80	40	40	80	160	160	
TEST VIRUSES														
B/Maevatanana/5151/2012	1A	2012-11-05	MDCK1/MDCK3	2560	<	20	20	40	80	40	20	40	80	
B/Tsiroanomandidy/3810/2012	1A	2012-07-03	MDCK1/MDCK1	2560	10	20	20	40	40	40	20	40	80	
B/Antsirabe/4171/2012	1A	2012-07-31	MDCK1/MDCK1	2560	10	40	40	80	80	40	20	80	80	
B/Ireland/87829/2012	1A	2012-11-19	MDCK4	2560	10	20	20	40	40	40	20	80	80	
B/Catalonia/5479S/2012		2012-12-04	MDCK1	2560	10	10	10	20	40	40	10	20	40	
B/Tehran/1236/2012	1A	2012-12-05	MDCK3/MDCK1	2560	10	20	20	80	80	40	20	80	80	
B/Tehran/1236/2012	1A	2012-12-05	E2/E1	1280	80	320	320	80	40	40	320	320	320	197/199 SET
B/England/22/2013	1A	2013-01-02	MDCK1/MDCK1	5120	<	20	20	40	40	80	20	40	80	
B/Slovenia/84/2013	1A	2013-01-08	MDCK1/MDCK1	2560	<	20	40	80	80	80	20	80	80	
B/Slovenia/176/2013		2013-01-15	MDCKx/MDCK1	2560	<	40	40	80	80	80	20	80	80	
B/Slovenia/221/2013		2013-01-17	MDCK/MDCK1	2560	<	20	40	80	40	40	20	40	40	
B/Slovenia/473/2013	1A	2013-01-28	MDCK/MDCK1	2560	<	20	40	80	80	40	20	40	80	
B/England/226/2013	1A	2013-02-01	MDCK1/MDCK1	5120	<	40	40	80	80	80	40	80	80	
B/Slovenia/527/2013	1A	2013-02-03	MDCK/MDCK1	2560	<	20	40	80	40	40	20	40	40	
B/Slovenia/1190/2013	1A	2013-03-11	MDCK/MDCK1	2560	<	20	40	80	80	40	20	40	40	
B/Valladolid/93/2013	1A	2013-03-13	MDCK2	5120	40	80	80	80	80	80	80	160	160	
B/St. Petersburg/144/2013	1A	2013-03-14	C1/MDCK1	2560	<	20	40	80	40	40	20	40	80	
B/Samara/5/2013		2013-03-14	C2/MDCK1	2560	<	20	40	80	40	40	20	40	80	
B/Samara/11/2013	1A	2013-03-14	C3/MDCK1	2560	<	20	40	80	40	40	20	40	80	
B/Zamora/102/2013	1A	2013-03-19	MDCK2	5120	40	80	40	80	80	80	40	80	320	
B/Leon/116/2013	1A	2013-03-26	MDCK2	5120	10	80	40	40	80	80	40	80	160	

1. < = <10; 2. hyperimmune sheep serum

Vaccine*

Sequences in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 10-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-06-11 + 2013-06-19 + 2013-06-26

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre										
					Post infection ferret sera[1]									
				B/Bris[2] 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Eng 393/08 F05/11	B/Bris 60/08 F22/12	B/Paris 1762/08 F07/11	B/HK 514/09 F13/10	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F33/11	B/Jhb 3964/12 F01/13	B/For V2367/12 F04/13	
	Genetic group					1A	1A	1A	1B	1B	1A	1A	1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	640	40	80	10	<	<	80	160	80	
B/England/393/2008	1A	2008-08-29	E1/E2	2560	80	320	320	80	40	40	160	320	160	
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	20	160	320	40	20	40	160	160	160	
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	10	40	40	80	40	40	20	40	80	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	5120	<	20	40	160	80	80	20	80	80	
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	2560	40	160	160	40	40	40	80	160	160	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	160	320	320	80	40	40	320	320	320	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E1	2560	40	320	320	40	40	40	320	640	320	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK2	2560	10	80	160	80	40	40	80	160	160	
TEST VIRUSES														
B/Georgia/22/2013		2013-01-04	MDCK3	5120	40	40	80	80	80	80	20	80	80	
B/Belgrade/658/2013	1A	2013-02-05	C1/MDCK1	1280	<	20	20	80	40	40	20	40	40	
B/Kraljevo/899/2013	1A	2013-02-08	C0/MDCK1	2560	<	40	40	80	80	80	40	80	80	
B/Kutaisi/384/2013	1A	2013-02-14	MDCK2	5120	40	80	80	160	80	80	40	80	160	
B/Ozirgeti/542/2013	1A	2013-02-19	MDCK2	5120	20	40	40	80	40	80	20	40	80	
B/Tbilisi/524/2013	1A	2013-02-20	MDCK3	5120	160	160	160	40	40	40	160	160	160	197/199 NEI
B/Hungary/06/2013		2013-02-25	MDCK2/E2/MDCK2	2560	160	160	320	80	40	40	160	320	320	
B/Hungary/07/2013	1A	2013-02-27	MDCK2/E3/MDCK3	2560	40	320	320	80	40	40	160	320	320	197/199 NEA
B/Hungary/05/2013	1A	2013-02-27	MDCK2/E2/MDCK1	5120	40	40	40	80	80	80	20	40	80	
B/Hungary/13/2013		2013-03-13	MDCK2/E2/MDCK1	2560	80	160	160	40	20	20	160	320	160	
B/Hungary/23/2013	1A	2013-04-02	E2/E1	1280	160	80	160	40	20	40	80	160	160	197/199 NEA
B/Hungary/26/2013	1A	2013-04-08	MDCK1/E1/MDCK2	1280	80	160	160	40	20	20	160	160	320	197/199 XEX
							Vaccine*							

1. < = <10; 2. hyperimmune sheep serum

Sequences in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 10-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-07-05 + 2013-07-12 + 2013-07-19 + 2013-07-24

				Haemagglutination inhibition titre										
					Post infection ferret sera[1]									
Viruses	Genetic group	Collection date	Passage History	B/Bris[2] 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Eng 393/08 F05/11 1A	B/Bris 60/08 F22/12 1A	B/Paris 1762/08 F07/11 1A	B/HK 514/09 F13/10 1B	B/Odessa 3886/10 F19/11 1B	B/Malta 636714/11 F33/11 1A	B/Jhb 3964/12 F01/13 1A	B/For V2367/12 F04/13 1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	320	40	80	<	<	<	80	80	80	
B/England/393/2008	1A	2008-08-29	E1/E2	2560	40	320	320	40	40	80	320	320	320	
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	320	320	80	40	80	320	320	320	
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	10	20	40	80	40	80	40	40	80	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	2560	20	10	20	80	80	160	20	20	80	
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	2560	20	80	80	40	40	80	80	160	160	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	320	320	40	40	80	320	320	320	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	160	640	640	80	80	80	640	640	640	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK2	2560	10	40	40	40	40	80	40	80	160	
TEST VIRUSES														
B/Firenze/1/2013		2013-01-16	MDCK2/MDCK1	2560	<	20	40	80	40	40	10	<	80	
B/Firenze/3/2013	1A	2013-01-21	MDCK2/MDCK2	2560	80	80	160	40	20	20	80	160	160	197/199 NEX
B/Hong Kong/20/2013	1A	2013-01-24	MDCK2/MDCK1	2560	20	20	40	80	80	160	40	40	80	
B/Lisboa PT/13/2013	1A	2013-01-28	SIAT1/MDCK1	5120	<	40	80	160	80	80	20	40	80	
B/Madeira PT/109/2013		2013-01-28	SIAT1/MDCK2	2560	20	10	20	80	80	80	20	<	40	
B/Lisboa PT/50/2013		2013-02-13	SIAT2/MDCK2	2560	20	10	20	80	80	80	20	<	40	
B/Milano/101/2013	1A	2013-02-19	MDCK1/MDCK1	2560	<	20	40	80	40	40	20	40	80	
B/Lisboa PT/74/2013	1A	2013-02-19	SIAT1/MDCK1	2560	10	20	<	80	40	40	10	20	40	
B/Portalegre PT/101/2013	1A	2013-02-24	SIAT1/MDCK1	2560	10	20	<	80	40	40	20	40	80	
B/Castelo Branco PT/176/2013		2013-02-28	SIAT1/MDCK1	2560	20	20	40	80	80	80	40	<	80	
B/Castelo Branco PT/177/2013	1A	2013-03-04	SIAT1/MDCK1	5120	20	20	40	80	80	160	40	40	40	
B/Trnava/783/2013	1A	2013-03-05	MDCK1/MDCK1	5120	<	20	40	160	80	80	20	40	80	
B/Lisboa PT/232/2013		2013-03-17	SIAT1/MDCK1	2560	<	20	40	80	80	80	40	40	40	
B/Portalegre PT/233/2013	1A	2013-03-18	SIAT1/MDCK1	2560	20	10	20	80	80	80	20	<	40	
B/Bratislava/911/2013	1A	2013-03-20	MDCK1/MDCK1	2560	<	20	40	80	40	40	20	40	80	
B/Norway/1958/2013	1A	2013-03-25	MDCK1/MDCK1	1280	<	<	10	40	40	80	10	<	40	
B/Vitebsk/1624/2013	1A	2013-03-31	MDCK2/MDCK1	2560	<	10	20	40	40	80	20	<	80	
B/Norway/2090/2013	1A	2013-04-14	MDCK1/MDCK1	1280	10	<	10	80	40	80	10	<	40	

1. < = <10; 2. hyperimmune sheep serum

Vaccine*

Sequences in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 10-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-08-14 + 2013-08-16

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre / Post infection ferret sera[1]									
				B/Bris[2] 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Eng 393/08 F05/11	B/Bris 60/08 F22/12	B/Paris 1762/08 F07/11	B/HK 514/09 F13/10	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F33/11	B/Jhb 3964/12 F01/13	B/For V2367/12 F04/13
						1A	1A	1A	1B	1B	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	320	40	80	<	<	<	80	80	80
B/England/393/2008	1A	2008-08-29	E1/E2	1280	40	320	320	40	40	80	320	320	320
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	80	320	320	40	40	80	320	320	320
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	<	<	40	40	40	80	40	<	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	2560	<	<	40	80	80	160	20	<	40
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	1280	20	80	80	40	40	80	80	80	160
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	160	320	40	40	80	320	320	160
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	2560	160	320	640	80	80	80	640	1280	640
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	80	160	40	40	40	160	160	320
TEST VIRUSES													
B/Jordan/30040/2013		2013-01-12	SIATx/MDCK1	2560	<	<	40	80	40	80	40	<	80
B/Jordan/30067/2013		2013-01-19	SIAT1/MDCK1	2560	<	<	40	80	80	160	40	<	80
B/Jordan/30100/2013	1A	2013-02-09	SIAT1/MDCK1	2560	<	<	40	80	80	160	40	<	80
B/Jordan/30141/2013		2013-02-17	SIATx/MDCK1	2560	<	<	40	80	80	160	40	<	80
B/Jordan/30138/2013	1A	2013-02-20	SIATx/MDCK1	2560	<	<	20	80	40	80	20	<	40
B/Moscow/29/2013		2013-02-25	MDCK1/MDCK1	2560	<	20	40	80	40	80	20	<	40
B/Austria/721922/2013	1A	2013-02-26	SIAT1/MDCK1	2560	20	<	10	40	40	40	<	<	20
B/Moscow/44/2013	1A	2013-03-05	MDCK2/MDCK1	2560	<	20	40	80	80	80	20	<	80
B/Moscow/25/2013		2013-03-07	MDCK1/MDCK1	2560	<	10	40	80	40	80	20	<	40
B/Austria/723800/2013	1A	2013-03-07	SIAT1/MDCK1	2560	20	<	20	40	40	80	10	<	20
B/Moscow/27/2013		2013-03-11	MDCK1/MDCK1	2560	<	10	40	80	80	80	20	<	40
B/Moscow/33/2013	1A	2013-03-13	MDCK1/MDCK1	1280	<	10	20	40	40	80	20	<	40
B/Estonia/77361/2013	1A	2013-04-05	MDCK2/MDCK1	5120	40	40	40	80	80	160	40	80	80
B/Ghana/FS-718/2013	1A	2013-07-09	MDCK2	2560	<	20	40	40	40	80	20	<	80

1. < = <10; 2. hyperimmune sheep serum

Vaccine* (B/Bris 60/08 F22/12 column)

Sequences in phylogenetic trees (highlighted viruses)

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 10-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-08-20

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre										
				Post infection ferret sera[1]										
				B/Bris[2] 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Eng 393/08 F31/08	B/Bris 60/08 F22/12	B/Paris 1762/08 F07/11	B/HK 514/09 F9/13	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F33/11	B/Jhb 3964/12 F01/13	B/For V2367/12 F04/13	B/S.Australia 81/12 F2574 AUS
						1A	1A	1A	1B	1B	1A	1A	1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	640	80	160	10	20	<	80	160	80	160
B/England/393/2008	1A	2008-08-29	E1/E2	2560	80	320	320	40	80	80	320	320	320	640
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	160	320	640	80	160	160	320	320	640	640
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	10	<	40	40	40	40	20	<	80	320
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	2560	<	<	20	80	80	160	20	<	80	320
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	2560	40	80	160	40	80	80	80	160	160	640
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	320	80	80	80	320	320	320	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	640	640	160	160	80	320	640	640	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	40	80	160	40	80	40	80	160	320	640
TEST VIRUSES														
B/South Australia/36/2012		2012-09-26	E3/E1	1280	40	<	160	40	40	80	160	160	160	320
B/South Australia/81/2012	1A	2012-11-28	E4/E1	640	40	<	160	40	40	80	80	160	160	320
B/Jordan/30175/2013		2013-03-03	SIATx/MDCK1	2560	10	<	40	80	80	80	20	40	80	320
B/Jordan/30187/2013	1A	2013-03-11	SI/MDCK1AT1	2560	10	<	40	80	80	80	40	<	80	320
B/Jordan/30204/2013		2013-03-19	SIATx/MDCK1	5120	10	<	40	80	80	80	40	80	80	640
B/Jordan/30206/2013		2013-03-20	SIAT1/MDCK1	5120	10	<	80	80	80	80	40	80	160	640
B/Jordan/30208/2013		2013-03-23	SIATx/MDCK1	5120	10	<	80	80	80	160	80	80	160	640
B/Jordan/30209/2013		2013-03-24	SIAT1/MDCK1	5120	20	<	160	80	80	160	80	80	160	640
B/Jordan/30231/2013		2013-03-26	SIAT1/MDCK1	2560	20	<	80	80	80	160	80	80	160	640
B/Jordan/30236/2013		2013-03-30	SIAT1/MDCK1	2560	10	<	80	80	80	80	40	40	80	640
B/Jordan/30237/2013		2013-03-30	SIAT1/MDCK1	2560	20	<	80	80	80	160	40	80	160	640
B/Jordan/30238/2013		2013-03-30	SIAT1/MDCK1	2560	10	<	40	80	80	80	40	40	80	1280
B/Jordan/30239/2013		2013-03-30	SIAT1/MDCK1	5120	20	<	80	80	80	80	40	40	160	640
B/Jordan/30240/2013		2013-03-31	SIAT1/MDCK1	2560	10	<	80	80	80	80	20	40	80	640
B/Jordan/30241/2013		2013-03-31	SIAT1/MDCK1	2560	20	<	80	80	80	80	80	80	160	640
B/Jordan/30242/2013	1A	2013-03-31	SIAT1/MDCK1	5120	10	<	80	80	80	80	40	80	160	640
B/Jordan/30244/2013	1A	2013-04-01	SIAT1/MDCK1	2560	10	<	80	80	80	80	40	80	160	640
B/Jordan/30249/2013		2013-04-04	SIAT1/MDCK1	5120	10	<	40	80	80	160	40	<	80	640

1. < = <10; 2. hyperimmune sheep serum

Vaccine*

Sequences in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 10-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-08-23 +2013-09-04

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre									
					Post infection ferret sera[1]								
				B/Bris[2] 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Eng 393/08 F31/08	B/Bris 60/08 F22/12	B/Paris 1762/08 F07/11	B/HK 514/09 F9/13	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F33/11	B/Jhb 3964/12 F01/13	B/For V2367/12 F04/13
	Genetic group					1A	1A	1A	1B	1B	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	640	80	80	20	20	<	80	160	80
B/England/393/2008	1A	2008-08-29	E1/E2	2560	80	160	320	40	80	80	320	320	320
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	160	320	40	80	80	320	320	320
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	<	10	20	20	40	40	20	20	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	20	20	80	80	160	20	80	80
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	2560	40	80	80	40	80	80	80	160	160
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	160	320	40	80	80	320	320	320
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	640	320	160	160	80	640	1280	640
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	40	80	160	40	40	40	160	160	320
TEST VIRUSES													
B/Jordan/30250/2013	1A	2013-04-06	SIAT1/MDCK1	2560	<	20	40	40	40	80	20	40	80
B/Jordan/30251/2013		2013-04-06	SIAT1/MDCK1	2560	20	40	40	80	80	80	20	80	80
B/Jordan/30262/2013		2013-04-07	SIAT1/MDCK1	2560	10	20	40	80	40	80	40	80	80
B/Jordan/30266/2013	1A	2013-04-10	SIAT1/MDCK1	5120	10	40	40	80	80	160	40	80	80
B/Jordan/30267/2013		2013-04-10	SIAT1/MDCK1	5120	10	20	40	80	80	160	20	80	80
B/Jordan/30269/2013		2013-04-10	SIAT1/MDCK1	5120	10	40	40	80	80	160	40	80	80
B/Jordan/30277/2013		2013-04-13	SIAT1/MDCK1	2560	10	40	40	80	80	160	40	80	80
B/Jordan/30278/2013	1A	2013-04-17	SIAT1/MDCK1	2560	10	20	40	80	80	80	40	80	80
B/Jordan/30300/2013		2013-04-22	SIAT1/MDCK1	2560	10	20	40	80	80	80	40	80	80
B/Jordan/30305/2013		2013-04-24	SIAT1/MDCK1	2560	10	20	40	80	80	160	40	80	80
B/Jordan/30306/2013	1A	2013-04-25	SIAT1/MDCK1	2560	<	20	40	80	80	160	20	80	80

1. < = <10; 2. hyperimmune sheep serum

Vaccine*

Sequences in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 10-8. Antigenic analyses of influenza B viruses (Victoria lineage) 2013-08-27 + 2013-08-30 + 2013-09-04

				Haemagglutination inhibition titre										
					Post infection ferret sera[1]									
Viruses	Genetic group	Collection date	Passage History	B/Bris[2] 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Eng 393/08 F31/08 1A	B/Bris 60/08 F26/13 1A	B/Paris 1762/08 F07/11 1A	B/HK 514/09 F9/13 1B	B/Odessa 3886/10 F19/11 1B	B/Malta 636714/11 F29/13 1A	B/Jhb 3964/12 F01/13 1A	B/For V2367/12 F04/13 1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	640	40	80	<	10	<	80	160	80	
B/England/393/2008	1A	2008-08-29	E1/E2	1280	80	160	320	40	80	80	320	320	320	
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	160	160	640	80	80	80	640	320	320	
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	10	20	80	40	40	40	40	40	80	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	2560	<	10	80	40	80	160	40	40	40	
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	2560	40	80	160	40	40	80	160	160	160	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	160	320	40	80	80	640	320	320	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	640	1280	80	160	80	1280	1280	1280	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	80	320	40	40	40	160	160	160	
TEST VIRUSES														
B/Jordan/20349/2013		2013-01-04	SIAT1/MDCK1	2560	20	20	160	80	80	80	80	80	80	
B/Dakar/03/2013		2013-01-09	C1/MDCK1	2560	20	20	160	80	80	160	40	80	80	
B/Texas/02/2013	1A	2013-01-09	MDCK1/C2/MDCK1	2560	20	<	80	40	40	80	20	40	80	
B/Texas/02/2013	1A	2013-01-09	E5/E1	1280	80	80	320	40	80	80	160	320	160	197/199 XEX
B/Dakar/02/2013		2013-01-15	C1/MDCK1	2560	10	20	80	80	80	80	20	40	80	
B/Jordan/20259/2013		2013-02-03	SIAT1/MDCK1	2560	10	20	40	40	40	80	40	40	80	
B/Dakar/01/2013		2013-02-05	C1/MDCK1	2560	10	20	80	40	40	80	20	40	80	
B/Dakar/04/2013		2013-02-11	C1/MDCK1	1280	10	20	80	80	80	80	20	40	80	
B/Dakar/05/2013	1A	2013-02-27	C1/MDCK1	2560	10	20	80	40	40	80	20	40	80	
B/Dakar/06/2013		2013-02-28	C1/MDCK1	2560	10	10	80	40	40	80	20	40	80	
B/Dakar/08/2013	1A	2013-03-01	C1/MDCK1	1280	10	10	80	40	40	80	20	40	80	
B/Jordan/20286/2013		2013-03-09	SIATx/MDCK1	2560	20	20	160	80	80	80	80	80	80	
B/Jordan/20267/2013		2013-03-10	SIAT1/MDCK1	2560	20	20	80	80	80	80	80	80	80	
B/Jordan/20274/2013		2013-03-11	SIAT1/MDCK1	2560	20	20	80	80	80	80	80	80	80	
B/Dakar/09/2013		2013-03-15	C1/MDCK1	1280	10	10	80	40	40	80	20	40	40	
B/Dakar/10/2013	1A	2013-03-18	C1/MDCK1	1280	10	20	40	40	40	80	10	20	40	
B/Dakar/11/2013	1A	2013-03-21	C1/MDCK1	1280	10	20	80	40	40	80	20	40	80	
B/Dakar/12/2013		2013-03-25	C2/MDCK1	2560	10	<	80	40	40	80	20	40	80	
B/Jordan/20264/2013		2013-04-03	SIATx/MDCK1	2560	40	40	160	40	80	80	80	160	160	
B/Dakar/14/2013		2013-04-08	C1/MDCK1	2560	10	20	80	80	80	80	20	40	80	
B/Dakar/15/2013	1A	2013-04-17	C1/MDCK1	2560	20	20	80	80	80	80	20	80	80	
B/Dakar/16/2013		2013-04-23	C2/MDCK1	2560	10	10	80	80	40	80	20	80	80	
B/Dakar/18/2013	1A	2013-04-26	C1/MDCK1	2560	10	10	80	40	40	80	20	40	80	
B/Dakar/19/2013	1A	2013-04-30	C1/MDCK1	2560	10	20	80	40	40	80	20	40	80	
B/Jordan/30323/2013		2013-05-18	SIAT1/MDCK1	2560	10	20	40	40	40	80	40	40	80	
B/Jordan/30330/2013	1A	2013-05-26	SIAT1/MDCK1	2560	10	20	80	40	40	80	80	80	80	
							Vaccine*							

1. < = <10; 2. hyperimmune sheep serum

Sequences in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Figure 17. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

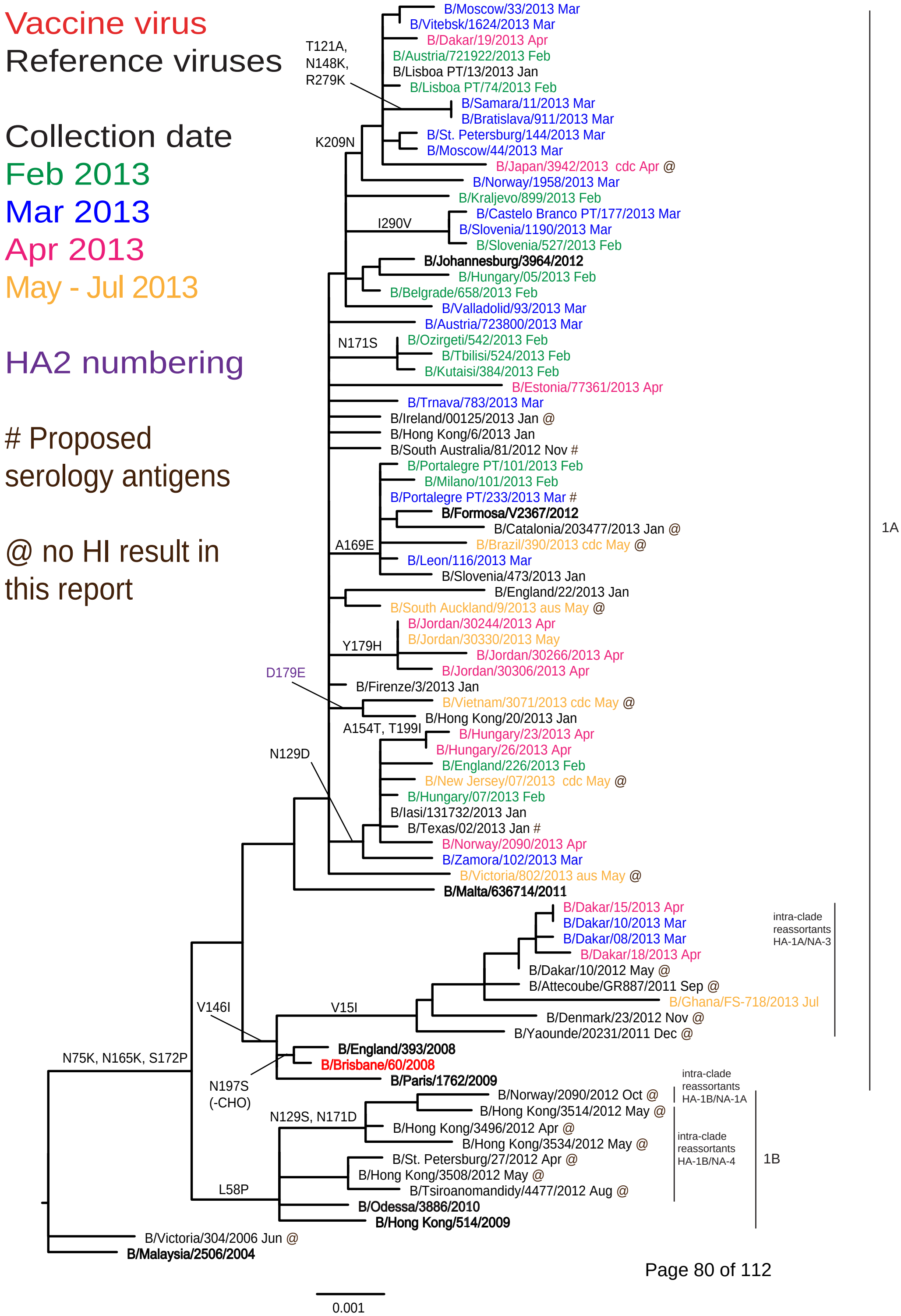


Figure 18. HA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                       10        20        30        40        50        60        70        80        90       100
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSHFANLKGTETRGKLCPKCLNCTDLDVALGRPKCTGNIPSARVSILHEVRPVTSGCFPIMHD
B/Hong Kong/514/2009          .........................................................P................K......................... 1B
B/Odessa/3886/2010            .........................................................P................K......................... 1B
B/Tsiroanomandidy/4477/2012   .........................................................P................K......................... 1B
B/Norway/2090/2012            .........................................................P................K...........I............. 1B
B/Paris/1762/2009             ..........................................................................K......................... 1A
B/Brisbane/60/2008            ..........................................................................K......................... 1A
B/England/393/2008            ..........................................................................K......................... 1A
B/Dakar/15/2013               ..............I...........................................................K......................... 1A
B/Dakar/18/2013               ..............I...........................................................K....K.................... 1A
B/Malta/636714/2011           ..........................................................................K......................... 1A
B/Norway/2090/2013            ..........................................S...............................K......................... 1A
B/Texas/02/2013               ..........................................................................K......................... 1A
B/Hungary/26/2013             ..........................................................................K......................... 1A
B/Jordan/30330/2013           ..........................................................................K......................... 1A
B/Leon/116/2013               ..........................................................................K......................... 1A
B/Formosa/V2367/2012          .................................A........................................K......................... 1A
B/Portalegre PT/233/2013      ..........................................................................K......................... 1A
B/South Australia/81/2012     ..........................................................................K......................... 1A
B/Trnava/783/2013             ..........................................................................K......................... 1A
B/Estonia/77361/2013          ..........................................................................K......................... 1A
B/Tbilisi/524/2013            ..........................................................................K......................... 1A
B/Austria/723800/2013         ..........................................................................K......................... 1A
B/Johannesburg/3964/2012      ..............................................A...........................K......................... 1A
B/Slovenia/1190/2013          ..........................................................................K......................... 1A
B/Norway/1958/2013            ..........................................................................K......................... 1A
B/Moscow/44/2013              ..........................................................................K......................... 1A
B/Bratislava/911/2013         ..........................................................................K......................... 1A
B/Dakar/19/2013               ..........................................................................K......................... 1A
B/Vitebsk/1624/2013           ..........................................................................K......................... 1A

                                      110       120       130       140       150       160       170       180       190       200
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          RTKIRQLPNLLRGYEHIRLSTHNVINAENAPGGPYKIGTSGSCPNVTNGNGFFATMAWAVPKNDNNKTATNSLTIEVPYICTEGEDQITVWGFHSDNEIQ
B/Hong Kong/514/2009          ................................................................K......P..........................T.
B/Odessa/3886/2010            ................................................................K......P..........................T.
B/Tsiroanomandidy/4477/2012   ................................................................K......P..........................T.
B/Norway/2090/2012            .....................L......S...................................K.....DP..........................T.
B/Paris/1762/2009             .............................................I..................K......P.................I........T.
B/Brisbane/60/2008            ................................................................K......P........................S.T.
B/England/393/2008            ................................................................K......P........................S.T.
B/Dakar/15/2013               .............................................I..................K......P..........................T.
B/Dakar/18/2013               .............................................I..................K......P..........................T.
B/Malta/636714/2011           ................................................................K......P........................S.T.
B/Norway/2090/2013            ............................D...................................K......P..........................T.
B/Texas/02/2013               ............................D...................................K......P........................I.T.
B/Hungary/26/2013             ............................D........................T..........K......P........................X.X.
B/Jordan/30330/2013           ................................................................K......P......H...................T.
B/Leon/116/2013               ................................................................K...E..P..........................T.
B/Formosa/V2367/2012          ................................................................K...E..P..........................T.
B/Portalegre PT/233/2013      ................................................................K...E..P..........................T.
B/South Australia/81/2012     ................................................................K......P..........................A.
B/Trnava/783/2013             ................................................................K......P..........................T.
B/Estonia/77361/2013          .....................................................T..........K......P..........................T.
B/Tbilisi/524/2013            ................................................................K.....SP............................
B/Austria/723800/2013         ................................................................K......P..........................T.
B/Johannesburg/3964/2012      ................................................................K......P........................X.T.
B/Slovenia/1190/2013          ................................................................K......P..........................T.
B/Norway/1958/2013            ................................................................K......P..........................T.
B/Moscow/44/2013              ................................................................K......P..........................T.
B/Bratislava/911/2013         ....................A..........................K................K......P..........................T.
B/Dakar/19/2013               ................................................................K......P..........................T.
B/Vitebsk/1624/2013           ................................................................K......P..........................T.

                                      210       220       230       240       250       260       270       280       290       300
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          MAKLYGDSKPQKFTSSANGVTTHYVSQIGGFPNQTEDGGLPQSGRIVVDYMVQKSGKTGTITYQRGILLPQKVWCASGRSKVIKGSLPLIGEADCLHEKY
B/Hong Kong/514/2009          ..............................................................................K.....................
B/Odessa/3886/2010            ....................................................................................................
B/Tsiroanomandidy/4477/2012   ....................................................................................................
B/Norway/2090/2012            ....................................................................................................
B/Paris/1762/2009             ........................I.............................P.............................................
B/Brisbane/60/2008            ....................................................................................................
B/England/393/2008            ....................................................................................................
B/Dakar/15/2013               ....................................................................................................
B/Dakar/18/2013               ....................................................................................................
B/Malta/636714/2011           ....................................................................................................
B/Norway/2090/2013            ....................................................................................................
B/Texas/02/2013               ....................................................................................................
B/Hungary/26/2013             ....................................................................................................
B/Jordan/30330/2013           ....................................................................................................
B/Leon/116/2013               ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/Portalegre PT/233/2013      ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Trnava/783/2013             ....................................................................................................
B/Estonia/77361/2013          ...........................................................................T........................
B/Tbilisi/524/2013            ....................................................................................................
B/Austria/723800/2013         ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Slovenia/1190/2013          .........................................................................................V..........
B/Norway/1958/2013            ........N...........................................................................................
B/Moscow/44/2013              ........N...........................................................................................
B/Bratislava/911/2013         ........N.....................................................................K.....................
B/Dakar/19/2013               ........N...........................................................................................
B/Vitebsk/1624/2013           ........N...........................................................................................

                                      310       320       330       340       350       360       370       380       390       400
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                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          GGLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLN
B/Hong Kong/514/2009          ....................................................................................................
B/Odessa/3886/2010            ....................................................................................................
B/Tsiroanomandidy/4477/2012   ......R.............................................................................................
B/Norway/2090/2012            ....................................................................................................
B/Paris/1762/2009             ....................................................................................................
B/Brisbane/60/2008            ....................................................................................................
B/England/393/2008            ....................................................................................................
B/Dakar/15/2013               ....................................................................................................
B/Dakar/18/2013               ....................................................................................................
B/Malta/636714/2011           ....................................................................................................
B/Norway/2090/2013            ....................................................................................................
B/Texas/02/2013               ....................................................................................................
B/Hungary/26/2013             ....................................................................................................
B/Jordan/30330/2013           ....................................................................................................
B/Leon/116/2013               ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/Portalegre PT/233/2013      ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Trnava/783/2013             ....................................................................................................
B/Estonia/77361/2013          ....................................................................................................
B/Tbilisi/524/2013            ...........................................I........................................................
B/Austria/723800/2013         ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Slovenia/1190/2013          ....................................................................................................
B/Norway/1958/2013            ....................................................................................................
B/Moscow/44/2013              ....................................................................................................
B/Bratislava/911/2013         ....................................................................................................
B/Dakar/19/2013               ....................................................................................................
B/Vitebsk/1624/2013           ....................................................................................................

                                      410       420       430       440       450       460       470       480       490       500
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          SLSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIA
B/Hong Kong/514/2009          ....................................................................................................
B/Odessa/3886/2010            ....................................................................................................
B/Tsiroanomandidy/4477/2012   ....................................................................................................
B/Norway/2090/2012            ....................................................................................................
B/Paris/1762/2009             ............................D.......................................................................
B/Brisbane/60/2008            ....................................................................................................
B/England/393/2008            ....................................................................................................
B/Dakar/15/2013               ....................................................................................................
B/Dakar/18/2013               ....................................................................................................
B/Malta/636714/2011           ....................................................................................................
B/Norway/2090/2013            ....................................................................................................
B/Texas/02/2013               ....................................................................................................
B/Hungary/26/2013             ....................................................................................................
B/Jordan/30330/2013           ....................................................................................................
B/Leon/116/2013               ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/Portalegre PT/233/2013      ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Trnava/783/2013             ....................................................................................................
B/Estonia/77361/2013          ....................................................................................................
B/Tbilisi/524/2013            ....................................................................................................
B/Austria/723800/2013         ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Slovenia/1190/2013          ....................................................................................................
B/Norway/1958/2013            ....................................................................................................
B/Moscow/44/2013              ....................................................................................................
B/Bratislava/911/2013         ....................................................................................................
B/Dakar/19/2013               ....................................................................................................
B/Vitebsk/1624/2013           ....................................................................................................

                                      510       520       530       540       550       560       570
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          AGTFDAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMIAIFVVYMVSRDNVSCSICL
B/Hong Kong/514/2009          ......................................................................
B/Odessa/3886/2010            ......................................................................
B/Tsiroanomandidy/4477/2012   ......................................................................
B/Norway/2090/2012            ......................................................................
B/Paris/1762/2009             ......................................................................
B/Brisbane/60/2008            ......................................................................
B/England/393/2008            ......................................................................
B/Dakar/15/2013               ......................................................................
B/Dakar/18/2013               ......................................................................
B/Malta/636714/2011           ......................................................................
B/Norway/2090/2013            ......................................................................
B/Texas/02/2013               ......................................................................
B/Hungary/26/2013             ......................................................................
B/Jordan/30330/2013           ......................................................................
B/Leon/116/2013               ......................................................................
B/Formosa/V2367/2012          ......................................................................
B/Portalegre PT/233/2013      ......................................................................
B/South Australia/81/2012     ......................................................................
B/Trnava/783/2013             ......................................................................
B/Estonia/77361/2013          ......................................................................
B/Tbilisi/524/2013            ......................................................................
B/Austria/723800/2013         ......................................................................
B/Johannesburg/3964/2012      ......................................................................
B/Slovenia/1190/2013          ......................................................................
B/Norway/1958/2013            ......................................................................
B/Moscow/44/2013              ......................................................................
B/Bratislava/911/2013         ......................................................................
B/Dakar/19/2013               ......................................................................
B/Vitebsk/1624/2013           ......................................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 19. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

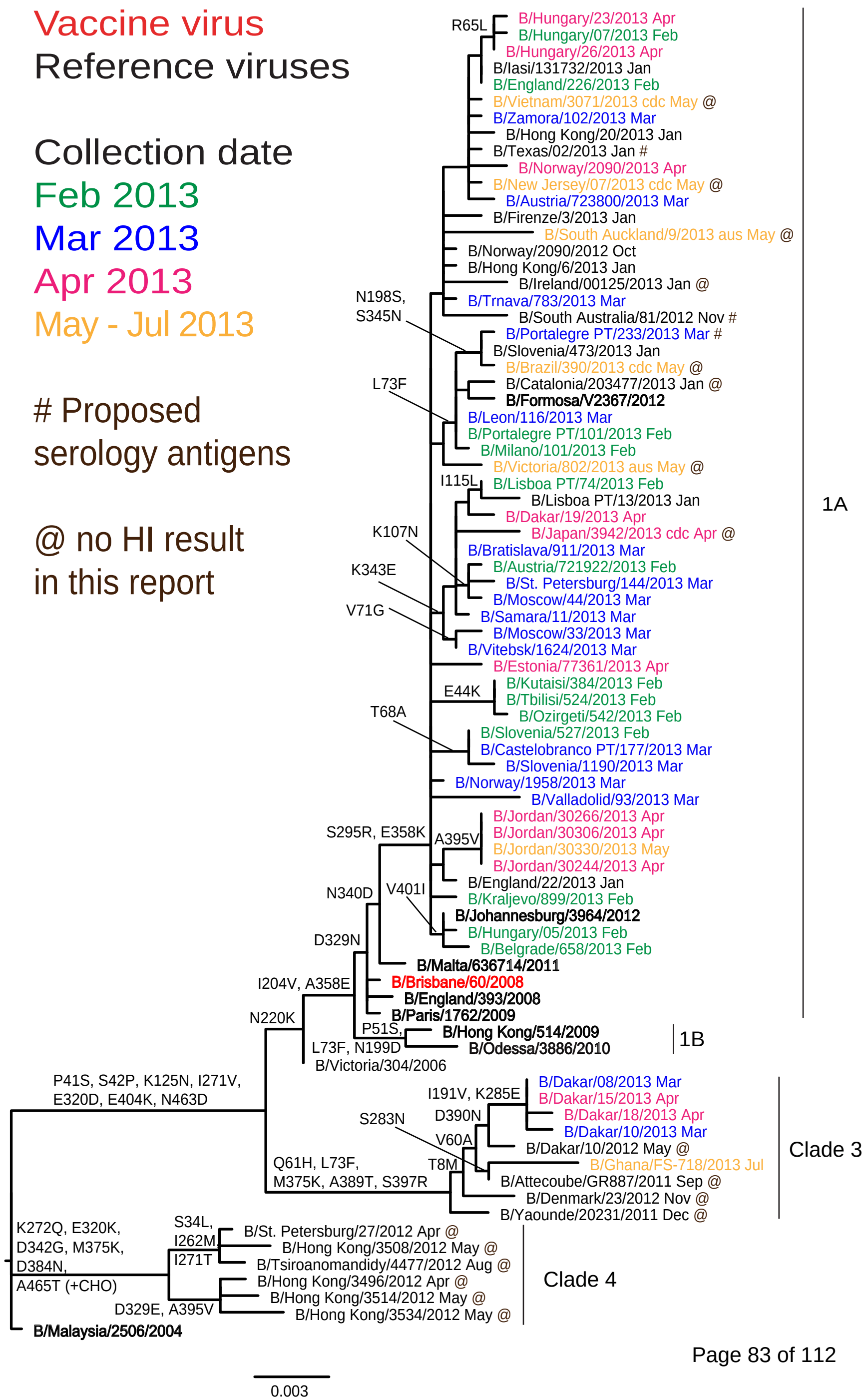

Figure 20. NA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                       10        20        30        40        50        60        70        80        90       100
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFPSTEITAPTMPLDCANASNVQAVNRSATKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Hong Kong/514/2009         ........................................SP........S.....................F........................... 1B
B/Odessa/3886/2010           ..........................L.............SP........S.....................F........................... 1B
B/Tsiroanomandidy/4477/2012  .................................L.................................................................. 4
B/Norway/2090/2012           ........................................SP.......................................................... 1A
B/Paris/1762/2009            ........................................SP.......T.................................................. 1A
B/Brisbane/60/2008           ........................................SP.......................................................... 1A
B/England/393/2008           ........................................SP.......................................................... 1A
B/Dakar/15/2013              .......M................................SP................AH............F........................... 3
B/Dakar/18/2013              .......M................................SP................AH............F........................... 3
B/Malta/636714/2011          ........................................SP......A................................................... 1A
B/Norway/2090/2013           ........................................SP.......................................................... 1A
B/Texas/02/2013              ........................................SP.......................................................... 1A
B/Hungary/26/2013            ........................................SP......................L................................... 1A
B/Jordan/30330/2013          ........................................SP.......................................................... 1A
B/Leon/116/2013              ........................................SP..............................F........................... 1A
B/Formosa/V2367/2012         ........................................SP..............................F........................... 1A
B/Portalegre PT/233/2013     ........................................SP..............................F........................... 1A
B/South Australia/81/2012    ........................................SP........................................................N. 1A
B/Trnava/783/2013            ........................................SP.......................................................... 1A
B/Estonia/77361/2013         ........................................SP..M....................................................... 1A
B/Tbilisi/524/2013           ........................................SP.K........................................................ 1A
B/Austria/723800/2013        ........................................SP.......................................................... 1A
B/Johannesburg/3964/2012     ........................................SP.......................................................... 1A
B/Slovenia/1190/2013         ........................................SP.........................A................................ 1A
B/Norway/1958/2013           ........................................SP.......................................................... 1A
B/Moscow/44/2013             ........................................SP........L................................................. 1A
B/Bratislava/911/2013        ........................................SP.......................................................... 1A
B/Dakar/19/2013              ........................................SP.......................................................... 1A
B/Vitebsk/1624/2013          ........................................SP............................G............................. 1A

                                      110       120       130       140       150       160       170       180       190       200
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         HRFGETKGNSAPLIIREPFIACGPKECKHFALTHYAAQPGGYYNGTRGDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDNNA
B/Hong Kong/514/2009         ........................N.........................................................................D.
B/Odessa/3886/2010           ........................N.........................................................................D.
B/Tsiroanomandidy/4477/2012  ..........................................................................................V.........
B/Norway/2090/2012           ........................N...........................................................................
B/Paris/1762/2009            ........................N...........................................................................
B/Brisbane/60/2008           ........................N...........................................................................
B/England/393/2008           ........................N...........................................................................
B/Dakar/15/2013              ........................N.................................................................V.........
B/Dakar/18/2013              ........................N.................................................................V.........
B/Malta/636714/2011          ........................N...........................................................................
B/Norway/2090/2013           ........................N...........................................................................
B/Texas/02/2013              ........................N...........................................................................
B/Hungary/26/2013            ........................N...........................................................................
B/Jordan/30330/2013          ........................N...........................................................................
B/Leon/116/2013              ........................N...........................................................................
B/Formosa/V2367/2012         ........................N...........................................................................
B/Portalegre PT/233/2013     ........................N........................................................................S..
B/South Australia/81/2012    ........................N...........................................................................
B/Trnava/783/2013            ........................N...........................................................................
B/Estonia/77361/2013         ........................N...........................................................................
B/Tbilisi/524/2013           ........................N...........................................................................
B/Austria/723800/2013        ........................N...........................................................................
B/Johannesburg/3964/2012     ........................N...........................................................................
B/Slovenia/1190/2013         ........................N...........................................................................
B/Norway/1958/2013           ........................N...........................................................................
B/Moscow/44/2013             ......N.................N...........................................................................
B/Bratislava/911/2013        ........................N...........................................................................
B/Dakar/19/2013              ........................N...........................................................................
B/Vitebsk/1624/2013          ........................N...........................................................................

                                      210       220       230       240       250       260       270       280       290       300
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         LLKIKYGEAYTDTYHSYANNILRTQESACNCIGGNCYLMITDGSASGVSECRFLKIREGRIIKEIFPTGRIKHTEECTCGFASNKTIECACRDNSYTAKR
B/Hong Kong/514/2009         ...V...............K..................................................V.............................
B/Odessa/3886/2010           ...V...............K..................................................V.............................
B/Tsiroanomandidy/4477/2012  .............................................................M........TQ............................
B/Norway/2090/2012           ...V...............K..................................................V.......................R.....
B/Paris/1762/2009            ...V...............K..................................................V.............................
B/Brisbane/60/2008           ...V...............K..................................................V.............................
B/England/393/2008           ...V...............K..................................................V.............................
B/Dakar/15/2013              ......................................................................V.............E...............
B/Dakar/18/2013              ......................................................................V.............E...............
B/Malta/636714/2011          ...V...............K..................................................V.............................
B/Norway/2090/2013           ...V...............K..................................................V.......................R.....
B/Texas/02/2013              ...V...............K..................................................V.......................R.....
B/Hungary/26/2013            ...V...............K..................................................V.......................R.....
B/Jordan/30330/2013          ...V...............K..................................................V.......................R.....
B/Leon/116/2013              ...V...............K..................................................V.......................R.....
B/Formosa/V2367/2012         ...V...............K..................................................V.......................R.....
B/Portalegre PT/233/2013     ...V...............K...............................................I..V.......................R.....
B/South Australia/81/2012    ...V...............K..................................................V.......................R.....
B/Trnava/783/2013            ...V...............K..................................................V.......................R.....
B/Estonia/77361/2013         ...V...............K..................................................V.......................R.....
B/Tbilisi/524/2013           ...V...............K..................................................V.......................R.....
B/Austria/723800/2013        ...V...............K..................................................V.......................R.....
B/Johannesburg/3964/2012     ...V...............K..................................................V.......................R.....
B/Slovenia/1190/2013         ...V...............K..................................................V.......................R.....
B/Norway/1958/2013           ...V...............K..................................................V.......................R.....
B/Moscow/44/2013             ...V...............K..................................................V.......................R.....
B/Bratislava/911/2013        ...V...............K..................................................V.......................R.....
B/Dakar/19/2013              ...V...............K..................................................V.......................R.....
B/Vitebsk/1624/2013          ...V...............K..................................................V.......................R.....
```

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                                     310       320       330       340       350       360       370       380       390       400
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          PFVKLNVETDTAEIRLMCTETYLDTPRPDDGSITGPCESNGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWADSDALAFSGVM
B/Hong Kong/514/2009          ...................D.....................................E..............E...........................
B/Odessa/3886/2010            ...................D.....................................E..............E...........................
B/Tsiroanomandidy/4477/2012   ...................K.....................G..............................E.K........N................
B/Norway/2090/2012            ...................D........N..........D.................K..............E...........................
B/Paris/1762/2009             ...................D........N............................E..............E...........................
B/Brisbane/60/2008            ...................D........N............................E..............E...........................
B/England/393/2008            ...................D........N............................E..............E...........................
B/Dakar/15/2013               ...................D....................................................E.K.............TN......R...
B/Dakar/18/2013               ...................D....................................................E.K.............TN......R...
B/Malta/636714/2011           ...................D........N..........D.................E..............E...........................
B/Norway/2090/2013            ...................D........N..........D.................K..............E...........................
B/Texas/02/2013               ...................D........N..........D.................K..............E...........................
B/Hungary/26/2013             ...................D........N..........D.................K..............E...........................
B/Jordan/30330/2013           ...................D........N..........D.................K..............E.....................V.....
B/Leon/116/2013               ...................D........N..........D.................K..............E...........................
B/Formosa/V2367/2012          ...................D........N..........D.................K..............E...........................
B/Portalegre PT/233/2013      ...................D........N..........D....N............K..............E...........................
B/South Australia/81/2012     ...................D........N..........D.................K..............E...........................
B/Trnava/783/2013             ...................D........N..........D.................K..............E...........................
B/Estonia/77361/2013          ...................D........N..........D.................K..............E...........................
B/Tbilisi/524/2013            ...................D........N..........D.................K..............E...........................
B/Austria/723800/2013         ...................D........N..........D.................K..............E...........................
B/Johannesburg/3964/2012      ...................D........N..........D.................K..............E...........................
B/Slovenia/1190/2013          ...................D........N..........D.................K..............E...........................
B/Norway/1958/2013            ...................D........N..........D.................K..............E...........................
B/Moscow/44/2013              ...................D........N..........D..E..............K..............E...........................
B/Bratislava/911/2013         ...................D........N..........D..E..............K..............E...........................
B/Dakar/19/2013               ...................D........N..........D..E..............K..............E...........................
B/Vitebsk/1624/2013           ...................D........N..........D..E..............K..............E...........................

                                     410       420       430       440       450       460
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Malaysia/2506/2004          VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKETWHSAATAIYCLMGSGQLLWDTVTGVNMAL
B/Hong Kong/514/2009          ...K..........................................................D...
B/Odessa/3886/2010            ...K..........................................................D...
B/Tsiroanomandidy/4477/2012   ................................................................T.
B/Norway/2090/2012            ...K..........................................................D...
B/Paris/1762/2009             ...K..........................................................D...
B/Brisbane/60/2008            ...K..........................................................D...
B/England/393/2008            ...K..........................................................D...
B/Dakar/15/2013               ...K..........................................................D...
B/Dakar/18/2013               ...K..........................................................D...
B/Malta/636714/2011           ...K..........................................................D...
B/Norway/2090/2013            ...K..........................................................D...
B/Texas/02/2013               ...K..........................................................D...
B/Hungary/26/2013             ...K..........................................................D...
B/Jordan/30330/2013           ...K..........................................................D...
B/Leon/116/2013               ...K..........................................................D...
B/Formosa/V2367/2012          ...K..........................................................D...
B/Portalegre PT/233/2013      ...K..........................................................D...
B/South Australia/81/2012     ...K..........................................................D...
B/Trnava/783/2013             ...K..........................................................D...
B/Estonia/77361/2013          ...K..........................................................D...
B/Tbilisi/524/2013            ...K..........................................................D...
B/Austria/723800/2013         ...K..........................................................D...
B/Johannesburg/3964/2012      I..K..........................................................D...
B/Slovenia/1190/2013          ...K..........................................................D...
B/Norway/1958/2013            ...K..........................................................D...
B/Moscow/44/2013              ...K..........................................................D...
B/Bratislava/911/2013         ...K..........................................................D...
B/Dakar/19/2013               ..VK..........................................................D...
B/Vitebsk/1624/2013           ...K..........................................................D...
```

Vaccine virus: **Reference virus**: **Genetic group**: **Potential N-linked glycosylation site**

Table 11. Summary of influenza B (Yamagata-lineage) samples analysed, collected after 2013-01-31

MONTH	B Yamagata lineage				
Country	Number received	Number propagated[1]	≤2 fold[2]	4 fold[2]	≥8 fold[2]
FEBRUARY					
Austria	1	1			1
Belgium	8	8	4	4	
Bulgaria	12	12		9	3
Czech Republic	4	4	3	1	
Georgia	1	1		1	
Hong Kong	2	2	2		
Hungary	3	2	1	1	
Iceland	1	1		1	
Italy	9	9		7	2
Jordan	3	3		3	
Lithuania	3	3		2	1
Madagascar	2	1	1		
Poland	4	4		3	1
Portugal	2	2			2
Romania	5	5	1	4	
Russia	7	6	1	1	4
Serbia	10	10	10		
Slovakia	6	6		5	1
Slovenia	5	4	1	3	
Sweden	4	4		1	3
Ukraine	10	10	9	1	
United Kingdom	2	2		1	1
MARCH					
Austria	5	5		2	3
Belgium	8	8	2	6	
Bulgaria	1	1		1	
Czech Republic	1	1	1		
Estonia	5	5	1	3	1
Georgia	1	1		1	
Hungary	9	3	1	1	1
Iceland	2	2		2	
Italy	1	1		1	
Jordan	7	7	4	1	2
Lithuania	4	4		3	1
Luxembourg	1	1		1	
Madagascar	8	6	5	1	
Norway	6	6	3	3	
Poland	6	6		3	3
Portugal	5	5		4	1
Romania	3	3	2	1	
Russia	7	6	3	1	2
Senegal	1	1		1	
Serbia	2	2	2		
Slovakia	4	4		3	1
Slovenia	1	1	1		
Spain	2	1	1		
Ukraine	19	19	15	4	
APRIL					
Belgium	1	1	1		
Estonia	8	8	1	7	
Hungary	1	1		1	
Iceland	1	1		1	
Jordan	10	10	1	8	1
Lithuania	1	1		1	
Madagascar	4	4	3	1	
Norway	8	8	2	6	
Portugal	1	1		1	
Romania	1	1	1		
Senegal	2	2		2	
Slovakia	1	1		1	
Slovenia	1	1		1	
Spain	1	1	1		
Ukraine	9	9	6	2	1
MAY					
Austria	1	1			1
Iceland	2	2		1	1
Jordan	4	4		3	1
Norway	1	1			1
South Africa	1	1			1
JUNE					
South Africa	2	2			2
28 Countries	274	260	90 35%	127 49%	43 16%

1. Propagated to sufficient titre to perform HI assay

2. Reduction in HI titre with ferret antiserum raised against B/Massachusetts/02/2012 (egg grown vaccine virus)

Table 12-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-02-13

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre Post infection ferret sera B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F21/07	B/Bris[2] 3/07 F21/12	B/Wis[1] 1/10 F26/10	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Serbia[2] 1894/11 F25/11	B/Stock[2] 12/11 T/C F8/12	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13
				1	1	2	3	3	2	3	3	3	2	2	2
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	320	640	320	20	640	40	320	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	640	160	640	320	<	320	80	320	1280	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	2560	320	320	640	640	40	40	640	160	160	640	80
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	80	160	160	320	10	20	80	40	40	320	40
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	40	160	40	160	1280	80	80	160	1280	320	640
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	2560	<	160	160	320	320	320	320	320	640	320	320
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK2	2560	40	80	80	160	80	160	160	320	320	320	80
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	1280	40	160	80	320	160	160	160	320	320	320	160
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK3	5120	320	320	160	320	1280	320	320	320	1280	640	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E2	2560	320	320	160	320	640	<	320	20	160	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	2560	160	320	40	160	320	40	80	40	1280	640	320
TEST VIRUSES															
B/Guyane/297/2012		2012-09-17	MDCK1/MDCK1	1280	20	160	40	160	640	20	40	40	160	320	160
B/Guyane/302/2012		2012-10-16	MDCK1/MDCK1	2560	40	320	40	320	1280	40	80	80	1280	640	320
B/Norway/2363/2012		2012-11-10	MDCK1/MDCK1	5120	40	160	80	320	640	80	80	80	1280	320	320
B/Guyane/308/2012	2	2012-11-12	MDCK1/MDCK1	2560	40	160	40	320	640	40	80	40	640	640	320
B/Blida/143/2012	3	2012-11-23	C0/MDCK1	2560	80	160	160	640	80	160	320	160	320	640	160
B/Norway/2387/2012		2012-11-24	MDCK1/MDCK1	2560	20	160	40	320	640	40	80	80	640	320	320
B/Hong Kong/3611/2012	2	2012-11-26	MDCK2/MDCK1	2560	160	160	80	320	640	40	80	80	1280	320	320
B/Pays de Loire/1825/2012		2012-11-27	MDCK1/MDCK1	2560	40	320	80	320	640	80	80	80	640	640	320
B/Paris/1828/2012	3	2012-11-29	MDCK1/MDCK1	2560	20	160	160	640	160	320	320	320	320	640	320
B/Netherlands/521/12	2	2012-11-29	MDCK2/MDCK1	5120	320	320	80	320	1280	80	80	80	1280	640	640
B/Norway/2365/2012		2012-11-30	MDCK2/MDCK1	2560	40	160	80	320	640	40	80	80	1280	320	320
B/Norway/2453/2012		2012-12-03	LLC-MK2 2/MDCK1/MDCK1	2560	40	160	80	160	1280	160	80	160	1280	320	320
B/Norway/2476/2012	2	2012-12-10	MDCK1/MDCK1	2560	40	160	40	320	640	80	80	80	640	320	320
B/Trieste/02/2012	3	2012-12-11	Cx/MDCK1	2560	40	160	160	640	80	160	320	160	160	640	160
B/Parma/02/2013		2012-12-27	MDCK1/MDCK1	2560	40	160	80	320	1280	40	80	80	640	320	320
B/Netherlands/039/13	2	2013-01-04	MDCK2/MDCK1	2560	160	320	80	320	640	40	80	80	1280	640	320
B/Hong Kong/3/2013		2013-01-07	MDCK2/MDCK1	2560	160	160	80	320	640	80	160	160	640	320	640
B/Athens GR/11/2013	2	2013-01-08	MDCK4	5120	640	320	160	640	1280	640	320	320	1280	640	1280
B/Parma/01/2013	2	2013-01-09	MDCK1/MDCK1	5120	80	320	80	320	1280	80	160	80	640	640	640
B/Iasi/131604/2013		2013-01-12	MDCK1/MDCK1	2560	160	160	40	160	640	40	80	80	640	320	320
B/Trieste/01/2013	2	2013-01-14	Cx/MDCK1	2560	320	160	80	320	1280	40	80	80	640	640	320
B/Bucuresti/131574/2013	2	2013-01-14	MDCK1/MDCK1	5120	320	320	80	320	1280	40	80	80	1280	640	320
B/Hong Kong/12/2013	2	2013-01-19	MDCK2/MDCK1	5120	320	320	160	640	1280	160	320	160	1280	640	1280
							Vaccine 2012-13 2013							Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Table 12-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-02-19

Viruses	Genetic Group	Collection date	Passage History	B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F21/07	B/Bris[2] 3/07 F24/07	B/Wis[1] 1/10 F26/10	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Serbia[2] 1894/11 F25/11	B/Stock[2] 12/11 T/C F8/12	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13
				Haemagglution Inhibition Titre											
				Post infection ferret sera											
Genetic Group				1	1	2	3	3	2	3	3	3	2	2	2
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7	5120	640	640	320	640	160	20	640	40	320	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	640	160	320	160	10	320	40	320	1280	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	1280	160	320	320	640	20	40	320	80	80	320	40
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	40	80	160	160	<	20	320	40	40	160	40
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	80	80	20	80	640	40	80	80	640	160	160
B/Serbia/1894/2011	3	2011-03-08	MDCK1/MDCK4	1280	40	80	80	160	160	160	160	160	160	160	160
B/Stockholm/12/2011	3	2011-03-28	Cx/MDCK2	640	<	40	40	80	40	40	80	80	80	160	80
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	1280	40	80	80	160	80	80	80	160	160	160	80
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	160	80	40	80	640	80	80	80	640	320	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E2	5120	320	320	160	320	80	<	160	20	160	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	1280	80	80	20	80	160	<	40	20	320	160	160
TEST VIRUSES															
B/Cameroon/12V-4394/2012		2012-09-22	MDCK2/MDCK1	5120	160	160	160	320	320	320	320	320	320	320	640
B/Cameroon/12V-4657/2012		2012-09-24	MDCK1/MDCK1	1280	80	80	80	160	80	80	160	160	160	160	80
B/Cameroon/12V-4437/2012	3	2012-10-03	MDCK1/MDCK1	2560	80	80	80	160	80	80	160	160	160	320	80
B/Cameroon/12V-4766/2012		2012-10-10	MDCK1/MDCK1	2560	80	80	160	160	80	160	160	160	160	320	80
B/Cameroon/12V-4798/2012		2012-10-16	MDCK1/MDCK1	1280	40	80	80	80	40	80	80	80	80	160	80
B/Cameroon/12V-4801/2012	3	2012-10-17	MDCK1/MDCK1	1280	40	80	80	160	40	80	160	80	160	320	80
B/Cameroon/12V-5202/2012		2012-10-24	MDCK1/MDCK1	1280	<	80	40	160	40	40	160	80	80	160	40
B/Cameroon/12V-5392/2012		2012-10-31	MDCK1/MDCK1	1280	<	80	80	160	20	80	160	80	80	160	40
B/Cameroon/12V-5391/2012	3	2012-11-01	MDCK1/MDCK1	1280	40	80	80	160	40	80	160	80	80	160	80
B/Cameroon/12V-5317/2012		2012-11-02	MDCK1/MDCK1	1280	<	40	80	80	40	80	80	80	80	160	80
B/Cameroon/12V-5506/2012		2012-11-08	MDCK1/MDCK1	1280	<	40	40	80	20	40	80	80	40	80	40
B/Cameroon/12V-5700/2012		2012-11-13	MDCK1/MDCK1	1280	<	80	40	80	20	40	80	80	80	160	40
B/Cameroon/12V-6101/2012		2012-11-16	MDCK1/MDCK1	1280	<	40	40	80	40	40	80	80	80	80	40
B/Cameroon/12V-5826/2012		2012-11-17	MDCK1/MDCK1	2560	80	80	80	160	160	160	160	160	160	160	160
B/Cameroon/12V-5637/2012	3	2012-11-19	MDCK1/MDCK1	1280	<	40	40	80	20	40	80	80	80	160	40
B/Cameroon/12V-6100/2012		2012-11-21	MDCK1/MDCK1	2560	40	160	80	160	80	80	160	160	160	160	160
B/Cameroon/12V-6102/2012		2012-11-26	MDCK1/MDCK1	2560	80	80	40	80	320	10	40	40	320	160	160
B/Cameroon/12V-6096/2012		2012-11-27	MDCK1/MDCK1	1280	40	80	80	160	40	80	160	80	80	160	80
B/Cameroon/12V-6089/2012		2012-11-28	MDCK1/MDCK1	1280	<	80	40	80	20	40	80	80	80	160	40
B/Cameroon/12V-6091/2012		2012-11-28	MDCK1/MDCK1	1280	<	80	80	80	40	40	160	80	80	320	80
B/Cameroon/12V-6095/2012		2012-11-28	MDCK1/MDCK1	1280	40	80	40	80	40	40	160	80	80	160	80
B/Cameroon/12V-5907/2012		2012-11-30	MDCK1/MDCK1	1280	80	80	80	80	80	160	160	160	160	160	160
B/Cameroon/12V-6088/2012	3	2012-12-04	MDCK1/MDCK1	1280	40	80	40	80	40	40	160	80	80	160	80
B/Cameroon/12V-6090/2012		2012-12-04	MDCK1/MDCK1	1280	40	80	80	80	40	80	160	80	160	160	80
B/Cameroon/12V-6092/2012	2	2012-12-05	MDCK1/MDCK1	1280	80	160	20	80	320	10	40	80	640	320	160
B/Cameroon/12V-6144/2012		2012-12-05	MDCK2/MDCK1	1280	40	40	40	80	40	80	80	80	80	80	80
B/Cameroon/12V-6127/2012	3	2012-12-08	MDCK2/MDCK1	2560	160	160	160	320	160	320	320	320	320	160	320
B/Cameroon/12V-6179/2012		2012-12-17	MDCK1/MDCK1	2560	80	80	20	80	320	20	40	40	320	160	160
B/Cameroon/12V-6180/2012	2	2012-12-17	MDCK1/MDCK1	2560	160	80	40	80	320	40	80	80	320	160	320
B/Cameroon/12V-6181/2012		2012-12-17	MDCK1/MDCK1	2560	40	80	80	160	80	160	160	160	160	160	160
B/Cameroon/13V-013/2012	3	2012-12-20	MDCK1/MDCK1	1280	40	80	80	160	40	80	160	80	80	160	80
B/Pais Vasco/13005/2012	3	2012-12-31	E1/E2	320	<	20	20	40	20	40	40	40	40	40	40
							Vaccine 2012-13 2013							Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Table 12-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-02-27

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre									
				Post infection ferret sera									
				B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F21/07	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F26/10	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13
	Genetic Group			1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E3/E4	5120	1280	1280	640	1280	320	80	640	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	640	640	640	320	40	640	1280	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	2560	320	320	1280	1280	40	160	160	1280	80
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	80	160	160	320	10	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	40	160	40	40	640	80	1280	320	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	1280	40	160	160	160	80	160	160	640	80
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK3	2560	160	160	160	160	1280	160	1280	640	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E2	2560	640	640	320	320	80	40	160	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	2560	80	320	80	320	320	80	1280	640	320
TEST VIRUSES													
B/Belgium/H20/2012	3	2012-02-06	MDCK2	1280	<	80	40	80	40	80	80	320	80
B/Belgium/G933/2012		2012-12-10	MDCK2	2560	320	320	80	160	1280	80	1280	640	320
B/Belgium/G938/2012		2012-12-10	MDCK2	2560	320	320	80	320	1280	80	1280	640	320
B/Belgium/G939/2012		2012-12-10	MDCK2	2560	160	320	40	160	640	40	640	320	320
B/Belgium/G943/2012		2012-12-10	MDCK2	2560	160	160	80	320	640	80	640	640	320
B/Belgium/G947/2012	2	2012-12-12	MDCK2	2560	80	160	40	160	640	80	640	640	320
B/Belgium/G1021/2012		2012-12-20	MDCK2	2560	80	320	40	160	1280	80	1280	640	320
B/Switzerland/7628604/2012		2012-12-21	MDCK2	2560	160	320	80	320	1280	80	1280	640	320
B/Switzerland/7659215/2012	3	2012-12-21	MDCK2	2560	40	160	80	160	80	160	160	320	80
B/Belgium/G1022/2012	2	2012-12-24	MDCK2	2560	160	320	40	160	640	80	1280	640	320
B/Belgium/S0310/2012		2012-12-24	MDCK2	2560	160	320	40	160	1280	80	1280	640	320
B/Belgium/S0314/2012	2	2012-12-25	MDCK2	1280	40	80	20	80	640	40	640	320	160
B/Switzerland/7659230/2012	2	2012-12-28	MDCK2	5120	320	320	80	320	1280	80	1280	1280	640
B/Belgium/G1023/2012		2012-12-31	MDCK2	2560	160	320	40	160	1280	80	1280	640	320
B/Malta/MV14117/2013	2	2013-01-08	MDCK2	2560	160	320	80	160	640	80	640	640	320
B/Malta/MV14378/2013	2	2013-01-14	MDCK2	5120	320	320	80	320	1280	80	1280	640	320
B/Malta/MV14443/2013	2	2013-01-15	MDCK2	2560	320	320	80	320	640	80	1280	640	320
B/Malta/MV14519/2013	2	2013-01-15	MDCK2	1280	160	160	40	160	320	40	640	320	160
B/Malta/MV720825/2013	2	2013-01-15	MDCK2	5120	320	320	160	640	1280	320	1280	640	640
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Table 12-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-03-13

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre – Post infection ferret sera: B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F26/10	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13
			Genetic Group	1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E3/E3	5120	1280	1280	640	640	320	80	640	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	1280	1280	320	640	320	80	640	1280	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	2560	640	640	640	640	40	80	80	640	80
B/Stockholm/12/2011	3	2007-08-07	E4/E2	1280	320	160	160	320	10	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	5120	320	320	160	320	1280	320	1280	640	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	5120	320	320	160	320	320	640	640	640	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	5120	320	320	160	320	1280	320	1280	320	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E2	2560	320	640	160	320	80	40	160	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	5120	640	640	160	320	640	160	1280	1280	640
TEST VIRUSES													
B/Antananarivo/3703/2012		2012-06-20	MDCK1/MDCK1	2560	640	320	320	640	40	160	160	640	80
B/Mahajanga/3849/2012	3	2012-07-03	MDCK1/MDCK1	2560	640	320	320	320	80	320	320	640	160
B/Moramanga/3972/2012		2012-07-16	MDCK1/MDCK3	2560	320	160	80	160	640	80	640	320	320
B/Moramanga/4313/2012		2012-08-14	MDCK1/MDCK1	2560	320	160	160	320	320	640	640	320	640
B/Antsirabe/4724/2012		2012-09-14	MDCK1/MDCK1	2560	320	160	320	320	80	320	320	640	160
B/Moramanga/4761/2012	3	2012-09-26	MDCK1/MDCK1	2560	320	160	160	320	80	320	320	320	160
B/Toamasina/5040/2012	3	2012-10-19	MDCK1/MDCK1	2560	160	160	160	320	80	320	80	320	160
B/Antananarivo/5020/2012	3	2012-10-23	MDCK1/MDCK1	2560	320	320	320	320	320	640	640	640	640
B/Antsirabe/5074/2012	3	2012-10-23	MDCK1/MDCK1	2560	320	160	160	320	40	160	160	320	80
B/Mahajanga/5196/2012	3	2012-11-06	MDCK1/MDCK1	5120	640	320	320	640	320	640	640	640	1280
B/Nosy Be/5203/2012		2012-11-07	MDCK1/MDCK1	2560	320	320	160	320	320	640	640	320	640
B/Antsirabe/5290/2012		2012-11-12	MDCK1/MDCK1	1280	320	320	160	320	40	160	160	320	80
B/Mananjary/5388/2012	3	2012-11-14	MDCK1/MDCK1	1280	320	160	160	320	40	160	160	320	80
B/Antananarivo/5413/2012		2012-11-20	MDCK1/MDCK1	2560	320	160	160	320	80	160	320	320	160
B/Ireland/89450/2012	2	2012-11-22	MDCK2	2560	320	320	80	160	640	80	640	320	160
B/Ireland/90999/2012		2012-11-25	MDCK2	2560	320	320	80	320	1280	80	640	640	320
B/Ireland/90800/2012		2012-11-26	MDCK2	2560	320	320	160	320	640	160	640	320	640
B/Cameroon/12V-5783/2012	3	2012-11-26	MDCK2	5120	320	160	160	320	80	320	320	320	320
B/Taolagnaro/5547/2012		2012-11-26	MDCK1/MDCK1	2560	320	320	160	640	640	320	640	640	320
B/Ireland/91035/2012		2012-11-28	MDCK2	2560	320	320	80	320	640	80	640	640	320
B/Maevatanana/5552/2012		2012-11-28	MDCK1/MDCK1	2560	320	160	160	320	160	320	320	320	320
B/Mahajanga/5701/2012		2012-12-10	MDCK1/MDCK1	5120	320	320	320	640	640	640	640	320	640
B/Nosy Be/5719/2012		2012-12-10	MDCK1/MDCK1	5120	320	320	320	640	640	640	640	640	640
B/Catalonia/5485S/2012		2012-12-10	P0/MDCK1	2560	320	160	80	320	640	80	640	320	320
B/Ukraine/5621/2012	2	2012-12-12	MDCK2	2560	320	160	80	160	640	80	640	320	320
B/Ireland/94775/2012		2012-12-12	MDCK2	2560	160	160	40	160	320	40	320	320	320
B/Ireland/94979/2012	3	2012-12-12	MDCK2	2560	320	160	80	320	40	160	80	320	80
B/Catalonia/2061945NS/2012		2012-12-13	P0/MDCK1	5120	320	320	80	320	640	80	640	320	320
B/Catalonia/5536S/2012		2012-12-16	P0/MDCK1	5120	640	320	80	320	1280	160	1280	320	640
B/Catalonia/2062653NS/2012		2012-12-17	P0/MDCK1	5120	320	320	80	320	1280	80	1280	640	640
B/Ghom/26191/2012	3	2012-12-26	MDCK2/MDCK1	2560	160	160	80	320	160	640	320	320	320
B/Ireland/98426/2012		2012-12-27	MDCK3	2560	320	320	80	320	640	80	1280	640	320
B/Ireland/98719/2012		2012-12-28	MDCK3	2560	160	160	40	80	320	40	320	320	160
B/Ireland/00329/2013	2	2013-01-02	MDCK2	2560	320	160	80	320	1280	80	640	640	320
B/Ireland/00217/2013	3	2013-01-02	MDCK3	640	160	80	80	160	10	80	40	160	20
B/Belgium/S0039/2013		2013-01-05	MDCK2	5120	320	320	80	320	1280	80	1280	640	320
B/Catalonia/2066914NS/2013	3	2013-01-07	P0/MDCK1	1280	160	80	40	80	20	80	80	160	40
B/Belgium/S0052/2013		2013-01-08	MDCK2	5120	320	320	80	160	640	80	640	640	320
B/Catalonia/5595S/2013		2013-01-08	P0/MDCK1	2560	160	160	40	160	320	80	320	320	160
B/Ireland/02490/2013	2	2013-01-08	MDCK3	1280	160	160	40	80	320	20	320	320	160
B/Belgium/S0064/2013		2013-01-10	MDCK2	5120	640	640	320	640	1280	320	1280	1280	640
B/Valencia/7S/2013	2	2013-01-10	P0/MDCK1	2560	320	160	80	160	1280	160	1280	320	640
B/Belgium/S0081/2013	2	2013-01-12	MDCK2	2560	160	160	40	160	320	40	640	320	160
B/Belgium/S0092/2013	2	2013-01-14	MDCK2	5120	320	320	160	320	640	160	640	320	640
B/Catalonia/5653S/2013		2013-01-14	P0/MDCK1	2560	320	160	80	160	640	80	640	320	320
B/Belgium/S0084/2013		2013-01-15	MDCK2	2560	160	160	40	160	640	40	640	320	160
B/Catalonia/2071966NS/2013		2013-01-28	P0/MDCK1	2560	320	160	80	160	640	80	640	320	320
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Table 12-5. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-04-16

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre									
				Post infection ferret sera									
				B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F26/10	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13
	Genetic Group			1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7	5120	1280	1280	320	640	160	160	320	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E1	5120	640	640	320	640	160	40	320	1280	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	5120	640	320	640	640	20	80	80	640	80
B/Stockholm/12/2011	3	2007-08-07	Ex/E2	5120	320	160	320	320	<	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	5120	160	80	40	80	640	80	640	160	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	5120	160	160	80	320	160	320	320	320	320
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	5120	160	80	40	160	640	160	640	320	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	5120	640	1280	320	640	80	20	320	1280	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	320	320	80	320	320	80	640	640	320
TEST VIRUSES													
B/Cameroon/12V-5756/2012		2012-11-22	MDCK2	2560	130	160	80	320	80	80	160	320	160
B/England/286/2013	2	2013-02-03	MDCK1/MDCK1	5120	640	160	160	320	640	320	640	320	640
B/England/213/2013	2	2013-02-04	SIAT1/MDCK1	5120	160	80	20	80	640	40	640	160	160
B/Bulgaria/173/2013		2013-02-04	C1/MDCK1	2560	320	320	80	320	1280	80	1280	320	320
B/Bulgaria/178/2013	2	2013-02-04	C1/MDCK1	1280	320	320	80	320	1280	80	1280	640	640
B/Bulgaria/340/2013		2013-02-08	C1/MDCK1	2560	320	160	80	320	1280	160	1280	320	640
B/Bulgaria/327/2013		2013-02-11	C1/MDCK1	2560	320	160	160	320	160	320	320	320	160
B/Bulgaria/297/2013		2013-02-12	C1/MDCK1	1280	320	160	40	160	1280	40	1280	320	320
B/Bulgaria/298/2013	2	2013-02-12	C1/MDCK1	2560	320	160	40	160	1280	40	640	320	320
B/Bulgaria/351/2013		2013-02-12	C1/MDCK1	2560	320	320	80	320	1280	80	1280	320	640
B/Bulgaria/348/2013	2	2013-02-13	C1/MDCK1	2560	320	320	80	320	1280	80	1280	640	320
B/Bulgaria/337/2013		2013-02-14	C1/MDCK1	2560	640	320	160	640	1280	160	1280	640	1280
B/Bulgaria/365/2013		2013-02-14	C1/MDCK1	1280	160	160	80	160	1280	80	1280	320	640
B/Novosibirsk/5/2013	2	2013-02-18	C3/MDCK2	5120	160	80	40	80	320	80	320	160	320
B/St. Petersburg/23/2013	2	2013-02-18	C2/MDCK1	5120	160	80	40	80	320	20	320	160	160
B/Bulgaria/402/2013	2	2013-02-20	C1/MDCK1	1280	160	160	40	160	640	40	640	320	320
B/Bulgaria/505/2013	2	2013-02-23	C2/MDCK1	2560	320	160	80	160	1280	160	1280	320	640
B/Bulgaria/608/2013	2	2013-03-04	C1/MDCK1	2560	320	160	80	320	1280	160	1280	320	640
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-6. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-04-24 + 2013-04-30

Viruses	Genetic Group	Collection date	Passage History	B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F26/10	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13	
				Haemagglution Inhibition Titre										
				Post infection ferret sera										
				1	1	2	3	3	2	3	2	2	2	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E3	5120	1280	1280	640	1280	640	80	640	1280	320	
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	320	40	640	1280	160	
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	1280	320	160	320	640	10	40	40	320	40	
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	160	160	160	320	10	40	40	320	20	
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	320	160	80	160	1280	160	1280	320	640	
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	5120	320	160	160	320	320	320	640	320	320	
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	320	160	80	160	1280	160	1280	320	1280	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	5120	640	640	320	320	160	40	320	1280	160	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	2560	320	160	40	80	320	40	320	320	160	
TEST VIRUSES														
B/Cameroon/12V-5323/2012		2012-11-05	MDCK3	1280	80	40	40	80	20	80	80	160	40	
B/Moscow/93/2012	3	2012-11-22	Cx/MDCK1	2560	160	80	80	320	160	320	320	160	320	
B/Cameroon/12V-5780/2012		2012-11-23	MDCK3	1280	80	80	40	160	40	80	80	160	80	
B/Luxembourg/650/2012	2	2012-12-18	MDCK2	5120	320	640	320	1280	2560	80	640	320	640	
B/Astana/14/2013	2	2013-01-08	MDCK2	2560	320	160	80	160	1280	80	640	320	320	
B/Cairo/161/2013	2	2013-01-09	C1/MDCK2	2560	160	160	20	80	640	40	640	320	160	
B/Almaty/2845/2013	2	2013-01-13	MDCK2	2560	320	160	40	160	640	40	640	320	320	
B/Almaty/2845/2013	2	2013-01-13	Ex/E1	1280	160	160	80	80	80	20	80	320	40	196/198 DKT
B/Cairo/164/2013	2	2013-01-14	C1/MDCK2	2560	160	160	20	80	640	40	640	320	160	
B/Luxembourg/83/2013	2	2013-01-16	MDCK1/MDCK1	1280	80	40	20	80	640	40	320	160	160	
B/Luxembourg/92/2013	2	2013-01-17	MDCK1/MDCK1	1280	80	80	20	40	640	20	320	160	160	
B/Luxembourg/101/2013	2	2013-01-17	MDCK1/MDCK1	1280	80	80	40	80	320	40	320	160	160	
B/Karaganda/2914/2013	2	2013-01-21	MDCK2	2560	320	160	40	160	1280	40	640	320	320	
B/Estonia/74786/2013		2013-01-24	MDCK3	2560	160	160	160	320	160	320	320	320	160	
B/Luxembourg/177/2013	2	2013-01-24	MDCK1/MDCK1	640	80	80	20	40	320	20	320	160	160	
B/Estonia/74903/2013	3	2013-01-28	MDCK2	5120	320	160	160	320	160	320	320	320	160	
B/Luxembourg/724/2013	2	2013-03-07	MDCK2	2560	320	1280	320	1280	2560	320	640	320	1280	
							Vaccine 2012-13 2013					Vaccine 2013-14		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-7. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-05-14 + 2013-05-29

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre / Post infection ferret sera: B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13	
Genetic Group				1	1	2	3	3	2	3	2	2	2	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	320	640	320	80	640	1280	160	
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	640	320	640	160	40	320	1280	160	
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	1280	640	320	320	640	20	80	80	640	40	
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	160	160	160	320	<	40	40	320	20	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK2	1280	160	80	80	80	640	80	640	160	320	
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	320	160	320	320	160	320	320	320	640	
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK2	2560	320	160	160	160	640	160	640	320	640	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	320	320	160	40	320	640	80	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	320	160	320	320	80	640	640	320	
TEST VIRUSES														
B/Ghom/28441/2012	2	2012-12-08	E1/E1	1280	160	80	20	20	320	20	640	160	80	
B/Yazd/9800/2012	2	2012-12-18	E1/E1	1280	320	320	80	160	80	40	160	320	80	196/198 XKT
B/Slovenia/319/2013		2013-01-21	MDCKx/MDCK1	2560	160	160	80	160	640	80	640	320	320	
B/Slovenia/312/2013	3	2013-01-22	MDCK	1280	160	160	160	320	80	320	160	160	160	
B/Slovenia/386/2013		2013-01-25	MDCKx/MDCK1	2560	320	160	160	320	1280	160	640	320	640	
B/Ostrava/1/2013	2	2013-02-08	MDCK2/MDCK1	2560	160	160	80	160	640	160	640	160	320	
B/Slovenia/651/2013	2	2013-02-08	MDCKx/MDCK1	1280	160	160	80	80	640	40	640	160	160	
B/Slovenia/699/2013		2013-02-12	MDCKx/MDCK1	2560	320	160	80	160	640	80	640	160	320	
B/Ostrava/59/2013	2	2013-02-17	MDCK2/MDCK1	2560	320	320	160	320	640	80	640	320	320	
B/Slovenia/828/2013	2	2013-02-20	MDCKx/MDCK1	1280	160	80	80	80	1280	40	640	320	160	
B/Ostrava/58/2013		2013-02-22	MDCK2/MDCK1	1280	160	160	80	160	640	80	640	320	160	
B/St. Petersburg/91/2013		2013-02-25	C3/MDCK1	2560	160	160	80	160	640	80	640	320	160	
B/Slovenia/942/2013		2013-02-25	MDCKx/MDCK1	2560	160	160	80	160	640	40	640	160	160	
B/Plzen/52/2013	2	2013-02-28	MDCK2/MDCK1	2560	320	160	80	160	640	80	640	320	160	
B/Plzen/53/2013	2	2013-03-05	MDCK2/MDCK1	2560	320	160	160	320	640	160	640	320	320	
B/Slovenia/1281/2013	2	2013-03-19	MDCKx/MDCK1	2560	160	160	80	80	640	40	640	320	160	
B/Omsk/31/2013		2013-03-28	C3/MDCK1	2560	320	160	160	320	160	320	160	320	160	
B/Omsk/35/2013	2	2013-03-28	C3/MDCK1	2560	160	160	160	160	640	160	640	160	640	
B/Novosibirsk/13/2013	2	2013-03-28	C2/MDCK1	2560	320	160	160	320	1280	160	640	320	640	
B/Leon/120/2013		2013-03-30	MDCK2	5120	320	160	40	160	80	80	80	320	80	
B/Leon/121/2013	3	2013-04-02	MDCK2	2560	160	160	80	160	40	80	80	320	80	
B/Slovenia/1613/2013	3	2013-04-15	MDCKx/MDCK1	2560	160	160	160	160	160	160	160	160	80	
							Vaccine 2012-13 2013					Vaccine 2013-14		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-8. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-06-04

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre: Post infection ferret sera									
				B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13
	Genetic Group			1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	320	320	640	160	40	320	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	160	320	320	160	40	320	1280	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	1280	640	320	320	320	20	80	80	320	40
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	320	160	160	320	10	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	160	40	80	80	640	80	640	160	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	1280	320	80	160	160	160	320	320	160	320
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4/MDCK1	2560	320	80	80	160	640	160	640	160	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	160	160	320	80	20	160	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	80	160	320	320	80	640	640	320
TEST VIRUSES													
B/Bucuresti/133317/2013	2	2013-02-01	MDCK1/MDCK1	1280	160	20	40	80	640	40	320	160	160
B/Sibiu/133676/2013		2013-02-04	MDCK1/MDCK1	5120	320	80	160	320	640	160	640	320	640
B/Suceava/133820/2013	2	2013-02-06	MDCK1/MDCK1	1280	160	40	80	80	640	80	320	160	320
B/Iasi/135846/2013		2013-02-27	MDCK1/MDCK1	1280	320	40	80	80	320	80	640	160	320
B/Bihor/135892/2013	2	2013-02-27	MDCK1/MDCK1	1280	320	40	80	10	640	40	320	160	320
B/Mures/136729/2013	2	2013-03-06	MDCK1/MDCK1	1280	160	40	40	160	640	80	320	160	320
B/Iasi/138319/2013	2	2013-03-26	MDCK1/MDCK1	2560	320	80	160	160	640	160	640	320	640
B/Vaslui/139688/2013	2	2013-04-08	MDCK1/MDCK1	2560	640	80	160	320	640	160	640	320	640
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-9. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-06-11

Viruses	Genetic Group	Collection date	Passage History	B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F02/13	B/Mass[2] 2/12 T/C F03/13	
				Haemagglution Inhibition Titre										
				Post infection ferret sera										
				1	1	2	3	3	2	3	2	2	2	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	320	320	160	40	320	1280	80	
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	320	640	320	40	320	1280	160	
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	1280	320	320	320	320	10	80	80	320	40	
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	160	160	160	160	<	40	40	320	20	
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	640	80	80	40	40	640	80	640	160	160	
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	160	160	160	320	320	320	320	320	320	
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4/MDCK1	2560	160	160	80	160	1280	160	640	320	640	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	640	160	80	40	40	160	20	320	160	160	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	320	640	160	320	160	40	320	640	160	
TEST VIRUSES														
B/Serbia/NS-196/2013		2013-01-08	MDCK2	1280	80	80	80	40	320	40	320	160	160	
B/Belgrade/132/2013		2013-01-16	C0/MDCK1	1280	160	80	80	80	640	40	640	160	160	
B/Sabac/259/2013		2013-01-21	C1/MDCK1	1280	80	80	80	80	320	40	320	160	160	
B/Serbia/NS-217/2013		2013-01-21	MDCK2	1280	80	80	80	80	320	40	640	160	160	
B/Serbia/NS-219/2013	2	2013-01-23	MDCK2	1280	80	80	80	40	320	40	320	160	80	
B/Ukraine/5729/2013		2013-01-23	C2/MDCK1	1280	160	160	160	80	640	80	640	320	320	
B/Valjevo/470/2013		2013-01-29	C1/MDCK1	1280	160	160	80	80	640	40	320	160	160	
B/Serbia/NS-249/2013	3	2013-01-31	MDCK2	1280	80	80	80	40	640	40	320	160	160	
B/Belgrade/660/2013		2013-02-06	C1/MDCK1	1280	160	160	80	80	640	40	320	160	160	
B/Ukraine/5733/2013		2013-02-06	C2/MDCK1	1280	80	80	160	160	80	160	160	160	160	
B/Ukraine/5732/2013	3	2013-02-07	C2/MDCK1	1280	80	80	160	80	40	80	80	160	40	
B/Ukraine/5734/2013	3	2013-02-12	C2/MDCK1	1280	80	80	80	160	40	80	80	160	40	
B/Leskovac/904/2013		2013-02-13	C0/MDCK1	1280	160	80	80	80	640	40	320	160	160	
B/Leskovac/905/2013	2	2013-02-13	C1/MDCK1	1280	160	160	80	160	640	40	320	160	160	
B/Ukraine/5735/2013		2013-02-13	C2/MDCK1	1280	160	80	80	160	40	80	160	160	80	
B/Belgrade/898/2013		2013-02-14	C1/MDCK1	1280	160	80	80	80	640	40	320	160	160	
B/Belgrade/959/2013		2013-02-14	C0/MDCK1	1280	80	80	80	80	320	40	320	160	160	
B/Belgrade/1051/2013		2013-02-18	C0/MDCK1	1280	160	80	80	80	320	40	320	160	160	
B/Obrenovac/1056/2013		2013-02-20	C0/MDCK1	1280	80	80	80	80	320	40	320	160	160	
B/Cuprija/1097/2013	2	2013-02-21	C1/MDCK1	1280	160	160	80	80	640	40	320	160	160	
B/Nis/1197/2013		2013-02-25	C0/MDCK1	1280	80	80	80	80	320	40	320	160	160	
B/Belgrade/1316/2013		2013-02-27	C0/MDCK1	1280	160	160	80	160	640	80	320	160	160	
B/Ukraine/5835/2013		2013-02-28	C2/MDCK1	1280	80	80	80	160	320	80	320	160	160	
B/Belgrade/1317/2013		2013-03-01	C0/MDCK1	1280	160	160	80	80	320	40	320	160	160	
B/Ukraine/5813/2013		2013-03-04	C2/MDCK1	1280	80	80	80	80	640	80	320	160	320	
B/Ukraine/5836/2013		2013-03-05	C2/MDCK1	1280	160	80	80	160	320	40	320	160	160	
B/Belgrade/1481/2013	2	2013-03-07	C1/MDCK1	1280	160	80	80	80	320	40	320	160	160	
B/Ukraine/5815/2013		2013-03-10	C2/MDCK1	1280	160	80	80	80	320	40	320	160	80	
B/Ukraine/5814/2013	2	2013-03-12	C2/MDCK1	640	160	80	80	80	320	40	320	160	160	
B/St. Petersburg/60/2013	2	2013-03-14	E2/E1	320	80	40	10	20	80	<	80	80	20	196/198 NKK
B/Iasi/138316/2013		2013-03-26	MDCK1/MDCK1	1280	160	160	80	160	640	80	640	160	160	
B/Ukraine/5834/2013	2	2013-04-28	C0/MDCK1	640	80	80	80	40	320	40	320	160	160	
							Vaccine 2012-13 2013					Vaccine 2013-14		

1. < = <40;   2. < = <10;   3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-10. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-06-19 + 2013-06-26

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre Post infection ferret sera B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F07/13	B/Mass[2] 2/12 T/C F08/13	
	Genetic Group			1	1	2	3	3	2	3	2	2	2	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	1280	640	1280	320	80	640	1280	320	
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	640	640	320	80	640	1280	160	
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	2560	640	640	640	1280	40	160	80	640	160	
B/Stockholm/12/2011	3	2007-08-07	E4/E1	2560	320	160	320	320	10	40	40	320	40	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	5120	320	160	160	320	1280	320	1280	320	1280	
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	5120	640	320	640	1280	640	640	640	640	1280	
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	5120	320	320	320	640	1280	320	1280	320	1280	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	5120	640	640	320	640	160	40	320	640	160	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	5120	1280	640	320	640	1280	320	1280	640	1280	
TEST VIRUSES														
B/Tbilisi/253/2013	2	2013-02-08	MDCK3	2560	320	160	160	160	1280	80	640	160	320	
B/Hungary/04/2013		2013-02-15	MDCK1/E3/MDCK2	1280	160	160	80	160	320	40	320	160	80	
B/Ukraine/5886/2013		2013-02-19	C2/MDCK1	5120	640	320	640	640	640	640	1280	640	1280	
B/Ukraine/5891/2013		2013-02-23	C2/MDCK1	5120	640	320	320	640	1280	640	1280	640	1280	
B/Ukraine/5894/2013	2	2013-02-23	C2/MDCK1	5120	320	320	320	640	1280	320	1280	320	1280	
B/Ukraine/5895/2013		2013-02-26	C2/MDCK1	5120	640	640	640	1280	1280	1280	1280	640	1280	
B/Ukraine/227/2013	2	2013-02-26	MDCK1/MDCK1	1280	160	160	80	80	640	20	640	160	80	
B/Ukraine/5934/2013	2	2013-03-01	C4/MDCK1	5120	320	320	320	320	1280	320	1280	320	1280	
B/Ukraine/5902/2013		2013-03-03	C2/MDCK1	5120	640	640	640	1280	1280	640	1280	640	1280	
B/Ukraine/5897/2013	2	2013-03-04	C2/MDCK1	2560	320	320	320	320	1280	80	1280	320	320	
B/Ukraine/5900/2013		2013-03-04	C2/MDCK1	5120	320	320	320	640	640	640	640	160	1280	
B/Hungary/15/2013		2013-03-04	MDCK2/E2/MDCK2	2560	160	160	80	160	1280	80	640	160	320	
B/Tbilisi/755/2013	2	2013-03-05	MDCK3	1280	160	160	80	160	640	80	640	160	160	
B/Ukraine/5904/2013	3	2013-03-05	C2/MDCK1	2560	320	160	320	320	160	320	160	160	320	
B/Ukraine/5907/2013		2013-03-05	C2/MDCK1	5120	320	320	320	640	1280	320	1280	320	640	
B/Ukraine/5905/2013		2013-03-06	C2/MDCK1	5120	640	640	320	640	1280	640	1280	320	1280	
B/Ukraine/5938/2013		2013-03-15	C3/MDCK1	5120	640	320	320	640	1280	320	1280	640	1280	
B/Hungary/22/2013	2	2013-03-26	E3/E1	320	80	40	10	20	40	<	40	80	20	196/198 NKN
B/Ukraine/5940/2013	2	2013-03-28	C3/MDCK1	5120	640	320	320	640	1280	640	1280	640	1280	
B/Ukraine/5941/2013	2	2013-04-03	C2/MDCK1	5120	640	320	320	640	1280	320	1280	640	1280	
B/Ukraine/5943/2013	2	2013-04-12	C2/MDCK2	1280	160	80	80	160	640	80	640	80	320	
							Vaccine 2012-13 2013					Vaccine 2013-14		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-11. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-07-05

Viruses	Genetic Group	Collection date	Passage History	B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F2/13	B/Mass[2] 2/12 T/C F3/13	
				Haemagglutinion Inhibition Titre										
				Post infection ferret sera										
				1	1	2	3	3	2	3	2	2	2	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	1280	640	1280	320	80	640	1280	160	
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	640	1280	320	80	640	1280	160	
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	2560	640	640	640	1280	40	160	160	640	80	
B/Stockholm/12/2011	3	2007-08-07	E4/E1	2560	160	320	320	640	10	80	80	320	40	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	2560	160	160	160	320	640	320	1280	320	640	
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	5120	160	320	320	640	640	640	640	640	640	
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	5120	160	320	320	640	1280	320	1280	640	640	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	5120	1280	1280	640	640	160	40	640	1280	160	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	2560	160	320	160	320	320	80	640	640	320	
TEST VIRUSES														
B/Iceland/29/2012		2012-12-25	MDCK1/MDCK1	2560	320	160	160	320	640	80	640	320	320	
B/Pavia/7/2013		2013-01-15	Cx/MDCK1	2560	320	160	160	320	640	80	640	320	320	
B/Milano/72/2013		2013-01-28	MDCK1/MDCK1	1280	160	160	80	160	640	40	640	320	320	
B/Parma/50/2013		2013-01-29	MDCK2/MDCK2	2560	320	160	160	320	1280	160	1280	320	640	
B/Perugia/6/2013		2013-01-30	MDCK1/MDCK1	2560	320	160	160	160	320	40	640	320	160	
B/Firenze/6/2013	3	2013-01-31	MDCK2/MDCK1	2560	320	160	320	320	160	320	320	320	320	
B/Trieste/06/2013		2013-02-04	Cx/MDCK1	2560	320	320	160	320	640	80	1280	320	320	
B/Pavia/52/2013		2013-02-13	Cx/MDCK1	2560	320	160	160	160	1280	160	1280	320	640	
B/Trieste/11/2013		2013-02-17	Cx/MDCK1	2560	160	160	80	160	640	80	640	320	320	
B/Trieste/15/2013		2013-02-19	Cx/MDCK1	2560	320	160	160	320	640	80	640	320	320	
B/Perugia/19/2013		2013-02-20	MDCK1/MDCK1	2560	320	160	320	320	320	160	640	320	160	
B/Roma/14/2013	2	2013-02-21	MDCK2/MDCK1	1280	160	160	80	160	640	80	640	160	320	
B/Pavia/76/2013		2013-02-21	Cx/MDCK1	1280	160	160	80	160	640	160	640	320	640	
B/Parma/129/2013		2013-02-22	MDCK1/MDCK1	2560	320	160	320	320	320	320	640	160	640	
B/Iceland/37/2013	2	2013-02-25	MDCK1/MDCK1	1280	160	160	40	80	80	20	160	320	80	196/198 NKA
B/Hungary/08/2013		2013-02-27	MDCK1/E3/MDCK3	5120	1280	640	320	640	320	160	640	640	640	
B/Perugia/25/2013		2013-02-27	MDCK1/MDCK1	1280	160	160	160	160	320	40	640	320	320	
B/Iceland/39/2013		2013-03-04	MDCK1/MDCK1	2560	160	160	160	320	640	80	640	320	320	
B/Perugia/35/2013	2	2013-03-06	MDCK1/MDCK1	2560	160	160	160	160	640	80	640	320	320	
B/Ukraine/272/2013	3	2013-03-09	MDCK1/MDCK1	2560	320	160	160	320	40	160	160	320	80	
B/Hungary/09/2013		2013-03-11	MDCK1/E2/MDCK3	5120	320	320	320	640	1280	640	1280	640	1280	
B/Ukraine/294/2013		2013-03-13	MDCK1/MDCK1	5120	320	320	160	640	640	160	640	640	320	
B/Ukraine/346/2013		2013-03-20	MDCK1/MDCK1	5120	640	640	640	640	640	80	640	640	320	
B/Iceland/44/2013		2013-03-20	MDCK1/MDCK1	2560	160	160	160	160	640	40	640	320	160	
B/Ukraine/320/2013		2013-03-26	MDCK1/MDCK1	5120	320	320	320	320	640	80	640	640	320	
B/Ukraine/323/2013		2013-03-28	MDCK1/MDCK1	2560	320	320	160	320	640	80	640	640	320	
B/Dnipropetrovsk/367/2013	2	2013-03-29	MDCK1/MDCK1	5120	320	320	320	320	640	80	640	320	320	
B/Ukraine/340/2013		2013-04-01	MDCK1/MDCK1	5120	320	320	160	320	640	80	640	640	320	
B/Ukraine/374/2013		2013-04-02	MDCK1/MDCK1	5120	640	320	320	640	1280	160	1280	640	640	
B/Ukraine/378/2013		2013-04-03	MDCK1/MDCK1	5120	320	320	320	320	640	80	640	640	320	
B/Ukraine/385/2013	3	2013-04-08	MDCK1/MDCK1	2560	320	160	640	320	160	160	160	320	160	
B/Ukraine/406/2013	2	2013-04-10	MDCK1/MDCK1	5120	320	320	320	320	640	80	1280	640	320	
B/Iceland/52/2013		2013-04-10	MDCK1/MDCK1	2560	160	160	160	160	640	40	640	320	160	
B/Ukraine/474/2013	3	2013-04-18	MDCK1/MDCK1	5120	320	160	320	320	80	160	160	320	160	
B/Iceland/63/2013	2	2013-05-16	MDCK1/MDCK1	1280	160	80	80	160	320	40	320	160	160	
B/Iceland/64/2013	2	2013-05-17	MDCK1/MDCK1	2560	160	160	160	320	640	80	640	320	320	
B/Firenze/11/2013		unknown	MDCK2/MDCK1	2560	320	160	160	320	640	80	640	320	160	
							Vaccine 2012-13 2013					Vaccine 2013-14		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-12. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-07-12

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre / Post infection ferret sera: B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F2/13	B/Mass[2] 2/12 T/C F3/13	
	Genetic Group			1	1	2	3	3	2	3	2	2	2	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	160	640	160	40	320	640	80	
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	160	40	320	1280	160	
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	2560	640	320	320	640	20	80	80	640	40	
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	160	160	160	320	<	40	40	320	20	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	2560	160	160	80	160	640	160	640	320	320	
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	160	160	160	320	160	320	320	320	320	
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	160	160	80	160	640	160	640	320	320	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	320	640	160	40	320	1280	80	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	5120	640	320	160	320	640	160	640	640	640	
TEST VIRUSES														
B/Iceland/26/2012		2012-10-17	MDCK1/MDCK1	5120	160	160	160	320	160	320	320	160	320	
B/Stockholm/6/2013		2013-01-01	MDCK1/MDCK1	1280	160	80	40	80	640	40	320	160	160	
B/Stockholm/1/2013	2	2013-01-02	MDCK0/MDCK1	1280	160	80	40	80	640	40	320	160	160	
B/Stockholm/5/2013		2013-01-03	MDCK1/MDCK1	2560	320	160	80	160	640	40	640	320	160	
B/Stockholm/2/2013	2	2013-01-06	MDCK0/MDCK1	1280	160	80	40	80	640	40	320	160	80	
B/Stockholm/7/2013		2013-01-07	MDCK0/MDCK1	2560	160	160	80	160	640	40	640	320	160	
B/Lisboa PT/8/2013		2013-01-18	SIAT3/MDCK1	1280	160	80	40	80	640	40	320	160	160	
B/Porto PT/12/2013		2013-01-21	SIAT1/MDCK1	2560	160	160	160	160	80	320	160	160	160	
B/Braga PT/17/2013		2013-01-22	SIAT1/MDCK1	1280	160	160	80	160	640	40	640	320	160	
B/Vila Real PT/15/2013		2013-01-25	SIAT1/MDCK1	2560	160	160	80	160	640	40	640	320	160	
B/Braga PT/16/2013	3	2013-01-26	SIAT1/MDCK1	1280	160	80	80	160	40	80	80	160	40	
B/Pirot/540/2013	2	2013-01-31	C1/MDCK1	1280	160	80	160	160	40	160	80	160	80	
B/Stockholm/3/2013	2	2013-02-04	MDCK0/MDCK1	1280	160	80	40	80	320	40	320	160	80	
B/Nitra/359/2013		2013-02-05	MDCK1/MDCK1	2560	160	160	80	160	640	40	640	320	160	
B/Poprad/415/2013	2	2013-02-08	MDCK1/MDCK1	1280	160	80	80	80	640	40	640	320	160	
B/Bratislava/451/2013		2013-02-11	MDCK1/MDCK1	2560	160	160	80	160	640	40	640	320	160	
B/Trnava/486/2013		2013-02-13	MDCK1/MDCK1	1280	160	160	80	80	640	40	640	320	160	
B/Lisboa PT/72/2013	2	2013-02-13	SIAT1/MDCK1	1280	160	80	80	80	640	80	320	160	320	
B/Bratislava/501/2013		2013-02-14	MDCK1/MDCK1	1280	160	80	80	80	640	40	640	320	160	
B/Stockholm/12/2013	2	2013-02-16	MDCK1/MDCK1	640	80	80	40	40	320	40	320	160	160	
B/Stockholm/11/2013	2	2013-02-18	MDCK0/MDCK1	1280	80	80	40	80	320	40	320	160	80	
B/Stockholm/9/2013	2	2013-02-21	MDCK1/MDCK1	1280	160	160	80	80	640	40	320	320	160	
B/Trnava/636/2013	3	2013-02-25	MDCK1/MDCK1	1280	80	80	80	80	20	80	80	160	40	
B/Lisboa PT/135/2013		2013-02-28	SIAT1/MDCK1	1280	160	80	80	80	640	80	640	160	320	
B/Lisboa PT/136/2013		2013-03-06	SIAT1/MDCK1	1280	160	80	80	80	320	80	320	160	160	
B/Poprad/801/2013		2013-03-07	MDCK1/MDCK1	1280	160	80	80	80	320	40	320	320	160	
B/Trnava/832/2013	2	2013-03-08	MDCK1/MDCK1	1280	160	80	80	80	640	40	640	320	160	
B/Komarno/926/2013	2	2013-03-22	MDCK1/MDCK1	2560	160	160	80	80	640	40	640	320	160	
B/Bratislava/939/2013		2013-03-28	MDCK1/MDCK1	1280	160	80	80	80	640	80	640	160	320	
B/Hungary/27/2013	2	2013-04-03	MDCK1/E1/MDCK1	1280	160	160	40	80	80	20	160	320	80	196/198 NKX
B/Bratislava/955/2013	2	2013-04-05	MDCK1/MDCK1	2560	160	160	80	80	640	40	640	320	160	
							Vaccine 2012-13 2013					Vaccine 2013-14		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-13. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-07-19 + 2013-07-24

Viruses	Genetic Group	Collection date	Passage History	B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F2/13	B/Mass[2] 2/12 T/C F3/13
				Haemagglution Inhibition Titre									
				Post infection ferret sera									
	Genetic Group			1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7	5120	1280	1280	320	640	320	80	640	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	320	40	320	1280	80
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	2560	640	320	320	640	20	80	80	640	40
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	160	160	80	320	10	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	2560	160	160	80	160	640	160	640	160	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	160	160	160	320	160	320	320	160	320
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	1280	160	80	80	160	640	80	640	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	160	320	160	20	160	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	5120	640	320	160	320	640	160	640	640	640
TEST VIRUSES													
B/Belgium/G0525/2013		2013-01-28	MDCK3	1280	160	80	40	80	640	80	320	160	160
B/Belgium/G0537/2013		2013-01-29	MDCK2	1280	160	80	80	80	640	80	640	160	160
B/Belgium/S0283/2013		2013-02-02	MDCK2	1280	160	80	40	80	320	40	320	160	160
B/Belgium/G0506/2013	2	2013-02-04	MDCK2	1280	160	80	40	80	320	40	320	160	80
B/Belgium/G0509/2013	2	2013-02-04	MDCK2	1280	80	80	40	80	320	80	320	160	80
B/Belgium/G0532/2013		2013-02-04	MDCK2	2560	160	160	80	160	640	80	640	320	160
B/Belgium/G0535/2013		2013-02-04	MDCK2	2560	160	160	80	160	640	80	640	320	160
B/Belgium/G0536/2013		2013-02-04	MDCK2	1280	160	80	80	80	640	80	640	320	160
B/Belgium/G0538/2013	2	2013-02-05	MDCK2	1280	80	80	40	40	320	40	320	160	80
B/Belgium/S0327/2013		2013-02-06	MDCK2	2560	160	160	80	160	640	40	640	320	160
B/Hong Kong/40/2013	2	2013-02-24	MDCK2/MDCK1	2560	320	160	160	320	640	160	640	320	640
B/Hong Kong/41/2013	2	2013-02-28	MDCK2/MDCK1	2560	320	160	160	320	640	160	640	320	320
B/Lisboa PT/138/2013		2013-03-04	SIAT1/MDCK1	2560	160	160	80	320	640	160	640	160	640
B/Norway/1677/2013		2013-03-08	MDCK1/MDCK1	1280	160	80	40	80	640	80	640	160	320
B/Norway/1654/2013		2013-03-11	MDCK1/MDCK1	2560	320	160	160	320	1280	160	640	320	640
B/Castelo Branco PT/178/2013		2013-03-14	SIAT1/MDCK1	2560	160	160	160	320	160	160	160	160	160
B/Lisboa PT/210/2013	2	2013-03-14	SIAT1/MDCK1	1280	80	80	40	40	320	40	320	160	160
B/Belgium/G1044/2013	2	2013-03-19	MDCK2	640	80	80	40	80	320	40	320	160	80
B/Castelo Branco PT/179/2013	3	2013-03-21	SIAT1/MDCK1	2560	160	160	160	160	160	320	320	160	320
B/Belgium/G1062/2013		2013-03-21	MDCK2	2560	160	160	80	80	640	40	640	320	160
B/Norway/1887/2013		2013-03-22	MDCK1/MDCK1	2560	160	160	160	320	640	160	640	160	640
B/Norway/2020/2013		2013-03-22	MDCK1/MDCK1	1280	160	80	80	80	640	80	320	160	320
B/Belgium/G1076/2013		2013-03-23	MDCK2	1280	160	160	40	80	320	40	320	160	160
B/Belgium/G1064/2013	2	2013-03-25	MDCK2	1280	160	160	80	80	640	40	640	320	160
B/Belgium/G1068/2013		2013-03-25	MDCK2	1280	160	80	40	80	320	40	320	160	160
B/Belgium/G1082/2013		2013-03-25	MDCK2	1280	160	80	40	80	320	40	320	160	160
B/Belgium/G1086/2013	3	2013-03-25	MDCK2	1280	160	80	80	160	80	160	80	160	80
B/Norway/1924/2013	2	2013-03-26	MDCK1/MDCK1	2560	160	160	80	160	640	160	640	320	640
B/Belgium/G1077/2013	2	2013-03-26	MDCK2	1280	160	80	40	80	320	20	320	160	160
B/Portalegre PT/229/2013	2	2013-04-01	SIAT1/MDCK1	2560	160	80	80	80	640	80	640	160	160
B/Belgium/S0929/2013	2	2013-04-05	MDCK2	2560	160	160	80	160	320	80	320	320	160
B/Norway/2057/2013		2013-04-07	MDCK2/MDCK1	2560	160	80	80	80	640	80	640	160	320
B/Norway/2103/2013	2	2013-04-07	MDCK1/MDCK1	2560	160	80	80	160	320	160	640	160	320
B/Norway/2161/2013	2	2013-04-08	MDCK1/MDCK1	2560	160	80	80	160	640	80	320	160	320
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-14. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-08-02

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre / Post infection ferret sera: B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F34/11	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F2/13	B/Mass[2] 2/12 T/C F3/13
			Genetic Group	1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	1280	640	640	320	80	640	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	320	640	160	40	320	1280	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	5120	1280	640	640	640	40	160	160	640	80
B/Stockholm/12/2011	3	2007-08-07	E4/E1	2560	320	160	160	640	10	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	160	160	80	320	640	80	640	320	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	640	160	80	160	320	40	160	160	160	40
B/Hong Kong/3577/2012	2	2012-06-13	MDCK3	1280	160	160	80	320	640	40	640	320	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	1280	320	640	160	40	320	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	5120	640	640	320	640	640	160	640	640	640
TEST VIRUSES													
B/Tsiroanomandidy/485/2013	3	2013-02-18	MDCK1/MDCK1	1280	320	160	160	640	80	160	160	320	80
B/Mahajanga/697/2013	3	2013-03-05	MDCK1/MDCK1	1280	320	160	160	640	80	160	160	320	80
B/Estonia/76502/2013		2013-03-08	MDCK1/MDCK1	1280	160	160	160	320	40	80	80	160	40
B/Estonia/76615/2013	3	2013-03-11	MDCK1/MDCK1	1280	160	160	160	320	40	80	80	160	40
B/Moramanga/1070/2013	3	2013-03-20	MDCK1/MDCK1	1280	160	160	160	320	40	160	160	320	80
B/Antananarivo/1077/2013	3	2013-03-22	MDCK1/MDCK1	1280	320	160	160	640	80	160	160	320	80
B/Estonia/77011/2013		2013-03-22	MDCK2/MDCK1	1280	160	80	80	160	320	40	320	160	160
B/Norway/1946/2013	2	2013-03-23	MDCK1/MDCK2	2560	160	160	80	320	640	80	640	320	320
B/Antananarivo/1081/2013		2013-03-25	MDCK1/MDCK2	1280	160	160	160	320	40	80	80	160	80
B/Tsiroanomandidy/1124/2013	3	2013-03-26	MDCK1/MDCK2	5120	320	320	160	640	320	640	640	320	640
B/Antsirabe/1153/2013		2013-03-27	MDCK1/MDCK2	1280	160	160	160	320	80	160	160	320	80
B/Estonia/77112/2013	2	2013-03-27	MDCK2/SIAT1	2560	320	320	160	640	640	80	640	320	320
B/Estonia/77128/2013		2013-03-27	MDCK2/SIAT1	1280	80	80	40	80	160	40	160	80	160
B/Antananarivo/1226/2013	3	2013-04-03	MDCK1/MDCK1	1280	160	160	160	320	80	160	160	160	80
B/Estonia/77368/2013		2013-04-08	MDCK1/MDCK1	1280	160	80	80	160	320	40	320	160	160
B/Estonia/77387/2013		2013-04-08	MDCK2/MDCK1	2560	160	160	80	320	320	40	320	160	160
B/Estonia/77391/2013	2	2013-04-08	MDCK1/MDCK1	1280	160	80	80	320	320	40	320	160	160
B/Estonia/77427/2013	2	2013-04-08	MDCK2/MDCK1	5120	320	160	160	320	1280	320	640	320	640
B/Estonia/77447/2013	2	2013-04-09	MDCK1/MDCK1	1280	160	160	80	320	320	40	320	160	160
B/Norway/2162/2013		2013-04-10	MDCK1/MDCK1	1280	160	160	160	320	160	80	80	160	40
B/Estonia/77471/2013		2013-04-10	MDCK2/MDCK1	1280	160	160	80	320	320	40	320	160	160
B/Estonia/77557/2013		2013-04-12	MDCK1/MDCK1	1280	160	160	80	320	320	40	320	160	160
B/Norway/2531/2013	3	2013-04-17	MDCK1/MDCK1	1280	160	80	80	640	40	80	80	160	40
B/Estonia/77676/2013		2013-04-18	MDCK2/MDCK1	2560	160	160	80	320	320	40	320	160	160
B/Norway/2254/2013		2013-04-21	MDCK1/MDCK1	5120	320	320	320	640	320	640	640	320	640
B/Madagascar/01481/2013	2	2013-04-22	MDCK2	2560	160	160	160	320	640	80	320	320	320
B/Madagascar/01534/2013	2	2013-04-23	MDCK2	2560	160	160	160	320	640	80	640	320	320
B/Norway/2277/2013		2013-04-28	MDCK1/MDCK1	5120	320	320	160	320	640	160	640	320	640
B/Norway/2281/2013	2	2013-04-29	MDCK1/MDCK1	1280	160	160	80	320	320	40	320	160	160
B/Madagascar/01609/2013	2	2013-04-29	MDCK2	2560	160	160	80	320	640	80	320	320	320
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-15. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-08-06 + 2013-08-14

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre: Post infection ferret sera: B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F2/13	B/Mass[2] 2/12 T/C F3/13
Genetic Group				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	1280	640	640	320	80	640	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	160	80	320	1280	80
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	5120	640	320	640	640	40	160	80	640	80
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	320	160	160	320	10	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	40	160	640	40	320	160	160
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	640	80	80	160	320	80	160	80	160	40
B/Hong Kong/3577/2012	2	2012-06-13	MDCK3	1280	80	80	80	160	640	40	320	160	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	320	640	160	40	320	1280	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	1280	320	320	80	320	320	40	320	640	160
TEST VIRUSES													
B/Poland/4/2013		2013-01-28	MDCKx/MDCK1	2560	320	80	80	320	640	40	640	320	160
B/Austria/717998/2013		2013-02-01	SIAT1/MDCK1	2560	160	80	80	320	320	80	320	160	160
B/Moscow/13/2013		2013-02-13	MDCK2/MDCK1	2560	80	80	80	320	640	80	640	160	160
B/Poland/2/2013		2013-02-20	MDCKx/MDCK1	2560	320	80	160	320	640	320	640	320	320
B/Tomsk/34/2013	2	2013-02-27	MDCK2/MDCK1	1280	40	80	80	160	320	80	320	160	160
B/Austria/723028/2013	2	2013-03-04	SIAT1/MDCK1	2560	80	80	80	160	320	40	320	160	160
B/Austria/723027/2013	2	2013-03-04	SIAT1/MDCK2	2560	160	80	80	320	320	40	320	160	160
B/Poland/1/2013	3	2013-03-04	MDCKx/MDCK1	2560	160	80	160	320	160	320	320	160	320
B/Poland/3/2013		2013-03-04	MDCKx/MDCK1	2560	160	80	80	320	640	80	640	160	160
B/Poland/5/2013	2	2013-03-04	MDCKx/MDCK1	1280	80	80	80	160	640	40	320	160	160
B/Austria/726751/2013	2	2013-03-21	SIAT1/MDCK1	1280	160	160	40	320	640	40	640	320	160
B/Moscow/64/2013	2	2013-03-22	MDCK2/MDCK1	2560	160	160	80	320	640	80	640	160	320
B/Tomsk/36/2013	2	2013-03-22	MDCK2/MDCK1	2560	160	160	80	160	320	40	320	160	160
B/Austria/727096/2013		2013-03-26	SIAT2/MDCK1	2560	160	160	40	160	320	160	320	160	320
B/Austria/727343/2013	2	2013-03-27	SIAT1/MDCK1	2560	320	320	160	320	320	40	320	320	320
B/Austria/735346/2013	2	2013-05-13	SIAT1/MDCK1	2560	160	160	80	320	320	80	320	160	320
B/South Africa/2800/2013	2	2013-05-20	MDCK1/MDCK1	1280	160	80	80	320	320	20	320	160	80
B/Norway/2377/2013	2	2013-05-21	MDCK2	1280	160	80	80	160	320	40	320	160	160
B/South Africa/3786/2013	2	2013-06-03	MDCK1/MDCK1	1280	80	80	80	320	640	40	320	160	160
B/South Africa/4437/2013	2	2013-06-14	MDCK1/MDCK1	1280	80	80	40	320	160	20	320	160	80
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-16. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-08-16 + 2013-08-20 + 2013-08-13

Viruses	Genetic Group	Collection date	Passage History	B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F34/11	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F2/13	B/Mass[2] 2/12 T/C F3/13
				Haemagglution Inhibition Titre									
				Post infection ferret sera									
Genetic Group				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	640	640	320	320	160	40	320	640	80
B/Brisbane/3/2007	2	2006-12-15	E2/E2	1280	640	640	160	320	160	40	320	640	80
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	640	320	160	160	320	<	40	40	320	20
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	160	160	80	320	<	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	40	160	320	40	160	80	80
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	640	80	80	80	320	40	80	80	160	40
B/Hong Kong/3577/2012	2	2012-06-13	MDCK3	640	80	80	40	160	640	40	320	160	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	160	320	160	40	320	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK2	1280	160	160	80	320	320	40	320	320	160
TEST VIRUSES													
B/Jordan/30045/2013		2013-01-13	SIAT1/MDCK1	1280	160	160	80	320	320	20	160	160	80
B/Poland/7/2013		2013-02-04	MDCKx/MDCK1	1280	160	160	80	320	640	40	320	160	160
B/Poland/8/2013	2	2013-02-11	MDCKx/MDCK1	1280	160	80	80	160	640	80	640	80	320
B/Poland/10/2013		2013-02-14	MDCKx/MDCK1	1280	160	80	40	160	320	80	320	160	320
B/Jordan/30158/2013		2013-02-20	SIATx/MDCK1	1280	80	80	40	160	320	20	320	160	160
B/Jordan/30154/2013		2013-02-23	SIATx/MDCK1	640	80	80	40	160	320	20	320	160	160
B/Veliky Novgorod/28/2013	2	2013-02-27	MDCK3/MDCK2	1280	80	80	40	160	640	40	320	160	160
B/Jordan/30169/2013	2	2013-02-28	SIAT1/MDCK1	1280	160	80	40	320	320	40	320	160	160
B/Poland/9/2013		2013-03-04	MDCKx/MDCK1	1280	80	80	40	160	640	40	320	160	160
B/Jordan/30186/2013	2	2013-03-11	SIAT1/MDCK1	1280	160	160	80	160	320	40	320	320	160
B/Poland/6/2013	2	2013-03-12	MDCKx/MDCK1	1280	160	80	80	320	80	160	160	160	80
B/Jordan/30213/2013		2013-03-20	SIATx/MDCK1	2560	160	160	160	320	640	40	640	320	160
B/Jordan/30216/2013		2013-03-20	SIAT1/MDCK1	2560	40	160	80	320	640	40	640	320	160
B/Poland/12/2013		2013-03-25	MDCKx/MDCK1	1280	160	80	80	160	640	160	320	160	640
B/Jordan/30232/2013		2013-03-26	SIAT1/MDCK1	2560	160	160	160	320	640	40	640	320	160
B/Jordan/30246/2013		2013-04-02	SIAT1/MDCK1	2560	40	160	80	320	640	40	640	320	160
B/Jordan/30257/2013	2	2013-04-03	SIAT1/MDCK1	1280	160	80	80	160	320	40	320	160	80
B/Jordan/30261/2013		2013-04-07	SIAT1/MDCK1	1280	160	160	80	160	320	40	320	160	80
B/Jordan/30263/2013		2013-04-07	SIAT1/MDCK1	1280	160	80	80	160	320	40	320	160	80
B/Jordan/30286/2013	2	2013-04-16	SIAT1/MDCK1	1280	80	160	40	160	320	40	320	160	160
B/Jordan/30279/2013		2013-04-17	SIAT1/MDCK1	1280	160	160	80	320	640	40	320	160	160
B/Jordan/30297/2013		2013-04-20	SIAT1/MDCK1	1280	80	160	40	160	320	40	320	160	160
B/Jordan/30303/2013		2013-04-23	SIAT1/MDCK1	1280	80	160	40	160	320	40	320	160	160
B/Jordan/30314/2013	2	2013-05-01	SIAT1/MDCK1	1280	160	160	80	320	320	80	320	160	160
B/Victoria/502/2013	2	2013-05-12	E3/E1	1280	40	320	40	160	80	20	80	320	40
B/Jordan/30322/2013	2	2013-05-12	SIAT1/MDCK1	1280	80	160	80	160	320	40	320	160	160
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 12-17. Antigenic analyses of influenza B viruses (Yamagata lineage) 2013-08-27 + 2013-08-30 + 2013-09-04 + 2013-09-06 + 2013-09-10

				Haemagglution Inhibition Titre									
				Post infection ferret sera									
Viruses	Genetic Group	Collection date	Passage History	B/Fl[3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F24/12	B/Stock[2] 12/11 F34/11	B/Estonia[2] 55669/11 F27/13	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 2/12 Egg F28/13	B/Mass[2] 2/12 T/C F3/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	320	320	160	40	320	640	80
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	320	320	160	40	320	640	80
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	640	320	160	320	320	10	40	40	320	20
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	160	80	80	320	10	40	40	320	20
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	160	80	80	160	640	40	320	160	160
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	160	160	160	320	80	160	160	160	80
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	160	160	160	320	640	80	320	160	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	320	160	320	160	40	160	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	2560	320	320	160	320	320	40	640	320	320
TEST VIRUSES													
B/Hawaii/01/2013	2	2013-01-05	C1/MDCK1	2560	160	160	80	160	320	80	640	160	320
B/Lithuania/3693/2013		2013-01-31	MDCK2/MDCK1	2560	160	160	80	320	640	160	640	160	320
B/Lithuania/3726/2013		2013-02-01	MDCK2/MDCK1	1280	160	80	80	320	640	80	640	160	160
B/Lithuania/6530/2013		2013-02-25	MDCK3	1280	80	40	40	320	20	40	40	<	20
B/Lithuania/6453/2013		2013-02-28	MDCK2	2560	320	160	80	320	640	80	1280	160	320
B/Jordan/20276/2013		2013-03-03	SIAT1/MDCK1	1280	80	80	40	160	320	20	320	80	80
B/Lithuania/6935/2013		2013-03-04	MDCK3	1280	160	160	80	320	640	40	640	40	160
B/Lithuania/6942/2013		2013-03-04	MDCK2	5120	320	320	320	1280	160	160	160	160	80
B/Dakar/07/2013	2	2013-03-04	C1/MDCK1	1280	160	160	80	320	640	320	640	160	640
B/Lithuania/6931/2013		2013-03-05	MDCK2	2560	320	160	160	320	320	80	640	160	160
B/Jordan/20312/2013		2013-03-16	SIAT1/MDCK1	2560	160	160	80	320	640	40	640	160	160
B/Jordan/20308/2013	2	2013-03-16	SIATx/MDCK1	1280	80	80	40	320	640	40	320	80	80
B/Lithuania/9556/2013	3	2013-03-28	MDCK2/MDCK1	2560	160	160	160	320	320	320	640	160	640
B/Lithuania/9521/2013	3	2013-04-02	MDCK2/MDCK1	1280	160	160	160	320	320	320	320	160	320
B/Dakar/13/2013	2	2013-04-03	C1/MDCK1	1280	160	160	80	320	640	80	640	160	160
B/Jordan/20355/2013		2013-04-07	SIAT1/MDCK1	1280	80	80	40	160	640	20	320	80	80
B/Jordan/20367/2013	2	2013-04-11	SIAT1/MDCK1	2560	160	160	80	320	640	40	640	160	160
B/Dakar/17/2013	2	2013-04-20	C1/MDCK1	1280	160	160	80	320	640	40	320	160	160
B/Fukui/24/2013	2	2013-05-07	MDCK1/MDCK1/MDCK1	1280	160	80	40	160	320	20	320	80	80
B/Sapporo/21/2013	2	2013-05-07	MDCK1/MDCK1/MDCK1	2560	160	160	160	160	320	40	640	160	160
B/Jordan/30324/2013		2013-05-22	SIAT1/MDCK1	1280	160	80	40	160	320	40	320	80	80
B/Jordan/30332/2013	2	2013-05-30	SIAT1/MDCK1	2560	160	160	80	320	640	40	640	160	160
							Vaccine 2012-13 2013					Vaccine 2013-14	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Figure 21. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

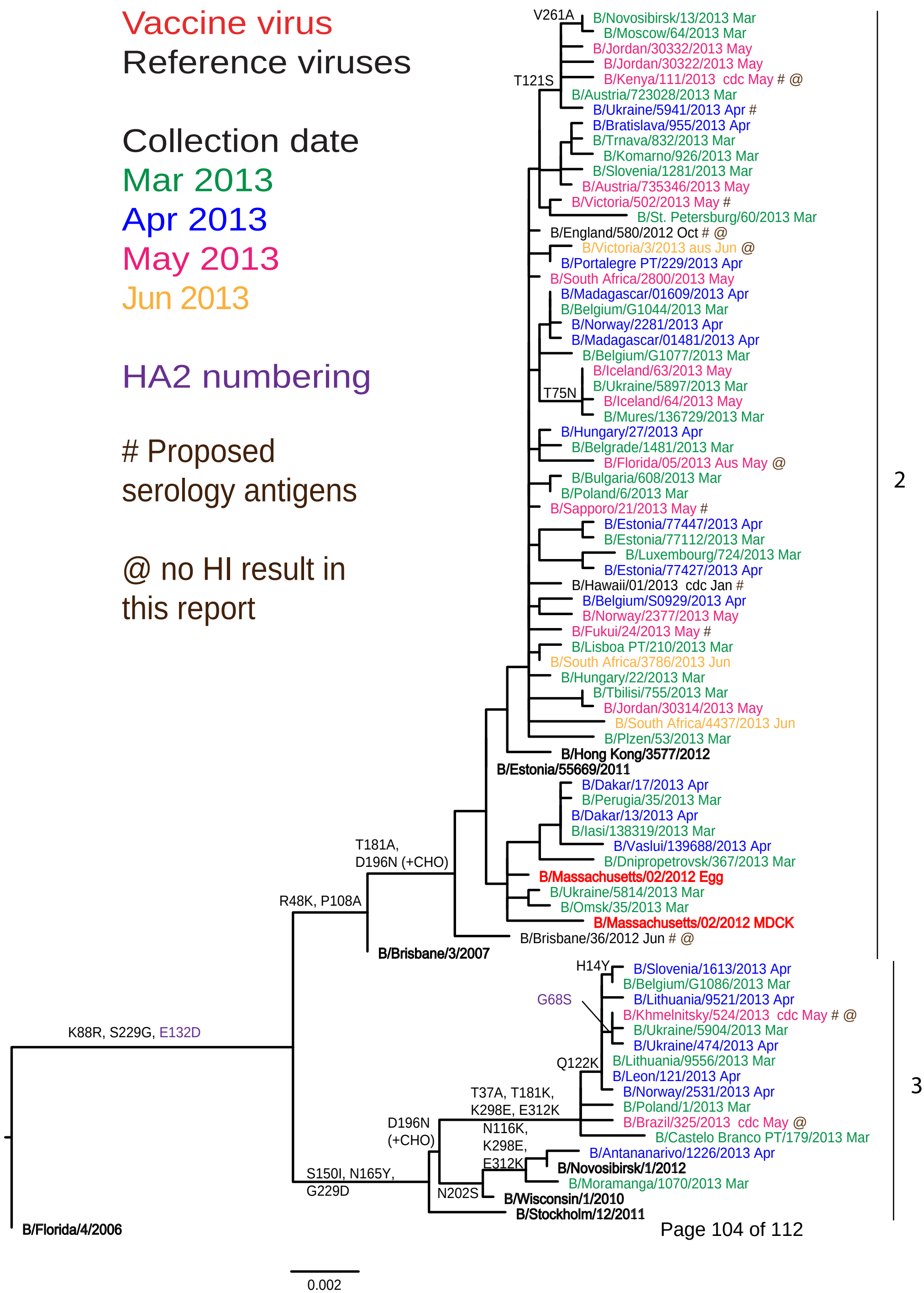

Figure 22. HA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                 10        20        30        40        50        60        70        80        90       100
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006         DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSYFANLKGTRTRGKLCPDCLNCTDLDVALGRPMCVGTTPSAKASILHEVKPVTSGCFPIMHD
B/Stockholm/12/2011      ............................A..........................................................R............ 3
B/Wisconsin/1/2010       .......................................................................................R............ 3
B/Moramanga/1070/2013    ......................................................................................AR............ 3
B/Novosibirsk/1/2012     .......................................................................................R............ 3
B/Antananarivo/1226/2013 .......................................................................................R............ 3
B/Norway/2531/2013       ....................................A..................................................R............ 3
B/Leon/121/2013          ....................................A..................................................R............ 3
B/Ukraine/474/2013       ....................................A..................................................R............ 3
B/Lithuania/9521/2013    ....................................A..................................................R............ 3
B/Slovenia/1613/2013     .............Y......................A..................................................R............ 3
B/Brisbane/3/2007        ...............................................K.......................................R............ 2
B/Massachusetts/02/2012  ...............................................K.......................................R............ 2
B/Vaslui/139688/2013     ...............................................K.......................................R............ 2
B/Dakar/13/2013          ...............................................K.......................................R............ 2
B/Estonia/55669/2011     ...............................................K.......................................R............ 2
B/Hong Kong/3577/2012    ...............................................K.......................................R............ 2
B/South Africa/4437/2013 ...............................................K.......................................R............ 2
B/Jordan/30314/2013      ...............................................K.......................................R............ 2
B/Fukui/24/2013          ...............................................K.......................................R............ 2
B/Norway/2377/2013       ...............................................K.......................................R............ 2
B/Sapporo/21/2013        ...............................................K.......................................R............ 2
B/Iceland/64/2013        ...............................................K..........................N............R............ 2
B/Madagascar/01481/2013  ...............................................K.......................................R............ 2
B/South Africa/2800/2013 ...............................................K.......................................R............ 2
B/Austria/735346/2013    ...............................................K.......................................R............ 2
B/Bratislava/955/2013    ...............................................K.......................................R............ 2
B/Ukraine/5941/2013      ...............................................K.......................................R............ 2
B/Jordan/30332/2013      ...............................................K.......................................R............ 2
B/Moscow/64/2013         ...............................................K.......................................R............ 2

                                110       120       130       140       150       160       170       180       190       200
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006         RTKIRQLPNLLRGYENIRLSTQNVIDAEKAPGGPYRLGTSGSCPNATSKSGFFATMAWAVPKDNNKNATNPLTVEVPYICTEGEDQITVWGFHSDDKTQM
B/Stockholm/12/2011      .................................................I..............Y......Q............................
B/Wisconsin/1/2010       .................................................I..............Y..............................N....
B/Moramanga/1070/2013    ...............K.................................I..............Y..............................N....
B/Novosibirsk/1/2012     ...............K.................................I..............Y..............................N....
B/Antananarivo/1226/2013 ...............K.................................I..............Y......................I.......N....
B/Norway/2531/2013       .....................K...........................I..............Y...............K..............N....
B/Leon/121/2013          .....................K...........................I..............Y...............K..............N....
B/Ukraine/474/2013       .....................K...........................I..............Y...............K..............N....
B/Lithuania/9521/2013    .....................K...........................I..............Y...............K..............N....
B/Slovenia/1613/2013     .....................K...........................I..............Y...............K..............N....
B/Brisbane/3/2007        .......A............................................................................................
B/Massachusetts/02/2012  .......A........................................................................A...................
B/Vaslui/139688/2013     .......A.........K..............................................................A..............N....
B/Dakar/13/2013          .......A........................................................................A..............N....
B/Estonia/55669/2011     .......A........................................................................A..............N....
B/Hong Kong/3577/2012    .......A........................................................................A..............N....
B/South Africa/4437/2013 .......A.................N......................................................A..............N....
B/Jordan/30314/2013      .......A.................N......................................................A..............N....
B/Fukui/24/2013          .......A........................................................................A..............N....
B/Norway/2377/2013       .......A........................................................................A..............N....
B/Sapporo/21/2013        .......A........................................................................A..............N....
B/Iceland/64/2013        .......A........................................................................A..............N....
B/Madagascar/01481/2013  .......A........................................................................A..............N....
B/South Africa/2800/2013 .......A........................................................................A..............N....
B/Austria/735346/2013    .......A........................................................................A..............N....
B/Bratislava/955/2013    .......A........................................................................A..............N....
B/Ukraine/5941/2013      .......A............S...........................................................A..............N....
B/Jordan/30332/2013      .......A............S...........................................................A..............N....
B/Moscow/64/2013         .......A............S...........................................................A..............N....

                                210       220       230       240       250       260       270       280       290       300
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006         KNLYGDSNPQKFTSSANGVTTHYVSQIGSFPDQTEDGGLPQSGRIVVDYMMQKPGKTGTIVYQRGVLLPQKVWCASGRSKVIKGSLPLIGEADCLHEKYG
B/Stockholm/12/2011      ............................D.....................V.................................................
B/Wisconsin/1/2010       .S..........................D.......................................................................
B/Moramanga/1070/2013    .S..........................D..........................R.........................................E..
B/Novosibirsk/1/2012     .S..........................D....................................................................E..
B/Antananarivo/1226/2013 .S..........................D....................................................................E..
B/Norway/2531/2013       ............................D....................................................................E..
B/Leon/121/2013          ............................D....................................................................E..
B/Ukraine/474/2013       ............................D....................................................................E..
B/Lithuania/9521/2013    ............................D....................................................................E..
B/Slovenia/1613/2013     ............................D....................................................................E..
B/Brisbane/3/2007        ............................G.......................................................................
B/Massachusetts/02/2012  ............................G.......................................................................
B/Vaslui/139688/2013     ............................G.......................................................................
B/Dakar/13/2013          ............................G.......................................................................
B/Estonia/55669/2011     ............................G.......................................................................
B/Hong Kong/3577/2012    ............................G.......................................................................
B/South Africa/4437/2013 ............................G.......................................................................
B/Jordan/30314/2013      ............................G.......................................................................
B/Fukui/24/2013          ............................G.......................................................................
B/Norway/2377/2013       ............................G.......................................................................
B/Sapporo/21/2013        ............................G.......................................................................
B/Iceland/64/2013        ............................G.......................................................................
B/Madagascar/01481/2013  ............................G.......................................................................
B/South Africa/2800/2013 ............................G.......................................................................
B/Austria/735346/2013    ............................G.......................................................................
B/Bratislava/955/2013    ............................G.......................................................................
B/Ukraine/5941/2013      ............................G.....................L.................................................
B/Jordan/30332/2013      ............................G.......................................................................
B/Moscow/64/2013         ............................G...............................A.......................................
```

```
                                     310       320       330       340       350       360       370       380       390       400
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006             GLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLNS
B/Stockholm/12/2011          ....................................................................................................
B/Wisconsin/1/2010           ....................................................................................................
B/Moramanga/1070/2013        ...........K........................................................................................
B/Novosibirsk/1/2012         ...........K........................................................................................
B/Antananarivo/1226/2013     ...........K........................................................................................
B/Norway/2531/2013           ...........K........................................................................................
B/Leon/121/2013              ...........K........................................................................................
B/Ukraine/474/2013           ...........K........................................................................................
B/Lithuania/9521/2013        ...........K........................................................................................
B/Slovenia/1613/2013         ...........K........................................................................................
B/Brisbane/3/2007            ....................................................................................................
B/Massachusetts/02/2012      ....................................................................................................
B/Vaslui/139688/2013         .....R..............................................................................................
B/Dakar/13/2013              ....................................................................................................
B/Estonia/55669/2011         ....................................................................................................
B/Hong Kong/3577/2012        ....................................................................................................
B/South Africa/4437/2013     ....................................................................................................
B/Jordan/30314/2013          ....................................................................................................
B/Fukui/24/2013              ....................................................................................................
B/Norway/2377/2013           ....................................................................................................
B/Sapporo/21/2013            ....................................................................................................
B/Iceland/64/2013            ....................................................................................................
B/Madagascar/01481/2013      ....................................................................................................
B/South Africa/2800/2013     ....................................................................................................
B/Austria/735346/2013        ....................................................................................................
B/Bratislava/955/2013        ....................................................................................................
B/Ukraine/5941/2013          ....................................................................................................
B/Jordan/30332/2013          ...............T....................................................................................
B/Moscow/64/2013             ....................................................................................................

                                     410       420       430       440       450       460       470       480       490       500
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006             LSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIAA
B/Stockholm/12/2011          .............................................................................D......................
B/Wisconsin/1/2010           .............................................................................D......................
B/Moramanga/1070/2013        .............................................................................D......................
B/Novosibirsk/1/2012         .............................................................................D......................
B/Antananarivo/1226/2013     .............................................................................D......................
B/Norway/2531/2013           .............................................................................D......................
B/Leon/121/2013              .............................................................................D......................
B/Ukraine/474/2013           .............S...............................................................D......................
B/Lithuania/9521/2013        .............................................................................D......................
B/Slovenia/1613/2013         .............................................................................D......................
B/Brisbane/3/2007            .............................................................................D......................
B/Massachusetts/02/2012      .............................................................................D......................
B/Vaslui/139688/2013         .............................................................................D......................
B/Dakar/13/2013              .............................................................................D......................
B/Estonia/55669/2011         .............................................................................D......................
B/Hong Kong/3577/2012        .............................................................................D......................
B/South Africa/4437/2013     .............................................................................D......................
B/Jordan/30314/2013          .............................................................................D......................
B/Fukui/24/2013              .............................................................................D......................
B/Norway/2377/2013           .............................................................................D......................
B/Sapporo/21/2013            .............................................................................D......................
B/Iceland/64/2013            .............................................................................D......................
B/Madagascar/01481/2013      .............................................................................D......................
B/South Africa/2800/2013     .............................................................................D......................
B/Austria/735346/2013        ..........K..................................................................D......................
B/Bratislava/955/2013        .............................................................................D......................
B/Ukraine/5941/2013          .............................................................................D......................
B/Jordan/30332/2013          .............................................................................D......................
B/Moscow/64/2013             .............................................................................D......................

                                     510       520       530       540       550       560
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....
B/Florida/4/2006             GTFNAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMLAIFIVYMVSRDNVSCSICL
B/Stockholm/12/2011          .....................................................................
B/Wisconsin/1/2010           .....................................................................
B/Moramanga/1070/2013        .....................................................................
B/Novosibirsk/1/2012         .....................................................................
B/Antananarivo/1226/2013     .....................................................................
B/Norway/2531/2013           .....................................................................
B/Leon/121/2013              .....................................................................
B/Ukraine/474/2013           .....................................................................
B/Lithuania/9521/2013        .....................................................................
B/Slovenia/1613/2013         .....................................................................
B/Brisbane/3/2007            .....................................................................
B/Massachusetts/02/2012      .....................................................................
B/Vaslui/139688/2013         .....................................................................
B/Dakar/13/2013              .....................................................................
B/Estonia/55669/2011         .....................................................................
B/Hong Kong/3577/2012        .....................................................................
B/South Africa/4437/2013     .....................................................................
B/Jordan/30314/2013          .....................................................................
B/Fukui/24/2013              .....................................................................
B/Norway/2377/2013           .....................................................................
B/Sapporo/21/2013            .....................................................................
B/Iceland/64/2013            .....................................................................
B/Madagascar/01481/2013      .....................................................................
B/South Africa/2800/2013     .....................................................................
B/Austria/735346/2013        .....................................................................
B/Bratislava/955/2013        .....................................................................
B/Ukraine/5941/2013          .....................................................................
B/Jordan/30332/2013          .....................................................................
B/Moscow/64/2013             .....................................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 23. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

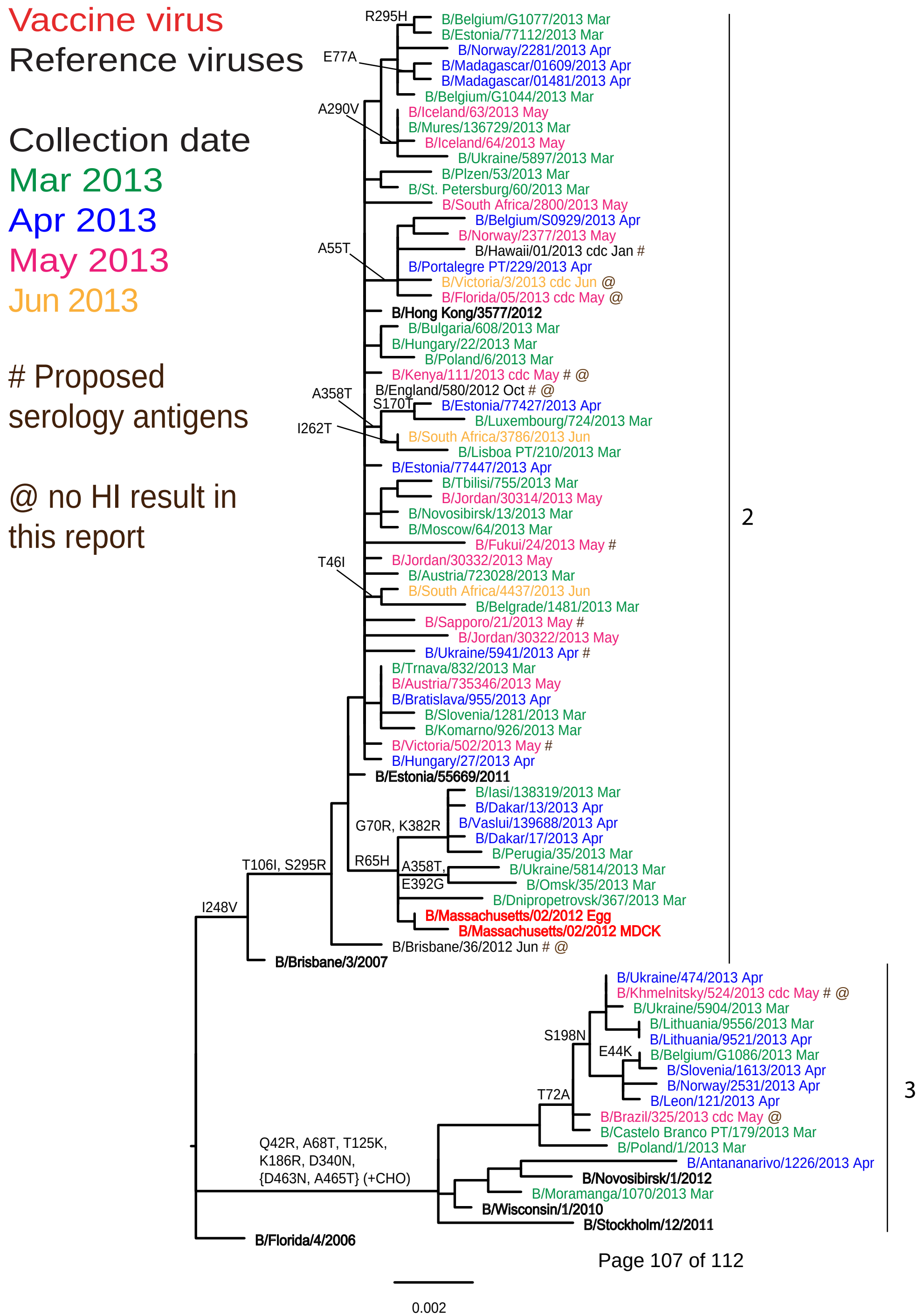

Figure 24. NA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                  10        20        30        40        50        60        70        80        90       100
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006          MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFSQTEITAPIMPLDCANASNVQAVNRSAAKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Stockholm/12/2011       .........................................R.........................T....P........................... 3
B/Wisconsin/1/2010        .........................................R.........................T................................ 3
B/Moramanga/1070/2013     .........................................R.........................T................................ 3
B/Novosibirsk/1/2012      .........................................R.........................T................................ 3
B/Antananarivo/1226/2013  .........................................R.........................T.RG............................. 3
B/Norway/2531/2013        .........................................R.........................T...A............................ 3
B/Leon/121/2013           .........................................R.........................T...A............................ 3
B/Ukraine/474/2013        .........................................R.........................T...A............................ 3
B/Lithuania/9521/2013     .........................................R.........................T...A............................ 3
B/Slovenia/1613/2013      .........................................R.K.......................T...A............................ 3
B/Brisbane/3/2007         .......K............................................................................................ 2
B/Massachusetts/02/2012   ................................................................H................................... 2
B/Vaslui/139688/2013      ................................................................H....R.............................. 2
B/Dakar/13/2013           .....................................................Y..........H....R.............................. 2
B/Estonia/55669/2011      .................................................................................................... 2
B/Hong Kong/3577/2012     .................................................................................................... 2
B/South Africa/4437/2013  .............................................I...................................................... 2
B/Jordan/30314/2013       .................................................................................................... 2
B/Fukui/24/2013           ...........................................................F........................................ 2
B/Norway/2377/2013        ......................................................T............................................. 2
B/Sapporo/21/2013         ................................................T................................................... 2
B/Iceland/64/2013         .................................................................................................... 2
B/Madagascar/01481/2013   ............................................................................A....................... 2
B/South Africa/2800/2013  .................................................I.................................................. 2
B/Austria/735346/2013     .................................................................................................... 2
B/Bratislava/955/2013     .................................................................................................... 2
B/Ukraine/5941/2013       .................................................................................................... 2
B/Jordan/30332/2013       .................................................................................................... 2
B/Moscow/64/2013          .................................................................................................... 2

                                 110       120       130       140       150       160       170       180       190       200
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006          HRFGETKGNSAPLIIREPFIACGPTECKHFALTHYAAQPGGYYNGTREDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDSNA
B/Stockholm/12/2011       ........................K............................................................R..............
B/Wisconsin/1/2010        ........................K............................................................R..............
B/Moramanga/1070/2013     ........................K............................................................R..............
B/Novosibirsk/1/2012      ........................K............................................................R..............
B/Antananarivo/1226/2013  ........................R..................K..............................T..........R..............
B/Norway/2531/2013        ........................K............................................................R...........N..
B/Leon/121/2013           ........................K............................................................R...........N..
B/Ukraine/474/2013        ........................K............................................................R...........N..
B/Lithuania/9521/2013     ........................K............................................................R...........N..
B/Slovenia/1613/2013      ........................K............................................................R...........N..
B/Brisbane/3/2007         ....................................................................................................
B/Massachusetts/02/2012   .....I..............................................................................................
B/Vaslui/139688/2013      .....I..............................................................................................
B/Dakar/13/2013           .....I..............................................................................................
B/Estonia/55669/2011      .....I..............................................................................................
B/Hong Kong/3577/2012     .....I..............................................................................................
B/South Africa/4437/2013  .....I..............................................................................................
B/Jordan/30314/2013       .....I..............................................................................................
B/Fukui/24/2013           .....I..............................................................................................
B/Norway/2377/2013        .....I..............................................................................................
B/Sapporo/21/2013         .....I..............................................................................................
B/Iceland/64/2013         .....I............................................................................................D.
B/Madagascar/01481/2013   .....I..............................................................................................
B/South Africa/2800/2013  .....I..............................................................................................
B/Austria/735346/2013     .....I..............................................................................................
B/Bratislava/955/2013     .....I..............................................................................................
B/Ukraine/5941/2013       .....I..............................................................................................
B/Jordan/30332/2013       .....I..............................................................................................
B/Moscow/64/2013          .....I..............................................................................................

                                 210       220       230       240       250       260       270       280       290       300
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006          LLKIKYGEAYTDTYHSYAKNILRTQESACNCIGGDCYLMITDGPASGISECRFLKIREGRIIKEIFPTGRVKHTEECTCGFASNKTIECACRDNSYTAKR
B/Stockholm/12/2011       ....................................................................................................
B/Wisconsin/1/2010        ....................................................................................................
B/Moramanga/1070/2013     ....................................................................................................
B/Novosibirsk/1/2012      ....................................................................................................
B/Antananarivo/1226/2013  ....................................................................................................
B/Norway/2531/2013        ....................................................................................................
B/Leon/121/2013           ....................................................................................................
B/Ukraine/474/2013        ....................................................................................................
B/Lithuania/9521/2013     ....................................................................................................
B/Slovenia/1613/2013      ....................................................................................................
B/Brisbane/3/2007         ...............................................V....................................................
B/Massachusetts/02/2012   ...............................................V..............................................R.....
B/Vaslui/139688/2013      ...............................................V..............................................R.....
B/Dakar/13/2013           ...............................................V..............................................R.....
B/Estonia/55669/2011      ...............................................V..............................................R.....
B/Hong Kong/3577/2012     ...............................................V..............................................R.....
B/South Africa/4437/2013  ...............................................V..............................................R.....
B/Jordan/30314/2013       ...............................................V..............................................R.....
B/Fukui/24/2013           ...............................................V..............................................R.....
B/Norway/2377/2013        ...............................................V..............................................R.....
B/Sapporo/21/2013         ...............................................V..............................................R.....
B/Iceland/64/2013         ...............................................V.........................................V....R.....
B/Madagascar/01481/2013   ...............................................V..............................................R.....
B/South Africa/2800/2013  ...............................................V..............................................R.....
B/Austria/735346/2013     ...............................................V..............................................R.....
B/Bratislava/955/2013     ...............................................V..............................................R.....
B/Ukraine/5941/2013       ...............................................V..............................................R.....
B/Jordan/30332/2013       ...............................................V..............................................R.....
B/Moscow/64/2013          ...............................................V..............................................R.....
```

```
                                   310       320       330       340       350       360       370       380       390       400
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006            PFVKLNVETDTAEIRLMCTETYLDTPRPNDGSITGPCESDGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWTDSEALALSGVM
B/Stockholm/12/2011         .......................................N..E.........................................................
B/Wisconsin/1/2010          .......................................N............................................................
B/Moramanga/1070/2013       .......................................N............................................................
B/Novosibirsk/1/2012        .......................................N............................................................
B/Antananarivo/1226/2013    .......................................N............................................................
B/Norway/2531/2013          .......................................N............................................................
B/Leon/121/2013             .......................................N............................................................
B/Ukraine/474/2013          .......................................N............................................................
B/Lithuania/9521/2013       .......................................N............................................................
B/Slovenia/1613/2013        .......................................N............................................................
B/Brisbane/3/2007           ....................................................................................................
B/Massachusetts/02/2012     ....................................................................................................
B/Vaslui/139688/2013        .................................................................................R..................
B/Dakar/13/2013             .................................................................................R..................
B/Estonia/55669/2011        ....................................................................................................
B/Hong Kong/3577/2012       ....................................................................................................
B/South Africa/4437/2013    ....................................................................................................
B/Jordan/30314/2013         ....................................................................................................
B/Fukui/24/2013             ....................................................................................................
B/Norway/2377/2013          ....................................................................................................
B/Sapporo/21/2013           ...................K................................................................................
B/Iceland/64/2013           ....................................................................................................
B/Madagascar/01481/2013     ....................................................................................................
B/South Africa/2800/2013    ....................................................................................................
B/Austria/735346/2013       ....................................................................................................
B/Bratislava/955/2013       ....................................................................................................
B/Ukraine/5941/2013         ........X...........................................................................................
B/Jordan/30332/2013         ....................................................................................................
B/Moscow/64/2013            ....................................................................................................

                                   410       420       430       440       450       460
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Florida/4/2006            VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKTTWHSAATAIYCLMGSGQLLWDTVTGVDMAL
B/Stockholm/12/2011         ..............................................................N.T.
B/Wisconsin/1/2010          ..............................................................N.T.
B/Moramanga/1070/2013       ..............................................................N.T.
B/Novosibirsk/1/2012        ..............................................................N.T.
B/Antananarivo/1226/2013    ..............................................................N.T.
B/Norway/2531/2013          ..............................................................N.T.
B/Leon/121/2013             ..............................................................N.T.
B/Ukraine/474/2013          ..............................................................N.T.
B/Lithuania/9521/2013       ..............................................................N.T.
B/Slovenia/1613/2013        ..............................................................N.T.
B/Brisbane/3/2007           ..................................................................
B/Massachusetts/02/2012     ..................................................................
B/Vaslui/139688/2013        ..................................................................
B/Dakar/13/2013             ..................................................................
B/Estonia/55669/2011        ..................................................................
B/Hong Kong/3577/2012       ..................................................................
B/South Africa/4437/2013    ..................................................................
B/Jordan/30314/2013         ..................................................................
B/Fukui/24/2013             ..................................................................
B/Norway/2377/2013          ..................................................................
B/Sapporo/21/2013           ..................................................................
B/Iceland/64/2013           ..................................................................
B/Madagascar/01481/2013     ..................................................................
B/South Africa/2800/2013    ..................................................................
B/Austria/735346/2013       ..................................................................
B/Bratislava/955/2013       ..................................................................
B/Ukraine/5941/2013         ..................................................................
B/Jordan/30332/2013         ..................................................................
B/Moscow/64/2013            ................................................................V.
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 13. Antiviral (NI) testing of isolates received since 2013-01-31

Month of Collection	A(H1N1)pdm09		A(H3N2)		Flu B		Total
	Total	RI/HRI[1]	Total	RI/HRI[2]	Total	RI/HRI	
2012							
February	1		1		1		2
March			4	1			5
April			15	1	1		17
May			3				3
June			9		2		11
July	1		2		13		15
August	2		5		6		11
September			23		7		30
October	6		69		15		84
November	23		37		71		108
December	82		69		83		152
2013							
January	167	1	93		90		183
Total	282	1	330	2	289	0	904
February	103		49		116		165
March	83		52		119		171
April	14		17		54		71
May	5				6		6
June			10		2		12
July			2		1		3
Total	205	0	130	0	298	0	633
Grand Total	**487**	**1**	**460**	**2**	**587**	**0**	**1537**

All viruses were subjected to phenotypic testing against oseltamivir and zanamivir

[1] A(H1N1)pdm09 viruses carried H275Y substitutions

[2] A(H3N2) viruses carried E119V substitutions

Figure 25. Phylogenetic comparison of influenza A(H1N1)pdm09 M genes

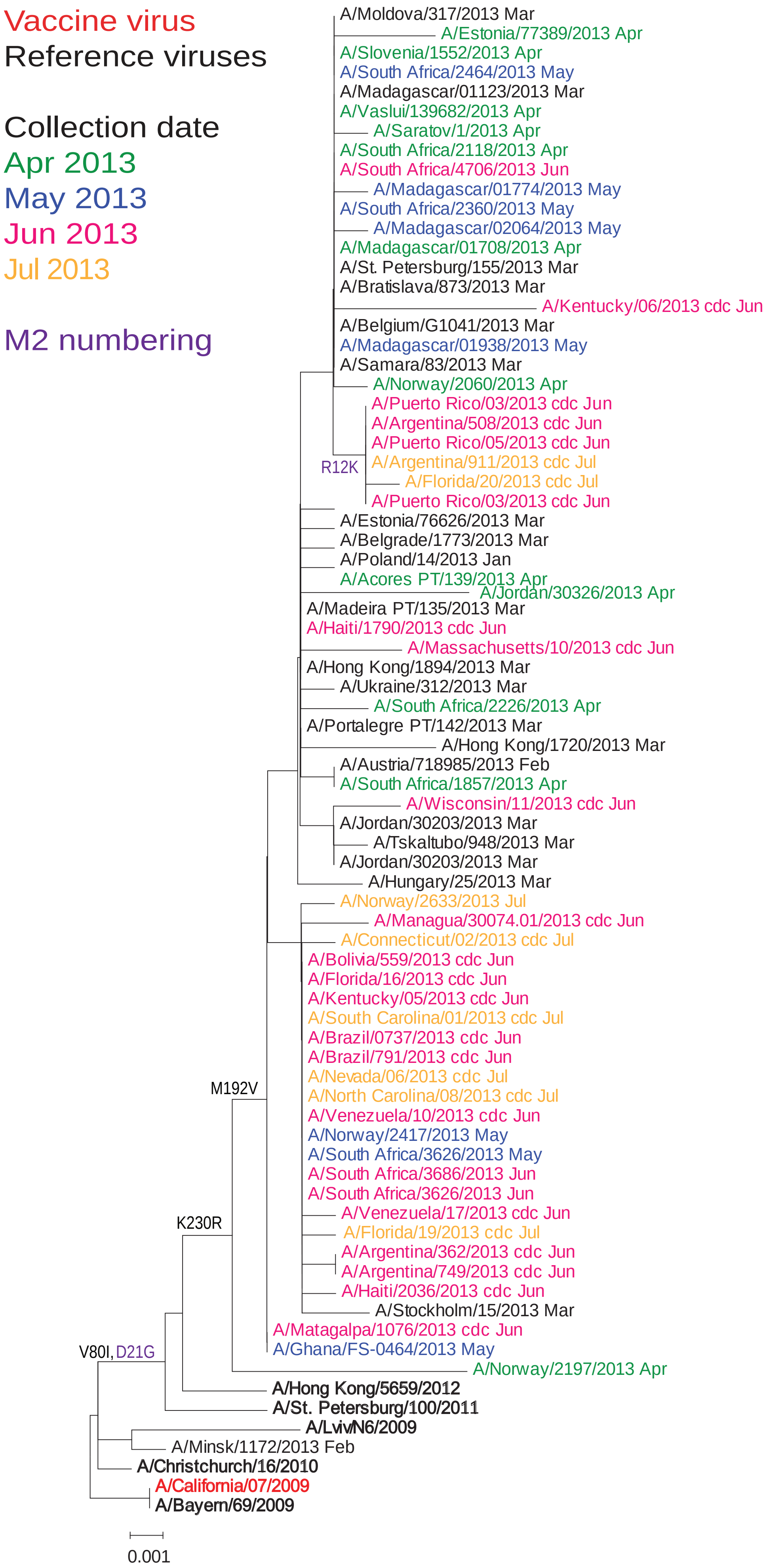

Figure 26. Phylogenetic comparison of influenza A(H3N2) M genes